AEROSPACE OEM ENGINEERING COMPENDIUM — VOLUME 4

Covers 25 Certified Machine Models (Models #76 to #100)
350 Full-Page Fault SOPs | 125 Pages PLC Registers | 75 Pages BOM & PM Matrices

INCLUDED MACHINE MODELS IN THIS VOLUME:

• Model #76: Grob-Aerospace-G400 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #77: Grob-Aerospace-G450 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #78: Grob-Aerospace-G500 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #79: Grob-Aerospace-G550 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #80: Grob-Aerospace-G600 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #81: Hermle-Performance-C150 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #82: Hermle-Performance-C200 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #83: Hermle-Performance-C250 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #84: Hermle-Performance-C300 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #85: Hermle-Performance-C350 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #86: Hermle-Performance-C400 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #87: Hermle-Performance-C450 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #88: Hermle-Performance-C500 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #89: Hermle-Performance-C550 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #90: Hermle-Performance-C600 (5-Axis Impeller & Blisk Machining Center) — Controller: Heidenhain TNC7 — Prefix: BLADE-MILL-XXXX
• Model #91: Zeiss-Prismo-Z150 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #92: Zeiss-Prismo-Z200 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #93: Zeiss-Prismo-Z250 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #94: Zeiss-Prismo-Z300 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #95: Zeiss-Prismo-Z350 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #96: Zeiss-Prismo-Z400 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #97: Zeiss-Prismo-Z450 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #98: Zeiss-Prismo-Z500 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #99: Zeiss-Prismo-Z550 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller — Prefix: CMM-INSP-XXXX
• Model #100: Zeiss-Prismo-Z600 (Sub-Micron Coordinate Metrology Gantry) — Controller: Zeiss C99m CAA Controller —

Prefix: CMM-INSP-XXXX

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 1/9)

MACHINE MODEL: Grob-Aerospace-G400 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G400"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

MACHINE MODEL: Grob-Aerospace-G450 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G450"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

MACHINE MODEL: Grob-Aerospace-G500 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G500"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 2/9)

MACHINE MODEL: Grob-Aerospace-G550 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G550"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

MACHINE MODEL: Grob-Aerospace-G600 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G600"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Hermle-Performance-C150 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C150"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 3/9)

MACHINE MODEL: Hermle-Performance-C200 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C200"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

MACHINE MODEL: Hermle-Performance-C250 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C250"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

MACHINE MODEL: Hermle-Performance-C300 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C300"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 4/9)

MACHINE MODEL: Hermle-Performance-C350 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C350"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

MACHINE MODEL: Hermle-Performance-C400 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C400"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

MACHINE MODEL: Hermle-Performance-C450 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C450"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 5/9)

MACHINE MODEL: Hermle-Performance-C500 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C500"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

MACHINE MODEL: Hermle-Performance-C550 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C550"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

MACHINE MODEL: Hermle-Performance-C600 (Asset Prefix: BLADE-MILL)

• RAG Metadata Filter Value: "Hermle-Performance-C600"
• Equipment Classification: 5-Axis Impeller & Blisk Machining Center | Manufacturer: Hermle-Performance
• CNC / PLC Control System: Heidenhain TNC7
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 6/9)

MACHINE MODEL: Zeiss-Prismo-Z150 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z150"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Zeiss-Prismo-Z200 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z200"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

MACHINE MODEL: Zeiss-Prismo-Z250 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z250"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 7/9)

MACHINE MODEL: Zeiss-Prismo-Z300 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z300"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

MACHINE MODEL: Zeiss-Prismo-Z350 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z350"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

MACHINE MODEL: Zeiss-Prismo-Z400 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z400"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 8/9)

MACHINE MODEL: Zeiss-Prismo-Z450 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z450"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

MACHINE MODEL: Zeiss-Prismo-Z500 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z500"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

MACHINE MODEL: Zeiss-Prismo-Z550 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z550"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 9/9)

MACHINE MODEL: Zeiss-Prismo-Z600 (Asset Prefix: CMM-INSP)

• RAG Metadata Filter Value: "Zeiss-Prismo-Z600"
• Equipment Classification: Sub-Micron Coordinate Metrology Gantry | Manufacturer: Zeiss-Prismo
• CNC / PLC Control System: Zeiss C99m CAA Controller
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-SPN-1

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO400-1
- PLC Hex Register Address: DB20.DBW_0x10D88
- Nominal Operating Range: 15 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-BRG-CER-7601
- Overhaul Seal & Consumable Kit: KIT-GRO400-SPN-1
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-SPN-1 at PLC register 0x10D88.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-BRG-CER-7601) with seal kit (KIT-GRO400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-SPN-1 on Grob-Aerospace-G400 by installing PN-GRO400-BRG-CER-7601 & kit KIT-GRO400-SPN-1. Fasteners torqued to 29 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-ATC-2

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO400-2
- PLC Hex Register Address: DB20.DBW_0x10D90
- Nominal Operating Range: 18 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-CAM-GRIP-7602
- Overhaul Seal & Consumable Kit: KIT-GRO400-ATC-2
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-ATC-2 at PLC register 0x10D90.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-CAM-GRIP-7602) with seal kit (KIT-GRO400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-ATC-2 on Grob-Aerospace-G400 by installing PN-GRO400-CAM-GRIP-7602 & kit KIT-GRO400-ATC-2. Fasteners torqued to 38 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-HYD-3

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO400-3
- PLC Hex Register Address: DB20.DBW_0x10D98
- Nominal Operating Range: 21 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-VLV-PROP-7603
- Overhaul Seal & Consumable Kit: KIT-GRO400-HYD-3
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-HYD-3 at PLC register 0x10D98.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-VLV-PROP-7603) with seal kit (KIT-GRO400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-HYD-3 on Grob-Aerospace-G400 by installing PN-GRO400-VLV-PROP-7603 & kit KIT-GRO400-HYD-3. Fasteners torqued to 47 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-CLN-4

Target Machine Model: Grob-Aerospace-G400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO400-4
- PLC Hex Register Address: DB20.DBW_0x10DA0
- Nominal Operating Range: 24 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-PMP-PLG-7604
- Overhaul Seal & Consumable Kit: KIT-GRO400-CLN-4
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-CLN-4 at PLC register 0x10DA0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-PMP-PLG-7604) with seal kit (KIT-GRO400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-CLN-4 on Grob-Aerospace-G400 by installing PN-GRO400-PMP-PLG-7604 & kit KIT-GRO400-CLN-4. Fasteners torqued to 56 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-XAX-5

Target Machine Model: Grob-Aerospace-G400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO400-5
- PLC Hex Register Address: DB20.DBW_0x10DA8
- Nominal Operating Range: 27 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-LIN-COIL-7605
- Overhaul Seal & Consumable Kit: KIT-GRO400-XAX-5
- Critical Fastener Torque Value: 65 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-XAX-5 at PLC register 0x10DA8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-LIN-COIL-7605) with seal kit (KIT-GRO400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 65 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-XAX-5 on Grob-Aerospace-G400 by installing PN-GRO400-LIN-COIL-7605 & kit KIT-GRO400-XAX-5. Fasteners torqued to 65 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-YAX-6

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO400-6
- PLC Hex Register Address: DB20.DBW_0x10DB0
- Nominal Operating Range: 30 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-BSW-NUT-7606
- Overhaul Seal & Consumable Kit: KIT-GRO400-YAX-6
- Critical Fastener Torque Value: 74 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-YAX-6 at PLC register 0x10DB0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-BSW-NUT-7606) with seal kit (KIT-GRO400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 74 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-YAX-6 on Grob-Aerospace-G400 by installing PN-GRO400-BSW-NUT-7606 & kit KIT-GRO400-YAX-6. Fasteners torqued to 74 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-ZAX-7

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO400-7
- PLC Hex Register Address: DB20.DBW_0x10DB8
- Nominal Operating Range: 33 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-CYL-SEAL-7607
- Overhaul Seal & Consumable Kit: KIT-GRO400-ZAX-7
- Critical Fastener Torque Value: 83 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-ZAX-7 at PLC register 0x10DB8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-CYL-SEAL-7607) with seal kit (KIT-GRO400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 83 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-ZAX-7 on Grob-Aerospace-G400 by installing PN-GRO400-CYL-SEAL-7607 & kit KIT-GRO400-ZAX-7. Fasteners torqued to 83 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-BAX-8

Target Machine Model: Grob-Aerospace-G400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO400-8
- PLC Hex Register Address: DB20.DBW_0x10DC0
- Nominal Operating Range: 36 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-TRQ-STAT-7608
- Overhaul Seal & Consumable Kit: KIT-GRO400-BAX-8
- Critical Fastener Torque Value: 92 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-BAX-8 at PLC register 0x10DC0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-TRQ-STAT-7608) with seal kit (KIT-GRO400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 92 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-BAX-8 on Grob-Aerospace-G400 by installing PN-GRO400-TRQ-STAT-7608 & kit KIT-GRO400-BAX-8. Fasteners torqued to 92 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-CAX-9

Target Machine Model: Grob-Aerospace-G400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO400-9
- PLC Hex Register Address: DB20.DBW_0x10DC8
- Nominal Operating Range: 39 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-BRK-DISC-7609
- Overhaul Seal & Consumable Kit: KIT-GRO400-CAX-9
- Critical Fastener Torque Value: 101 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-CAX-9 at PLC register 0x10DC8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-BRK-DISC-7609) with seal kit (KIT-GRO400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 101 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-CAX-9 on Grob-Aerospace-G400 by installing PN-GRO400-BRK-DISC-7609 & kit KIT-GRO400-CAX-9. Fasteners torqued to 101 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-OPT-10

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO400-10
- PLC Hex Register Address: DB20.DBW_0x10DD0
- Nominal Operating Range: 42 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-GLASS-RD-7610
- Overhaul Seal & Consumable Kit: KIT-GRO400-OPT-10
- Critical Fastener Torque Value: 110 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-OPT-10 at PLC register 0x10DD0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-GLASS-RD-7610) with seal kit (KIT-GRO400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 110 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-OPT-10 on Grob-Aerospace-G400 by installing PN-GRO400-GLASS-RD-7610 & kit KIT-GRO400-OPT-10. Fasteners torqued to 110 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-PLC-11

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO400-11
- PLC Hex Register Address: DB20.DBW_0x10DD8
- Nominal Operating Range: 45 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-IGBT-MOD-7611
- Overhaul Seal & Consumable Kit: KIT-GRO400-PLC-11
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-PLC-11 at PLC register 0x10DD8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-IGBT-MOD-7611) with seal kit (KIT-GRO400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-PLC-11 on Grob-Aerospace-G400 by installing PN-GRO400-IGBT-MOD-7611 & kit KIT-GRO400-PLC-11. Fasteners torqued to 119 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-LUB-12

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO400-12
- PLC Hex Register Address: DB20.DBW_0x10DE0
- Nominal Operating Range: 48 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-INJ-MET-7612
- Overhaul Seal & Consumable Kit: KIT-GRO400-LUB-12
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-LUB-12 at PLC register 0x10DE0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-INJ-MET-7612) with seal kit (KIT-GRO400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-LUB-12 on Grob-Aerospace-G400 by installing PN-GRO400-INJ-MET-7612 & kit KIT-GRO400-LUB-12. Fasteners torqued to 128 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-VAC-13

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO400-13
- PLC Hex Register Address: DB20.DBW_0x10DE8
- Nominal Operating Range: 51 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-GASK-VIT-7613
- Overhaul Seal & Consumable Kit: KIT-GRO400-VAC-13
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-VAC-13 at PLC register 0x10DE8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-GASK-VIT-7613) with seal kit (KIT-GRO400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-VAC-13 on Grob-Aerospace-G400 by installing PN-GRO400-GASK-VIT-7613 & kit KIT-GRO400-VAC-13. Fasteners torqued to 137 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO400-THM-14

Target Machine Model: Grob-Aerospace-G400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G400 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO400-14
- PLC Hex Register Address: DB20.DBW_0x10DF0
- Nominal Operating Range: 54 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO400-HEX-CORE-7614
- Overhaul Seal & Consumable Kit: KIT-GRO400-THM-14
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G400 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO400-THM-14 at PLC register 0x10DF0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO400-HEX-CORE-7614) with seal kit (KIT-GRO400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G400, technician must log: "Resolved ERR-GRO400-THM-14 on Grob-Aerospace-G400 by installing PN-GRO400-HEX-CORE-7614 & kit KIT-GRO400-THM-14. Fasteners torqued to 146 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-SPN-1

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO450-1
- PLC Hex Register Address: DB21.DBW_0x110A8
- Nominal Operating Range: 15 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-BRG-CER-7701
- Overhaul Seal & Consumable Kit: KIT-GRO450-SPN-1
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-SPN-1 at PLC register 0x110A8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-BRG-CER-7701) with seal kit (KIT-GRO450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-SPN-1 on Grob-Aerospace-G450 by installing PN-GRO450-BRG-CER-7701 & kit KIT-GRO450-SPN-1. Fasteners torqued to 169 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-ATC-2

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO450-2
- PLC Hex Register Address: DB21.DBW_0x110B0
- Nominal Operating Range: 18 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-CAM-GRIP-7702
- Overhaul Seal & Consumable Kit: KIT-GRO450-ATC-2
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-ATC-2 at PLC register 0x110B0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-CAM-GRIP-7702) with seal kit (KIT-GRO450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-ATC-2 on Grob-Aerospace-G450 by installing PN-GRO450-CAM-GRIP-7702 & kit KIT-GRO450-ATC-2. Fasteners torqued to 178 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-HYD-3

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO450-3
- PLC Hex Register Address: DB21.DBW_0x110B8
- Nominal Operating Range: 21 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-VLV-PROP-7703
- Overhaul Seal & Consumable Kit: KIT-GRO450-HYD-3
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-HYD-3 at PLC register 0x110B8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-VLV-PROP-7703) with seal kit (KIT-GRO450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-HYD-3 on Grob-Aerospace-G450 by installing PN-GRO450-VLV-PROP-7703 & kit KIT-GRO450-HYD-3. Fasteners torqued to 187 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-CLN-4

Target Machine Model: Grob-Aerospace-G450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO450-4
- PLC Hex Register Address: DB21.DBW_0x110C0
- Nominal Operating Range: 24 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-PMP-PLG-7704
- Overhaul Seal & Consumable Kit: KIT-GRO450-CLN-4
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-CLN-4 at PLC register 0x110C0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-PMP-PLG-7704) with seal kit (KIT-GRO450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-CLN-4 on Grob-Aerospace-G450 by installing PN-GRO450-PMP-PLG-7704 & kit KIT-GRO450-CLN-4. Fasteners torqued to 196 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-XAX-5

Target Machine Model: Grob-Aerospace-G450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO450-5
- PLC Hex Register Address: DB21.DBW_0x110C8
- Nominal Operating Range: 27 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-LIN-COIL-7705
- Overhaul Seal & Consumable Kit: KIT-GRO450-XAX-5
- Critical Fastener Torque Value: 205 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-XAX-5 at PLC register 0x110C8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-LIN-COIL-7705) with seal kit (KIT-GRO450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 205 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-XAX-5 on Grob-Aerospace-G450 by installing PN-GRO450-LIN-COIL-7705 & kit KIT-GRO450-XAX-5. Fasteners torqued to 205 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-YAX-6

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO450-6
- PLC Hex Register Address: DB21.DBW_0x110D0
- Nominal Operating Range: 30 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-BSW-NUT-7706
- Overhaul Seal & Consumable Kit: KIT-GRO450-YAX-6
- Critical Fastener Torque Value: 24 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-YAX-6 at PLC register 0x110D0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-BSW-NUT-7706) with seal kit (KIT-GRO450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 24 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-YAX-6 on Grob-Aerospace-G450 by installing PN-GRO450-BSW-NUT-7706 & kit KIT-GRO450-YAX-6. Fasteners torqued to 24 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-ZAX-7

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO450-7
- PLC Hex Register Address: DB21.DBW_0x110D8
- Nominal Operating Range: 33 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-CYL-SEAL-7707
- Overhaul Seal & Consumable Kit: KIT-GRO450-ZAX-7
- Critical Fastener Torque Value: 33 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-ZAX-7 at PLC register 0x110D8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-CYL-SEAL-7707) with seal kit (KIT-GRO450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 33 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-ZAX-7 on Grob-Aerospace-G450 by installing PN-GRO450-CYL-SEAL-7707 & kit KIT-GRO450-ZAX-7. Fasteners torqued to 33 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-BAX-8

Target Machine Model: Grob-Aerospace-G450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO450-8
- PLC Hex Register Address: DB21.DBW_0x110E0
- Nominal Operating Range: 36 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-TRQ-STAT-7708
- Overhaul Seal & Consumable Kit: KIT-GRO450-BAX-8
- Critical Fastener Torque Value: 42 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-BAX-8 at PLC register 0x110E0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-TRQ-STAT-7708) with seal kit (KIT-GRO450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 42 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-BAX-8 on Grob-Aerospace-G450 by installing PN-GRO450-TRQ-STAT-7708 & kit KIT-GRO450-BAX-8. Fasteners torqued to 42 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-CAX-9

Target Machine Model: Grob-Aerospace-G450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO450-9
- PLC Hex Register Address: DB21.DBW_0x110E8
- Nominal Operating Range: 39 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-BRK-DISC-7709
- Overhaul Seal & Consumable Kit: KIT-GRO450-CAX-9
- Critical Fastener Torque Value: 51 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-CAX-9 at PLC register 0x110E8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-BRK-DISC-7709) with seal kit (KIT-GRO450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 51 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-CAX-9 on Grob-Aerospace-G450 by installing PN-GRO450-BRK-DISC-7709 & kit KIT-GRO450-CAX-9. Fasteners torqued to 51 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-OPT-10

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO450-10
- PLC Hex Register Address: DB21.DBW_0x110F0
- Nominal Operating Range: 42 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-GLASS-RD-7710
- Overhaul Seal & Consumable Kit: KIT-GRO450-OPT-10
- Critical Fastener Torque Value: 60 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-OPT-10 at PLC register 0x110F0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-GLASS-RD-7710) with seal kit (KIT-GRO450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 60 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-OPT-10 on Grob-Aerospace-G450 by installing PN-GRO450-GLASS-RD-7710 & kit KIT-GRO450-OPT-10. Fasteners torqued to 60 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-PLC-11

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO450-11
- PLC Hex Register Address: DB21.DBW_0x110F8
- Nominal Operating Range: 45 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-IGBT-MOD-7711
- Overhaul Seal & Consumable Kit: KIT-GRO450-PLC-11
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-PLC-11 at PLC register 0x110F8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-IGBT-MOD-7711) with seal kit (KIT-GRO450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-PLC-11 on Grob-Aerospace-G450 by installing PN-GRO450-IGBT-MOD-7711 & kit KIT-GRO450-PLC-11. Fasteners torqued to 69 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-LUB-12

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO450-12
- PLC Hex Register Address: DB21.DBW_0x11100
- Nominal Operating Range: 48 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-INJ-MET-7712
- Overhaul Seal & Consumable Kit: KIT-GRO450-LUB-12
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-LUB-12 at PLC register 0x11100.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-INJ-MET-7712) with seal kit (KIT-GRO450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-LUB-12 on Grob-Aerospace-G450 by installing PN-GRO450-INJ-MET-7712 & kit KIT-GRO450-LUB-12. Fasteners torqued to 78 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-VAC-13

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO450-13
- PLC Hex Register Address: DB21.DBW_0x11108
- Nominal Operating Range: 51 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-GASK-VIT-7713
- Overhaul Seal & Consumable Kit: KIT-GRO450-VAC-13
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-VAC-13 at PLC register 0x11108.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-GASK-VIT-7713) with seal kit (KIT-GRO450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-VAC-13 on Grob-Aerospace-G450 by installing PN-GRO450-GASK-VIT-7713 & kit KIT-GRO450-VAC-13. Fasteners torqued to 87 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO450-THM-14

Target Machine Model: Grob-Aerospace-G450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G450 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO450-14
- PLC Hex Register Address: DB21.DBW_0x11110
- Nominal Operating Range: 54 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO450-HEX-CORE-7714
- Overhaul Seal & Consumable Kit: KIT-GRO450-THM-14
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G450 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO450-THM-14 at PLC register 0x11110.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO450-HEX-CORE-7714) with seal kit (KIT-GRO450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G450, technician must log: "Resolved ERR-GRO450-THM-14 on Grob-Aerospace-G450 by installing PN-GRO450-HEX-CORE-7714 & kit KIT-GRO450-THM-14. Fasteners torqued to 96 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-SPN-1

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO500-1
- PLC Hex Register Address: DB22.DBW_0x113C8
- Nominal Operating Range: 15 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-BRG-CER-7801
- Overhaul Seal & Consumable Kit: KIT-GRO500-SPN-1
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-SPN-1 at PLC register 0x113C8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-BRG-CER-7801) with seal kit (KIT-GRO500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-SPN-1 on Grob-Aerospace-G500 by installing PN-GRO500-BRG-CER-7801 & kit KIT-GRO500-SPN-1. Fasteners torqued to 119 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-ATC-2

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO500-2
- PLC Hex Register Address: DB22.DBW_0x113D0
- Nominal Operating Range: 18 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-CAM-GRIP-7802
- Overhaul Seal & Consumable Kit: KIT-GRO500-ATC-2
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-ATC-2 at PLC register 0x113D0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-CAM-GRIP-7802) with seal kit (KIT-GRO500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-ATC-2 on Grob-Aerospace-G500 by installing PN-GRO500-CAM-GRIP-7802 & kit KIT-GRO500-ATC-2. Fasteners torqued to 128 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-HYD-3

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO500-3
- PLC Hex Register Address: DB22.DBW_0x113D8
- Nominal Operating Range: 21 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-VLV-PROP-7803
- Overhaul Seal & Consumable Kit: KIT-GRO500-HYD-3
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-HYD-3 at PLC register 0x113D8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-VLV-PROP-7803) with seal kit (KIT-GRO500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-HYD-3 on Grob-Aerospace-G500 by installing PN-GRO500-VLV-PROP-7803 & kit KIT-GRO500-HYD-3. Fasteners torqued to 137 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-CLN-4

Target Machine Model: Grob-Aerospace-G500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO500-4
- PLC Hex Register Address: DB22.DBW_0x113E0
- Nominal Operating Range: 24 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-PMP-PLG-7804
- Overhaul Seal & Consumable Kit: KIT-GRO500-CLN-4
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-CLN-4 at PLC register 0x113E0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-PMP-PLG-7804) with seal kit (KIT-GRO500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-CLN-4 on Grob-Aerospace-G500 by installing PN-GRO500-PMP-PLG-7804 & kit KIT-GRO500-CLN-4. Fasteners torqued to 146 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-XAX-5

Target Machine Model: Grob-Aerospace-G500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO500-5
- PLC Hex Register Address: DB22.DBW_0x113E8
- Nominal Operating Range: 27 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-LIN-COIL-7805
- Overhaul Seal & Consumable Kit: KIT-GRO500-XAX-5
- Critical Fastener Torque Value: 155 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-XAX-5 at PLC register 0x113E8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-LIN-COIL-7805) with seal kit (KIT-GRO500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 155 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-XAX-5 on Grob-Aerospace-G500 by installing PN-GRO500-LIN-COIL-7805 & kit KIT-GRO500-XAX-5. Fasteners torqued to 155 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-YAX-6

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO500-6
- PLC Hex Register Address: DB22.DBW_0x113F0
- Nominal Operating Range: 30 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-BSW-NUT-7806
- Overhaul Seal & Consumable Kit: KIT-GRO500-YAX-6
- Critical Fastener Torque Value: 164 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-YAX-6 at PLC register 0x113F0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-BSW-NUT-7806) with seal kit (KIT-GRO500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 164 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-YAX-6 on Grob-Aerospace-G500 by installing PN-GRO500-BSW-NUT-7806 & kit KIT-GRO500-YAX-6. Fasteners torqued to 164 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-ZAX-7

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO500-7
- PLC Hex Register Address: DB22.DBW_0x113F8
- Nominal Operating Range: 33 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-CYL-SEAL-7807
- Overhaul Seal & Consumable Kit: KIT-GRO500-ZAX-7
- Critical Fastener Torque Value: 173 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-ZAX-7 at PLC register 0x113F8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-CYL-SEAL-7807) with seal kit (KIT-GRO500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 173 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-ZAX-7 on Grob-Aerospace-G500 by installing PN-GRO500-CYL-SEAL-7807 & kit KIT-GRO500-ZAX-7. Fasteners torqued to 173 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-BAX-8

Target Machine Model: Grob-Aerospace-G500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO500-8
- PLC Hex Register Address: DB22.DBW_0x11400
- Nominal Operating Range: 36 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-TRQ-STAT-7808
- Overhaul Seal & Consumable Kit: KIT-GRO500-BAX-8
- Critical Fastener Torque Value: 182 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-BAX-8 at PLC register 0x11400.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-TRQ-STAT-7808) with seal kit (KIT-GRO500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 182 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-BAX-8 on Grob-Aerospace-G500 by installing PN-GRO500-TRQ-STAT-7808 & kit KIT-GRO500-BAX-8. Fasteners torqued to 182 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-CAX-9

Target Machine Model: Grob-Aerospace-G500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO500-9
- PLC Hex Register Address: DB22.DBW_0x11408
- Nominal Operating Range: 39 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-BRK-DISC-7809
- Overhaul Seal & Consumable Kit: KIT-GRO500-CAX-9
- Critical Fastener Torque Value: 191 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-CAX-9 at PLC register 0x11408.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-BRK-DISC-7809) with seal kit (KIT-GRO500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 191 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-CAX-9 on Grob-Aerospace-G500 by installing PN-GRO500-BRK-DISC-7809 & kit KIT-GRO500-CAX-9. Fasteners torqued to 191 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-OPT-10

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO500-10
- PLC Hex Register Address: DB22.DBW_0x11410
- Nominal Operating Range: 42 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-GLASS-RD-7810
- Overhaul Seal & Consumable Kit: KIT-GRO500-OPT-10
- Critical Fastener Torque Value: 200 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-OPT-10 at PLC register 0x11410.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-GLASS-RD-7810) with seal kit (KIT-GRO500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 200 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-OPT-10 on Grob-Aerospace-G500 by installing PN-GRO500-GLASS-RD-7810 & kit KIT-GRO500-OPT-10. Fasteners torqued to 200 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-PLC-11

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO500-11
- PLC Hex Register Address: DB22.DBW_0x11418
- Nominal Operating Range: 45 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-IGBT-MOD-7811
- Overhaul Seal & Consumable Kit: KIT-GRO500-PLC-11
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-PLC-11 at PLC register 0x11418.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-IGBT-MOD-7811) with seal kit (KIT-GRO500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-PLC-11 on Grob-Aerospace-G500 by installing PN-GRO500-IGBT-MOD-7811 & kit KIT-GRO500-PLC-11. Fasteners torqued to 209 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-LUB-12

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO500-12
- PLC Hex Register Address: DB22.DBW_0x11420
- Nominal Operating Range: 48 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-INJ-MET-7812
- Overhaul Seal & Consumable Kit: KIT-GRO500-LUB-12
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-LUB-12 at PLC register 0x11420.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-INJ-MET-7812) with seal kit (KIT-GRO500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-LUB-12 on Grob-Aerospace-G500 by installing PN-GRO500-INJ-MET-7812 & kit KIT-GRO500-LUB-12. Fasteners torqued to 28 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-VAC-13

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO500-13
- PLC Hex Register Address: DB22.DBW_0x11428
- Nominal Operating Range: 51 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-GASK-VIT-7813
- Overhaul Seal & Consumable Kit: KIT-GRO500-VAC-13
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-VAC-13 at PLC register 0x11428.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-GASK-VIT-7813) with seal kit (KIT-GRO500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-VAC-13 on Grob-Aerospace-G500 by installing PN-GRO500-GASK-VIT-7813 & kit KIT-GRO500-VAC-13. Fasteners torqued to 37 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO500-THM-14

Target Machine Model: Grob-Aerospace-G500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G500 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO500-14
- PLC Hex Register Address: DB22.DBW_0x11430
- Nominal Operating Range: 54 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO500-HEX-CORE-7814
- Overhaul Seal & Consumable Kit: KIT-GRO500-THM-14
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G500 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO500-THM-14 at PLC register 0x11430.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO500-HEX-CORE-7814) with seal kit (KIT-GRO500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G500, technician must log: "Resolved ERR-GRO500-THM-14 on Grob-Aerospace-G500 by installing PN-GRO500-HEX-CORE-7814 & kit KIT-GRO500-THM-14. Fasteners torqued to 46 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-SPN-1

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO550-1
- PLC Hex Register Address: DB23.DBW_0x116E8
- Nominal Operating Range: 15 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-BRG-CER-7901
- Overhaul Seal & Consumable Kit: KIT-GRO550-SPN-1
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-SPN-1 at PLC register 0x116E8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-BRG-CER-7901) with seal kit (KIT-GRO550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-SPN-1 on Grob-Aerospace-G550 by installing PN-GRO550-BRG-CER-7901 & kit KIT-GRO550-SPN-1. Fasteners torqued to 69 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-ATC-2

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO550-2
- PLC Hex Register Address: DB23.DBW_0x116F0
- Nominal Operating Range: 18 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-CAM-GRIP-7902
- Overhaul Seal & Consumable Kit: KIT-GRO550-ATC-2
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-ATC-2 at PLC register 0x116F0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-CAM-GRIP-7902) with seal kit (KIT-GRO550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-ATC-2 on Grob-Aerospace-G550 by installing PN-GRO550-CAM-GRIP-7902 & kit KIT-GRO550-ATC-2. Fasteners torqued to 78 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-HYD-3

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO550-3
- PLC Hex Register Address: DB23.DBW_0x116F8
- Nominal Operating Range: 21 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-VLV-PROP-7903
- Overhaul Seal & Consumable Kit: KIT-GRO550-HYD-3
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-HYD-3 at PLC register 0x116F8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-VLV-PROP-7903) with seal kit (KIT-GRO550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-HYD-3 on Grob-Aerospace-G550 by installing PN-GRO550-VLV-PROP-7903 & kit KIT-GRO550-HYD-3. Fasteners torqued to 87 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-CLN-4

Target Machine Model: Grob-Aerospace-G550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO550-4
- PLC Hex Register Address: DB23.DBW_0x11700
- Nominal Operating Range: 24 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-PMP-PLG-7904
- Overhaul Seal & Consumable Kit: KIT-GRO550-CLN-4
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-CLN-4 at PLC register 0x11700.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-PMP-PLG-7904) with seal kit (KIT-GRO550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-CLN-4 on Grob-Aerospace-G550 by installing PN-GRO550-PMP-PLG-7904 & kit KIT-GRO550-CLN-4. Fasteners torqued to 96 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-XAX-5

Target Machine Model: Grob-Aerospace-G550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO550-5
- PLC Hex Register Address: DB23.DBW_0x11708
- Nominal Operating Range: 27 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-LIN-COIL-7905
- Overhaul Seal & Consumable Kit: KIT-GRO550-XAX-5
- Critical Fastener Torque Value: 105 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-XAX-5 at PLC register 0x11708.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-LIN-COIL-7905) with seal kit (KIT-GRO550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 105 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-XAX-5 on Grob-Aerospace-G550 by installing PN-GRO550-LIN-COIL-7905 & kit KIT-GRO550-XAX-5. Fasteners torqued to 105 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-YAX-6

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO550-6
- PLC Hex Register Address: DB23.DBW_0x11710
- Nominal Operating Range: 30 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-BSW-NUT-7906
- Overhaul Seal & Consumable Kit: KIT-GRO550-YAX-6
- Critical Fastener Torque Value: 114 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-YAX-6 at PLC register 0x11710.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-BSW-NUT-7906) with seal kit (KIT-GRO550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 114 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-YAX-6 on Grob-Aerospace-G550 by installing PN-GRO550-BSW-NUT-7906 & kit KIT-GRO550-YAX-6. Fasteners torqued to 114 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-ZAX-7

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO550-7
- PLC Hex Register Address: DB23.DBW_0x11718
- Nominal Operating Range: 33 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-CYL-SEAL-7907
- Overhaul Seal & Consumable Kit: KIT-GRO550-ZAX-7
- Critical Fastener Torque Value: 123 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-ZAX-7 at PLC register 0x11718.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-CYL-SEAL-7907) with seal kit (KIT-GRO550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 123 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-ZAX-7 on Grob-Aerospace-G550 by installing PN-GRO550-CYL-SEAL-7907 & kit KIT-GRO550-ZAX-7. Fasteners torqued to 123 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-BAX-8

Target Machine Model: Grob-Aerospace-G550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO550-8
- PLC Hex Register Address: DB23.DBW_0x11720
- Nominal Operating Range: 36 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-TRQ-STAT-7908
- Overhaul Seal & Consumable Kit: KIT-GRO550-BAX-8
- Critical Fastener Torque Value: 132 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-BAX-8 at PLC register 0x11720.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-TRQ-STAT-7908) with seal kit (KIT-GRO550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 132 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-BAX-8 on Grob-Aerospace-G550 by installing PN-GRO550-TRQ-STAT-7908 & kit KIT-GRO550-BAX-8. Fasteners torqued to 132 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-CAX-9

Target Machine Model: Grob-Aerospace-G550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO550-9
- PLC Hex Register Address: DB23.DBW_0x11728
- Nominal Operating Range: 39 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-BRK-DISC-7909
- Overhaul Seal & Consumable Kit: KIT-GRO550-CAX-9
- Critical Fastener Torque Value: 141 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-CAX-9 at PLC register 0x11728.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-BRK-DISC-7909) with seal kit (KIT-GRO550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 141 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-CAX-9 on Grob-Aerospace-G550 by installing PN-GRO550-BRK-DISC-7909 & kit KIT-GRO550-CAX-9. Fasteners torqued to 141 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-OPT-10

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO550-10
- PLC Hex Register Address: DB23.DBW_0x11730
- Nominal Operating Range: 42 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-GLASS-RD-7910
- Overhaul Seal & Consumable Kit: KIT-GRO550-OPT-10
- Critical Fastener Torque Value: 150 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-OPT-10 at PLC register 0x11730.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-GLASS-RD-7910) with seal kit (KIT-GRO550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 150 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-OPT-10 on Grob-Aerospace-G550 by installing PN-GRO550-GLASS-RD-7910 & kit KIT-GRO550-OPT-10. Fasteners torqued to 150 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-PLC-11

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO550-11
- PLC Hex Register Address: DB23.DBW_0x11738
- Nominal Operating Range: 45 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-IGBT-MOD-7911
- Overhaul Seal & Consumable Kit: KIT-GRO550-PLC-11
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-PLC-11 at PLC register 0x11738.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-IGBT-MOD-7911) with seal kit (KIT-GRO550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-PLC-11 on Grob-Aerospace-G550 by installing PN-GRO550-IGBT-MOD-7911 & kit KIT-GRO550-PLC-11. Fasteners torqued to 159 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-LUB-12

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO550-12
- PLC Hex Register Address: DB23.DBW_0x11740
- Nominal Operating Range: 48 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-INJ-MET-7912
- Overhaul Seal & Consumable Kit: KIT-GRO550-LUB-12
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-LUB-12 at PLC register 0x11740.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-INJ-MET-7912) with seal kit (KIT-GRO550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-LUB-12 on Grob-Aerospace-G550 by installing PN-GRO550-INJ-MET-7912 & kit KIT-GRO550-LUB-12. Fasteners torqued to 168 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-VAC-13

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO550-13
- PLC Hex Register Address: DB23.DBW_0x11748
- Nominal Operating Range: 51 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-GASK-VIT-7913
- Overhaul Seal & Consumable Kit: KIT-GRO550-VAC-13
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-VAC-13 at PLC register 0x11748.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-GASK-VIT-7913) with seal kit (KIT-GRO550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-VAC-13 on Grob-Aerospace-G550 by installing PN-GRO550-GASK-VIT-7913 & kit KIT-GRO550-VAC-13. Fasteners torqued to 177 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO550-THM-14

Target Machine Model: Grob-Aerospace-G550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G550 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO550-14
- PLC Hex Register Address: DB23.DBW_0x11750
- Nominal Operating Range: 54 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO550-HEX-CORE-7914
- Overhaul Seal & Consumable Kit: KIT-GRO550-THM-14
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G550 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO550-THM-14 at PLC register 0x11750.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO550-HEX-CORE-7914) with seal kit (KIT-GRO550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G550, technician must log: "Resolved ERR-GRO550-THM-14 on Grob-Aerospace-G550 by installing PN-GRO550-HEX-CORE-7914 & kit KIT-GRO550-THM-14. Fasteners torqued to 186 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-SPN-1

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO600-1
- PLC Hex Register Address: DB24.DBW_0x11A08
- Nominal Operating Range: 15 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-BRG-CER-8001
- Overhaul Seal & Consumable Kit: KIT-GRO600-SPN-1
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-SPN-1 at PLC register 0x11A08.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-BRG-CER-8001) with seal kit (KIT-GRO600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-SPN-1 on Grob-Aerospace-G600 by installing PN-GRO600-BRG-CER-8001 & kit KIT-GRO600-SPN-1. Fasteners torqued to 209 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-ATC-2

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO600-2
- PLC Hex Register Address: DB24.DBW_0x11A10
- Nominal Operating Range: 18 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-CAM-GRIP-8002
- Overhaul Seal & Consumable Kit: KIT-GRO600-ATC-2
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-ATC-2 at PLC register 0x11A10.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-CAM-GRIP-8002) with seal kit (KIT-GRO600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-ATC-2 on Grob-Aerospace-G600 by installing PN-GRO600-CAM-GRIP-8002 & kit KIT-GRO600-ATC-2. Fasteners torqued to 28 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-HYD-3

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO600-3
- PLC Hex Register Address: DB24.DBW_0x11A18
- Nominal Operating Range: 21 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-VLV-PROP-8003
- Overhaul Seal & Consumable Kit: KIT-GRO600-HYD-3
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-HYD-3 at PLC register 0x11A18.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-VLV-PROP-8003) with seal kit (KIT-GRO600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-HYD-3 on Grob-Aerospace-G600 by installing PN-GRO600-VLV-PROP-8003 & kit KIT-GRO600-HYD-3. Fasteners torqued to 37 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-CLN-4

Target Machine Model: Grob-Aerospace-G600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO600-4
- PLC Hex Register Address: DB24.DBW_0x11A20
- Nominal Operating Range: 24 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-PMP-PLG-8004
- Overhaul Seal & Consumable Kit: KIT-GRO600-CLN-4
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-CLN-4 at PLC register 0x11A20.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-PMP-PLG-8004) with seal kit (KIT-GRO600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-CLN-4 on Grob-Aerospace-G600 by installing PN-GRO600-PMP-PLG-8004 & kit KIT-GRO600-CLN-4. Fasteners torqued to 46 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-XAX-5

Target Machine Model: Grob-Aerospace-G600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO600-5
- PLC Hex Register Address: DB24.DBW_0x11A28
- Nominal Operating Range: 27 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-LIN-COIL-8005
- Overhaul Seal & Consumable Kit: KIT-GRO600-XAX-5
- Critical Fastener Torque Value: 55 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-XAX-5 at PLC register 0x11A28.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-LIN-COIL-8005) with seal kit (KIT-GRO600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 55 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-XAX-5 on Grob-Aerospace-G600 by installing PN-GRO600-LIN-COIL-8005 & kit KIT-GRO600-XAX-5. Fasteners torqued to 55 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-YAX-6

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO600-6
- PLC Hex Register Address: DB24.DBW_0x11A30
- Nominal Operating Range: 30 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-BSW-NUT-8006
- Overhaul Seal & Consumable Kit: KIT-GRO600-YAX-6
- Critical Fastener Torque Value: 64 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-YAX-6 at PLC register 0x11A30.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-BSW-NUT-8006) with seal kit (KIT-GRO600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 64 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-YAX-6 on Grob-Aerospace-G600 by installing PN-GRO600-BSW-NUT-8006 & kit KIT-GRO600-YAX-6. Fasteners torqued to 64 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-ZAX-7

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO600-7
- PLC Hex Register Address: DB24.DBW_0x11A38
- Nominal Operating Range: 33 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-CYL-SEAL-8007
- Overhaul Seal & Consumable Kit: KIT-GRO600-ZAX-7
- Critical Fastener Torque Value: 73 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-ZAX-7 at PLC register 0x11A38.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-CYL-SEAL-8007) with seal kit (KIT-GRO600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 73 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-ZAX-7 on Grob-Aerospace-G600 by installing PN-GRO600-CYL-SEAL-8007 & kit KIT-GRO600-ZAX-7. Fasteners torqued to 73 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-BAX-8

Target Machine Model: Grob-Aerospace-G600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO600-8
- PLC Hex Register Address: DB24.DBW_0x11A40
- Nominal Operating Range: 36 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-TRQ-STAT-8008
- Overhaul Seal & Consumable Kit: KIT-GRO600-BAX-8
- Critical Fastener Torque Value: 82 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-BAX-8 at PLC register 0x11A40.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-TRQ-STAT-8008) with seal kit (KIT-GRO600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 82 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-BAX-8 on Grob-Aerospace-G600 by installing PN-GRO600-TRQ-STAT-8008 & kit KIT-GRO600-BAX-8. Fasteners torqued to 82 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-CAX-9

Target Machine Model: Grob-Aerospace-G600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO600-9
- PLC Hex Register Address: DB24.DBW_0x11A48
- Nominal Operating Range: 39 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-BRK-DISC-8009
- Overhaul Seal & Consumable Kit: KIT-GRO600-CAX-9
- Critical Fastener Torque Value: 91 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-CAX-9 at PLC register 0x11A48.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-BRK-DISC-8009) with seal kit (KIT-GRO600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 91 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-CAX-9 on Grob-Aerospace-G600 by installing PN-GRO600-BRK-DISC-8009 & kit KIT-GRO600-CAX-9. Fasteners torqued to 91 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-OPT-10

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO600-10
- PLC Hex Register Address: DB24.DBW_0x11A50
- Nominal Operating Range: 42 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-GLASS-RD-8010
- Overhaul Seal & Consumable Kit: KIT-GRO600-OPT-10
- Critical Fastener Torque Value: 100 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-OPT-10 at PLC register 0x11A50.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-GLASS-RD-8010) with seal kit (KIT-GRO600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 100 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-OPT-10 on Grob-Aerospace-G600 by installing PN-GRO600-GLASS-RD-8010 & kit KIT-GRO600-OPT-10. Fasteners torqued to 100 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-PLC-11

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO600-11
- PLC Hex Register Address: DB24.DBW_0x11A58
- Nominal Operating Range: 45 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-IGBT-MOD-8011
- Overhaul Seal & Consumable Kit: KIT-GRO600-PLC-11
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-PLC-11 at PLC register 0x11A58.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-IGBT-MOD-8011) with seal kit (KIT-GRO600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-PLC-11 on Grob-Aerospace-G600 by installing PN-GRO600-IGBT-MOD-8011 & kit KIT-GRO600-PLC-11. Fasteners torqued to 109 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-LUB-12

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO600-12
- PLC Hex Register Address: DB24.DBW_0x11A60
- Nominal Operating Range: 48 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-INJ-MET-8012
- Overhaul Seal & Consumable Kit: KIT-GRO600-LUB-12
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-LUB-12 at PLC register 0x11A60.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-INJ-MET-8012) with seal kit (KIT-GRO600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-LUB-12 on Grob-Aerospace-G600 by installing PN-GRO600-INJ-MET-8012 & kit KIT-GRO600-LUB-12. Fasteners torqued to 118 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-VAC-13

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO600-13
- PLC Hex Register Address: DB24.DBW_0x11A68
- Nominal Operating Range: 51 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-GASK-VIT-8013
- Overhaul Seal & Consumable Kit: KIT-GRO600-VAC-13
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-VAC-13 at PLC register 0x11A68.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-GASK-VIT-8013) with seal kit (KIT-GRO600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-VAC-13 on Grob-Aerospace-G600 by installing PN-GRO600-GASK-VIT-8013 & kit KIT-GRO600-VAC-13. Fasteners torqued to 127 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO600-THM-14

Target Machine Model: Grob-Aerospace-G600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G600 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO600-14
- PLC Hex Register Address: DB24.DBW_0x11A70
- Nominal Operating Range: 54 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO600-HEX-CORE-8014
- Overhaul Seal & Consumable Kit: KIT-GRO600-THM-14
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G600 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO600-THM-14 at PLC register 0x11A70.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO600-HEX-CORE-8014) with seal kit (KIT-GRO600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G600, technician must log: "Resolved ERR-GRO600-THM-14 on Grob-Aerospace-G600 by installing PN-GRO600-HEX-CORE-8014 & kit KIT-GRO600-THM-14. Fasteners torqued to 136 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-SPN-1

Target Machine Model: Hermle-Performance-C150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER150-1
- PLC Hex Register Address: DB25.DBW_0x11D28
- Nominal Operating Range: 15 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-BRG-CER-8101
- Overhaul Seal & Consumable Kit: KIT-HER150-SPN-1
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-SPN-1 at PLC register 0x11D28.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-BRG-CER-8101) with seal kit (KIT-HER150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-SPN-1 on Hermle-Performance-C150 by installing PN-HER150-BRG-CER-8101 & kit KIT-HER150-SPN-1. Fasteners torqued to 159 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-ATC-2

Target Machine Model: Hermle-Performance-C150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER150-2
- PLC Hex Register Address: DB25.DBW_0x11D30
- Nominal Operating Range: 18 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-CAM-GRIP-8102
- Overhaul Seal & Consumable Kit: KIT-HER150-ATC-2
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-ATC-2 at PLC register 0x11D30.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-CAM-GRIP-8102) with seal kit (KIT-HER150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-ATC-2 on Hermle-Performance-C150 by installing PN-HER150-CAM-GRIP-8102 & kit KIT-HER150-ATC-2. Fasteners torqued to 168 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-HYD-3

Target Machine Model: Hermle-Performance-C150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER150-3
- PLC Hex Register Address: DB25.DBW_0x11D38
- Nominal Operating Range: 21 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-VLV-PROP-8103
- Overhaul Seal & Consumable Kit: KIT-HER150-HYD-3
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-HYD-3 at PLC register 0x11D38.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-VLV-PROP-8103) with seal kit (KIT-HER150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-HYD-3 on Hermle-Performance-C150 by installing PN-HER150-VLV-PROP-8103 & kit KIT-HER150-HYD-3. Fasteners torqued to 177 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-CLN-4

Target Machine Model: Hermle-Performance-C150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER150-4
- PLC Hex Register Address: DB25.DBW_0x11D40
- Nominal Operating Range: 24 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-PMP-PLG-8104
- Overhaul Seal & Consumable Kit: KIT-HER150-CLN-4
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-CLN-4 at PLC register 0x11D40.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-PMP-PLG-8104) with seal kit (KIT-HER150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-CLN-4 on Hermle-Performance-C150 by installing PN-HER150-PMP-PLG-8104 & kit KIT-HER150-CLN-4. Fasteners torqued to 186 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-XAX-5

Target Machine Model: Hermle-Performance-C150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER150-5
- PLC Hex Register Address: DB25.DBW_0x11D48
- Nominal Operating Range: 27 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-LIN-COIL-8105
- Overhaul Seal & Consumable Kit: KIT-HER150-XAX-5
- Critical Fastener Torque Value: 195 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-XAX-5 at PLC register 0x11D48.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-LIN-COIL-8105) with seal kit (KIT-HER150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 195 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-XAX-5 on Hermle-Performance-C150 by installing PN-HER150-LIN-COIL-8105 & kit KIT-HER150-XAX-5. Fasteners torqued to 195 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-YAX-6

Target Machine Model: Hermle-Performance-C150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER150-6
- PLC Hex Register Address: DB25.DBW_0x11D50
- Nominal Operating Range: 30 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-BSW-NUT-8106
- Overhaul Seal & Consumable Kit: KIT-HER150-YAX-6
- Critical Fastener Torque Value: 204 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-YAX-6 at PLC register 0x11D50.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-BSW-NUT-8106) with seal kit (KIT-HER150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 204 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-YAX-6 on Hermle-Performance-C150 by installing PN-HER150-BSW-NUT-8106 & kit KIT-HER150-YAX-6. Fasteners torqued to 204 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-ZAX-7

Target Machine Model: Hermle-Performance-C150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER150-7
- PLC Hex Register Address: DB25.DBW_0x11D58
- Nominal Operating Range: 33 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-CYL-SEAL-8107
- Overhaul Seal & Consumable Kit: KIT-HER150-ZAX-7
- Critical Fastener Torque Value: 23 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-ZAX-7 at PLC register 0x11D58.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-CYL-SEAL-8107) with seal kit (KIT-HER150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 23 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-ZAX-7 on Hermle-Performance-C150 by installing PN-HER150-CYL-SEAL-8107 & kit KIT-HER150-ZAX-7. Fasteners torqued to 23 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-BAX-8

Target Machine Model: Hermle-Performance-C150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER150-8
- PLC Hex Register Address: DB25.DBW_0x11D60
- Nominal Operating Range: 36 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-TRQ-STAT-8108
- Overhaul Seal & Consumable Kit: KIT-HER150-BAX-8
- Critical Fastener Torque Value: 32 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-BAX-8 at PLC register 0x11D60.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-TRQ-STAT-8108) with seal kit (KIT-HER150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 32 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-BAX-8 on Hermle-Performance-C150 by installing PN-HER150-TRQ-STAT-8108 & kit KIT-HER150-BAX-8. Fasteners torqued to 32 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-CAX-9

Target Machine Model: Hermle-Performance-C150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER150-9
- PLC Hex Register Address: DB25.DBW_0x11D68
- Nominal Operating Range: 39 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-BRK-DISC-8109
- Overhaul Seal & Consumable Kit: KIT-HER150-CAX-9
- Critical Fastener Torque Value: 41 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-CAX-9 at PLC register 0x11D68.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-BRK-DISC-8109) with seal kit (KIT-HER150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 41 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-CAX-9 on Hermle-Performance-C150 by installing PN-HER150-BRK-DISC-8109 & kit KIT-HER150-CAX-9. Fasteners torqued to 41 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-OPT-10

Target Machine Model: Hermle-Performance-C150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER150-10
- PLC Hex Register Address: DB25.DBW_0x11D70
- Nominal Operating Range: 42 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-GLASS-RD-8110
- Overhaul Seal & Consumable Kit: KIT-HER150-OPT-10
- Critical Fastener Torque Value: 50 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-OPT-10 at PLC register 0x11D70.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-GLASS-RD-8110) with seal kit (KIT-HER150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 50 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-OPT-10 on Hermle-Performance-C150 by installing PN-HER150-GLASS-RD-8110 & kit KIT-HER150-OPT-10. Fasteners torqued to 50 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-PLC-11

Target Machine Model: Hermle-Performance-C150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER150-11
- PLC Hex Register Address: DB25.DBW_0x11D78
- Nominal Operating Range: 45 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-IGBT-MOD-8111
- Overhaul Seal & Consumable Kit: KIT-HER150-PLC-11
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-PLC-11 at PLC register 0x11D78.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-IGBT-MOD-8111) with seal kit (KIT-HER150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-PLC-11 on Hermle-Performance-C150 by installing PN-HER150-IGBT-MOD-8111 & kit KIT-HER150-PLC-11. Fasteners torqued to 59 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-LUB-12

Target Machine Model: Hermle-Performance-C150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER150-12
- PLC Hex Register Address: DB25.DBW_0x11D80
- Nominal Operating Range: 48 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-INJ-MET-8112
- Overhaul Seal & Consumable Kit: KIT-HER150-LUB-12
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-LUB-12 at PLC register 0x11D80.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-INJ-MET-8112) with seal kit (KIT-HER150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-LUB-12 on Hermle-Performance-C150 by installing PN-HER150-INJ-MET-8112 & kit KIT-HER150-LUB-12. Fasteners torqued to 68 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-VAC-13

Target Machine Model: Hermle-Performance-C150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER150-13
- PLC Hex Register Address: DB25.DBW_0x11D88
- Nominal Operating Range: 51 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-GASK-VIT-8113
- Overhaul Seal & Consumable Kit: KIT-HER150-VAC-13
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-VAC-13 at PLC register 0x11D88.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-GASK-VIT-8113) with seal kit (KIT-HER150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-VAC-13 on Hermle-Performance-C150 by installing PN-HER150-GASK-VIT-8113 & kit KIT-HER150-VAC-13. Fasteners torqued to 77 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER150-THM-14

Target Machine Model: Hermle-Performance-C150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C150 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER150-14
- PLC Hex Register Address: DB25.DBW_0x11D90
- Nominal Operating Range: 54 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER150-HEX-CORE-8114
- Overhaul Seal & Consumable Kit: KIT-HER150-THM-14
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C150 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER150-THM-14 at PLC register 0x11D90.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER150-HEX-CORE-8114) with seal kit (KIT-HER150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C150, technician must log: "Resolved ERR-HER150-THM-14 on Hermle-Performance-C150 by installing PN-HER150-HEX-CORE-8114 & kit KIT-HER150-THM-14. Fasteners torqued to 86 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-SPN-1

Target Machine Model: Hermle-Performance-C200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER200-1
- PLC Hex Register Address: DB26.DBW_0x12048
- Nominal Operating Range: 15 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-BRG-CER-8201
- Overhaul Seal & Consumable Kit: KIT-HER200-SPN-1
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-SPN-1 at PLC register 0x12048.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-BRG-CER-8201) with seal kit (KIT-HER200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-SPN-1 on Hermle-Performance-C200 by installing PN-HER200-BRG-CER-8201 & kit KIT-HER200-SPN-1. Fasteners torqued to 109 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-ATC-2

Target Machine Model: Hermle-Performance-C200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER200-2
- PLC Hex Register Address: DB26.DBW_0x12050
- Nominal Operating Range: 18 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-CAM-GRIP-8202
- Overhaul Seal & Consumable Kit: KIT-HER200-ATC-2
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-ATC-2 at PLC register 0x12050.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-CAM-GRIP-8202) with seal kit (KIT-HER200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-ATC-2 on Hermle-Performance-C200 by installing PN-HER200-CAM-GRIP-8202 & kit KIT-HER200-ATC-2. Fasteners torqued to 118 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-HYD-3

Target Machine Model: Hermle-Performance-C200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER200-3
- PLC Hex Register Address: DB26.DBW_0x12058
- Nominal Operating Range: 21 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-VLV-PROP-8203
- Overhaul Seal & Consumable Kit: KIT-HER200-HYD-3
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-HYD-3 at PLC register 0x12058.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-VLV-PROP-8203) with seal kit (KIT-HER200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-HYD-3 on Hermle-Performance-C200 by installing PN-HER200-VLV-PROP-8203 & kit KIT-HER200-HYD-3. Fasteners torqued to 127 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-CLN-4

Target Machine Model: Hermle-Performance-C200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER200-4
- PLC Hex Register Address: DB26.DBW_0x12060
- Nominal Operating Range: 24 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-PMP-PLG-8204
- Overhaul Seal & Consumable Kit: KIT-HER200-CLN-4
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-CLN-4 at PLC register 0x12060.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-PMP-PLG-8204) with seal kit (KIT-HER200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-CLN-4 on Hermle-Performance-C200 by installing PN-HER200-PMP-PLG-8204 & kit KIT-HER200-CLN-4. Fasteners torqued to 136 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-XAX-5

Target Machine Model: Hermle-Performance-C200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER200-5
- PLC Hex Register Address: DB26.DBW_0x12068
- Nominal Operating Range: 27 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-LIN-COIL-8205
- Overhaul Seal & Consumable Kit: KIT-HER200-XAX-5
- Critical Fastener Torque Value: 145 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-XAX-5 at PLC register 0x12068.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-LIN-COIL-8205) with seal kit (KIT-HER200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 145 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-XAX-5 on Hermle-Performance-C200 by installing PN-HER200-LIN-COIL-8205 & kit KIT-HER200-XAX-5. Fasteners torqued to 145 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-YAX-6

Target Machine Model: Hermle-Performance-C200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER200-6
- PLC Hex Register Address: DB26.DBW_0x12070
- Nominal Operating Range: 30 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-BSW-NUT-8206
- Overhaul Seal & Consumable Kit: KIT-HER200-YAX-6
- Critical Fastener Torque Value: 154 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-YAX-6 at PLC register 0x12070.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-BSW-NUT-8206) with seal kit (KIT-HER200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 154 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-YAX-6 on Hermle-Performance-C200 by installing PN-HER200-BSW-NUT-8206 & kit KIT-HER200-YAX-6. Fasteners torqued to 154 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-ZAX-7

Target Machine Model: Hermle-Performance-C200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER200-7
- PLC Hex Register Address: DB26.DBW_0x12078
- Nominal Operating Range: 33 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-CYL-SEAL-8207
- Overhaul Seal & Consumable Kit: KIT-HER200-ZAX-7
- Critical Fastener Torque Value: 163 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-ZAX-7 at PLC register 0x12078.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-CYL-SEAL-8207) with seal kit (KIT-HER200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 163 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-ZAX-7 on Hermle-Performance-C200 by installing PN-HER200-CYL-SEAL-8207 & kit KIT-HER200-ZAX-7. Fasteners torqued to 163 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-BAX-8

Target Machine Model: Hermle-Performance-C200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER200-8
- PLC Hex Register Address: DB26.DBW_0x12080
- Nominal Operating Range: 36 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-TRQ-STAT-8208
- Overhaul Seal & Consumable Kit: KIT-HER200-BAX-8
- Critical Fastener Torque Value: 172 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-BAX-8 at PLC register 0x12080.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-TRQ-STAT-8208) with seal kit (KIT-HER200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 172 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-BAX-8 on Hermle-Performance-C200 by installing PN-HER200-TRQ-STAT-8208 & kit KIT-HER200-BAX-8. Fasteners torqued to 172 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-CAX-9

Target Machine Model: Hermle-Performance-C200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER200-9
- PLC Hex Register Address: DB26.DBW_0x12088
- Nominal Operating Range: 39 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-BRK-DISC-8209
- Overhaul Seal & Consumable Kit: KIT-HER200-CAX-9
- Critical Fastener Torque Value: 181 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-CAX-9 at PLC register 0x12088.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-BRK-DISC-8209) with seal kit (KIT-HER200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 181 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-CAX-9 on Hermle-Performance-C200 by installing PN-HER200-BRK-DISC-8209 & kit KIT-HER200-CAX-9. Fasteners torqued to 181 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-OPT-10

Target Machine Model: Hermle-Performance-C200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER200-10
- PLC Hex Register Address: DB26.DBW_0x12090
- Nominal Operating Range: 42 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-GLASS-RD-8210
- Overhaul Seal & Consumable Kit: KIT-HER200-OPT-10
- Critical Fastener Torque Value: 190 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-OPT-10 at PLC register 0x12090.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-GLASS-RD-8210) with seal kit (KIT-HER200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 190 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-OPT-10 on Hermle-Performance-C200 by installing PN-HER200-GLASS-RD-8210 & kit KIT-HER200-OPT-10. Fasteners torqued to 190 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-PLC-11

Target Machine Model: Hermle-Performance-C200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER200-11
- PLC Hex Register Address: DB26.DBW_0x12098
- Nominal Operating Range: 45 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-IGBT-MOD-8211
- Overhaul Seal & Consumable Kit: KIT-HER200-PLC-11
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-PLC-11 at PLC register 0x12098.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-IGBT-MOD-8211) with seal kit (KIT-HER200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-PLC-11 on Hermle-Performance-C200 by installing PN-HER200-IGBT-MOD-8211 & kit KIT-HER200-PLC-11. Fasteners torqued to 199 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-LUB-12

Target Machine Model: Hermle-Performance-C200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER200-12
- PLC Hex Register Address: DB26.DBW_0x120A0
- Nominal Operating Range: 48 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-INJ-MET-8212
- Overhaul Seal & Consumable Kit: KIT-HER200-LUB-12
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-LUB-12 at PLC register 0x120A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-INJ-MET-8212) with seal kit (KIT-HER200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-LUB-12 on Hermle-Performance-C200 by installing PN-HER200-INJ-MET-8212 & kit KIT-HER200-LUB-12. Fasteners torqued to 208 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-VAC-13

Target Machine Model: Hermle-Performance-C200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER200-13
- PLC Hex Register Address: DB26.DBW_0x120A8
- Nominal Operating Range: 51 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-GASK-VIT-8213
- Overhaul Seal & Consumable Kit: KIT-HER200-VAC-13
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-VAC-13 at PLC register 0x120A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-GASK-VIT-8213) with seal kit (KIT-HER200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-VAC-13 on Hermle-Performance-C200 by installing PN-HER200-GASK-VIT-8213 & kit KIT-HER200-VAC-13. Fasteners torqued to 27 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER200-THM-14

Target Machine Model: Hermle-Performance-C200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C200 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER200-14
- PLC Hex Register Address: DB26.DBW_0x120B0
- Nominal Operating Range: 54 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER200-HEX-CORE-8214
- Overhaul Seal & Consumable Kit: KIT-HER200-THM-14
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C200 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER200-THM-14 at PLC register 0x120B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER200-HEX-CORE-8214) with seal kit (KIT-HER200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C200, technician must log: "Resolved ERR-HER200-THM-14 on Hermle-Performance-C200 by installing PN-HER200-HEX-CORE-8214 & kit KIT-HER200-THM-14. Fasteners torqued to 36 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-SPN-1

Target Machine Model: Hermle-Performance-C250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER250-1
- PLC Hex Register Address: DB27.DBW_0x12368
- Nominal Operating Range: 15 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-BRG-CER-8301
- Overhaul Seal & Consumable Kit: KIT-HER250-SPN-1
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-SPN-1 at PLC register 0x12368.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-BRG-CER-8301) with seal kit (KIT-HER250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-SPN-1 on Hermle-Performance-C250 by installing PN-HER250-BRG-CER-8301 & kit KIT-HER250-SPN-1. Fasteners torqued to 59 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-ATC-2

Target Machine Model: Hermle-Performance-C250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER250-2
- PLC Hex Register Address: DB27.DBW_0x12370
- Nominal Operating Range: 18 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-CAM-GRIP-8302
- Overhaul Seal & Consumable Kit: KIT-HER250-ATC-2
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-ATC-2 at PLC register 0x12370.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-CAM-GRIP-8302) with seal kit (KIT-HER250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-ATC-2 on Hermle-Performance-C250 by installing PN-HER250-CAM-GRIP-8302 & kit KIT-HER250-ATC-2. Fasteners torqued to 68 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-HYD-3

Target Machine Model: Hermle-Performance-C250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER250-3
- PLC Hex Register Address: DB27.DBW_0x12378
- Nominal Operating Range: 21 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-VLV-PROP-8303
- Overhaul Seal & Consumable Kit: KIT-HER250-HYD-3
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-HYD-3 at PLC register 0x12378.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-VLV-PROP-8303) with seal kit (KIT-HER250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-HYD-3 on Hermle-Performance-C250 by installing PN-HER250-VLV-PROP-8303 & kit KIT-HER250-HYD-3. Fasteners torqued to 77 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-CLN-4

Target Machine Model: Hermle-Performance-C250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER250-4
- PLC Hex Register Address: DB27.DBW_0x12380
- Nominal Operating Range: 24 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-PMP-PLG-8304
- Overhaul Seal & Consumable Kit: KIT-HER250-CLN-4
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-CLN-4 at PLC register 0x12380.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-PMP-PLG-8304) with seal kit (KIT-HER250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-CLN-4 on Hermle-Performance-C250 by installing PN-HER250-PMP-PLG-8304 & kit KIT-HER250-CLN-4. Fasteners torqued to 86 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-XAX-5

Target Machine Model: Hermle-Performance-C250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER250-5
- PLC Hex Register Address: DB27.DBW_0x12388
- Nominal Operating Range: 27 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-LIN-COIL-8305
- Overhaul Seal & Consumable Kit: KIT-HER250-XAX-5
- Critical Fastener Torque Value: 95 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-XAX-5 at PLC register 0x12388.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-LIN-COIL-8305) with seal kit (KIT-HER250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 95 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-XAX-5 on Hermle-Performance-C250 by installing PN-HER250-LIN-COIL-8305 & kit KIT-HER250-XAX-5. Fasteners torqued to 95 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-YAX-6

Target Machine Model: Hermle-Performance-C250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER250-6
- PLC Hex Register Address: DB27.DBW_0x12390
- Nominal Operating Range: 30 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-BSW-NUT-8306
- Overhaul Seal & Consumable Kit: KIT-HER250-YAX-6
- Critical Fastener Torque Value: 104 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-YAX-6 at PLC register 0x12390.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-BSW-NUT-8306) with seal kit (KIT-HER250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 104 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-YAX-6 on Hermle-Performance-C250 by installing PN-HER250-BSW-NUT-8306 & kit KIT-HER250-YAX-6. Fasteners torqued to 104 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-ZAX-7

Target Machine Model: Hermle-Performance-C250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER250-7
- PLC Hex Register Address: DB27.DBW_0x12398
- Nominal Operating Range: 33 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-CYL-SEAL-8307
- Overhaul Seal & Consumable Kit: KIT-HER250-ZAX-7
- Critical Fastener Torque Value: 113 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-ZAX-7 at PLC register 0x12398.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-CYL-SEAL-8307) with seal kit (KIT-HER250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 113 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-ZAX-7 on Hermle-Performance-C250 by installing PN-HER250-CYL-SEAL-8307 & kit KIT-HER250-ZAX-7. Fasteners torqued to 113 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-BAX-8

Target Machine Model: Hermle-Performance-C250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER250-8
- PLC Hex Register Address: DB27.DBW_0x123A0
- Nominal Operating Range: 36 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-TRQ-STAT-8308
- Overhaul Seal & Consumable Kit: KIT-HER250-BAX-8
- Critical Fastener Torque Value: 122 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-BAX-8 at PLC register 0x123A0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-TRQ-STAT-8308) with seal kit (KIT-HER250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 122 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-BAX-8 on Hermle-Performance-C250 by installing PN-HER250-TRQ-STAT-8308 & kit KIT-HER250-BAX-8. Fasteners torqued to 122 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-CAX-9

Target Machine Model: Hermle-Performance-C250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER250-9
- PLC Hex Register Address: DB27.DBW_0x123A8
- Nominal Operating Range: 39 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-BRK-DISC-8309
- Overhaul Seal & Consumable Kit: KIT-HER250-CAX-9
- Critical Fastener Torque Value: 131 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-CAX-9 at PLC register 0x123A8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-BRK-DISC-8309) with seal kit (KIT-HER250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 131 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-CAX-9 on Hermle-Performance-C250 by installing PN-HER250-BRK-DISC-8309 & kit KIT-HER250-CAX-9. Fasteners torqued to 131 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-OPT-10

Target Machine Model: Hermle-Performance-C250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER250-10
- PLC Hex Register Address: DB27.DBW_0x123B0
- Nominal Operating Range: 42 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-GLASS-RD-8310
- Overhaul Seal & Consumable Kit: KIT-HER250-OPT-10
- Critical Fastener Torque Value: 140 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-OPT-10 at PLC register 0x123B0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-GLASS-RD-8310) with seal kit (KIT-HER250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 140 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-OPT-10 on Hermle-Performance-C250 by installing PN-HER250-GLASS-RD-8310 & kit KIT-HER250-OPT-10. Fasteners torqued to 140 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-PLC-11

Target Machine Model: Hermle-Performance-C250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER250-11
- PLC Hex Register Address: DB27.DBW_0x123B8
- Nominal Operating Range: 45 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-IGBT-MOD-8311
- Overhaul Seal & Consumable Kit: KIT-HER250-PLC-11
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-PLC-11 at PLC register 0x123B8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-IGBT-MOD-8311) with seal kit (KIT-HER250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-PLC-11 on Hermle-Performance-C250 by installing PN-HER250-IGBT-MOD-8311 & kit KIT-HER250-PLC-11. Fasteners torqued to 149 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-LUB-12

Target Machine Model: Hermle-Performance-C250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER250-12
- PLC Hex Register Address: DB27.DBW_0x123C0
- Nominal Operating Range: 48 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-INJ-MET-8312
- Overhaul Seal & Consumable Kit: KIT-HER250-LUB-12
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-LUB-12 at PLC register 0x123C0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-INJ-MET-8312) with seal kit (KIT-HER250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-LUB-12 on Hermle-Performance-C250 by installing PN-HER250-INJ-MET-8312 & kit KIT-HER250-LUB-12. Fasteners torqued to 158 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-VAC-13

Target Machine Model: Hermle-Performance-C250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER250-13
- PLC Hex Register Address: DB27.DBW_0x123C8
- Nominal Operating Range: 51 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-GASK-VIT-8313
- Overhaul Seal & Consumable Kit: KIT-HER250-VAC-13
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-VAC-13 at PLC register 0x123C8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-GASK-VIT-8313) with seal kit (KIT-HER250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-VAC-13 on Hermle-Performance-C250 by installing PN-HER250-GASK-VIT-8313 & kit KIT-HER250-VAC-13. Fasteners torqued to 167 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER250-THM-14

Target Machine Model: Hermle-Performance-C250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C250 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER250-14
- PLC Hex Register Address: DB27.DBW_0x123D0
- Nominal Operating Range: 54 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER250-HEX-CORE-8314
- Overhaul Seal & Consumable Kit: KIT-HER250-THM-14
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C250 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER250-THM-14 at PLC register 0x123D0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER250-HEX-CORE-8314) with seal kit (KIT-HER250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C250, technician must log: "Resolved ERR-HER250-THM-14 on Hermle-Performance-C250 by installing PN-HER250-HEX-CORE-8314 & kit KIT-HER250-THM-14. Fasteners torqued to 176 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-SPN-1

Target Machine Model: Hermle-Performance-C300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER300-1
- PLC Hex Register Address: DB28.DBW_0x12688
- Nominal Operating Range: 15 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-BRG-CER-8401
- Overhaul Seal & Consumable Kit: KIT-HER300-SPN-1
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-SPN-1 at PLC register 0x12688.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-BRG-CER-8401) with seal kit (KIT-HER300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-SPN-1 on Hermle-Performance-C300 by installing PN-HER300-BRG-CER-8401 & kit KIT-HER300-SPN-1. Fasteners torqued to 199 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-ATC-2

Target Machine Model: Hermle-Performance-C300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER300-2
- PLC Hex Register Address: DB28.DBW_0x12690
- Nominal Operating Range: 18 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-CAM-GRIP-8402
- Overhaul Seal & Consumable Kit: KIT-HER300-ATC-2
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-ATC-2 at PLC register 0x12690.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-CAM-GRIP-8402) with seal kit (KIT-HER300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-ATC-2 on Hermle-Performance-C300 by installing PN-HER300-CAM-GRIP-8402 & kit KIT-HER300-ATC-2. Fasteners torqued to 208 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-HYD-3

Target Machine Model: Hermle-Performance-C300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER300-3
- PLC Hex Register Address: DB28.DBW_0x12698
- Nominal Operating Range: 21 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-VLV-PROP-8403
- Overhaul Seal & Consumable Kit: KIT-HER300-HYD-3
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-HYD-3 at PLC register 0x12698.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-VLV-PROP-8403) with seal kit (KIT-HER300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-HYD-3 on Hermle-Performance-C300 by installing PN-HER300-VLV-PROP-8403 & kit KIT-HER300-HYD-3. Fasteners torqued to 27 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-CLN-4

Target Machine Model: Hermle-Performance-C300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER300-4
- PLC Hex Register Address: DB28.DBW_0x126A0
- Nominal Operating Range: 24 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-PMP-PLG-8404
- Overhaul Seal & Consumable Kit: KIT-HER300-CLN-4
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-CLN-4 at PLC register 0x126A0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-PMP-PLG-8404) with seal kit (KIT-HER300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-CLN-4 on Hermle-Performance-C300 by installing PN-HER300-PMP-PLG-8404 & kit KIT-HER300-CLN-4. Fasteners torqued to 36 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-XAX-5

Target Machine Model: Hermle-Performance-C300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER300-5
- PLC Hex Register Address: DB28.DBW_0x126A8
- Nominal Operating Range: 27 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-LIN-COIL-8405
- Overhaul Seal & Consumable Kit: KIT-HER300-XAX-5
- Critical Fastener Torque Value: 45 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-XAX-5 at PLC register 0x126A8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-LIN-COIL-8405) with seal kit (KIT-HER300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 45 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-XAX-5 on Hermle-Performance-C300 by installing PN-HER300-LIN-COIL-8405 & kit KIT-HER300-XAX-5. Fasteners torqued to 45 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-YAX-6

Target Machine Model: Hermle-Performance-C300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER300-6
- PLC Hex Register Address: DB28.DBW_0x126B0
- Nominal Operating Range: 30 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-BSW-NUT-8406
- Overhaul Seal & Consumable Kit: KIT-HER300-YAX-6
- Critical Fastener Torque Value: 54 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-YAX-6 at PLC register 0x126B0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-BSW-NUT-8406) with seal kit (KIT-HER300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 54 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-YAX-6 on Hermle-Performance-C300 by installing PN-HER300-BSW-NUT-8406 & kit KIT-HER300-YAX-6. Fasteners torqued to 54 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-ZAX-7

Target Machine Model: Hermle-Performance-C300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER300-7
- PLC Hex Register Address: DB28.DBW_0x126B8
- Nominal Operating Range: 33 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-CYL-SEAL-8407
- Overhaul Seal & Consumable Kit: KIT-HER300-ZAX-7
- Critical Fastener Torque Value: 63 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-ZAX-7 at PLC register 0x126B8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-CYL-SEAL-8407) with seal kit (KIT-HER300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 63 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-ZAX-7 on Hermle-Performance-C300 by installing PN-HER300-CYL-SEAL-8407 & kit KIT-HER300-ZAX-7. Fasteners torqued to 63 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-BAX-8

Target Machine Model: Hermle-Performance-C300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER300-8
- PLC Hex Register Address: DB28.DBW_0x126C0
- Nominal Operating Range: 36 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-TRQ-STAT-8408
- Overhaul Seal & Consumable Kit: KIT-HER300-BAX-8
- Critical Fastener Torque Value: 72 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-BAX-8 at PLC register 0x126C0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-TRQ-STAT-8408) with seal kit (KIT-HER300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 72 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-BAX-8 on Hermle-Performance-C300 by installing PN-HER300-TRQ-STAT-8408 & kit KIT-HER300-BAX-8. Fasteners torqued to 72 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-CAX-9

Target Machine Model: Hermle-Performance-C300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER300-9
- PLC Hex Register Address: DB28.DBW_0x126C8
- Nominal Operating Range: 39 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-BRK-DISC-8409
- Overhaul Seal & Consumable Kit: KIT-HER300-CAX-9
- Critical Fastener Torque Value: 81 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-CAX-9 at PLC register 0x126C8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-BRK-DISC-8409) with seal kit (KIT-HER300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 81 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-CAX-9 on Hermle-Performance-C300 by installing PN-HER300-BRK-DISC-8409 & kit KIT-HER300-CAX-9. Fasteners torqued to 81 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-OPT-10

Target Machine Model: Hermle-Performance-C300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER300-10
- PLC Hex Register Address: DB28.DBW_0x126D0
- Nominal Operating Range: 42 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-GLASS-RD-8410
- Overhaul Seal & Consumable Kit: KIT-HER300-OPT-10
- Critical Fastener Torque Value: 90 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-OPT-10 at PLC register 0x126D0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-GLASS-RD-8410) with seal kit (KIT-HER300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 90 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-OPT-10 on Hermle-Performance-C300 by installing PN-HER300-GLASS-RD-8410 & kit KIT-HER300-OPT-10. Fasteners torqued to 90 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-PLC-11

Target Machine Model: Hermle-Performance-C300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER300-11
- PLC Hex Register Address: DB28.DBW_0x126D8
- Nominal Operating Range: 45 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-IGBT-MOD-8411
- Overhaul Seal & Consumable Kit: KIT-HER300-PLC-11
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-PLC-11 at PLC register 0x126D8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-IGBT-MOD-8411) with seal kit (KIT-HER300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-PLC-11 on Hermle-Performance-C300 by installing PN-HER300-IGBT-MOD-8411 & kit KIT-HER300-PLC-11. Fasteners torqued to 99 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-LUB-12

Target Machine Model: Hermle-Performance-C300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER300-12
- PLC Hex Register Address: DB28.DBW_0x126E0
- Nominal Operating Range: 48 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-INJ-MET-8412
- Overhaul Seal & Consumable Kit: KIT-HER300-LUB-12
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-LUB-12 at PLC register 0x126E0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-INJ-MET-8412) with seal kit (KIT-HER300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-LUB-12 on Hermle-Performance-C300 by installing PN-HER300-INJ-MET-8412 & kit KIT-HER300-LUB-12. Fasteners torqued to 108 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-VAC-13

Target Machine Model: Hermle-Performance-C300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER300-13
- PLC Hex Register Address: DB28.DBW_0x126E8
- Nominal Operating Range: 51 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-GASK-VIT-8413
- Overhaul Seal & Consumable Kit: KIT-HER300-VAC-13
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-VAC-13 at PLC register 0x126E8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-GASK-VIT-8413) with seal kit (KIT-HER300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-VAC-13 on Hermle-Performance-C300 by installing PN-HER300-GASK-VIT-8413 & kit KIT-HER300-VAC-13. Fasteners torqued to 117 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER300-THM-14

Target Machine Model: Hermle-Performance-C300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C300 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER300-14
- PLC Hex Register Address: DB28.DBW_0x126F0
- Nominal Operating Range: 54 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER300-HEX-CORE-8414
- Overhaul Seal & Consumable Kit: KIT-HER300-THM-14
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C300 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER300-THM-14 at PLC register 0x126F0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER300-HEX-CORE-8414) with seal kit (KIT-HER300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C300, technician must log: "Resolved ERR-HER300-THM-14 on Hermle-Performance-C300 by installing PN-HER300-HEX-CORE-8414 & kit KIT-HER300-THM-14. Fasteners torqued to 126 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-SPN-1

Target Machine Model: Hermle-Performance-C350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER350-1
- PLC Hex Register Address: DB29.DBW_0x129A8
- Nominal Operating Range: 15 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-BRG-CER-8501
- Overhaul Seal & Consumable Kit: KIT-HER350-SPN-1
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-SPN-1 at PLC register 0x129A8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-BRG-CER-8501) with seal kit (KIT-HER350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-SPN-1 on Hermle-Performance-C350 by installing PN-HER350-BRG-CER-8501 & kit KIT-HER350-SPN-1. Fasteners torqued to 149 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-ATC-2

Target Machine Model: Hermle-Performance-C350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER350-2
- PLC Hex Register Address: DB29.DBW_0x129B0
- Nominal Operating Range: 18 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-CAM-GRIP-8502
- Overhaul Seal & Consumable Kit: KIT-HER350-ATC-2
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-ATC-2 at PLC register 0x129B0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-CAM-GRIP-8502) with seal kit (KIT-HER350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-ATC-2 on Hermle-Performance-C350 by installing PN-HER350-CAM-GRIP-8502 & kit KIT-HER350-ATC-2. Fasteners torqued to 158 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-HYD-3

Target Machine Model: Hermle-Performance-C350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER350-3
- PLC Hex Register Address: DB29.DBW_0x129B8
- Nominal Operating Range: 21 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-VLV-PROP-8503
- Overhaul Seal & Consumable Kit: KIT-HER350-HYD-3
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-HYD-3 at PLC register 0x129B8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-VLV-PROP-8503) with seal kit (KIT-HER350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-HYD-3 on Hermle-Performance-C350 by installing PN-HER350-VLV-PROP-8503 & kit KIT-HER350-HYD-3. Fasteners torqued to 167 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-CLN-4

Target Machine Model: Hermle-Performance-C350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER350-4
- PLC Hex Register Address: DB29.DBW_0x129C0
- Nominal Operating Range: 24 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-PMP-PLG-8504
- Overhaul Seal & Consumable Kit: KIT-HER350-CLN-4
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-CLN-4 at PLC register 0x129C0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-PMP-PLG-8504) with seal kit (KIT-HER350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-CLN-4 on Hermle-Performance-C350 by installing PN-HER350-PMP-PLG-8504 & kit KIT-HER350-CLN-4. Fasteners torqued to 176 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-XAX-5

Target Machine Model: Hermle-Performance-C350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER350-5
- PLC Hex Register Address: DB29.DBW_0x129C8
- Nominal Operating Range: 27 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-LIN-COIL-8505
- Overhaul Seal & Consumable Kit: KIT-HER350-XAX-5
- Critical Fastener Torque Value: 185 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-XAX-5 at PLC register 0x129C8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-LIN-COIL-8505) with seal kit (KIT-HER350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 185 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-XAX-5 on Hermle-Performance-C350 by installing PN-HER350-LIN-COIL-8505 & kit KIT-HER350-XAX-5. Fasteners torqued to 185 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-YAX-6

Target Machine Model: Hermle-Performance-C350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER350-6
- PLC Hex Register Address: DB29.DBW_0x129D0
- Nominal Operating Range: 30 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-BSW-NUT-8506
- Overhaul Seal & Consumable Kit: KIT-HER350-YAX-6
- Critical Fastener Torque Value: 194 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-YAX-6 at PLC register 0x129D0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-BSW-NUT-8506) with seal kit (KIT-HER350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 194 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-YAX-6 on Hermle-Performance-C350 by installing PN-HER350-BSW-NUT-8506 & kit KIT-HER350-YAX-6. Fasteners torqued to 194 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-ZAX-7

Target Machine Model: Hermle-Performance-C350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER350-7
- PLC Hex Register Address: DB29.DBW_0x129D8
- Nominal Operating Range: 33 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-CYL-SEAL-8507
- Overhaul Seal & Consumable Kit: KIT-HER350-ZAX-7
- Critical Fastener Torque Value: 203 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-ZAX-7 at PLC register 0x129D8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-CYL-SEAL-8507) with seal kit (KIT-HER350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 203 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-ZAX-7 on Hermle-Performance-C350 by installing PN-HER350-CYL-SEAL-8507 & kit KIT-HER350-ZAX-7. Fasteners torqued to 203 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-BAX-8

Target Machine Model: Hermle-Performance-C350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER350-8
- PLC Hex Register Address: DB29.DBW_0x129E0
- Nominal Operating Range: 36 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-TRQ-STAT-8508
- Overhaul Seal & Consumable Kit: KIT-HER350-BAX-8
- Critical Fastener Torque Value: 22 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-BAX-8 at PLC register 0x129E0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-TRQ-STAT-8508) with seal kit (KIT-HER350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 22 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-BAX-8 on Hermle-Performance-C350 by installing PN-HER350-TRQ-STAT-8508 & kit KIT-HER350-BAX-8. Fasteners torqued to 22 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-CAX-9

Target Machine Model: Hermle-Performance-C350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER350-9
- PLC Hex Register Address: DB29.DBW_0x129E8
- Nominal Operating Range: 39 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-BRK-DISC-8509
- Overhaul Seal & Consumable Kit: KIT-HER350-CAX-9
- Critical Fastener Torque Value: 31 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-CAX-9 at PLC register 0x129E8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-BRK-DISC-8509) with seal kit (KIT-HER350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 31 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-CAX-9 on Hermle-Performance-C350 by installing PN-HER350-BRK-DISC-8509 & kit KIT-HER350-CAX-9. Fasteners torqued to 31 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-OPT-10

Target Machine Model: Hermle-Performance-C350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER350-10
- PLC Hex Register Address: DB29.DBW_0x129F0
- Nominal Operating Range: 42 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-GLASS-RD-8510
- Overhaul Seal & Consumable Kit: KIT-HER350-OPT-10
- Critical Fastener Torque Value: 40 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-OPT-10 at PLC register 0x129F0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-GLASS-RD-8510) with seal kit (KIT-HER350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 40 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-OPT-10 on Hermle-Performance-C350 by installing PN-HER350-GLASS-RD-8510 & kit KIT-HER350-OPT-10. Fasteners torqued to 40 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-PLC-11

Target Machine Model: Hermle-Performance-C350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER350-11
- PLC Hex Register Address: DB29.DBW_0x129F8
- Nominal Operating Range: 45 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-IGBT-MOD-8511
- Overhaul Seal & Consumable Kit: KIT-HER350-PLC-11
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-PLC-11 at PLC register 0x129F8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-IGBT-MOD-8511) with seal kit (KIT-HER350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-PLC-11 on Hermle-Performance-C350 by installing PN-HER350-IGBT-MOD-8511 & kit KIT-HER350-PLC-11. Fasteners torqued to 49 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-LUB-12

Target Machine Model: Hermle-Performance-C350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER350-12
- PLC Hex Register Address: DB29.DBW_0x12A00
- Nominal Operating Range: 48 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-INJ-MET-8512
- Overhaul Seal & Consumable Kit: KIT-HER350-LUB-12
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-LUB-12 at PLC register 0x12A00.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-INJ-MET-8512) with seal kit (KIT-HER350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-LUB-12 on Hermle-Performance-C350 by installing PN-HER350-INJ-MET-8512 & kit KIT-HER350-LUB-12. Fasteners torqued to 58 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-VAC-13

Target Machine Model: Hermle-Performance-C350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER350-13
- PLC Hex Register Address: DB29.DBW_0x12A08
- Nominal Operating Range: 51 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-GASK-VIT-8513
- Overhaul Seal & Consumable Kit: KIT-HER350-VAC-13
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-VAC-13 at PLC register 0x12A08.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-GASK-VIT-8513) with seal kit (KIT-HER350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-VAC-13 on Hermle-Performance-C350 by installing PN-HER350-GASK-VIT-8513 & kit KIT-HER350-VAC-13. Fasteners torqued to 67 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER350-THM-14

Target Machine Model: Hermle-Performance-C350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C350 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER350-14
- PLC Hex Register Address: DB29.DBW_0x12A10
- Nominal Operating Range: 54 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER350-HEX-CORE-8514
- Overhaul Seal & Consumable Kit: KIT-HER350-THM-14
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C350 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER350-THM-14 at PLC register 0x12A10.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER350-HEX-CORE-8514) with seal kit (KIT-HER350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C350, technician must log: "Resolved ERR-HER350-THM-14 on Hermle-Performance-C350 by installing PN-HER350-HEX-CORE-8514 & kit KIT-HER350-THM-14. Fasteners torqued to 76 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-SPN-1

Target Machine Model: Hermle-Performance-C400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER400-1
- PLC Hex Register Address: DB30.DBW_0x12CC8
- Nominal Operating Range: 15 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-BRG-CER-8601
- Overhaul Seal & Consumable Kit: KIT-HER400-SPN-1
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-SPN-1 at PLC register 0x12CC8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-BRG-CER-8601) with seal kit (KIT-HER400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-SPN-1 on Hermle-Performance-C400 by installing PN-HER400-BRG-CER-8601 & kit KIT-HER400-SPN-1. Fasteners torqued to 99 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-ATC-2

Target Machine Model: Hermle-Performance-C400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER400-2
- PLC Hex Register Address: DB30.DBW_0x12CD0
- Nominal Operating Range: 18 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-CAM-GRIP-8602
- Overhaul Seal & Consumable Kit: KIT-HER400-ATC-2
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-ATC-2 at PLC register 0x12CD0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-CAM-GRIP-8602) with seal kit (KIT-HER400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-ATC-2 on Hermle-Performance-C400 by installing PN-HER400-CAM-GRIP-8602 & kit KIT-HER400-ATC-2. Fasteners torqued to 108 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-HYD-3

Target Machine Model: Hermle-Performance-C400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER400-3
- PLC Hex Register Address: DB30.DBW_0x12CD8
- Nominal Operating Range: 21 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-VLV-PROP-8603
- Overhaul Seal & Consumable Kit: KIT-HER400-HYD-3
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-HYD-3 at PLC register 0x12CD8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-VLV-PROP-8603) with seal kit (KIT-HER400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-HYD-3 on Hermle-Performance-C400 by installing PN-HER400-VLV-PROP-8603 & kit KIT-HER400-HYD-3. Fasteners torqued to 117 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-CLN-4

Target Machine Model: Hermle-Performance-C400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER400-4
- PLC Hex Register Address: DB30.DBW_0x12CE0
- Nominal Operating Range: 24 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-PMP-PLG-8604
- Overhaul Seal & Consumable Kit: KIT-HER400-CLN-4
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-CLN-4 at PLC register 0x12CE0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-PMP-PLG-8604) with seal kit (KIT-HER400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-CLN-4 on Hermle-Performance-C400 by installing PN-HER400-PMP-PLG-8604 & kit KIT-HER400-CLN-4. Fasteners torqued to 126 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-XAX-5

Target Machine Model: Hermle-Performance-C400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER400-5
- PLC Hex Register Address: DB30.DBW_0x12CE8
- Nominal Operating Range: 27 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-LIN-COIL-8605
- Overhaul Seal & Consumable Kit: KIT-HER400-XAX-5
- Critical Fastener Torque Value: 135 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-XAX-5 at PLC register 0x12CE8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-LIN-COIL-8605) with seal kit (KIT-HER400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 135 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-XAX-5 on Hermle-Performance-C400 by installing PN-HER400-LIN-COIL-8605 & kit KIT-HER400-XAX-5. Fasteners torqued to 135 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-YAX-6

Target Machine Model: Hermle-Performance-C400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER400-6
- PLC Hex Register Address: DB30.DBW_0x12CF0
- Nominal Operating Range: 30 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-BSW-NUT-8606
- Overhaul Seal & Consumable Kit: KIT-HER400-YAX-6
- Critical Fastener Torque Value: 144 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-YAX-6 at PLC register 0x12CF0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-BSW-NUT-8606) with seal kit (KIT-HER400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 144 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-YAX-6 on Hermle-Performance-C400 by installing PN-HER400-BSW-NUT-8606 & kit KIT-HER400-YAX-6. Fasteners torqued to 144 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-ZAX-7

Target Machine Model: Hermle-Performance-C400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER400-7
- PLC Hex Register Address: DB30.DBW_0x12CF8
- Nominal Operating Range: 33 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-CYL-SEAL-8607
- Overhaul Seal & Consumable Kit: KIT-HER400-ZAX-7
- Critical Fastener Torque Value: 153 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-ZAX-7 at PLC register 0x12CF8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-CYL-SEAL-8607) with seal kit (KIT-HER400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 153 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-ZAX-7 on Hermle-Performance-C400 by installing PN-HER400-CYL-SEAL-8607 & kit KIT-HER400-ZAX-7. Fasteners torqued to 153 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-BAX-8

Target Machine Model: Hermle-Performance-C400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER400-8
- PLC Hex Register Address: DB30.DBW_0x12D00
- Nominal Operating Range: 36 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-TRQ-STAT-8608
- Overhaul Seal & Consumable Kit: KIT-HER400-BAX-8
- Critical Fastener Torque Value: 162 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-BAX-8 at PLC register 0x12D00.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-TRQ-STAT-8608) with seal kit (KIT-HER400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 162 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-BAX-8 on Hermle-Performance-C400 by installing PN-HER400-TRQ-STAT-8608 & kit KIT-HER400-BAX-8. Fasteners torqued to 162 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-CAX-9

Target Machine Model: Hermle-Performance-C400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER400-9
- PLC Hex Register Address: DB30.DBW_0x12D08
- Nominal Operating Range: 39 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-BRK-DISC-8609
- Overhaul Seal & Consumable Kit: KIT-HER400-CAX-9
- Critical Fastener Torque Value: 171 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-CAX-9 at PLC register 0x12D08.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-BRK-DISC-8609) with seal kit (KIT-HER400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 171 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-CAX-9 on Hermle-Performance-C400 by installing PN-HER400-BRK-DISC-8609 & kit KIT-HER400-CAX-9. Fasteners torqued to 171 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-OPT-10

Target Machine Model: Hermle-Performance-C400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER400-10
- PLC Hex Register Address: DB30.DBW_0x12D10
- Nominal Operating Range: 42 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-GLASS-RD-8610
- Overhaul Seal & Consumable Kit: KIT-HER400-OPT-10
- Critical Fastener Torque Value: 180 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-OPT-10 at PLC register 0x12D10.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-GLASS-RD-8610) with seal kit (KIT-HER400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 180 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-OPT-10 on Hermle-Performance-C400 by installing PN-HER400-GLASS-RD-8610 & kit KIT-HER400-OPT-10. Fasteners torqued to 180 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-PLC-11

Target Machine Model: Hermle-Performance-C400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER400-11
- PLC Hex Register Address: DB30.DBW_0x12D18
- Nominal Operating Range: 45 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-IGBT-MOD-8611
- Overhaul Seal & Consumable Kit: KIT-HER400-PLC-11
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-PLC-11 at PLC register 0x12D18.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-IGBT-MOD-8611) with seal kit (KIT-HER400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-PLC-11 on Hermle-Performance-C400 by installing PN-HER400-IGBT-MOD-8611 & kit KIT-HER400-PLC-11. Fasteners torqued to 189 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-LUB-12

Target Machine Model: Hermle-Performance-C400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER400-12
- PLC Hex Register Address: DB30.DBW_0x12D20
- Nominal Operating Range: 48 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-INJ-MET-8612
- Overhaul Seal & Consumable Kit: KIT-HER400-LUB-12
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-LUB-12 at PLC register 0x12D20.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-INJ-MET-8612) with seal kit (KIT-HER400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-LUB-12 on Hermle-Performance-C400 by installing PN-HER400-INJ-MET-8612 & kit KIT-HER400-LUB-12. Fasteners torqued to 198 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-VAC-13

Target Machine Model: Hermle-Performance-C400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER400-13
- PLC Hex Register Address: DB30.DBW_0x12D28
- Nominal Operating Range: 51 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-GASK-VIT-8613
- Overhaul Seal & Consumable Kit: KIT-HER400-VAC-13
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-VAC-13 at PLC register 0x12D28.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-GASK-VIT-8613) with seal kit (KIT-HER400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-VAC-13 on Hermle-Performance-C400 by installing PN-HER400-GASK-VIT-8613 & kit KIT-HER400-VAC-13. Fasteners torqued to 207 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER400-THM-14

Target Machine Model: Hermle-Performance-C400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C400 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER400-14
- PLC Hex Register Address: DB30.DBW_0x12D30
- Nominal Operating Range: 54 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER400-HEX-CORE-8614
- Overhaul Seal & Consumable Kit: KIT-HER400-THM-14
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C400 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER400-THM-14 at PLC register 0x12D30.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER400-HEX-CORE-8614) with seal kit (KIT-HER400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C400, technician must log: "Resolved ERR-HER400-THM-14 on Hermle-Performance-C400 by installing PN-HER400-HEX-CORE-8614 & kit KIT-HER400-THM-14. Fasteners torqued to 26 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-SPN-1

Target Machine Model: Hermle-Performance-C450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER450-1
- PLC Hex Register Address: DB31.DBW_0x12FE8
- Nominal Operating Range: 15 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-BRG-CER-8701
- Overhaul Seal & Consumable Kit: KIT-HER450-SPN-1
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-SPN-1 at PLC register 0x12FE8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-BRG-CER-8701) with seal kit (KIT-HER450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-SPN-1 on Hermle-Performance-C450 by installing PN-HER450-BRG-CER-8701 & kit KIT-HER450-SPN-1. Fasteners torqued to 49 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-ATC-2

Target Machine Model: Hermle-Performance-C450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER450-2
- PLC Hex Register Address: DB31.DBW_0x12FF0
- Nominal Operating Range: 18 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-CAM-GRIP-8702
- Overhaul Seal & Consumable Kit: KIT-HER450-ATC-2
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-ATC-2 at PLC register 0x12FF0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-CAM-GRIP-8702) with seal kit (KIT-HER450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-ATC-2 on Hermle-Performance-C450 by installing PN-HER450-CAM-GRIP-8702 & kit KIT-HER450-ATC-2. Fasteners torqued to 58 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-HYD-3

Target Machine Model: Hermle-Performance-C450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER450-3
- PLC Hex Register Address: DB31.DBW_0x12FF8
- Nominal Operating Range: 21 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-VLV-PROP-8703
- Overhaul Seal & Consumable Kit: KIT-HER450-HYD-3
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-HYD-3 at PLC register 0x12FF8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-VLV-PROP-8703) with seal kit (KIT-HER450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-HYD-3 on Hermle-Performance-C450 by installing PN-HER450-VLV-PROP-8703 & kit KIT-HER450-HYD-3. Fasteners torqued to 67 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-CLN-4

Target Machine Model: Hermle-Performance-C450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER450-4
- PLC Hex Register Address: DB31.DBW_0x13000
- Nominal Operating Range: 24 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-PMP-PLG-8704
- Overhaul Seal & Consumable Kit: KIT-HER450-CLN-4
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-CLN-4 at PLC register 0x13000.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-PMP-PLG-8704) with seal kit (KIT-HER450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-CLN-4 on Hermle-Performance-C450 by installing PN-HER450-PMP-PLG-8704 & kit KIT-HER450-CLN-4. Fasteners torqued to 76 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-XAX-5

Target Machine Model: Hermle-Performance-C450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER450-5
- PLC Hex Register Address: DB31.DBW_0x13008
- Nominal Operating Range: 27 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-LIN-COIL-8705
- Overhaul Seal & Consumable Kit: KIT-HER450-XAX-5
- Critical Fastener Torque Value: 85 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-XAX-5 at PLC register 0x13008.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-LIN-COIL-8705) with seal kit (KIT-HER450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 85 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-XAX-5 on Hermle-Performance-C450 by installing PN-HER450-LIN-COIL-8705 & kit KIT-HER450-XAX-5. Fasteners torqued to 85 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-YAX-6

Target Machine Model: Hermle-Performance-C450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER450-6
- PLC Hex Register Address: DB31.DBW_0x13010
- Nominal Operating Range: 30 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-BSW-NUT-8706
- Overhaul Seal & Consumable Kit: KIT-HER450-YAX-6
- Critical Fastener Torque Value: 94 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-YAX-6 at PLC register 0x13010.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-BSW-NUT-8706) with seal kit (KIT-HER450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 94 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-YAX-6 on Hermle-Performance-C450 by installing PN-HER450-BSW-NUT-8706 & kit KIT-HER450-YAX-6. Fasteners torqued to 94 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-ZAX-7

Target Machine Model: Hermle-Performance-C450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER450-7
- PLC Hex Register Address: DB31.DBW_0x13018
- Nominal Operating Range: 33 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-CYL-SEAL-8707
- Overhaul Seal & Consumable Kit: KIT-HER450-ZAX-7
- Critical Fastener Torque Value: 103 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-ZAX-7 at PLC register 0x13018.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-CYL-SEAL-8707) with seal kit (KIT-HER450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 103 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-ZAX-7 on Hermle-Performance-C450 by installing PN-HER450-CYL-SEAL-8707 & kit KIT-HER450-ZAX-7. Fasteners torqued to 103 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-BAX-8

Target Machine Model: Hermle-Performance-C450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER450-8
- PLC Hex Register Address: DB31.DBW_0x13020
- Nominal Operating Range: 36 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-TRQ-STAT-8708
- Overhaul Seal & Consumable Kit: KIT-HER450-BAX-8
- Critical Fastener Torque Value: 112 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-BAX-8 at PLC register 0x13020.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-TRQ-STAT-8708) with seal kit (KIT-HER450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 112 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-BAX-8 on Hermle-Performance-C450 by installing PN-HER450-TRQ-STAT-8708 & kit KIT-HER450-BAX-8. Fasteners torqued to 112 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-CAX-9

Target Machine Model: Hermle-Performance-C450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER450-9
- PLC Hex Register Address: DB31.DBW_0x13028
- Nominal Operating Range: 39 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-BRK-DISC-8709
- Overhaul Seal & Consumable Kit: KIT-HER450-CAX-9
- Critical Fastener Torque Value: 121 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-CAX-9 at PLC register 0x13028.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-BRK-DISC-8709) with seal kit (KIT-HER450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 121 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-CAX-9 on Hermle-Performance-C450 by installing PN-HER450-BRK-DISC-8709 & kit KIT-HER450-CAX-9. Fasteners torqued to 121 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-OPT-10

Target Machine Model: Hermle-Performance-C450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER450-10
- PLC Hex Register Address: DB31.DBW_0x13030
- Nominal Operating Range: 42 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-GLASS-RD-8710
- Overhaul Seal & Consumable Kit: KIT-HER450-OPT-10
- Critical Fastener Torque Value: 130 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-OPT-10 at PLC register 0x13030.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-GLASS-RD-8710) with seal kit (KIT-HER450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 130 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-OPT-10 on Hermle-Performance-C450 by installing PN-HER450-GLASS-RD-8710 & kit KIT-HER450-OPT-10. Fasteners torqued to 130 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-PLC-11

Target Machine Model: Hermle-Performance-C450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER450-11
- PLC Hex Register Address: DB31.DBW_0x13038
- Nominal Operating Range: 45 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-IGBT-MOD-8711
- Overhaul Seal & Consumable Kit: KIT-HER450-PLC-11
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-PLC-11 at PLC register 0x13038.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-IGBT-MOD-8711) with seal kit (KIT-HER450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-PLC-11 on Hermle-Performance-C450 by installing PN-HER450-IGBT-MOD-8711 & kit KIT-HER450-PLC-11. Fasteners torqued to 139 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-LUB-12

Target Machine Model: Hermle-Performance-C450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER450-12
- PLC Hex Register Address: DB31.DBW_0x13040
- Nominal Operating Range: 48 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-INJ-MET-8712
- Overhaul Seal & Consumable Kit: KIT-HER450-LUB-12
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-LUB-12 at PLC register 0x13040.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-INJ-MET-8712) with seal kit (KIT-HER450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-LUB-12 on Hermle-Performance-C450 by installing PN-HER450-INJ-MET-8712 & kit KIT-HER450-LUB-12. Fasteners torqued to 148 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-VAC-13

Target Machine Model: Hermle-Performance-C450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER450-13
- PLC Hex Register Address: DB31.DBW_0x13048
- Nominal Operating Range: 51 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-GASK-VIT-8713
- Overhaul Seal & Consumable Kit: KIT-HER450-VAC-13
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-VAC-13 at PLC register 0x13048.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-GASK-VIT-8713) with seal kit (KIT-HER450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-VAC-13 on Hermle-Performance-C450 by installing PN-HER450-GASK-VIT-8713 & kit KIT-HER450-VAC-13. Fasteners torqued to 157 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER450-THM-14

Target Machine Model: Hermle-Performance-C450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C450 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER450-14
- PLC Hex Register Address: DB31.DBW_0x13050
- Nominal Operating Range: 54 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER450-HEX-CORE-8714
- Overhaul Seal & Consumable Kit: KIT-HER450-THM-14
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C450 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER450-THM-14 at PLC register 0x13050.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER450-HEX-CORE-8714) with seal kit (KIT-HER450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C450, technician must log: "Resolved ERR-HER450-THM-14 on Hermle-Performance-C450 by installing PN-HER450-HEX-CORE-8714 & kit KIT-HER450-THM-14. Fasteners torqued to 166 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-SPN-1

Target Machine Model: Hermle-Performance-C500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER500-1
- PLC Hex Register Address: DB32.DBW_0x13308
- Nominal Operating Range: 15 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-BRG-CER-8801
- Overhaul Seal & Consumable Kit: KIT-HER500-SPN-1
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-SPN-1 at PLC register 0x13308.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-BRG-CER-8801) with seal kit (KIT-HER500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-SPN-1 on Hermle-Performance-C500 by installing PN-HER500-BRG-CER-8801 & kit KIT-HER500-SPN-1. Fasteners torqued to 189 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-ATC-2

Target Machine Model: Hermle-Performance-C500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER500-2
- PLC Hex Register Address: DB32.DBW_0x13310
- Nominal Operating Range: 18 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-CAM-GRIP-8802
- Overhaul Seal & Consumable Kit: KIT-HER500-ATC-2
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-ATC-2 at PLC register 0x13310.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-CAM-GRIP-8802) with seal kit (KIT-HER500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-ATC-2 on Hermle-Performance-C500 by installing PN-HER500-CAM-GRIP-8802 & kit KIT-HER500-ATC-2. Fasteners torqued to 198 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-HYD-3

Target Machine Model: Hermle-Performance-C500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER500-3
- PLC Hex Register Address: DB32.DBW_0x13318
- Nominal Operating Range: 21 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-VLV-PROP-8803
- Overhaul Seal & Consumable Kit: KIT-HER500-HYD-3
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-HYD-3 at PLC register 0x13318.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-VLV-PROP-8803) with seal kit (KIT-HER500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-HYD-3 on Hermle-Performance-C500 by installing PN-HER500-VLV-PROP-8803 & kit KIT-HER500-HYD-3. Fasteners torqued to 207 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-CLN-4

Target Machine Model: Hermle-Performance-C500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER500-4
- PLC Hex Register Address: DB32.DBW_0x13320
- Nominal Operating Range: 24 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-PMP-PLG-8804
- Overhaul Seal & Consumable Kit: KIT-HER500-CLN-4
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-CLN-4 at PLC register 0x13320.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-PMP-PLG-8804) with seal kit (KIT-HER500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-CLN-4 on Hermle-Performance-C500 by installing PN-HER500-PMP-PLG-8804 & kit KIT-HER500-CLN-4. Fasteners torqued to 26 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-XAX-5

Target Machine Model: Hermle-Performance-C500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER500-5
- PLC Hex Register Address: DB32.DBW_0x13328
- Nominal Operating Range: 27 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-LIN-COIL-8805
- Overhaul Seal & Consumable Kit: KIT-HER500-XAX-5
- Critical Fastener Torque Value: 35 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-XAX-5 at PLC register 0x13328.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-LIN-COIL-8805) with seal kit (KIT-HER500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 35 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-XAX-5 on Hermle-Performance-C500 by installing PN-HER500-LIN-COIL-8805 & kit KIT-HER500-XAX-5. Fasteners torqued to 35 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-YAX-6

Target Machine Model: Hermle-Performance-C500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER500-6
- PLC Hex Register Address: DB32.DBW_0x13330
- Nominal Operating Range: 30 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-BSW-NUT-8806
- Overhaul Seal & Consumable Kit: KIT-HER500-YAX-6
- Critical Fastener Torque Value: 44 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-YAX-6 at PLC register 0x13330.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-BSW-NUT-8806) with seal kit (KIT-HER500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 44 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-YAX-6 on Hermle-Performance-C500 by installing PN-HER500-BSW-NUT-8806 & kit KIT-HER500-YAX-6. Fasteners torqued to 44 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-ZAX-7

Target Machine Model: Hermle-Performance-C500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER500-7
- PLC Hex Register Address: DB32.DBW_0x13338
- Nominal Operating Range: 33 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-CYL-SEAL-8807
- Overhaul Seal & Consumable Kit: KIT-HER500-ZAX-7
- Critical Fastener Torque Value: 53 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-ZAX-7 at PLC register 0x13338.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-CYL-SEAL-8807) with seal kit (KIT-HER500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 53 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-ZAX-7 on Hermle-Performance-C500 by installing PN-HER500-CYL-SEAL-8807 & kit KIT-HER500-ZAX-7. Fasteners torqued to 53 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-BAX-8

Target Machine Model: Hermle-Performance-C500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER500-8
- PLC Hex Register Address: DB32.DBW_0x13340
- Nominal Operating Range: 36 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-TRQ-STAT-8808
- Overhaul Seal & Consumable Kit: KIT-HER500-BAX-8
- Critical Fastener Torque Value: 62 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-BAX-8 at PLC register 0x13340.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-TRQ-STAT-8808) with seal kit (KIT-HER500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 62 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-BAX-8 on Hermle-Performance-C500 by installing PN-HER500-TRQ-STAT-8808 & kit KIT-HER500-BAX-8. Fasteners torqued to 62 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-CAX-9

Target Machine Model: Hermle-Performance-C500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER500-9
- PLC Hex Register Address: DB32.DBW_0x13348
- Nominal Operating Range: 39 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-BRK-DISC-8809
- Overhaul Seal & Consumable Kit: KIT-HER500-CAX-9
- Critical Fastener Torque Value: 71 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-CAX-9 at PLC register 0x13348.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-BRK-DISC-8809) with seal kit (KIT-HER500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 71 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-CAX-9 on Hermle-Performance-C500 by installing PN-HER500-BRK-DISC-8809 & kit KIT-HER500-CAX-9. Fasteners torqued to 71 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-OPT-10

Target Machine Model: Hermle-Performance-C500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER500-10
- PLC Hex Register Address: DB32.DBW_0x13350
- Nominal Operating Range: 42 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-GLASS-RD-8810
- Overhaul Seal & Consumable Kit: KIT-HER500-OPT-10
- Critical Fastener Torque Value: 80 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-OPT-10 at PLC register 0x13350.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-GLASS-RD-8810) with seal kit (KIT-HER500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 80 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-OPT-10 on Hermle-Performance-C500 by installing PN-HER500-GLASS-RD-8810 & kit KIT-HER500-OPT-10. Fasteners torqued to 80 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-PLC-11

Target Machine Model: Hermle-Performance-C500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER500-11
- PLC Hex Register Address: DB32.DBW_0x13358
- Nominal Operating Range: 45 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-IGBT-MOD-8811
- Overhaul Seal & Consumable Kit: KIT-HER500-PLC-11
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-PLC-11 at PLC register 0x13358.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-IGBT-MOD-8811) with seal kit (KIT-HER500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-PLC-11 on Hermle-Performance-C500 by installing PN-HER500-IGBT-MOD-8811 & kit KIT-HER500-PLC-11. Fasteners torqued to 89 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-LUB-12

Target Machine Model: Hermle-Performance-C500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER500-12
- PLC Hex Register Address: DB32.DBW_0x13360
- Nominal Operating Range: 48 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-INJ-MET-8812
- Overhaul Seal & Consumable Kit: KIT-HER500-LUB-12
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-LUB-12 at PLC register 0x13360.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-INJ-MET-8812) with seal kit (KIT-HER500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-LUB-12 on Hermle-Performance-C500 by installing PN-HER500-INJ-MET-8812 & kit KIT-HER500-LUB-12. Fasteners torqued to 98 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-VAC-13

Target Machine Model: Hermle-Performance-C500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER500-13
- PLC Hex Register Address: DB32.DBW_0x13368
- Nominal Operating Range: 51 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-GASK-VIT-8813
- Overhaul Seal & Consumable Kit: KIT-HER500-VAC-13
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-VAC-13 at PLC register 0x13368.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-GASK-VIT-8813) with seal kit (KIT-HER500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-VAC-13 on Hermle-Performance-C500 by installing PN-HER500-GASK-VIT-8813 & kit KIT-HER500-VAC-13. Fasteners torqued to 107 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER500-THM-14

Target Machine Model: Hermle-Performance-C500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C500 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER500-14
- PLC Hex Register Address: DB32.DBW_0x13370
- Nominal Operating Range: 54 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER500-HEX-CORE-8814
- Overhaul Seal & Consumable Kit: KIT-HER500-THM-14
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C500 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER500-THM-14 at PLC register 0x13370.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER500-HEX-CORE-8814) with seal kit (KIT-HER500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C500, technician must log: "Resolved ERR-HER500-THM-14 on Hermle-Performance-C500 by installing PN-HER500-HEX-CORE-8814 & kit KIT-HER500-THM-14. Fasteners torqued to 116 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-SPN-1

Target Machine Model: Hermle-Performance-C550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER550-1
- PLC Hex Register Address: DB33.DBW_0x13628
- Nominal Operating Range: 15 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-BRG-CER-8901
- Overhaul Seal & Consumable Kit: KIT-HER550-SPN-1
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-SPN-1 at PLC register 0x13628.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-BRG-CER-8901) with seal kit (KIT-HER550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-SPN-1 on Hermle-Performance-C550 by installing PN-HER550-BRG-CER-8901 & kit KIT-HER550-SPN-1. Fasteners torqued to 139 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-ATC-2

Target Machine Model: Hermle-Performance-C550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER550-2
- PLC Hex Register Address: DB33.DBW_0x13630
- Nominal Operating Range: 18 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-CAM-GRIP-8902
- Overhaul Seal & Consumable Kit: KIT-HER550-ATC-2
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-ATC-2 at PLC register 0x13630.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-CAM-GRIP-8902) with seal kit (KIT-HER550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-ATC-2 on Hermle-Performance-C550 by installing PN-HER550-CAM-GRIP-8902 & kit KIT-HER550-ATC-2. Fasteners torqued to 148 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-HYD-3

Target Machine Model: Hermle-Performance-C550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER550-3
- PLC Hex Register Address: DB33.DBW_0x13638
- Nominal Operating Range: 21 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-VLV-PROP-8903
- Overhaul Seal & Consumable Kit: KIT-HER550-HYD-3
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-HYD-3 at PLC register 0x13638.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-VLV-PROP-8903) with seal kit (KIT-HER550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-HYD-3 on Hermle-Performance-C550 by installing PN-HER550-VLV-PROP-8903 & kit KIT-HER550-HYD-3. Fasteners torqued to 157 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-CLN-4

Target Machine Model: Hermle-Performance-C550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER550-4
- PLC Hex Register Address: DB33.DBW_0x13640
- Nominal Operating Range: 24 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-PMP-PLG-8904
- Overhaul Seal & Consumable Kit: KIT-HER550-CLN-4
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-CLN-4 at PLC register 0x13640.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-PMP-PLG-8904) with seal kit (KIT-HER550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-CLN-4 on Hermle-Performance-C550 by installing PN-HER550-PMP-PLG-8904 & kit KIT-HER550-CLN-4. Fasteners torqued to 166 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-XAX-5

Target Machine Model: Hermle-Performance-C550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER550-5
- PLC Hex Register Address: DB33.DBW_0x13648
- Nominal Operating Range: 27 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-LIN-COIL-8905
- Overhaul Seal & Consumable Kit: KIT-HER550-XAX-5
- Critical Fastener Torque Value: 175 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-XAX-5 at PLC register 0x13648.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-LIN-COIL-8905) with seal kit (KIT-HER550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 175 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-XAX-5 on Hermle-Performance-C550 by installing PN-HER550-LIN-COIL-8905 & kit KIT-HER550-XAX-5. Fasteners torqued to 175 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-YAX-6

Target Machine Model: Hermle-Performance-C550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER550-6
- PLC Hex Register Address: DB33.DBW_0x13650
- Nominal Operating Range: 30 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-BSW-NUT-8906
- Overhaul Seal & Consumable Kit: KIT-HER550-YAX-6
- Critical Fastener Torque Value: 184 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-YAX-6 at PLC register 0x13650.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-BSW-NUT-8906) with seal kit (KIT-HER550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 184 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-YAX-6 on Hermle-Performance-C550 by installing PN-HER550-BSW-NUT-8906 & kit KIT-HER550-YAX-6. Fasteners torqued to 184 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-ZAX-7

Target Machine Model: Hermle-Performance-C550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER550-7
- PLC Hex Register Address: DB33.DBW_0x13658
- Nominal Operating Range: 33 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-CYL-SEAL-8907
- Overhaul Seal & Consumable Kit: KIT-HER550-ZAX-7
- Critical Fastener Torque Value: 193 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-ZAX-7 at PLC register 0x13658.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-CYL-SEAL-8907) with seal kit (KIT-HER550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 193 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-ZAX-7 on Hermle-Performance-C550 by installing PN-HER550-CYL-SEAL-8907 & kit KIT-HER550-ZAX-7. Fasteners torqued to 193 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-BAX-8

Target Machine Model: Hermle-Performance-C550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER550-8
- PLC Hex Register Address: DB33.DBW_0x13660
- Nominal Operating Range: 36 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-TRQ-STAT-8908
- Overhaul Seal & Consumable Kit: KIT-HER550-BAX-8
- Critical Fastener Torque Value: 202 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-BAX-8 at PLC register 0x13660.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-TRQ-STAT-8908) with seal kit (KIT-HER550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 202 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-BAX-8 on Hermle-Performance-C550 by installing PN-HER550-TRQ-STAT-8908 & kit KIT-HER550-BAX-8. Fasteners torqued to 202 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-CAX-9

Target Machine Model: Hermle-Performance-C550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER550-9
- PLC Hex Register Address: DB33.DBW_0x13668
- Nominal Operating Range: 39 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-BRK-DISC-8909
- Overhaul Seal & Consumable Kit: KIT-HER550-CAX-9
- Critical Fastener Torque Value: 21 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-CAX-9 at PLC register 0x13668.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-BRK-DISC-8909) with seal kit (KIT-HER550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 21 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-CAX-9 on Hermle-Performance-C550 by installing PN-HER550-BRK-DISC-8909 & kit KIT-HER550-CAX-9. Fasteners torqued to 21 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-OPT-10

Target Machine Model: Hermle-Performance-C550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER550-10
- PLC Hex Register Address: DB33.DBW_0x13670
- Nominal Operating Range: 42 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-GLASS-RD-8910
- Overhaul Seal & Consumable Kit: KIT-HER550-OPT-10
- Critical Fastener Torque Value: 30 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-OPT-10 at PLC register 0x13670.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-GLASS-RD-8910) with seal kit (KIT-HER550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 30 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-OPT-10 on Hermle-Performance-C550 by installing PN-HER550-GLASS-RD-8910 & kit KIT-HER550-OPT-10. Fasteners torqued to 30 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-PLC-11

Target Machine Model: Hermle-Performance-C550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER550-11
- PLC Hex Register Address: DB33.DBW_0x13678
- Nominal Operating Range: 45 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-IGBT-MOD-8911
- Overhaul Seal & Consumable Kit: KIT-HER550-PLC-11
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-PLC-11 at PLC register 0x13678.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-IGBT-MOD-8911) with seal kit (KIT-HER550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-PLC-11 on Hermle-Performance-C550 by installing PN-HER550-IGBT-MOD-8911 & kit KIT-HER550-PLC-11. Fasteners torqued to 39 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-LUB-12

Target Machine Model: Hermle-Performance-C550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER550-12
- PLC Hex Register Address: DB33.DBW_0x13680
- Nominal Operating Range: 48 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-INJ-MET-8912
- Overhaul Seal & Consumable Kit: KIT-HER550-LUB-12
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-LUB-12 at PLC register 0x13680.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-INJ-MET-8912) with seal kit (KIT-HER550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-LUB-12 on Hermle-Performance-C550 by installing PN-HER550-INJ-MET-8912 & kit KIT-HER550-LUB-12. Fasteners torqued to 48 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-VAC-13

Target Machine Model: Hermle-Performance-C550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER550-13
- PLC Hex Register Address: DB33.DBW_0x13688
- Nominal Operating Range: 51 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-GASK-VIT-8913
- Overhaul Seal & Consumable Kit: KIT-HER550-VAC-13
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-VAC-13 at PLC register 0x13688.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-GASK-VIT-8913) with seal kit (KIT-HER550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-VAC-13 on Hermle-Performance-C550 by installing PN-HER550-GASK-VIT-8913 & kit KIT-HER550-VAC-13. Fasteners torqued to 57 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER550-THM-14

Target Machine Model: Hermle-Performance-C550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C550 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER550-14
- PLC Hex Register Address: DB33.DBW_0x13690
- Nominal Operating Range: 54 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER550-HEX-CORE-8914
- Overhaul Seal & Consumable Kit: KIT-HER550-THM-14
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C550 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER550-THM-14 at PLC register 0x13690.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER550-HEX-CORE-8914) with seal kit (KIT-HER550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C550, technician must log: "Resolved ERR-HER550-THM-14 on Hermle-Performance-C550 by installing PN-HER550-HEX-CORE-8914 & kit KIT-HER550-THM-14. Fasteners torqued to 66 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-SPN-1

Target Machine Model: Hermle-Performance-C600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: PT-SPN-HER600-1
- PLC Hex Register Address: DB34.DBW_0x13948
- Nominal Operating Range: 15 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-BRG-CER-9001
- Overhaul Seal & Consumable Kit: KIT-HER600-SPN-1
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-SPN-1 at PLC register 0x13948.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-BRG-CER-9001) with seal kit (KIT-HER600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-SPN-1 on Hermle-Performance-C600 by installing PN-HER600-BRG-CER-9001 & kit KIT-HER600-SPN-1. Fasteners torqued to 89 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-ATC-2

Target Machine Model: Hermle-Performance-C600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: PX-ATC-HER600-2
- PLC Hex Register Address: DB34.DBW_0x13950
- Nominal Operating Range: 18 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-CAM-GRIP-9002
- Overhaul Seal & Consumable Kit: KIT-HER600-ATC-2
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-ATC-2 at PLC register 0x13950.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-CAM-GRIP-9002) with seal kit (KIT-HER600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-ATC-2 on Hermle-Performance-C600 by installing PN-HER600-CAM-GRIP-9002 & kit KIT-HER600-ATC-2. Fasteners torqued to 98 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-HYD-3

Target Machine Model: Hermle-Performance-C600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-HYD-HER600-3
- PLC Hex Register Address: DB34.DBW_0x13958
- Nominal Operating Range: 21 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-VLV-PROP-9003
- Overhaul Seal & Consumable Kit: KIT-HER600-HYD-3
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-HYD-3 at PLC register 0x13958.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-VLV-PROP-9003) with seal kit (KIT-HER600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-HYD-3 on Hermle-Performance-C600 by installing PN-HER600-VLV-PROP-9003 & kit KIT-HER600-HYD-3. Fasteners torqued to 107 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-CLN-4

Target Machine Model: Hermle-Performance-C600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-CLN-HER600-4
- PLC Hex Register Address: DB34.DBW_0x13960
- Nominal Operating Range: 24 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-PMP-PLG-9004
- Overhaul Seal & Consumable Kit: KIT-HER600-CLN-4
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-CLN-4 at PLC register 0x13960.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-PMP-PLG-9004) with seal kit (KIT-HER600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-CLN-4 on Hermle-Performance-C600 by installing PN-HER600-PMP-PLG-9004 & kit KIT-HER600-CLN-4. Fasteners torqued to 116 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-XAX-5

Target Machine Model: Hermle-Performance-C600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-XAX-HER600-5
- PLC Hex Register Address: DB34.DBW_0x13968
- Nominal Operating Range: 27 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-LIN-COIL-9005
- Overhaul Seal & Consumable Kit: KIT-HER600-XAX-5
- Critical Fastener Torque Value: 125 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-XAX-5 at PLC register 0x13968.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-LIN-COIL-9005) with seal kit (KIT-HER600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 125 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-XAX-5 on Hermle-Performance-C600 by installing PN-HER600-LIN-COIL-9005 & kit KIT-HER600-XAX-5. Fasteners torqued to 125 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-YAX-6

Target Machine Model: Hermle-Performance-C600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-YAX-HER600-6
- PLC Hex Register Address: DB34.DBW_0x13970
- Nominal Operating Range: 30 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-BSW-NUT-9006
- Overhaul Seal & Consumable Kit: KIT-HER600-YAX-6
- Critical Fastener Torque Value: 134 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-YAX-6 at PLC register 0x13970.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-BSW-NUT-9006) with seal kit (KIT-HER600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 134 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-YAX-6 on Hermle-Performance-C600 by installing PN-HER600-BSW-NUT-9006 & kit KIT-HER600-YAX-6. Fasteners torqued to 134 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-ZAX-7

Target Machine Model: Hermle-Performance-C600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-ZAX-HER600-7
- PLC Hex Register Address: DB34.DBW_0x13978
- Nominal Operating Range: 33 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-CYL-SEAL-9007
- Overhaul Seal & Consumable Kit: KIT-HER600-ZAX-7
- Critical Fastener Torque Value: 143 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-ZAX-7 at PLC register 0x13978.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-CYL-SEAL-9007) with seal kit (KIT-HER600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 143 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-ZAX-7 on Hermle-Performance-C600 by installing PN-HER600-CYL-SEAL-9007 & kit KIT-HER600-ZAX-7. Fasteners torqued to 143 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-BAX-8

Target Machine Model: Hermle-Performance-C600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: EN-BAX-HER600-8
- PLC Hex Register Address: DB34.DBW_0x13980
- Nominal Operating Range: 36 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-TRQ-STAT-9008
- Overhaul Seal & Consumable Kit: KIT-HER600-BAX-8
- Critical Fastener Torque Value: 152 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-BAX-8 at PLC register 0x13980.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-TRQ-STAT-9008) with seal kit (KIT-HER600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 152 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-BAX-8 on Hermle-Performance-C600 by installing PN-HER600-TRQ-STAT-9008 & kit KIT-HER600-BAX-8. Fasteners torqued to 152 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-CAX-9

Target Machine Model: Hermle-Performance-C600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: PR-CAX-HER600-9
- PLC Hex Register Address: DB34.DBW_0x13988
- Nominal Operating Range: 39 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-BRK-DISC-9009
- Overhaul Seal & Consumable Kit: KIT-HER600-CAX-9
- Critical Fastener Torque Value: 161 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-CAX-9 at PLC register 0x13988.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-BRK-DISC-9009) with seal kit (KIT-HER600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 161 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-CAX-9 on Hermle-Performance-C600 by installing PN-HER600-BRK-DISC-9009 & kit KIT-HER600-CAX-9. Fasteners torqued to 161 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-OPT-10

Target Machine Model: Hermle-Performance-C600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: OP-SCL-HER600-10
- PLC Hex Register Address: DB34.DBW_0x13990
- Nominal Operating Range: 42 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-GLASS-RD-9010
- Overhaul Seal & Consumable Kit: KIT-HER600-OPT-10
- Critical Fastener Torque Value: 170 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-OPT-10 at PLC register 0x13990.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-GLASS-RD-9010) with seal kit (KIT-HER600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 170 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-OPT-10 on Hermle-Performance-C600 by installing PN-HER600-GLASS-RD-9010 & kit KIT-HER600-OPT-10. Fasteners torqued to 170 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-PLC-11

Target Machine Model: Hermle-Performance-C600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: VT-DRV-HER600-11
- PLC Hex Register Address: DB34.DBW_0x13998
- Nominal Operating Range: 45 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-IGBT-MOD-9011
- Overhaul Seal & Consumable Kit: KIT-HER600-PLC-11
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-PLC-11 at PLC register 0x13998.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-IGBT-MOD-9011) with seal kit (KIT-HER600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-PLC-11 on Hermle-Performance-C600 by installing PN-HER600-IGBT-MOD-9011 & kit KIT-HER600-PLC-11. Fasteners torqued to 179 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-LUB-12

Target Machine Model: Hermle-Performance-C600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: FL-LUB-HER600-12
- PLC Hex Register Address: DB34.DBW_0x139A0
- Nominal Operating Range: 48 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-INJ-MET-9012
- Overhaul Seal & Consumable Kit: KIT-HER600-LUB-12
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-LUB-12 at PLC register 0x139A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-INJ-MET-9012) with seal kit (KIT-HER600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-LUB-12 on Hermle-Performance-C600 by installing PN-HER600-INJ-MET-9012 & kit KIT-HER600-LUB-12. Fasteners torqued to 188 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-VAC-13

Target Machine Model: Hermle-Performance-C600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: VC-MNF-HER600-13
- PLC Hex Register Address: DB34.DBW_0x139A8
- Nominal Operating Range: 51 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-GASK-VIT-9013
- Overhaul Seal & Consumable Kit: KIT-HER600-VAC-13
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-VAC-13 at PLC register 0x139A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-GASK-VIT-9013) with seal kit (KIT-HER600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-VAC-13 on Hermle-Performance-C600 by installing PN-HER600-GASK-VIT-9013 & kit KIT-HER600-VAC-13. Fasteners torqued to 197 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-HER600-THM-14

Target Machine Model: Hermle-Performance-C600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Hermle-Performance-C600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Hermle-Performance-C600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Hermle-Performance-C600 (Heidenhain TNC7)
- Telemetry Sensor ID: TM-CHL-HER600-14
- PLC Hex Register Address: DB34.DBW_0x139B0
- Nominal Operating Range: 54 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-HER600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-HER600-HEX-CORE-9014
- Overhaul Seal & Consumable Kit: KIT-HER600-THM-14
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Hermle-Performance-C600 (Heidenhain TNC7) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC7 and verify fault latch ERR-HER600-THM-14 at PLC register 0x139B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Hermle-Performance-C600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-HER600-HEX-CORE-9014) with seal kit (KIT-HER600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-HER600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Hermle-Performance-C600, technician must log: "Resolved ERR-HER600-THM-14 on Hermle-Performance-C600 by installing PN-HER600-HEX-CORE-9014 & kit KIT-HER600-THM-14. Fasteners torqued to 206 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-SPN-1

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI150-1
- PLC Hex Register Address: DB20.DBW_0x13C68
- Nominal Operating Range: 15 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-BRG-CER-9101
- Overhaul Seal & Consumable Kit: KIT-ZEI150-SPN-1
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-SPN-1 at PLC register 0x13C68.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-BRG-CER-9101) with seal kit (KIT-ZEI150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-SPN-1 on Zeiss-Prismo-Z150 by installing PN-ZEI150-BRG-CER-9101 & kit KIT-ZEI150-SPN-1. Fasteners torqued to 39 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-ATC-2

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI150-2
- PLC Hex Register Address: DB20.DBW_0x13C70
- Nominal Operating Range: 18 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-CAM-GRIP-9102
- Overhaul Seal & Consumable Kit: KIT-ZEI150-ATC-2
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-ATC-2 at PLC register 0x13C70.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-CAM-GRIP-9102) with seal kit (KIT-ZEI150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-ATC-2 on Zeiss-Prismo-Z150 by installing PN-ZEI150-CAM-GRIP-9102 & kit KIT-ZEI150-ATC-2. Fasteners torqued to 48 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-HYD-3

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI150-3
- PLC Hex Register Address: DB20.DBW_0x13C78
- Nominal Operating Range: 21 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-VLV-PROP-9103
- Overhaul Seal & Consumable Kit: KIT-ZEI150-HYD-3
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-HYD-3 at PLC register 0x13C78.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-VLV-PROP-9103) with seal kit (KIT-ZEI150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-HYD-3 on Zeiss-Prismo-Z150 by installing PN-ZEI150-VLV-PROP-9103 & kit KIT-ZEI150-HYD-3. Fasteners torqued to 57 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-CLN-4

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI150-4
- PLC Hex Register Address: DB20.DBW_0x13C80
- Nominal Operating Range: 24 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-PMP-PLG-9104
- Overhaul Seal & Consumable Kit: KIT-ZEI150-CLN-4
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-CLN-4 at PLC register 0x13C80.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-PMP-PLG-9104) with seal kit (KIT-ZEI150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-CLN-4 on Zeiss-Prismo-Z150 by installing PN-ZEI150-PMP-PLG-9104 & kit KIT-ZEI150-CLN-4. Fasteners torqued to 66 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-XAX-5

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI150-5
- PLC Hex Register Address: DB20.DBW_0x13C88
- Nominal Operating Range: 27 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-LIN-COIL-9105
- Overhaul Seal & Consumable Kit: KIT-ZEI150-XAX-5
- Critical Fastener Torque Value: 75 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-XAX-5 at PLC register 0x13C88.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-LIN-COIL-9105) with seal kit (KIT-ZEI150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 75 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-XAX-5 on Zeiss-Prismo-Z150 by installing PN-ZEI150-LIN-COIL-9105 & kit KIT-ZEI150-XAX-5. Fasteners torqued to 75 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-YAX-6

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI150-6
- PLC Hex Register Address: DB20.DBW_0x13C90
- Nominal Operating Range: 30 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-BSW-NUT-9106
- Overhaul Seal & Consumable Kit: KIT-ZEI150-YAX-6
- Critical Fastener Torque Value: 84 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-YAX-6 at PLC register 0x13C90.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-BSW-NUT-9106) with seal kit (KIT-ZEI150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 84 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-YAX-6 on Zeiss-Prismo-Z150 by installing PN-ZEI150-BSW-NUT-9106 & kit KIT-ZEI150-YAX-6. Fasteners torqued to 84 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-ZAX-7

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI150-7
- PLC Hex Register Address: DB20.DBW_0x13C98
- Nominal Operating Range: 33 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-CYL-SEAL-9107
- Overhaul Seal & Consumable Kit: KIT-ZEI150-ZAX-7
- Critical Fastener Torque Value: 93 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-ZAX-7 at PLC register 0x13C98.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-CYL-SEAL-9107) with seal kit (KIT-ZEI150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 93 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-ZAX-7 on Zeiss-Prismo-Z150 by installing PN-ZEI150-CYL-SEAL-9107 & kit KIT-ZEI150-ZAX-7. Fasteners torqued to 93 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-BAX-8

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI150-8
- PLC Hex Register Address: DB20.DBW_0x13CA0
- Nominal Operating Range: 36 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-TRQ-STAT-9108
- Overhaul Seal & Consumable Kit: KIT-ZEI150-BAX-8
- Critical Fastener Torque Value: 102 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-BAX-8 at PLC register 0x13CA0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-TRQ-STAT-9108) with seal kit (KIT-ZEI150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 102 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-BAX-8 on Zeiss-Prismo-Z150 by installing PN-ZEI150-TRQ-STAT-9108 & kit KIT-ZEI150-BAX-8. Fasteners torqued to 102 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-CAX-9

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI150-9
- PLC Hex Register Address: DB20.DBW_0x13CA8
- Nominal Operating Range: 39 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-BRK-DISC-9109
- Overhaul Seal & Consumable Kit: KIT-ZEI150-CAX-9
- Critical Fastener Torque Value: 111 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-CAX-9 at PLC register 0x13CA8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-BRK-DISC-9109) with seal kit (KIT-ZEI150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 111 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-CAX-9 on Zeiss-Prismo-Z150 by installing PN-ZEI150-BRK-DISC-9109 & kit KIT-ZEI150-CAX-9. Fasteners torqued to 111 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-OPT-10

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI150-10
- PLC Hex Register Address: DB20.DBW_0x13CB0
- Nominal Operating Range: 42 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-GLASS-RD-9110
- Overhaul Seal & Consumable Kit: KIT-ZEI150-OPT-10
- Critical Fastener Torque Value: 120 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-OPT-10 at PLC register 0x13CB0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-GLASS-RD-9110) with seal kit (KIT-ZEI150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 120 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-OPT-10 on Zeiss-Prismo-Z150 by installing PN-ZEI150-GLASS-RD-9110 & kit KIT-ZEI150-OPT-10. Fasteners torqued to 120 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-PLC-11

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI150-11
- PLC Hex Register Address: DB20.DBW_0x13CB8
- Nominal Operating Range: 45 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-IGBT-MOD-9111
- Overhaul Seal & Consumable Kit: KIT-ZEI150-PLC-11
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-PLC-11 at PLC register 0x13CB8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-IGBT-MOD-9111) with seal kit (KIT-ZEI150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-PLC-11 on Zeiss-Prismo-Z150 by installing PN-ZEI150-IGBT-MOD-9111 & kit KIT-ZEI150-PLC-11. Fasteners torqued to 129 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-LUB-12

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI150-12
- PLC Hex Register Address: DB20.DBW_0x13CC0
- Nominal Operating Range: 48 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-INJ-MET-9112
- Overhaul Seal & Consumable Kit: KIT-ZEI150-LUB-12
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-LUB-12 at PLC register 0x13CC0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-INJ-MET-9112) with seal kit (KIT-ZEI150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-LUB-12 on Zeiss-Prismo-Z150 by installing PN-ZEI150-INJ-MET-9112 & kit KIT-ZEI150-LUB-12. Fasteners torqued to 138 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-VAC-13

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI150-13
- PLC Hex Register Address: DB20.DBW_0x13CC8
- Nominal Operating Range: 51 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-GASK-VIT-9113
- Overhaul Seal & Consumable Kit: KIT-ZEI150-VAC-13
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-VAC-13 at PLC register 0x13CC8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-GASK-VIT-9113) with seal kit (KIT-ZEI150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-VAC-13 on Zeiss-Prismo-Z150 by installing PN-ZEI150-GASK-VIT-9113 & kit KIT-ZEI150-VAC-13. Fasteners torqued to 147 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI150-THM-14

Target Machine Model: Zeiss-Prismo-Z150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI150-14
- PLC Hex Register Address: DB20.DBW_0x13CD0
- Nominal Operating Range: 54 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI150-HEX-CORE-9114
- Overhaul Seal & Consumable Kit: KIT-ZEI150-THM-14
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z150 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI150-THM-14 at PLC register 0x13CD0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI150-HEX-CORE-9114) with seal kit (KIT-ZEI150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z150, technician must log: "Resolved ERR-ZEI150-THM-14 on Zeiss-Prismo-Z150 by installing PN-ZEI150-HEX-CORE-9114 & kit KIT-ZEI150-THM-14. Fasteners torqued to 156 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-SPN-1

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI200-1
- PLC Hex Register Address: DB21.DBW_0x13F88
- Nominal Operating Range: 15 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-BRG-CER-9201
- Overhaul Seal & Consumable Kit: KIT-ZEI200-SPN-1
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-SPN-1 at PLC register 0x13F88.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-BRG-CER-9201) with seal kit (KIT-ZEI200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-SPN-1 on Zeiss-Prismo-Z200 by installing PN-ZEI200-BRG-CER-9201 & kit KIT-ZEI200-SPN-1. Fasteners torqued to 179 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-ATC-2

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI200-2
- PLC Hex Register Address: DB21.DBW_0x13F90
- Nominal Operating Range: 18 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-CAM-GRIP-9202
- Overhaul Seal & Consumable Kit: KIT-ZEI200-ATC-2
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-ATC-2 at PLC register 0x13F90.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-CAM-GRIP-9202) with seal kit (KIT-ZEI200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-ATC-2 on Zeiss-Prismo-Z200 by installing PN-ZEI200-CAM-GRIP-9202 & kit KIT-ZEI200-ATC-2. Fasteners torqued to 188 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-HYD-3

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI200-3
- PLC Hex Register Address: DB21.DBW_0x13F98
- Nominal Operating Range: 21 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-VLV-PROP-9203
- Overhaul Seal & Consumable Kit: KIT-ZEI200-HYD-3
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-HYD-3 at PLC register 0x13F98.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-VLV-PROP-9203) with seal kit (KIT-ZEI200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-HYD-3 on Zeiss-Prismo-Z200 by installing PN-ZEI200-VLV-PROP-9203 & kit KIT-ZEI200-HYD-3. Fasteners torqued to 197 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-CLN-4

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI200-4
- PLC Hex Register Address: DB21.DBW_0x13FA0
- Nominal Operating Range: 24 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-PMP-PLG-9204
- Overhaul Seal & Consumable Kit: KIT-ZEI200-CLN-4
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-CLN-4 at PLC register 0x13FA0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-PMP-PLG-9204) with seal kit (KIT-ZEI200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-CLN-4 on Zeiss-Prismo-Z200 by installing PN-ZEI200-PMP-PLG-9204 & kit KIT-ZEI200-CLN-4. Fasteners torqued to 206 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-XAX-5

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI200-5
- PLC Hex Register Address: DB21.DBW_0x13FA8
- Nominal Operating Range: 27 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-LIN-COIL-9205
- Overhaul Seal & Consumable Kit: KIT-ZEI200-XAX-5
- Critical Fastener Torque Value: 25 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-XAX-5 at PLC register 0x13FA8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-LIN-COIL-9205) with seal kit (KIT-ZEI200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 25 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-XAX-5 on Zeiss-Prismo-Z200 by installing PN-ZEI200-LIN-COIL-9205 & kit KIT-ZEI200-XAX-5. Fasteners torqued to 25 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-YAX-6

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI200-6
- PLC Hex Register Address: DB21.DBW_0x13FB0
- Nominal Operating Range: 30 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-BSW-NUT-9206
- Overhaul Seal & Consumable Kit: KIT-ZEI200-YAX-6
- Critical Fastener Torque Value: 34 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-YAX-6 at PLC register 0x13FB0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-BSW-NUT-9206) with seal kit (KIT-ZEI200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 34 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-YAX-6 on Zeiss-Prismo-Z200 by installing PN-ZEI200-BSW-NUT-9206 & kit KIT-ZEI200-YAX-6. Fasteners torqued to 34 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-ZAX-7

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI200-7
- PLC Hex Register Address: DB21.DBW_0x13FB8
- Nominal Operating Range: 33 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-CYL-SEAL-9207
- Overhaul Seal & Consumable Kit: KIT-ZEI200-ZAX-7
- Critical Fastener Torque Value: 43 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-ZAX-7 at PLC register 0x13FB8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-CYL-SEAL-9207) with seal kit (KIT-ZEI200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 43 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-ZAX-7 on Zeiss-Prismo-Z200 by installing PN-ZEI200-CYL-SEAL-9207 & kit KIT-ZEI200-ZAX-7. Fasteners torqued to 43 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-BAX-8

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI200-8
- PLC Hex Register Address: DB21.DBW_0x13FC0
- Nominal Operating Range: 36 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-TRQ-STAT-9208
- Overhaul Seal & Consumable Kit: KIT-ZEI200-BAX-8
- Critical Fastener Torque Value: 52 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-BAX-8 at PLC register 0x13FC0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-TRQ-STAT-9208) with seal kit (KIT-ZEI200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 52 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-BAX-8 on Zeiss-Prismo-Z200 by installing PN-ZEI200-TRQ-STAT-9208 & kit KIT-ZEI200-BAX-8. Fasteners torqued to 52 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-CAX-9

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI200-9
- PLC Hex Register Address: DB21.DBW_0x13FC8
- Nominal Operating Range: 39 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-BRK-DISC-9209
- Overhaul Seal & Consumable Kit: KIT-ZEI200-CAX-9
- Critical Fastener Torque Value: 61 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-CAX-9 at PLC register 0x13FC8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-BRK-DISC-9209) with seal kit (KIT-ZEI200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 61 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-CAX-9 on Zeiss-Prismo-Z200 by installing PN-ZEI200-BRK-DISC-9209 & kit KIT-ZEI200-CAX-9. Fasteners torqued to 61 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-OPT-10

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI200-10
- PLC Hex Register Address: DB21.DBW_0x13FD0
- Nominal Operating Range: 42 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-GLASS-RD-9210
- Overhaul Seal & Consumable Kit: KIT-ZEI200-OPT-10
- Critical Fastener Torque Value: 70 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-OPT-10 at PLC register 0x13FD0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-GLASS-RD-9210) with seal kit (KIT-ZEI200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 70 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-OPT-10 on Zeiss-Prismo-Z200 by installing PN-ZEI200-GLASS-RD-9210 & kit KIT-ZEI200-OPT-10. Fasteners torqued to 70 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-PLC-11

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI200-11
- PLC Hex Register Address: DB21.DBW_0x13FD8
- Nominal Operating Range: 45 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-IGBT-MOD-9211
- Overhaul Seal & Consumable Kit: KIT-ZEI200-PLC-11
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-PLC-11 at PLC register 0x13FD8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-IGBT-MOD-9211) with seal kit (KIT-ZEI200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-PLC-11 on Zeiss-Prismo-Z200 by installing PN-ZEI200-IGBT-MOD-9211 & kit KIT-ZEI200-PLC-11. Fasteners torqued to 79 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-LUB-12

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI200-12
- PLC Hex Register Address: DB21.DBW_0x13FE0
- Nominal Operating Range: 48 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-INJ-MET-9212
- Overhaul Seal & Consumable Kit: KIT-ZEI200-LUB-12
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-LUB-12 at PLC register 0x13FE0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-INJ-MET-9212) with seal kit (KIT-ZEI200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-LUB-12 on Zeiss-Prismo-Z200 by installing PN-ZEI200-INJ-MET-9212 & kit KIT-ZEI200-LUB-12. Fasteners torqued to 88 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-VAC-13

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI200-13
- PLC Hex Register Address: DB21.DBW_0x13FE8
- Nominal Operating Range: 51 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-GASK-VIT-9213
- Overhaul Seal & Consumable Kit: KIT-ZEI200-VAC-13
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-VAC-13 at PLC register 0x13FE8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-GASK-VIT-9213) with seal kit (KIT-ZEI200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-VAC-13 on Zeiss-Prismo-Z200 by installing PN-ZEI200-GASK-VIT-9213 & kit KIT-ZEI200-VAC-13. Fasteners torqued to 97 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI200-THM-14

Target Machine Model: Zeiss-Prismo-Z200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI200-14
- PLC Hex Register Address: DB21.DBW_0x13FF0
- Nominal Operating Range: 54 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI200-HEX-CORE-9214
- Overhaul Seal & Consumable Kit: KIT-ZEI200-THM-14
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z200 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI200-THM-14 at PLC register 0x13FF0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI200-HEX-CORE-9214) with seal kit (KIT-ZEI200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z200, technician must log: "Resolved ERR-ZEI200-THM-14 on Zeiss-Prismo-Z200 by installing PN-ZEI200-HEX-CORE-9214 & kit KIT-ZEI200-THM-14. Fasteners torqued to 106 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-SPN-1

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI250-1
- PLC Hex Register Address: DB22.DBW_0x142A8
- Nominal Operating Range: 15 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-BRG-CER-9301
- Overhaul Seal & Consumable Kit: KIT-ZEI250-SPN-1
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-SPN-1 at PLC register 0x142A8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-BRG-CER-9301) with seal kit (KIT-ZEI250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-SPN-1 on Zeiss-Prismo-Z250 by installing PN-ZEI250-BRG-CER-9301 & kit KIT-ZEI250-SPN-1. Fasteners torqued to 129 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-ATC-2

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI250-2
- PLC Hex Register Address: DB22.DBW_0x142B0
- Nominal Operating Range: 18 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-CAM-GRIP-9302
- Overhaul Seal & Consumable Kit: KIT-ZEI250-ATC-2
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-ATC-2 at PLC register 0x142B0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-CAM-GRIP-9302) with seal kit (KIT-ZEI250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-ATC-2 on Zeiss-Prismo-Z250 by installing PN-ZEI250-CAM-GRIP-9302 & kit KIT-ZEI250-ATC-2. Fasteners torqued to 138 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-HYD-3

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI250-3
- PLC Hex Register Address: DB22.DBW_0x142B8
- Nominal Operating Range: 21 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-VLV-PROP-9303
- Overhaul Seal & Consumable Kit: KIT-ZEI250-HYD-3
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-HYD-3 at PLC register 0x142B8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-VLV-PROP-9303) with seal kit (KIT-ZEI250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-HYD-3 on Zeiss-Prismo-Z250 by installing PN-ZEI250-VLV-PROP-9303 & kit KIT-ZEI250-HYD-3. Fasteners torqued to 147 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-CLN-4

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI250-4
- PLC Hex Register Address: DB22.DBW_0x142C0
- Nominal Operating Range: 24 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-PMP-PLG-9304
- Overhaul Seal & Consumable Kit: KIT-ZEI250-CLN-4
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-CLN-4 at PLC register 0x142C0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-PMP-PLG-9304) with seal kit (KIT-ZEI250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-CLN-4 on Zeiss-Prismo-Z250 by installing PN-ZEI250-PMP-PLG-9304 & kit KIT-ZEI250-CLN-4. Fasteners torqued to 156 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-XAX-5

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI250-5
- PLC Hex Register Address: DB22.DBW_0x142C8
- Nominal Operating Range: 27 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-LIN-COIL-9305
- Overhaul Seal & Consumable Kit: KIT-ZEI250-XAX-5
- Critical Fastener Torque Value: 165 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-XAX-5 at PLC register 0x142C8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-LIN-COIL-9305) with seal kit (KIT-ZEI250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 165 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-XAX-5 on Zeiss-Prismo-Z250 by installing PN-ZEI250-LIN-COIL-9305 & kit KIT-ZEI250-XAX-5. Fasteners torqued to 165 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-YAX-6

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI250-6
- PLC Hex Register Address: DB22.DBW_0x142D0
- Nominal Operating Range: 30 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-BSW-NUT-9306
- Overhaul Seal & Consumable Kit: KIT-ZEI250-YAX-6
- Critical Fastener Torque Value: 174 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-YAX-6 at PLC register 0x142D0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-BSW-NUT-9306) with seal kit (KIT-ZEI250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 174 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-YAX-6 on Zeiss-Prismo-Z250 by installing PN-ZEI250-BSW-NUT-9306 & kit KIT-ZEI250-YAX-6. Fasteners torqued to 174 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-ZAX-7

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI250-7
- PLC Hex Register Address: DB22.DBW_0x142D8
- Nominal Operating Range: 33 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-CYL-SEAL-9307
- Overhaul Seal & Consumable Kit: KIT-ZEI250-ZAX-7
- Critical Fastener Torque Value: 183 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-ZAX-7 at PLC register 0x142D8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-CYL-SEAL-9307) with seal kit (KIT-ZEI250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 183 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-ZAX-7 on Zeiss-Prismo-Z250 by installing PN-ZEI250-CYL-SEAL-9307 & kit KIT-ZEI250-ZAX-7. Fasteners torqued to 183 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-BAX-8

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI250-8
- PLC Hex Register Address: DB22.DBW_0x142E0
- Nominal Operating Range: 36 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-TRQ-STAT-9308
- Overhaul Seal & Consumable Kit: KIT-ZEI250-BAX-8
- Critical Fastener Torque Value: 192 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-BAX-8 at PLC register 0x142E0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-TRQ-STAT-9308) with seal kit (KIT-ZEI250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 192 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-BAX-8 on Zeiss-Prismo-Z250 by installing PN-ZEI250-TRQ-STAT-9308 & kit KIT-ZEI250-BAX-8. Fasteners torqued to 192 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-CAX-9

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI250-9
- PLC Hex Register Address: DB22.DBW_0x142E8
- Nominal Operating Range: 39 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-BRK-DISC-9309
- Overhaul Seal & Consumable Kit: KIT-ZEI250-CAX-9
- Critical Fastener Torque Value: 201 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-CAX-9 at PLC register 0x142E8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-BRK-DISC-9309) with seal kit (KIT-ZEI250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 201 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-CAX-9 on Zeiss-Prismo-Z250 by installing PN-ZEI250-BRK-DISC-9309 & kit KIT-ZEI250-CAX-9. Fasteners torqued to 201 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-OPT-10

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI250-10
- PLC Hex Register Address: DB22.DBW_0x142F0
- Nominal Operating Range: 42 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-GLASS-RD-9310
- Overhaul Seal & Consumable Kit: KIT-ZEI250-OPT-10
- Critical Fastener Torque Value: 20 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-OPT-10 at PLC register 0x142F0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-GLASS-RD-9310) with seal kit (KIT-ZEI250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 20 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-OPT-10 on Zeiss-Prismo-Z250 by installing PN-ZEI250-GLASS-RD-9310 & kit KIT-ZEI250-OPT-10. Fasteners torqued to 20 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-PLC-11

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI250-11
- PLC Hex Register Address: DB22.DBW_0x142F8
- Nominal Operating Range: 45 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-IGBT-MOD-9311
- Overhaul Seal & Consumable Kit: KIT-ZEI250-PLC-11
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-PLC-11 at PLC register 0x142F8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-IGBT-MOD-9311) with seal kit (KIT-ZEI250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-PLC-11 on Zeiss-Prismo-Z250 by installing PN-ZEI250-IGBT-MOD-9311 & kit KIT-ZEI250-PLC-11. Fasteners torqued to 29 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-LUB-12

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI250-12
- PLC Hex Register Address: DB22.DBW_0x14300
- Nominal Operating Range: 48 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-INJ-MET-9312
- Overhaul Seal & Consumable Kit: KIT-ZEI250-LUB-12
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-LUB-12 at PLC register 0x14300.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-INJ-MET-9312) with seal kit (KIT-ZEI250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-LUB-12 on Zeiss-Prismo-Z250 by installing PN-ZEI250-INJ-MET-9312 & kit KIT-ZEI250-LUB-12. Fasteners torqued to 38 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-VAC-13

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI250-13
- PLC Hex Register Address: DB22.DBW_0x14308
- Nominal Operating Range: 51 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-GASK-VIT-9313
- Overhaul Seal & Consumable Kit: KIT-ZEI250-VAC-13
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-VAC-13 at PLC register 0x14308.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-GASK-VIT-9313) with seal kit (KIT-ZEI250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-VAC-13 on Zeiss-Prismo-Z250 by installing PN-ZEI250-GASK-VIT-9313 & kit KIT-ZEI250-VAC-13. Fasteners torqued to 47 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI250-THM-14

Target Machine Model: Zeiss-Prismo-Z250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI250-14
- PLC Hex Register Address: DB22.DBW_0x14310
- Nominal Operating Range: 54 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI250-HEX-CORE-9314
- Overhaul Seal & Consumable Kit: KIT-ZEI250-THM-14
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z250 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI250-THM-14 at PLC register 0x14310.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI250-HEX-CORE-9314) with seal kit (KIT-ZEI250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z250, technician must log: "Resolved ERR-ZEI250-THM-14 on Zeiss-Prismo-Z250 by installing PN-ZEI250-HEX-CORE-9314 & kit KIT-ZEI250-THM-14. Fasteners torqued to 56 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-SPN-1

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI300-1
- PLC Hex Register Address: DB23.DBW_0x145C8
- Nominal Operating Range: 15 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-BRG-CER-9401
- Overhaul Seal & Consumable Kit: KIT-ZEI300-SPN-1
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-SPN-1 at PLC register 0x145C8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-BRG-CER-9401) with seal kit (KIT-ZEI300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-SPN-1 on Zeiss-Prismo-Z300 by installing PN-ZEI300-BRG-CER-9401 & kit KIT-ZEI300-SPN-1. Fasteners torqued to 79 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-ATC-2

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI300-2
- PLC Hex Register Address: DB23.DBW_0x145D0
- Nominal Operating Range: 18 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-CAM-GRIP-9402
- Overhaul Seal & Consumable Kit: KIT-ZEI300-ATC-2
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-ATC-2 at PLC register 0x145D0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-CAM-GRIP-9402) with seal kit (KIT-ZEI300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-ATC-2 on Zeiss-Prismo-Z300 by installing PN-ZEI300-CAM-GRIP-9402 & kit KIT-ZEI300-ATC-2. Fasteners torqued to 88 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-HYD-3

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI300-3
- PLC Hex Register Address: DB23.DBW_0x145D8
- Nominal Operating Range: 21 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-VLV-PROP-9403
- Overhaul Seal & Consumable Kit: KIT-ZEI300-HYD-3
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-HYD-3 at PLC register 0x145D8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-VLV-PROP-9403) with seal kit (KIT-ZEI300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-HYD-3 on Zeiss-Prismo-Z300 by installing PN-ZEI300-VLV-PROP-9403 & kit KIT-ZEI300-HYD-3. Fasteners torqued to 97 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-CLN-4

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI300-4
- PLC Hex Register Address: DB23.DBW_0x145E0
- Nominal Operating Range: 24 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-PMP-PLG-9404
- Overhaul Seal & Consumable Kit: KIT-ZEI300-CLN-4
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-CLN-4 at PLC register 0x145E0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-PMP-PLG-9404) with seal kit (KIT-ZEI300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-CLN-4 on Zeiss-Prismo-Z300 by installing PN-ZEI300-PMP-PLG-9404 & kit KIT-ZEI300-CLN-4. Fasteners torqued to 106 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-XAX-5

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI300-5
- PLC Hex Register Address: DB23.DBW_0x145E8
- Nominal Operating Range: 27 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-LIN-COIL-9405
- Overhaul Seal & Consumable Kit: KIT-ZEI300-XAX-5
- Critical Fastener Torque Value: 115 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-XAX-5 at PLC register 0x145E8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-LIN-COIL-9405) with seal kit (KIT-ZEI300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 115 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-XAX-5 on Zeiss-Prismo-Z300 by installing PN-ZEI300-LIN-COIL-9405 & kit KIT-ZEI300-XAX-5. Fasteners torqued to 115 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-YAX-6

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI300-6
- PLC Hex Register Address: DB23.DBW_0x145F0
- Nominal Operating Range: 30 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-BSW-NUT-9406
- Overhaul Seal & Consumable Kit: KIT-ZEI300-YAX-6
- Critical Fastener Torque Value: 124 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-YAX-6 at PLC register 0x145F0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-BSW-NUT-9406) with seal kit (KIT-ZEI300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 124 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-YAX-6 on Zeiss-Prismo-Z300 by installing PN-ZEI300-BSW-NUT-9406 & kit KIT-ZEI300-YAX-6. Fasteners torqued to 124 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-ZAX-7

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI300-7
- PLC Hex Register Address: DB23.DBW_0x145F8
- Nominal Operating Range: 33 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-CYL-SEAL-9407
- Overhaul Seal & Consumable Kit: KIT-ZEI300-ZAX-7
- Critical Fastener Torque Value: 133 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-ZAX-7 at PLC register 0x145F8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-CYL-SEAL-9407) with seal kit (KIT-ZEI300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 133 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-ZAX-7 on Zeiss-Prismo-Z300 by installing PN-ZEI300-CYL-SEAL-9407 & kit KIT-ZEI300-ZAX-7. Fasteners torqued to 133 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-BAX-8

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI300-8
- PLC Hex Register Address: DB23.DBW_0x14600
- Nominal Operating Range: 36 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-TRQ-STAT-9408
- Overhaul Seal & Consumable Kit: KIT-ZEI300-BAX-8
- Critical Fastener Torque Value: 142 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-BAX-8 at PLC register 0x14600.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-TRQ-STAT-9408) with seal kit (KIT-ZEI300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 142 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-BAX-8 on Zeiss-Prismo-Z300 by installing PN-ZEI300-TRQ-STAT-9408 & kit KIT-ZEI300-BAX-8. Fasteners torqued to 142 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-CAX-9

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI300-9
- PLC Hex Register Address: DB23.DBW_0x14608
- Nominal Operating Range: 39 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-BRK-DISC-9409
- Overhaul Seal & Consumable Kit: KIT-ZEI300-CAX-9
- Critical Fastener Torque Value: 151 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-CAX-9 at PLC register 0x14608.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-BRK-DISC-9409) with seal kit (KIT-ZEI300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 151 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-CAX-9 on Zeiss-Prismo-Z300 by installing PN-ZEI300-BRK-DISC-9409 & kit KIT-ZEI300-CAX-9. Fasteners torqued to 151 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-OPT-10

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI300-10
- PLC Hex Register Address: DB23.DBW_0x14610
- Nominal Operating Range: 42 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-GLASS-RD-9410
- Overhaul Seal & Consumable Kit: KIT-ZEI300-OPT-10
- Critical Fastener Torque Value: 160 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-OPT-10 at PLC register 0x14610.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-GLASS-RD-9410) with seal kit (KIT-ZEI300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 160 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-OPT-10 on Zeiss-Prismo-Z300 by installing PN-ZEI300-GLASS-RD-9410 & kit KIT-ZEI300-OPT-10. Fasteners torqued to 160 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-PLC-11

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI300-11
- PLC Hex Register Address: DB23.DBW_0x14618
- Nominal Operating Range: 45 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-IGBT-MOD-9411
- Overhaul Seal & Consumable Kit: KIT-ZEI300-PLC-11
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-PLC-11 at PLC register 0x14618.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-IGBT-MOD-9411) with seal kit (KIT-ZEI300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-PLC-11 on Zeiss-Prismo-Z300 by installing PN-ZEI300-IGBT-MOD-9411 & kit KIT-ZEI300-PLC-11. Fasteners torqued to 169 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-LUB-12

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI300-12
- PLC Hex Register Address: DB23.DBW_0x14620
- Nominal Operating Range: 48 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-INJ-MET-9412
- Overhaul Seal & Consumable Kit: KIT-ZEI300-LUB-12
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-LUB-12 at PLC register 0x14620.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-INJ-MET-9412) with seal kit (KIT-ZEI300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-LUB-12 on Zeiss-Prismo-Z300 by installing PN-ZEI300-INJ-MET-9412 & kit KIT-ZEI300-LUB-12. Fasteners torqued to 178 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-VAC-13

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI300-13
- PLC Hex Register Address: DB23.DBW_0x14628
- Nominal Operating Range: 51 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-GASK-VIT-9413
- Overhaul Seal & Consumable Kit: KIT-ZEI300-VAC-13
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-VAC-13 at PLC register 0x14628.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-GASK-VIT-9413) with seal kit (KIT-ZEI300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-VAC-13 on Zeiss-Prismo-Z300 by installing PN-ZEI300-GASK-VIT-9413 & kit KIT-ZEI300-VAC-13. Fasteners torqued to 187 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI300-THM-14

Target Machine Model: Zeiss-Prismo-Z300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI300-14
- PLC Hex Register Address: DB23.DBW_0x14630
- Nominal Operating Range: 54 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI300-HEX-CORE-9414
- Overhaul Seal & Consumable Kit: KIT-ZEI300-THM-14
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z300 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI300-THM-14 at PLC register 0x14630.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI300-HEX-CORE-9414) with seal kit (KIT-ZEI300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z300, technician must log: "Resolved ERR-ZEI300-THM-14 on Zeiss-Prismo-Z300 by installing PN-ZEI300-HEX-CORE-9414 & kit KIT-ZEI300-THM-14. Fasteners torqued to 196 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-SPN-1

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI350-1
- PLC Hex Register Address: DB24.DBW_0x148E8
- Nominal Operating Range: 15 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-BRG-CER-9501
- Overhaul Seal & Consumable Kit: KIT-ZEI350-SPN-1
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-SPN-1 at PLC register 0x148E8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-BRG-CER-9501) with seal kit (KIT-ZEI350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-SPN-1 on Zeiss-Prismo-Z350 by installing PN-ZEI350-BRG-CER-9501 & kit KIT-ZEI350-SPN-1. Fasteners torqued to 29 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-ATC-2

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI350-2
- PLC Hex Register Address: DB24.DBW_0x148F0
- Nominal Operating Range: 18 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-CAM-GRIP-9502
- Overhaul Seal & Consumable Kit: KIT-ZEI350-ATC-2
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-ATC-2 at PLC register 0x148F0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-CAM-GRIP-9502) with seal kit (KIT-ZEI350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-ATC-2 on Zeiss-Prismo-Z350 by installing PN-ZEI350-CAM-GRIP-9502 & kit KIT-ZEI350-ATC-2. Fasteners torqued to 38 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-HYD-3

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI350-3
- PLC Hex Register Address: DB24.DBW_0x148F8
- Nominal Operating Range: 21 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-VLV-PROP-9503
- Overhaul Seal & Consumable Kit: KIT-ZEI350-HYD-3
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-HYD-3 at PLC register 0x148F8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-VLV-PROP-9503) with seal kit (KIT-ZEI350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-HYD-3 on Zeiss-Prismo-Z350 by installing PN-ZEI350-VLV-PROP-9503 & kit KIT-ZEI350-HYD-3. Fasteners torqued to 47 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-CLN-4

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI350-4
- PLC Hex Register Address: DB24.DBW_0x14900
- Nominal Operating Range: 24 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-PMP-PLG-9504
- Overhaul Seal & Consumable Kit: KIT-ZEI350-CLN-4
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-CLN-4 at PLC register 0x14900.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-PMP-PLG-9504) with seal kit (KIT-ZEI350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-CLN-4 on Zeiss-Prismo-Z350 by installing PN-ZEI350-PMP-PLG-9504 & kit KIT-ZEI350-CLN-4. Fasteners torqued to 56 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-XAX-5

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI350-5
- PLC Hex Register Address: DB24.DBW_0x14908
- Nominal Operating Range: 27 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-LIN-COIL-9505
- Overhaul Seal & Consumable Kit: KIT-ZEI350-XAX-5
- Critical Fastener Torque Value: 65 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-XAX-5 at PLC register 0x14908.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-LIN-COIL-9505) with seal kit (KIT-ZEI350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 65 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-XAX-5 on Zeiss-Prismo-Z350 by installing PN-ZEI350-LIN-COIL-9505 & kit KIT-ZEI350-XAX-5. Fasteners torqued to 65 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-YAX-6

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI350-6
- PLC Hex Register Address: DB24.DBW_0x14910
- Nominal Operating Range: 30 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-BSW-NUT-9506
- Overhaul Seal & Consumable Kit: KIT-ZEI350-YAX-6
- Critical Fastener Torque Value: 74 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-YAX-6 at PLC register 0x14910.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-BSW-NUT-9506) with seal kit (KIT-ZEI350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 74 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-YAX-6 on Zeiss-Prismo-Z350 by installing PN-ZEI350-BSW-NUT-9506 & kit KIT-ZEI350-YAX-6. Fasteners torqued to 74 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-ZAX-7

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI350-7
- PLC Hex Register Address: DB24.DBW_0x14918
- Nominal Operating Range: 33 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-CYL-SEAL-9507
- Overhaul Seal & Consumable Kit: KIT-ZEI350-ZAX-7
- Critical Fastener Torque Value: 83 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-ZAX-7 at PLC register 0x14918.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-CYL-SEAL-9507) with seal kit (KIT-ZEI350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 83 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-ZAX-7 on Zeiss-Prismo-Z350 by installing PN-ZEI350-CYL-SEAL-9507 & kit KIT-ZEI350-ZAX-7. Fasteners torqued to 83 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-BAX-8

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI350-8
- PLC Hex Register Address: DB24.DBW_0x14920
- Nominal Operating Range: 36 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-TRQ-STAT-9508
- Overhaul Seal & Consumable Kit: KIT-ZEI350-BAX-8
- Critical Fastener Torque Value: 92 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-BAX-8 at PLC register 0x14920.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-TRQ-STAT-9508) with seal kit (KIT-ZEI350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 92 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-BAX-8 on Zeiss-Prismo-Z350 by installing PN-ZEI350-TRQ-STAT-9508 & kit KIT-ZEI350-BAX-8. Fasteners torqued to 92 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-CAX-9

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI350-9
- PLC Hex Register Address: DB24.DBW_0x14928
- Nominal Operating Range: 39 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-BRK-DISC-9509
- Overhaul Seal & Consumable Kit: KIT-ZEI350-CAX-9
- Critical Fastener Torque Value: 101 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-CAX-9 at PLC register 0x14928.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-BRK-DISC-9509) with seal kit (KIT-ZEI350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 101 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-CAX-9 on Zeiss-Prismo-Z350 by installing PN-ZEI350-BRK-DISC-9509 & kit KIT-ZEI350-CAX-9. Fasteners torqued to 101 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-OPT-10

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI350-10
- PLC Hex Register Address: DB24.DBW_0x14930
- Nominal Operating Range: 42 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-GLASS-RD-9510
- Overhaul Seal & Consumable Kit: KIT-ZEI350-OPT-10
- Critical Fastener Torque Value: 110 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-OPT-10 at PLC register 0x14930.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-GLASS-RD-9510) with seal kit (KIT-ZEI350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 110 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-OPT-10 on Zeiss-Prismo-Z350 by installing PN-ZEI350-GLASS-RD-9510 & kit KIT-ZEI350-OPT-10. Fasteners torqued to 110 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-PLC-11

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI350-11
- PLC Hex Register Address: DB24.DBW_0x14938
- Nominal Operating Range: 45 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-IGBT-MOD-9511
- Overhaul Seal & Consumable Kit: KIT-ZEI350-PLC-11
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-PLC-11 at PLC register 0x14938.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-IGBT-MOD-9511) with seal kit (KIT-ZEI350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-PLC-11 on Zeiss-Prismo-Z350 by installing PN-ZEI350-IGBT-MOD-9511 & kit KIT-ZEI350-PLC-11. Fasteners torqued to 119 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-LUB-12

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI350-12
- PLC Hex Register Address: DB24.DBW_0x14940
- Nominal Operating Range: 48 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-INJ-MET-9512
- Overhaul Seal & Consumable Kit: KIT-ZEI350-LUB-12
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-LUB-12 at PLC register 0x14940.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-INJ-MET-9512) with seal kit (KIT-ZEI350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-LUB-12 on Zeiss-Prismo-Z350 by installing PN-ZEI350-INJ-MET-9512 & kit KIT-ZEI350-LUB-12. Fasteners torqued to 128 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-VAC-13

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI350-13
- PLC Hex Register Address: DB24.DBW_0x14948
- Nominal Operating Range: 51 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-GASK-VIT-9513
- Overhaul Seal & Consumable Kit: KIT-ZEI350-VAC-13
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-VAC-13 at PLC register 0x14948.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-GASK-VIT-9513) with seal kit (KIT-ZEI350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-VAC-13 on Zeiss-Prismo-Z350 by installing PN-ZEI350-GASK-VIT-9513 & kit KIT-ZEI350-VAC-13. Fasteners torqued to 137 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI350-THM-14

Target Machine Model: Zeiss-Prismo-Z350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI350-14
- PLC Hex Register Address: DB24.DBW_0x14950
- Nominal Operating Range: 54 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI350-HEX-CORE-9514
- Overhaul Seal & Consumable Kit: KIT-ZEI350-THM-14
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z350 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI350-THM-14 at PLC register 0x14950.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI350-HEX-CORE-9514) with seal kit (KIT-ZEI350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z350, technician must log: "Resolved ERR-ZEI350-THM-14 on Zeiss-Prismo-Z350 by installing PN-ZEI350-HEX-CORE-9514 & kit KIT-ZEI350-THM-14. Fasteners torqued to 146 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-SPN-1

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI400-1
- PLC Hex Register Address: DB25.DBW_0x14C08
- Nominal Operating Range: 15 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-BRG-CER-9601
- Overhaul Seal & Consumable Kit: KIT-ZEI400-SPN-1
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-SPN-1 at PLC register 0x14C08.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-BRG-CER-9601) with seal kit (KIT-ZEI400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-SPN-1 on Zeiss-Prismo-Z400 by installing PN-ZEI400-BRG-CER-9601 & kit KIT-ZEI400-SPN-1. Fasteners torqued to 169 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-ATC-2

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI400-2
- PLC Hex Register Address: DB25.DBW_0x14C10
- Nominal Operating Range: 18 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-CAM-GRIP-9602
- Overhaul Seal & Consumable Kit: KIT-ZEI400-ATC-2
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-ATC-2 at PLC register 0x14C10.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-CAM-GRIP-9602) with seal kit (KIT-ZEI400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-ATC-2 on Zeiss-Prismo-Z400 by installing PN-ZEI400-CAM-GRIP-9602 & kit KIT-ZEI400-ATC-2. Fasteners torqued to 178 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-HYD-3

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI400-3
- PLC Hex Register Address: DB25.DBW_0x14C18
- Nominal Operating Range: 21 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-VLV-PROP-9603
- Overhaul Seal & Consumable Kit: KIT-ZEI400-HYD-3
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-HYD-3 at PLC register 0x14C18.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-VLV-PROP-9603) with seal kit (KIT-ZEI400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-HYD-3 on Zeiss-Prismo-Z400 by installing PN-ZEI400-VLV-PROP-9603 & kit KIT-ZEI400-HYD-3. Fasteners torqued to 187 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-CLN-4

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI400-4
- PLC Hex Register Address: DB25.DBW_0x14C20
- Nominal Operating Range: 24 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-PMP-PLG-9604
- Overhaul Seal & Consumable Kit: KIT-ZEI400-CLN-4
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-CLN-4 at PLC register 0x14C20.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-PMP-PLG-9604) with seal kit (KIT-ZEI400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-CLN-4 on Zeiss-Prismo-Z400 by installing PN-ZEI400-PMP-PLG-9604 & kit KIT-ZEI400-CLN-4. Fasteners torqued to 196 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-XAX-5

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI400-5
- PLC Hex Register Address: DB25.DBW_0x14C28
- Nominal Operating Range: 27 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-LIN-COIL-9605
- Overhaul Seal & Consumable Kit: KIT-ZEI400-XAX-5
- Critical Fastener Torque Value: 205 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-XAX-5 at PLC register 0x14C28.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-LIN-COIL-9605) with seal kit (KIT-ZEI400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 205 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-XAX-5 on Zeiss-Prismo-Z400 by installing PN-ZEI400-LIN-COIL-9605 & kit KIT-ZEI400-XAX-5. Fasteners torqued to 205 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-YAX-6

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI400-6
- PLC Hex Register Address: DB25.DBW_0x14C30
- Nominal Operating Range: 30 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-BSW-NUT-9606
- Overhaul Seal & Consumable Kit: KIT-ZEI400-YAX-6
- Critical Fastener Torque Value: 24 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-YAX-6 at PLC register 0x14C30.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-BSW-NUT-9606) with seal kit (KIT-ZEI400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 24 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-YAX-6 on Zeiss-Prismo-Z400 by installing PN-ZEI400-BSW-NUT-9606 & kit KIT-ZEI400-YAX-6. Fasteners torqued to 24 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-ZAX-7

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI400-7
- PLC Hex Register Address: DB25.DBW_0x14C38
- Nominal Operating Range: 33 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-CYL-SEAL-9607
- Overhaul Seal & Consumable Kit: KIT-ZEI400-ZAX-7
- Critical Fastener Torque Value: 33 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-ZAX-7 at PLC register 0x14C38.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-CYL-SEAL-9607) with seal kit (KIT-ZEI400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 33 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-ZAX-7 on Zeiss-Prismo-Z400 by installing PN-ZEI400-CYL-SEAL-9607 & kit KIT-ZEI400-ZAX-7. Fasteners torqued to 33 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-BAX-8

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI400-8
- PLC Hex Register Address: DB25.DBW_0x14C40
- Nominal Operating Range: 36 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-TRQ-STAT-9608
- Overhaul Seal & Consumable Kit: KIT-ZEI400-BAX-8
- Critical Fastener Torque Value: 42 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-BAX-8 at PLC register 0x14C40.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-TRQ-STAT-9608) with seal kit (KIT-ZEI400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 42 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-BAX-8 on Zeiss-Prismo-Z400 by installing PN-ZEI400-TRQ-STAT-9608 & kit KIT-ZEI400-BAX-8. Fasteners torqued to 42 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-CAX-9

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI400-9
- PLC Hex Register Address: DB25.DBW_0x14C48
- Nominal Operating Range: 39 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-BRK-DISC-9609
- Overhaul Seal & Consumable Kit: KIT-ZEI400-CAX-9
- Critical Fastener Torque Value: 51 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-CAX-9 at PLC register 0x14C48.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-BRK-DISC-9609) with seal kit (KIT-ZEI400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 51 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-CAX-9 on Zeiss-Prismo-Z400 by installing PN-ZEI400-BRK-DISC-9609 & kit KIT-ZEI400-CAX-9. Fasteners torqued to 51 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-OPT-10

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI400-10
- PLC Hex Register Address: DB25.DBW_0x14C50
- Nominal Operating Range: 42 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-GLASS-RD-9610
- Overhaul Seal & Consumable Kit: KIT-ZEI400-OPT-10
- Critical Fastener Torque Value: 60 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-OPT-10 at PLC register 0x14C50.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-GLASS-RD-9610) with seal kit (KIT-ZEI400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 60 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-OPT-10 on Zeiss-Prismo-Z400 by installing PN-ZEI400-GLASS-RD-9610 & kit KIT-ZEI400-OPT-10. Fasteners torqued to 60 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-PLC-11

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI400-11
- PLC Hex Register Address: DB25.DBW_0x14C58
- Nominal Operating Range: 45 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-IGBT-MOD-9611
- Overhaul Seal & Consumable Kit: KIT-ZEI400-PLC-11
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-PLC-11 at PLC register 0x14C58.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-IGBT-MOD-9611) with seal kit (KIT-ZEI400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-PLC-11 on Zeiss-Prismo-Z400 by installing PN-ZEI400-IGBT-MOD-9611 & kit KIT-ZEI400-PLC-11. Fasteners torqued to 69 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-LUB-12

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI400-12
- PLC Hex Register Address: DB25.DBW_0x14C60
- Nominal Operating Range: 48 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-INJ-MET-9612
- Overhaul Seal & Consumable Kit: KIT-ZEI400-LUB-12
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-LUB-12 at PLC register 0x14C60.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-INJ-MET-9612) with seal kit (KIT-ZEI400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-LUB-12 on Zeiss-Prismo-Z400 by installing PN-ZEI400-INJ-MET-9612 & kit KIT-ZEI400-LUB-12. Fasteners torqued to 78 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-VAC-13

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI400-13
- PLC Hex Register Address: DB25.DBW_0x14C68
- Nominal Operating Range: 51 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-GASK-VIT-9613
- Overhaul Seal & Consumable Kit: KIT-ZEI400-VAC-13
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-VAC-13 at PLC register 0x14C68.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-GASK-VIT-9613) with seal kit (KIT-ZEI400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-VAC-13 on Zeiss-Prismo-Z400 by installing PN-ZEI400-GASK-VIT-9613 & kit KIT-ZEI400-VAC-13. Fasteners torqued to 87 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI400-THM-14

Target Machine Model: Zeiss-Prismo-Z400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI400-14
- PLC Hex Register Address: DB25.DBW_0x14C70
- Nominal Operating Range: 54 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI400-HEX-CORE-9614
- Overhaul Seal & Consumable Kit: KIT-ZEI400-THM-14
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z400 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI400-THM-14 at PLC register 0x14C70.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI400-HEX-CORE-9614) with seal kit (KIT-ZEI400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z400, technician must log: "Resolved ERR-ZEI400-THM-14 on Zeiss-Prismo-Z400 by installing PN-ZEI400-HEX-CORE-9614 & kit KIT-ZEI400-THM-14. Fasteners torqued to 96 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-SPN-1

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI450-1
- PLC Hex Register Address: DB26.DBW_0x14F28
- Nominal Operating Range: 15 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-BRG-CER-9701
- Overhaul Seal & Consumable Kit: KIT-ZEI450-SPN-1
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-SPN-1 at PLC register 0x14F28.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-BRG-CER-9701) with seal kit (KIT-ZEI450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-SPN-1 on Zeiss-Prismo-Z450 by installing PN-ZEI450-BRG-CER-9701 & kit KIT-ZEI450-SPN-1. Fasteners torqued to 119 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-ATC-2

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI450-2
- PLC Hex Register Address: DB26.DBW_0x14F30
- Nominal Operating Range: 18 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-CAM-GRIP-9702
- Overhaul Seal & Consumable Kit: KIT-ZEI450-ATC-2
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-ATC-2 at PLC register 0x14F30.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-CAM-GRIP-9702) with seal kit (KIT-ZEI450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-ATC-2 on Zeiss-Prismo-Z450 by installing PN-ZEI450-CAM-GRIP-9702 & kit KIT-ZEI450-ATC-2. Fasteners torqued to 128 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-HYD-3

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI450-3
- PLC Hex Register Address: DB26.DBW_0x14F38
- Nominal Operating Range: 21 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-VLV-PROP-9703
- Overhaul Seal & Consumable Kit: KIT-ZEI450-HYD-3
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-HYD-3 at PLC register 0x14F38.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-VLV-PROP-9703) with seal kit (KIT-ZEI450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-HYD-3 on Zeiss-Prismo-Z450 by installing PN-ZEI450-VLV-PROP-9703 & kit KIT-ZEI450-HYD-3. Fasteners torqued to 137 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-CLN-4

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI450-4
- PLC Hex Register Address: DB26.DBW_0x14F40
- Nominal Operating Range: 24 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-PMP-PLG-9704
- Overhaul Seal & Consumable Kit: KIT-ZEI450-CLN-4
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-CLN-4 at PLC register 0x14F40.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-PMP-PLG-9704) with seal kit (KIT-ZEI450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-CLN-4 on Zeiss-Prismo-Z450 by installing PN-ZEI450-PMP-PLG-9704 & kit KIT-ZEI450-CLN-4. Fasteners torqued to 146 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-XAX-5

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI450-5
- PLC Hex Register Address: DB26.DBW_0x14F48
- Nominal Operating Range: 27 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-LIN-COIL-9705
- Overhaul Seal & Consumable Kit: KIT-ZEI450-XAX-5
- Critical Fastener Torque Value: 155 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-XAX-5 at PLC register 0x14F48.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-LIN-COIL-9705) with seal kit (KIT-ZEI450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 155 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-XAX-5 on Zeiss-Prismo-Z450 by installing PN-ZEI450-LIN-COIL-9705 & kit KIT-ZEI450-XAX-5. Fasteners torqued to 155 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-YAX-6

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI450-6
- PLC Hex Register Address: DB26.DBW_0x14F50
- Nominal Operating Range: 30 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-BSW-NUT-9706
- Overhaul Seal & Consumable Kit: KIT-ZEI450-YAX-6
- Critical Fastener Torque Value: 164 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-YAX-6 at PLC register 0x14F50.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-BSW-NUT-9706) with seal kit (KIT-ZEI450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 164 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-YAX-6 on Zeiss-Prismo-Z450 by installing PN-ZEI450-BSW-NUT-9706 & kit KIT-ZEI450-YAX-6. Fasteners torqued to 164 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-ZAX-7

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI450-7
- PLC Hex Register Address: DB26.DBW_0x14F58
- Nominal Operating Range: 33 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-CYL-SEAL-9707
- Overhaul Seal & Consumable Kit: KIT-ZEI450-ZAX-7
- Critical Fastener Torque Value: 173 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-ZAX-7 at PLC register 0x14F58.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-CYL-SEAL-9707) with seal kit (KIT-ZEI450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 173 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-ZAX-7 on Zeiss-Prismo-Z450 by installing PN-ZEI450-CYL-SEAL-9707 & kit KIT-ZEI450-ZAX-7. Fasteners torqued to 173 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-BAX-8

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI450-8
- PLC Hex Register Address: DB26.DBW_0x14F60
- Nominal Operating Range: 36 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-TRQ-STAT-9708
- Overhaul Seal & Consumable Kit: KIT-ZEI450-BAX-8
- Critical Fastener Torque Value: 182 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-BAX-8 at PLC register 0x14F60.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-TRQ-STAT-9708) with seal kit (KIT-ZEI450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 182 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-BAX-8 on Zeiss-Prismo-Z450 by installing PN-ZEI450-TRQ-STAT-9708 & kit KIT-ZEI450-BAX-8. Fasteners torqued to 182 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-CAX-9

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI450-9
- PLC Hex Register Address: DB26.DBW_0x14F68
- Nominal Operating Range: 39 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-BRK-DISC-9709
- Overhaul Seal & Consumable Kit: KIT-ZEI450-CAX-9
- Critical Fastener Torque Value: 191 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-CAX-9 at PLC register 0x14F68.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-BRK-DISC-9709) with seal kit (KIT-ZEI450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 191 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-CAX-9 on Zeiss-Prismo-Z450 by installing PN-ZEI450-BRK-DISC-9709 & kit KIT-ZEI450-CAX-9. Fasteners torqued to 191 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-OPT-10

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI450-10
- PLC Hex Register Address: DB26.DBW_0x14F70
- Nominal Operating Range: 42 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-GLASS-RD-9710
- Overhaul Seal & Consumable Kit: KIT-ZEI450-OPT-10
- Critical Fastener Torque Value: 200 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-OPT-10 at PLC register 0x14F70.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-GLASS-RD-9710) with seal kit (KIT-ZEI450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 200 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-OPT-10 on Zeiss-Prismo-Z450 by installing PN-ZEI450-GLASS-RD-9710 & kit KIT-ZEI450-OPT-10. Fasteners torqued to 200 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-PLC-11

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI450-11
- PLC Hex Register Address: DB26.DBW_0x14F78
- Nominal Operating Range: 45 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-IGBT-MOD-9711
- Overhaul Seal & Consumable Kit: KIT-ZEI450-PLC-11
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-PLC-11 at PLC register 0x14F78.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-IGBT-MOD-9711) with seal kit (KIT-ZEI450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-PLC-11 on Zeiss-Prismo-Z450 by installing PN-ZEI450-IGBT-MOD-9711 & kit KIT-ZEI450-PLC-11. Fasteners torqued to 209 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-LUB-12

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI450-12
- PLC Hex Register Address: DB26.DBW_0x14F80
- Nominal Operating Range: 48 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-INJ-MET-9712
- Overhaul Seal & Consumable Kit: KIT-ZEI450-LUB-12
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-LUB-12 at PLC register 0x14F80.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-INJ-MET-9712) with seal kit (KIT-ZEI450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-LUB-12 on Zeiss-Prismo-Z450 by installing PN-ZEI450-INJ-MET-9712 & kit KIT-ZEI450-LUB-12. Fasteners torqued to 28 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-VAC-13

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI450-13
- PLC Hex Register Address: DB26.DBW_0x14F88
- Nominal Operating Range: 51 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-GASK-VIT-9713
- Overhaul Seal & Consumable Kit: KIT-ZEI450-VAC-13
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-VAC-13 at PLC register 0x14F88.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-GASK-VIT-9713) with seal kit (KIT-ZEI450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-VAC-13 on Zeiss-Prismo-Z450 by installing PN-ZEI450-GASK-VIT-9713 & kit KIT-ZEI450-VAC-13. Fasteners torqued to 37 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI450-THM-14

Target Machine Model: Zeiss-Prismo-Z450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI450-14
- PLC Hex Register Address: DB26.DBW_0x14F90
- Nominal Operating Range: 54 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI450-HEX-CORE-9714
- Overhaul Seal & Consumable Kit: KIT-ZEI450-THM-14
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z450 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI450-THM-14 at PLC register 0x14F90.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI450-HEX-CORE-9714) with seal kit (KIT-ZEI450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z450, technician must log: "Resolved ERR-ZEI450-THM-14 on Zeiss-Prismo-Z450 by installing PN-ZEI450-HEX-CORE-9714 & kit KIT-ZEI450-THM-14. Fasteners torqued to 46 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-SPN-1

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI500-1
- PLC Hex Register Address: DB27.DBW_0x15248
- Nominal Operating Range: 15 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-BRG-CER-9801
- Overhaul Seal & Consumable Kit: KIT-ZEI500-SPN-1
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-SPN-1 at PLC register 0x15248.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-BRG-CER-9801) with seal kit (KIT-ZEI500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-SPN-1 on Zeiss-Prismo-Z500 by installing PN-ZEI500-BRG-CER-9801 & kit KIT-ZEI500-SPN-1. Fasteners torqued to 69 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-ATC-2

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI500-2
- PLC Hex Register Address: DB27.DBW_0x15250
- Nominal Operating Range: 18 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-CAM-GRIP-9802
- Overhaul Seal & Consumable Kit: KIT-ZEI500-ATC-2
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-ATC-2 at PLC register 0x15250.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-CAM-GRIP-9802) with seal kit (KIT-ZEI500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-ATC-2 on Zeiss-Prismo-Z500 by installing PN-ZEI500-CAM-GRIP-9802 & kit KIT-ZEI500-ATC-2. Fasteners torqued to 78 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-HYD-3

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI500-3
- PLC Hex Register Address: DB27.DBW_0x15258
- Nominal Operating Range: 21 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-VLV-PROP-9803
- Overhaul Seal & Consumable Kit: KIT-ZEI500-HYD-3
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-HYD-3 at PLC register 0x15258.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-VLV-PROP-9803) with seal kit (KIT-ZEI500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-HYD-3 on Zeiss-Prismo-Z500 by installing PN-ZEI500-VLV-PROP-9803 & kit KIT-ZEI500-HYD-3. Fasteners torqued to 87 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-CLN-4

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI500-4
- PLC Hex Register Address: DB27.DBW_0x15260
- Nominal Operating Range: 24 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-PMP-PLG-9804
- Overhaul Seal & Consumable Kit: KIT-ZEI500-CLN-4
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-CLN-4 at PLC register 0x15260.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-PMP-PLG-9804) with seal kit (KIT-ZEI500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-CLN-4 on Zeiss-Prismo-Z500 by installing PN-ZEI500-PMP-PLG-9804 & kit KIT-ZEI500-CLN-4. Fasteners torqued to 96 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-XAX-5

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI500-5
- PLC Hex Register Address: DB27.DBW_0x15268
- Nominal Operating Range: 27 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-LIN-COIL-9805
- Overhaul Seal & Consumable Kit: KIT-ZEI500-XAX-5
- Critical Fastener Torque Value: 105 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-XAX-5 at PLC register 0x15268.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-LIN-COIL-9805) with seal kit (KIT-ZEI500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 105 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-XAX-5 on Zeiss-Prismo-Z500 by installing PN-ZEI500-LIN-COIL-9805 & kit KIT-ZEI500-XAX-5. Fasteners torqued to 105 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-YAX-6

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI500-6
- PLC Hex Register Address: DB27.DBW_0x15270
- Nominal Operating Range: 30 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-BSW-NUT-9806
- Overhaul Seal & Consumable Kit: KIT-ZEI500-YAX-6
- Critical Fastener Torque Value: 114 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-YAX-6 at PLC register 0x15270.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-BSW-NUT-9806) with seal kit (KIT-ZEI500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 114 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-YAX-6 on Zeiss-Prismo-Z500 by installing PN-ZEI500-BSW-NUT-9806 & kit KIT-ZEI500-YAX-6. Fasteners torqued to 114 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-ZAX-7

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI500-7
- PLC Hex Register Address: DB27.DBW_0x15278
- Nominal Operating Range: 33 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-CYL-SEAL-9807
- Overhaul Seal & Consumable Kit: KIT-ZEI500-ZAX-7
- Critical Fastener Torque Value: 123 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-ZAX-7 at PLC register 0x15278.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-CYL-SEAL-9807) with seal kit (KIT-ZEI500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 123 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-ZAX-7 on Zeiss-Prismo-Z500 by installing PN-ZEI500-CYL-SEAL-9807 & kit KIT-ZEI500-ZAX-7. Fasteners torqued to 123 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-BAX-8

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI500-8
- PLC Hex Register Address: DB27.DBW_0x15280
- Nominal Operating Range: 36 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-TRQ-STAT-9808
- Overhaul Seal & Consumable Kit: KIT-ZEI500-BAX-8
- Critical Fastener Torque Value: 132 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-BAX-8 at PLC register 0x15280.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-TRQ-STAT-9808) with seal kit (KIT-ZEI500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 132 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-BAX-8 on Zeiss-Prismo-Z500 by installing PN-ZEI500-TRQ-STAT-9808 & kit KIT-ZEI500-BAX-8. Fasteners torqued to 132 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-CAX-9

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI500-9
- PLC Hex Register Address: DB27.DBW_0x15288
- Nominal Operating Range: 39 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-BRK-DISC-9809
- Overhaul Seal & Consumable Kit: KIT-ZEI500-CAX-9
- Critical Fastener Torque Value: 141 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-CAX-9 at PLC register 0x15288.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-BRK-DISC-9809) with seal kit (KIT-ZEI500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 141 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-CAX-9 on Zeiss-Prismo-Z500 by installing PN-ZEI500-BRK-DISC-9809 & kit KIT-ZEI500-CAX-9. Fasteners torqued to 141 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-OPT-10

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI500-10
- PLC Hex Register Address: DB27.DBW_0x15290
- Nominal Operating Range: 42 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-GLASS-RD-9810
- Overhaul Seal & Consumable Kit: KIT-ZEI500-OPT-10
- Critical Fastener Torque Value: 150 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-OPT-10 at PLC register 0x15290.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-GLASS-RD-9810) with seal kit (KIT-ZEI500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 150 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-OPT-10 on Zeiss-Prismo-Z500 by installing PN-ZEI500-GLASS-RD-9810 & kit KIT-ZEI500-OPT-10. Fasteners torqued to 150 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-PLC-11

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI500-11
- PLC Hex Register Address: DB27.DBW_0x15298
- Nominal Operating Range: 45 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-IGBT-MOD-9811
- Overhaul Seal & Consumable Kit: KIT-ZEI500-PLC-11
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-PLC-11 at PLC register 0x15298.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-IGBT-MOD-9811) with seal kit (KIT-ZEI500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-PLC-11 on Zeiss-Prismo-Z500 by installing PN-ZEI500-IGBT-MOD-9811 & kit KIT-ZEI500-PLC-11. Fasteners torqued to 159 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-LUB-12

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI500-12
- PLC Hex Register Address: DB27.DBW_0x152A0
- Nominal Operating Range: 48 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-INJ-MET-9812
- Overhaul Seal & Consumable Kit: KIT-ZEI500-LUB-12
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-LUB-12 at PLC register 0x152A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-INJ-MET-9812) with seal kit (KIT-ZEI500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-LUB-12 on Zeiss-Prismo-Z500 by installing PN-ZEI500-INJ-MET-9812 & kit KIT-ZEI500-LUB-12. Fasteners torqued to 168 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-VAC-13

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI500-13
- PLC Hex Register Address: DB27.DBW_0x152A8
- Nominal Operating Range: 51 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-GASK-VIT-9813
- Overhaul Seal & Consumable Kit: KIT-ZEI500-VAC-13
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-VAC-13 at PLC register 0x152A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-GASK-VIT-9813) with seal kit (KIT-ZEI500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-VAC-13 on Zeiss-Prismo-Z500 by installing PN-ZEI500-GASK-VIT-9813 & kit KIT-ZEI500-VAC-13. Fasteners torqued to 177 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI500-THM-14

Target Machine Model: Zeiss-Prismo-Z500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI500-14
- PLC Hex Register Address: DB27.DBW_0x152B0
- Nominal Operating Range: 54 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI500-HEX-CORE-9814
- Overhaul Seal & Consumable Kit: KIT-ZEI500-THM-14
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z500 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI500-THM-14 at PLC register 0x152B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI500-HEX-CORE-9814) with seal kit (KIT-ZEI500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z500, technician must log: "Resolved ERR-ZEI500-THM-14 on Zeiss-Prismo-Z500 by installing PN-ZEI500-HEX-CORE-9814 & kit KIT-ZEI500-THM-14. Fasteners torqued to 186 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-SPN-1

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI550-1
- PLC Hex Register Address: DB28.DBW_0x15568
- Nominal Operating Range: 15 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-BRG-CER-9901
- Overhaul Seal & Consumable Kit: KIT-ZEI550-SPN-1
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-SPN-1 at PLC register 0x15568.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-BRG-CER-9901) with seal kit (KIT-ZEI550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-SPN-1 on Zeiss-Prismo-Z550 by installing PN-ZEI550-BRG-CER-9901 & kit KIT-ZEI550-SPN-1. Fasteners torqued to 209 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-ATC-2

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI550-2
- PLC Hex Register Address: DB28.DBW_0x15570
- Nominal Operating Range: 18 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-CAM-GRIP-9902
- Overhaul Seal & Consumable Kit: KIT-ZEI550-ATC-2
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-ATC-2 at PLC register 0x15570.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-CAM-GRIP-9902) with seal kit (KIT-ZEI550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-ATC-2 on Zeiss-Prismo-Z550 by installing PN-ZEI550-CAM-GRIP-9902 & kit KIT-ZEI550-ATC-2. Fasteners torqued to 28 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-HYD-3

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI550-3
- PLC Hex Register Address: DB28.DBW_0x15578
- Nominal Operating Range: 21 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-VLV-PROP-9903
- Overhaul Seal & Consumable Kit: KIT-ZEI550-HYD-3
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-HYD-3 at PLC register 0x15578.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-VLV-PROP-9903) with seal kit (KIT-ZEI550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-HYD-3 on Zeiss-Prismo-Z550 by installing PN-ZEI550-VLV-PROP-9903 & kit KIT-ZEI550-HYD-3. Fasteners torqued to 37 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-CLN-4

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI550-4
- PLC Hex Register Address: DB28.DBW_0x15580
- Nominal Operating Range: 24 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-PMP-PLG-9904
- Overhaul Seal & Consumable Kit: KIT-ZEI550-CLN-4
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-CLN-4 at PLC register 0x15580.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-PMP-PLG-9904) with seal kit (KIT-ZEI550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-CLN-4 on Zeiss-Prismo-Z550 by installing PN-ZEI550-PMP-PLG-9904 & kit KIT-ZEI550-CLN-4. Fasteners torqued to 46 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-XAX-5

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI550-5
- PLC Hex Register Address: DB28.DBW_0x15588
- Nominal Operating Range: 27 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-LIN-COIL-9905
- Overhaul Seal & Consumable Kit: KIT-ZEI550-XAX-5
- Critical Fastener Torque Value: 55 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-XAX-5 at PLC register 0x15588.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-LIN-COIL-9905) with seal kit (KIT-ZEI550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 55 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-XAX-5 on Zeiss-Prismo-Z550 by installing PN-ZEI550-LIN-COIL-9905 & kit KIT-ZEI550-XAX-5. Fasteners torqued to 55 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-YAX-6

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI550-6
- PLC Hex Register Address: DB28.DBW_0x15590
- Nominal Operating Range: 30 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-BSW-NUT-9906
- Overhaul Seal & Consumable Kit: KIT-ZEI550-YAX-6
- Critical Fastener Torque Value: 64 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-YAX-6 at PLC register 0x15590.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-BSW-NUT-9906) with seal kit (KIT-ZEI550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 64 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-YAX-6 on Zeiss-Prismo-Z550 by installing PN-ZEI550-BSW-NUT-9906 & kit KIT-ZEI550-YAX-6. Fasteners torqued to 64 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-ZAX-7

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI550-7
- PLC Hex Register Address: DB28.DBW_0x15598
- Nominal Operating Range: 33 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-CYL-SEAL-9907
- Overhaul Seal & Consumable Kit: KIT-ZEI550-ZAX-7
- Critical Fastener Torque Value: 73 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-ZAX-7 at PLC register 0x15598.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-CYL-SEAL-9907) with seal kit (KIT-ZEI550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 73 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-ZAX-7 on Zeiss-Prismo-Z550 by installing PN-ZEI550-CYL-SEAL-9907 & kit KIT-ZEI550-ZAX-7. Fasteners torqued to 73 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-BAX-8

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI550-8
- PLC Hex Register Address: DB28.DBW_0x155A0
- Nominal Operating Range: 36 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-TRQ-STAT-9908
- Overhaul Seal & Consumable Kit: KIT-ZEI550-BAX-8
- Critical Fastener Torque Value: 82 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-BAX-8 at PLC register 0x155A0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-TRQ-STAT-9908) with seal kit (KIT-ZEI550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 82 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-BAX-8 on Zeiss-Prismo-Z550 by installing PN-ZEI550-TRQ-STAT-9908 & kit KIT-ZEI550-BAX-8. Fasteners torqued to 82 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-CAX-9

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI550-9
- PLC Hex Register Address: DB28.DBW_0x155A8
- Nominal Operating Range: 39 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-BRK-DISC-9909
- Overhaul Seal & Consumable Kit: KIT-ZEI550-CAX-9
- Critical Fastener Torque Value: 91 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-CAX-9 at PLC register 0x155A8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-BRK-DISC-9909) with seal kit (KIT-ZEI550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 91 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-CAX-9 on Zeiss-Prismo-Z550 by installing PN-ZEI550-BRK-DISC-9909 & kit KIT-ZEI550-CAX-9. Fasteners torqued to 91 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-OPT-10

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI550-10
- PLC Hex Register Address: DB28.DBW_0x155B0
- Nominal Operating Range: 42 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-GLASS-RD-9910
- Overhaul Seal & Consumable Kit: KIT-ZEI550-OPT-10
- Critical Fastener Torque Value: 100 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-OPT-10 at PLC register 0x155B0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-GLASS-RD-9910) with seal kit (KIT-ZEI550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 100 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-OPT-10 on Zeiss-Prismo-Z550 by installing PN-ZEI550-GLASS-RD-9910 & kit KIT-ZEI550-OPT-10. Fasteners torqued to 100 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-PLC-11

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI550-11
- PLC Hex Register Address: DB28.DBW_0x155B8
- Nominal Operating Range: 45 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-IGBT-MOD-9911
- Overhaul Seal & Consumable Kit: KIT-ZEI550-PLC-11
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-PLC-11 at PLC register 0x155B8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-IGBT-MOD-9911) with seal kit (KIT-ZEI550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-PLC-11 on Zeiss-Prismo-Z550 by installing PN-ZEI550-IGBT-MOD-9911 & kit KIT-ZEI550-PLC-11. Fasteners torqued to 109 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-LUB-12

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI550-12
- PLC Hex Register Address: DB28.DBW_0x155C0
- Nominal Operating Range: 48 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-INJ-MET-9912
- Overhaul Seal & Consumable Kit: KIT-ZEI550-LUB-12
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-LUB-12 at PLC register 0x155C0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-INJ-MET-9912) with seal kit (KIT-ZEI550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-LUB-12 on Zeiss-Prismo-Z550 by installing PN-ZEI550-INJ-MET-9912 & kit KIT-ZEI550-LUB-12. Fasteners torqued to 118 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-VAC-13

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI550-13
- PLC Hex Register Address: DB28.DBW_0x155C8
- Nominal Operating Range: 51 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-GASK-VIT-9913
- Overhaul Seal & Consumable Kit: KIT-ZEI550-VAC-13
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-VAC-13 at PLC register 0x155C8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-GASK-VIT-9913) with seal kit (KIT-ZEI550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-VAC-13 on Zeiss-Prismo-Z550 by installing PN-ZEI550-GASK-VIT-9913 & kit KIT-ZEI550-VAC-13. Fasteners torqued to 127 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI550-THM-14

Target Machine Model: Zeiss-Prismo-Z550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI550-14
- PLC Hex Register Address: DB28.DBW_0x155D0
- Nominal Operating Range: 54 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI550-HEX-CORE-9914
- Overhaul Seal & Consumable Kit: KIT-ZEI550-THM-14
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z550 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI550-THM-14 at PLC register 0x155D0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI550-HEX-CORE-9914) with seal kit (KIT-ZEI550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z550, technician must log: "Resolved ERR-ZEI550-THM-14 on Zeiss-Prismo-Z550 by installing PN-ZEI550-HEX-CORE-9914 & kit KIT-ZEI550-THM-14. Fasteners torqued to 136 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-SPN-1

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PT-SPN-ZEI600-1
- PLC Hex Register Address: DB29.DBW_0x15888
- Nominal Operating Range: 15 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-BRG-CER-10001
- Overhaul Seal & Consumable Kit: KIT-ZEI600-SPN-1
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-SPN-1 at PLC register 0x15888.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-BRG-CER-10001) with seal kit (KIT-ZEI600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-SPN-1 on Zeiss-Prismo-Z600 by installing PN-ZEI600-BRG-CER-10001 & kit KIT-ZEI600-SPN-1. Fasteners torqued to 159 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-ATC-2

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PX-ATC-ZEI600-2
- PLC Hex Register Address: DB29.DBW_0x15890
- Nominal Operating Range: 18 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-CAM-GRIP-10002
- Overhaul Seal & Consumable Kit: KIT-ZEI600-ATC-2
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-ATC-2 at PLC register 0x15890.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-CAM-GRIP-10002) with seal kit (KIT-ZEI600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-ATC-2 on Zeiss-Prismo-Z600 by installing PN-ZEI600-CAM-GRIP-10002 & kit KIT-ZEI600-ATC-2. Fasteners torqued to 168 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-HYD-3

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-HYD-ZEI600-3
- PLC Hex Register Address: DB29.DBW_0x15898
- Nominal Operating Range: 21 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-VLV-PROP-10003
- Overhaul Seal & Consumable Kit: KIT-ZEI600-HYD-3
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-HYD-3 at PLC register 0x15898.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-VLV-PROP-10003) with seal kit (KIT-ZEI600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-HYD-3 on Zeiss-Prismo-Z600 by installing PN-ZEI600-VLV-PROP-10003 & kit KIT-ZEI600-HYD-3. Fasteners torqued to 177 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-CLN-4

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-CLN-ZEI600-4
- PLC Hex Register Address: DB29.DBW_0x158A0
- Nominal Operating Range: 24 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-PMP-PLG-10004
- Overhaul Seal & Consumable Kit: KIT-ZEI600-CLN-4
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-CLN-4 at PLC register 0x158A0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-PMP-PLG-10004) with seal kit (KIT-ZEI600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-CLN-4 on Zeiss-Prismo-Z600 by installing PN-ZEI600-PMP-PLG-10004 & kit KIT-ZEI600-CLN-4. Fasteners torqued to 186 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-XAX-5

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-XAX-ZEI600-5
- PLC Hex Register Address: DB29.DBW_0x158A8
- Nominal Operating Range: 27 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-LIN-COIL-10005
- Overhaul Seal & Consumable Kit: KIT-ZEI600-XAX-5
- Critical Fastener Torque Value: 195 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-XAX-5 at PLC register 0x158A8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-LIN-COIL-10005) with seal kit (KIT-ZEI600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 195 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-XAX-5 on Zeiss-Prismo-Z600 by installing PN-ZEI600-LIN-COIL-10005 & kit KIT-ZEI600-XAX-5. Fasteners torqued to 195 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-YAX-6

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-YAX-ZEI600-6
- PLC Hex Register Address: DB29.DBW_0x158B0
- Nominal Operating Range: 30 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-BSW-NUT-10006
- Overhaul Seal & Consumable Kit: KIT-ZEI600-YAX-6
- Critical Fastener Torque Value: 204 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-YAX-6 at PLC register 0x158B0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-BSW-NUT-10006) with seal kit (KIT-ZEI600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 204 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-YAX-6 on Zeiss-Prismo-Z600 by installing PN-ZEI600-BSW-NUT-10006 & kit KIT-ZEI600-YAX-6. Fasteners torqued to 204 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-ZAX-7

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-ZAX-ZEI600-7
- PLC Hex Register Address: DB29.DBW_0x158B8
- Nominal Operating Range: 33 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-CYL-SEAL-10007
- Overhaul Seal & Consumable Kit: KIT-ZEI600-ZAX-7
- Critical Fastener Torque Value: 23 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-ZAX-7 at PLC register 0x158B8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-CYL-SEAL-10007) with seal kit (KIT-ZEI600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 23 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-ZAX-7 on Zeiss-Prismo-Z600 by installing PN-ZEI600-CYL-SEAL-10007 & kit KIT-ZEI600-ZAX-7. Fasteners torqued to 23 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-BAX-8

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: EN-BAX-ZEI600-8
- PLC Hex Register Address: DB29.DBW_0x158C0
- Nominal Operating Range: 36 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-TRQ-STAT-10008
- Overhaul Seal & Consumable Kit: KIT-ZEI600-BAX-8
- Critical Fastener Torque Value: 32 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-BAX-8 at PLC register 0x158C0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-TRQ-STAT-10008) with seal kit (KIT-ZEI600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 32 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-BAX-8 on Zeiss-Prismo-Z600 by installing PN-ZEI600-TRQ-STAT-10008 & kit KIT-ZEI600-BAX-8. Fasteners torqued to 32 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-CAX-9

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: PR-CAX-ZEI600-9
- PLC Hex Register Address: DB29.DBW_0x158C8
- Nominal Operating Range: 39 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-BRK-DISC-10009
- Overhaul Seal & Consumable Kit: KIT-ZEI600-CAX-9
- Critical Fastener Torque Value: 41 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-CAX-9 at PLC register 0x158C8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-BRK-DISC-10009) with seal kit (KIT-ZEI600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 41 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-CAX-9 on Zeiss-Prismo-Z600 by installing PN-ZEI600-BRK-DISC-10009 & kit KIT-ZEI600-CAX-9. Fasteners torqued to 41 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-OPT-10

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: OP-SCL-ZEI600-10
- PLC Hex Register Address: DB29.DBW_0x158D0
- Nominal Operating Range: 42 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-GLASS-RD-10010
- Overhaul Seal & Consumable Kit: KIT-ZEI600-OPT-10
- Critical Fastener Torque Value: 50 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-OPT-10 at PLC register 0x158D0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-GLASS-RD-10010) with seal kit (KIT-ZEI600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 50 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-OPT-10 on Zeiss-Prismo-Z600 by installing PN-ZEI600-GLASS-RD-10010 & kit KIT-ZEI600-OPT-10. Fasteners torqued to 50 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-PLC-11

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VT-DRV-ZEI600-11
- PLC Hex Register Address: DB29.DBW_0x158D8
- Nominal Operating Range: 45 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-IGBT-MOD-10011
- Overhaul Seal & Consumable Kit: KIT-ZEI600-PLC-11
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-PLC-11 at PLC register 0x158D8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-IGBT-MOD-10011) with seal kit (KIT-ZEI600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-PLC-11 on Zeiss-Prismo-Z600 by installing PN-ZEI600-IGBT-MOD-10011 & kit KIT-ZEI600-PLC-11. Fasteners torqued to 59 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-LUB-12

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: FL-LUB-ZEI600-12
- PLC Hex Register Address: DB29.DBW_0x158E0
- Nominal Operating Range: 48 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-INJ-MET-10012
- Overhaul Seal & Consumable Kit: KIT-ZEI600-LUB-12
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-LUB-12 at PLC register 0x158E0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-INJ-MET-10012) with seal kit (KIT-ZEI600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-LUB-12 on Zeiss-Prismo-Z600 by installing PN-ZEI600-INJ-MET-10012 & kit KIT-ZEI600-LUB-12. Fasteners torqued to 68 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-VAC-13

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: VC-MNF-ZEI600-13
- PLC Hex Register Address: DB29.DBW_0x158E8
- Nominal Operating Range: 51 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-GASK-VIT-10013
- Overhaul Seal & Consumable Kit: KIT-ZEI600-VAC-13
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-VAC-13 at PLC register 0x158E8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-GASK-VIT-10013) with seal kit (KIT-ZEI600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-VAC-13 on Zeiss-Prismo-Z600 by installing PN-ZEI600-GASK-VIT-10013 & kit KIT-ZEI600-VAC-13. Fasteners torqued to 77 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-ZEI600-THM-14

Target Machine Model: Zeiss-Prismo-Z600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Zeiss-Prismo-Z600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller)
- Telemetry Sensor ID: TM-CHL-ZEI600-14
- PLC Hex Register Address: DB29.DBW_0x158F0
- Nominal Operating Range: 54 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-ZEI600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-ZEI600-HEX-CORE-10014
- Overhaul Seal & Consumable Kit: KIT-ZEI600-THM-14
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Zeiss-Prismo-Z600 (Zeiss C99m CAA Controller) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Zeiss C99m CAA Controller and verify fault latch ERR-ZEI600-THM-14 at PLC register 0x158F0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Zeiss-Prismo-Z600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-ZEI600-HEX-CORE-10014) with seal kit (KIT-ZEI600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-ZEI600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Zeiss-Prismo-Z600, technician must log: "Resolved ERR-ZEI600-THM-14 on Zeiss-Prismo-Z600 by installing PN-ZEI600-HEX-CORE-10014 & kit KIT-ZEI600-THM-14. Fasteners torqued to 86 Nm; alignment verified at ±0.0040 mm."

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G400 (BANK 1)

Machine Model: Grob-Aerospace-G400 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G400.

PRM_47621 [Grob-Aerospace-G400] — ATC_LOOP_GAIN_1 | Default: 87.400 | Linked Fault: ERR-GRO400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G400.

PRM_47622 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_2 | Default: 174.800 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

PRM_47623 [Grob-Aerospace-G400] — CLN_LOOP_GAIN_3 | Default: 12.200 | Linked Fault: ERR-GRO400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G400.

PRM_47624 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_4 | Default: 99.600 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

PRM_47625 [Grob-Aerospace-G400] — YAX_LOOP_GAIN_5 | Default: 187.000 | Linked Fault: ERR-GRO400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G400.

PRM_47626 [Grob-Aerospace-G400] — ZAX_LOOP_GAIN_6 | Default: 24.400 | Linked Fault: ERR-GRO400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400.

PRM_47627 [Grob-Aerospace-G400] — BAX_LOOP_GAIN_7 | Default: 111.800 | Linked Fault: ERR-GRO400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G400.

PRM_47628 [Grob-Aerospace-G400] — CAX_LOOP_GAIN_8 | Default: 199.200 | Linked Fault: ERR-GRO400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400.

PRM_47629 [Grob-Aerospace-G400] — OPT_LOOP_GAIN_9 | Default: 36.600 | Linked Fault: ERR-GRO400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G400.

PRM_47630 [Grob-Aerospace-G400] — PLC_LOOP_GAIN_10 | Default: 124.000 | Linked Fault: ERR-GRO400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400.

PRM_47631 [Grob-Aerospace-G400] — LUB_LOOP_GAIN_11 | Default: 211.400 | Linked Fault: ERR-GRO400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G400.

PRM_47632 [Grob-Aerospace-G400] — VAC_LOOP_GAIN_12 | Default: 48.800 | Linked Fault: ERR-GRO400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G400.

PRM_47633 [Grob-Aerospace-G400] — THM_LOOP_GAIN_13 | Default: 136.200 | Linked Fault: ERR-GRO400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G400.

PRM_47634 [Grob-Aerospace-G400] — SPN_LOOP_GAIN_14 | Default: 223.600 | Linked Fault: ERR-GRO400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G400.

PRM_47635 [Grob-Aerospace-G400] — ATC_LOOP_GAIN_15 | Default: 61.000 | Linked Fault: ERR-GRO400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G400.

PRM_47636 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_16 | Default: 148.400 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G400 (BANK 2)

Machine Model: Grob-Aerospace-G400 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G400.

PRM_47641 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_1 | Default: 174.800 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

PRM_47642 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_2 | Default: 99.600 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

PRM_47643 [Grob-Aerospace-G400] — ZAX_LOOP_GAIN_3 | Default: 24.400 | Linked Fault: ERR-GRO400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400.

PRM_47644 [Grob-Aerospace-G400] — CAX_LOOP_GAIN_4 | Default: 199.200 | Linked Fault: ERR-GRO400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400.

PRM_47645 [Grob-Aerospace-G400] — PLC_LOOP_GAIN_5 | Default: 124.000 | Linked Fault: ERR-GRO400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400.

PRM_47646 [Grob-Aerospace-G400] — VAC_LOOP_GAIN_6 | Default: 48.800 | Linked Fault: ERR-GRO400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G400.

PRM_47647 [Grob-Aerospace-G400] — SPN_LOOP_GAIN_7 | Default: 223.600 | Linked Fault: ERR-GRO400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G400.

PRM_47648 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_8 | Default: 148.400 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

PRM_47649 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_9 | Default: 73.200 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

PRM_47650 [Grob-Aerospace-G400] — ZAX_LOOP_GAIN_10 | Default: 248.000 | Linked Fault: ERR-GRO400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400.

PRM_47651 [Grob-Aerospace-G400] — CAX_LOOP_GAIN_11 | Default: 172.800 | Linked Fault: ERR-GRO400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400.

PRM_47652 [Grob-Aerospace-G400] — PLC_LOOP_GAIN_12 | Default: 97.600 | Linked Fault: ERR-GRO400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400.

PRM_47653 [Grob-Aerospace-G400] — VAC_LOOP_GAIN_13 | Default: 22.400 | Linked Fault: ERR-GRO400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G400.

PRM_47654 [Grob-Aerospace-G400] — SPN_LOOP_GAIN_14 | Default: 197.200 | Linked Fault: ERR-GRO400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G400.

PRM_47655 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_15 | Default: 122.000 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

PRM_47656 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_16 | Default: 46.800 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G400 (BANK 3)

Machine Model: Grob-Aerospace-G400 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G400.

PRM_47661 [Grob-Aerospace-G400] — CLN_LOOP_GAIN_1 | Default: 12.200 | Linked Fault: ERR-GRO400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G400.

PRM_47662 [Grob-Aerospace-G400] — ZAX_LOOP_GAIN_2 | Default: 24.400 | Linked Fault: ERR-GRO400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400.

PRM_47663 [Grob-Aerospace-G400] — OPT_LOOP_GAIN_3 | Default: 36.600 | Linked Fault: ERR-GRO400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G400.

PRM_47664 [Grob-Aerospace-G400] — VAC_LOOP_GAIN_4 | Default: 48.800 | Linked Fault: ERR-GRO400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G400.

PRM_47665 [Grob-Aerospace-G400] — ATC_LOOP_GAIN_5 | Default: 61.000 | Linked Fault: ERR-GRO400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G400.

PRM_47666 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_6 | Default: 73.200 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

PRM_47667 [Grob-Aerospace-G400] — BAX_LOOP_GAIN_7 | Default: 85.400 | Linked Fault: ERR-GRO400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G400.

PRM_47668 [Grob-Aerospace-G400] — PLC_LOOP_GAIN_8 | Default: 97.600 | Linked Fault: ERR-GRO400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400.

PRM_47669 [Grob-Aerospace-G400] — THM_LOOP_GAIN_9 | Default: 109.800 | Linked Fault: ERR-GRO400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G400.

PRM_47670 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_10 | Default: 122.000 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

PRM_47671 [Grob-Aerospace-G400] — YAX_LOOP_GAIN_11 | Default: 134.200 | Linked Fault: ERR-GRO400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G400.

PRM_47672 [Grob-Aerospace-G400] — CAX_LOOP_GAIN_12 | Default: 146.400 | Linked Fault: ERR-GRO400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400.

PRM_47673 [Grob-Aerospace-G400] — LUB_LOOP_GAIN_13 | Default: 158.600 | Linked Fault: ERR-GRO400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G400.

PRM_47674 [Grob-Aerospace-G400] — SPN_LOOP_GAIN_14 | Default: 170.800 | Linked Fault: ERR-GRO400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G400.

PRM_47675 [Grob-Aerospace-G400] — CLN_LOOP_GAIN_15 | Default: 183.000 | Linked Fault: ERR-GRO400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G400.

PRM_47676 [Grob-Aerospace-G400] — ZAX_LOOP_GAIN_16 | Default: 195.200 | Linked Fault: ERR-GRO400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G400 (BANK 4)

Machine Model: Grob-Aerospace-G400 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G400.

PRM_47681 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_1 | Default: 99.600 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

PRM_47682 [Grob-Aerospace-G400] — CAX_LOOP_GAIN_2 | Default: 199.200 | Linked Fault: ERR-GRO400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400.

PRM_47683 [Grob-Aerospace-G400] — VAC_LOOP_GAIN_3 | Default: 48.800 | Linked Fault: ERR-GRO400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G400.

PRM_47684 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_4 | Default: 148.400 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

PRM_47685 [Grob-Aerospace-G400] — ZAX_LOOP_GAIN_5 | Default: 248.000 | Linked Fault: ERR-GRO400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400.

PRM_47686 [Grob-Aerospace-G400] — PLC_LOOP_GAIN_6 | Default: 97.600 | Linked Fault: ERR-GRO400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400.

PRM_47687 [Grob-Aerospace-G400] — SPN_LOOP_GAIN_7 | Default: 197.200 | Linked Fault: ERR-GRO400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G400.

PRM_47688 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_8 | Default: 46.800 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

PRM_47689 [Grob-Aerospace-G400] — CAX_LOOP_GAIN_9 | Default: 146.400 | Linked Fault: ERR-GRO400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400.

PRM_47690 [Grob-Aerospace-G400] — VAC_LOOP_GAIN_10 | Default: 246.000 | Linked Fault: ERR-GRO400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G400.

PRM_47691 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_11 | Default: 95.600 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

PRM_47692 [Grob-Aerospace-G400] — ZAX_LOOP_GAIN_12 | Default: 195.200 | Linked Fault: ERR-GRO400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400.

PRM_47693 [Grob-Aerospace-G400] — PLC_LOOP_GAIN_13 | Default: 44.800 | Linked Fault: ERR-GRO400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400.

PRM_47694 [Grob-Aerospace-G400] — SPN_LOOP_GAIN_14 | Default: 144.400 | Linked Fault: ERR-GRO400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G400.

PRM_47695 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_15 | Default: 244.000 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

PRM_47696 [Grob-Aerospace-G400] — CAX_LOOP_GAIN_16 | Default: 93.600 | Linked Fault: ERR-GRO400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G400 (BANK 5)

Machine Model: Grob-Aerospace-G400 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G400.

PRM_47701 [Grob-Aerospace-G400] — YAX_LOOP_GAIN_1 | Default: 187.000 | Linked Fault: ERR-GRO400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G400.

PRM_47702 [Grob-Aerospace-G400] — PLC_LOOP_GAIN_2 | Default: 124.000 | Linked Fault: ERR-GRO400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400.

PRM_47703 [Grob-Aerospace-G400] — ATC_LOOP_GAIN_3 | Default: 61.000 | Linked Fault: ERR-GRO400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G400.

PRM_47704 [Grob-Aerospace-G400] — ZAX_LOOP_GAIN_4 | Default: 248.000 | Linked Fault: ERR-GRO400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G400.

PRM_47705 [Grob-Aerospace-G400] — LUB_LOOP_GAIN_5 | Default: 185.000 | Linked Fault: ERR-GRO400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G400.

PRM_47706 [Grob-Aerospace-G400] — HYD_LOOP_GAIN_6 | Default: 122.000 | Linked Fault: ERR-GRO400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G400.

PRM_47707 [Grob-Aerospace-G400] — BAX_LOOP_GAIN_7 | Default: 59.000 | Linked Fault: ERR-GRO400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G400.

PRM_47708 [Grob-Aerospace-G400] — VAC_LOOP_GAIN_8 | Default: 246.000 | Linked Fault: ERR-GRO400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G400.

PRM_47709 [Grob-Aerospace-G400] — CLN_LOOP_GAIN_9 | Default: 183.000 | Linked Fault: ERR-GRO400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G400.

PRM_47710 [Grob-Aerospace-G400] — CAX_LOOP_GAIN_10 | Default: 120.000 | Linked Fault: ERR-GRO400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G400.

PRM_47711 [Grob-Aerospace-G400] — THM_LOOP_GAIN_11 | Default: 57.000 | Linked Fault: ERR-GRO400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G400.

PRM_47712 [Grob-Aerospace-G400] — XAX_LOOP_GAIN_12 | Default: 244.000 | Linked Fault: ERR-GRO400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G400.

PRM_47713 [Grob-Aerospace-G400] — OPT_LOOP_GAIN_13 | Default: 181.000 | Linked Fault: ERR-GRO400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G400.

PRM_47714 [Grob-Aerospace-G400] — SPN_LOOP_GAIN_14 | Default: 118.000 | Linked Fault: ERR-GRO400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G400.

PRM_47715 [Grob-Aerospace-G400] — YAX_LOOP_GAIN_15 | Default: 55.000 | Linked Fault: ERR-GRO400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G400.

PRM_47716 [Grob-Aerospace-G400] — PLC_LOOP_GAIN_16 | Default: 242.000 | Linked Fault: ERR-GRO400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G400.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G450 (BANK 1)

Machine Model: Grob-Aerospace-G450 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G450.

PRM_47721 [Grob-Aerospace-G450] — ATC_LOOP_GAIN_1 | Default: 88.550 | Linked Fault: ERR-GRO450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G450.

PRM_47722 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_2 | Default: 177.100 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

PRM_47723 [Grob-Aerospace-G450] — CLN_LOOP_GAIN_3 | Default: 15.650 | Linked Fault: ERR-GRO450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G450.

PRM_47724 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_4 | Default: 104.200 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

PRM_47725 [Grob-Aerospace-G450] — YAX_LOOP_GAIN_5 | Default: 192.750 | Linked Fault: ERR-GRO450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G450.

PRM_47726 [Grob-Aerospace-G450] — ZAX_LOOP_GAIN_6 | Default: 31.300 | Linked Fault: ERR-GRO450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450.

PRM_47727 [Grob-Aerospace-G450] — BAX_LOOP_GAIN_7 | Default: 119.850 | Linked Fault: ERR-GRO450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G450.

PRM_47728 [Grob-Aerospace-G450] — CAX_LOOP_GAIN_8 | Default: 208.400 | Linked Fault: ERR-GRO450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450.

PRM_47729 [Grob-Aerospace-G450] — OPT_LOOP_GAIN_9 | Default: 46.950 | Linked Fault: ERR-GRO450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G450.

PRM_47730 [Grob-Aerospace-G450] — PLC_LOOP_GAIN_10 | Default: 135.500 | Linked Fault: ERR-GRO450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450.

PRM_47731 [Grob-Aerospace-G450] — LUB_LOOP_GAIN_11 | Default: 224.050 | Linked Fault: ERR-GRO450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G450.

PRM_47732 [Grob-Aerospace-G450] — VAC_LOOP_GAIN_12 | Default: 62.600 | Linked Fault: ERR-GRO450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G450.

PRM_47733 [Grob-Aerospace-G450] — THM_LOOP_GAIN_13 | Default: 151.150 | Linked Fault: ERR-GRO450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G450.

PRM_47734 [Grob-Aerospace-G450] — SPN_LOOP_GAIN_14 | Default: 239.700 | Linked Fault: ERR-GRO450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G450.

PRM_47735 [Grob-Aerospace-G450] — ATC_LOOP_GAIN_15 | Default: 78.250 | Linked Fault: ERR-GRO450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G450.

PRM_47736 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_16 | Default: 166.800 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G450 (BANK 2)

Machine Model: Grob-Aerospace-G450 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G450.

PRM_47741 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_1 | Default: 177.100 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

PRM_47742 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_2 | Default: 104.200 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

PRM_47743 [Grob-Aerospace-G450] — ZAX_LOOP_GAIN_3 | Default: 31.300 | Linked Fault: ERR-GRO450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450.

PRM_47744 [Grob-Aerospace-G450] — CAX_LOOP_GAIN_4 | Default: 208.400 | Linked Fault: ERR-GRO450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450.

PRM_47745 [Grob-Aerospace-G450] — PLC_LOOP_GAIN_5 | Default: 135.500 | Linked Fault: ERR-GRO450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450.

PRM_47746 [Grob-Aerospace-G450] — VAC_LOOP_GAIN_6 | Default: 62.600 | Linked Fault: ERR-GRO450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G450.

PRM_47747 [Grob-Aerospace-G450] — SPN_LOOP_GAIN_7 | Default: 239.700 | Linked Fault: ERR-GRO450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G450.

PRM_47748 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_8 | Default: 166.800 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

PRM_47749 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_9 | Default: 93.900 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

PRM_47750 [Grob-Aerospace-G450] — ZAX_LOOP_GAIN_10 | Default: 21.000 | Linked Fault: ERR-GRO450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450.

PRM_47751 [Grob-Aerospace-G450] — CAX_LOOP_GAIN_11 | Default: 198.100 | Linked Fault: ERR-GRO450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450.

PRM_47752 [Grob-Aerospace-G450] — PLC_LOOP_GAIN_12 | Default: 125.200 | Linked Fault: ERR-GRO450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450.

PRM_47753 [Grob-Aerospace-G450] — VAC_LOOP_GAIN_13 | Default: 52.300 | Linked Fault: ERR-GRO450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G450.

PRM_47754 [Grob-Aerospace-G450] — SPN_LOOP_GAIN_14 | Default: 229.400 | Linked Fault: ERR-GRO450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G450.

PRM_47755 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_15 | Default: 156.500 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

PRM_47756 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_16 | Default: 83.600 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G450 (BANK 3)

Machine Model: Grob-Aerospace-G450 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G450.

PRM_47761 [Grob-Aerospace-G450] — CLN_LOOP_GAIN_1 | Default: 15.650 | Linked Fault: ERR-GRO450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G450.

PRM_47762 [Grob-Aerospace-G450] — ZAX_LOOP_GAIN_2 | Default: 31.300 | Linked Fault: ERR-GRO450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450.

PRM_47763 [Grob-Aerospace-G450] — OPT_LOOP_GAIN_3 | Default: 46.950 | Linked Fault: ERR-GRO450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G450.

PRM_47764 [Grob-Aerospace-G450] — VAC_LOOP_GAIN_4 | Default: 62.600 | Linked Fault: ERR-GRO450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G450.

PRM_47765 [Grob-Aerospace-G450] — ATC_LOOP_GAIN_5 | Default: 78.250 | Linked Fault: ERR-GRO450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G450.

PRM_47766 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_6 | Default: 93.900 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

PRM_47767 [Grob-Aerospace-G450] — BAX_LOOP_GAIN_7 | Default: 109.550 | Linked Fault: ERR-GRO450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G450.

PRM_47768 [Grob-Aerospace-G450] — PLC_LOOP_GAIN_8 | Default: 125.200 | Linked Fault: ERR-GRO450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450.

PRM_47769 [Grob-Aerospace-G450] — THM_LOOP_GAIN_9 | Default: 140.850 | Linked Fault: ERR-GRO450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G450.

PRM_47770 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_10 | Default: 156.500 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

PRM_47771 [Grob-Aerospace-G450] — YAX_LOOP_GAIN_11 | Default: 172.150 | Linked Fault: ERR-GRO450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G450.

PRM_47772 [Grob-Aerospace-G450] — CAX_LOOP_GAIN_12 | Default: 187.800 | Linked Fault: ERR-GRO450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450.

PRM_47773 [Grob-Aerospace-G450] — LUB_LOOP_GAIN_13 | Default: 203.450 | Linked Fault: ERR-GRO450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G450.

PRM_47774 [Grob-Aerospace-G450] — SPN_LOOP_GAIN_14 | Default: 219.100 | Linked Fault: ERR-GRO450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G450.

PRM_47775 [Grob-Aerospace-G450] — CLN_LOOP_GAIN_15 | Default: 234.750 | Linked Fault: ERR-GRO450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G450.

PRM_47776 [Grob-Aerospace-G450] — ZAX_LOOP_GAIN_16 | Default: 0.400 | Linked Fault: ERR-GRO450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G450 (BANK 4)

Machine Model: Grob-Aerospace-G450 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G450.

PRM_47781 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_1 | Default: 104.200 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

PRM_47782 [Grob-Aerospace-G450] — CAX_LOOP_GAIN_2 | Default: 208.400 | Linked Fault: ERR-GRO450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450.

PRM_47783 [Grob-Aerospace-G450] — VAC_LOOP_GAIN_3 | Default: 62.600 | Linked Fault: ERR-GRO450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G450.

PRM_47784 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_4 | Default: 166.800 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

PRM_47785 [Grob-Aerospace-G450] — ZAX_LOOP_GAIN_5 | Default: 21.000 | Linked Fault: ERR-GRO450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450.

PRM_47786 [Grob-Aerospace-G450] — PLC_LOOP_GAIN_6 | Default: 125.200 | Linked Fault: ERR-GRO450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450.

PRM_47787 [Grob-Aerospace-G450] — SPN_LOOP_GAIN_7 | Default: 229.400 | Linked Fault: ERR-GRO450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G450.

PRM_47788 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_8 | Default: 83.600 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

PRM_47789 [Grob-Aerospace-G450] — CAX_LOOP_GAIN_9 | Default: 187.800 | Linked Fault: ERR-GRO450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450.

PRM_47790 [Grob-Aerospace-G450] — VAC_LOOP_GAIN_10 | Default: 42.000 | Linked Fault: ERR-GRO450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G450.

PRM_47791 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_11 | Default: 146.200 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

PRM_47792 [Grob-Aerospace-G450] — ZAX_LOOP_GAIN_12 | Default: 0.400 | Linked Fault: ERR-GRO450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450.

PRM_47793 [Grob-Aerospace-G450] — PLC_LOOP_GAIN_13 | Default: 104.600 | Linked Fault: ERR-GRO450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450.

PRM_47794 [Grob-Aerospace-G450] — SPN_LOOP_GAIN_14 | Default: 208.800 | Linked Fault: ERR-GRO450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G450.

PRM_47795 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_15 | Default: 63.000 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

PRM_47796 [Grob-Aerospace-G450] — CAX_LOOP_GAIN_16 | Default: 167.200 | Linked Fault: ERR-GRO450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G450 (BANK 5)

Machine Model: Grob-Aerospace-G450 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G450.

PRM_47801 [Grob-Aerospace-G450] — YAX_LOOP_GAIN_1 | Default: 192.750 | Linked Fault: ERR-GRO450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G450.

PRM_47802 [Grob-Aerospace-G450] — PLC_LOOP_GAIN_2 | Default: 135.500 | Linked Fault: ERR-GRO450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450.

PRM_47803 [Grob-Aerospace-G450] — ATC_LOOP_GAIN_3 | Default: 78.250 | Linked Fault: ERR-GRO450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G450.

PRM_47804 [Grob-Aerospace-G450] — ZAX_LOOP_GAIN_4 | Default: 21.000 | Linked Fault: ERR-GRO450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G450.

PRM_47805 [Grob-Aerospace-G450] — LUB_LOOP_GAIN_5 | Default: 213.750 | Linked Fault: ERR-GRO450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G450.

PRM_47806 [Grob-Aerospace-G450] — HYD_LOOP_GAIN_6 | Default: 156.500 | Linked Fault: ERR-GRO450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G450.

PRM_47807 [Grob-Aerospace-G450] — BAX_LOOP_GAIN_7 | Default: 99.250 | Linked Fault: ERR-GRO450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G450.

PRM_47808 [Grob-Aerospace-G450] — VAC_LOOP_GAIN_8 | Default: 42.000 | Linked Fault: ERR-GRO450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G450.

PRM_47809 [Grob-Aerospace-G450] — CLN_LOOP_GAIN_9 | Default: 234.750 | Linked Fault: ERR-GRO450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G450.

PRM_47810 [Grob-Aerospace-G450] — CAX_LOOP_GAIN_10 | Default: 177.500 | Linked Fault: ERR-GRO450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G450.

PRM_47811 [Grob-Aerospace-G450] — THM_LOOP_GAIN_11 | Default: 120.250 | Linked Fault: ERR-GRO450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G450.

PRM_47812 [Grob-Aerospace-G450] — XAX_LOOP_GAIN_12 | Default: 63.000 | Linked Fault: ERR-GRO450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G450.

PRM_47813 [Grob-Aerospace-G450] — OPT_LOOP_GAIN_13 | Default: 5.750 | Linked Fault: ERR-GRO450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G450.

PRM_47814 [Grob-Aerospace-G450] — SPN_LOOP_GAIN_14 | Default: 198.500 | Linked Fault: ERR-GRO450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G450.

PRM_47815 [Grob-Aerospace-G450] — YAX_LOOP_GAIN_15 | Default: 141.250 | Linked Fault: ERR-GRO450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G450.

PRM_47816 [Grob-Aerospace-G450] — PLC_LOOP_GAIN_16 | Default: 84.000 | Linked Fault: ERR-GRO450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G450.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G500 (BANK 1)

Machine Model: Grob-Aerospace-G500 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G500.

PRM_47821 [Grob-Aerospace-G500] — ATC_LOOP_GAIN_1 | Default: 89.700 | Linked Fault: ERR-GRO500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G500.

PRM_47822 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_2 | Default: 179.400 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

PRM_47823 [Grob-Aerospace-G500] — CLN_LOOP_GAIN_3 | Default: 19.100 | Linked Fault: ERR-GRO500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G500.

PRM_47824 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_4 | Default: 108.800 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

PRM_47825 [Grob-Aerospace-G500] — YAX_LOOP_GAIN_5 | Default: 198.500 | Linked Fault: ERR-GRO500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G500.

PRM_47826 [Grob-Aerospace-G500] — ZAX_LOOP_GAIN_6 | Default: 38.200 | Linked Fault: ERR-GRO500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500.

PRM_47827 [Grob-Aerospace-G500] — BAX_LOOP_GAIN_7 | Default: 127.900 | Linked Fault: ERR-GRO500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G500.

PRM_47828 [Grob-Aerospace-G500] — CAX_LOOP_GAIN_8 | Default: 217.600 | Linked Fault: ERR-GRO500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500.

PRM_47829 [Grob-Aerospace-G500] — OPT_LOOP_GAIN_9 | Default: 57.300 | Linked Fault: ERR-GRO500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G500.

PRM_47830 [Grob-Aerospace-G500] — PLC_LOOP_GAIN_10 | Default: 147.000 | Linked Fault: ERR-GRO500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500.

PRM_47831 [Grob-Aerospace-G500] — LUB_LOOP_GAIN_11 | Default: 236.700 | Linked Fault: ERR-GRO500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G500.

PRM_47832 [Grob-Aerospace-G500] — VAC_LOOP_GAIN_12 | Default: 76.400 | Linked Fault: ERR-GRO500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G500.

PRM_47833 [Grob-Aerospace-G500] — THM_LOOP_GAIN_13 | Default: 166.100 | Linked Fault: ERR-GRO500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G500.

PRM_47834 [Grob-Aerospace-G500] — SPN_LOOP_GAIN_14 | Default: 5.800 | Linked Fault: ERR-GRO500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G500.

PRM_47835 [Grob-Aerospace-G500] — ATC_LOOP_GAIN_15 | Default: 95.500 | Linked Fault: ERR-GRO500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G500.

PRM_47836 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_16 | Default: 185.200 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G500 (BANK 2)

Machine Model: Grob-Aerospace-G500 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G500.

PRM_47841 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_1 | Default: 179.400 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

PRM_47842 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_2 | Default: 108.800 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

PRM_47843 [Grob-Aerospace-G500] — ZAX_LOOP_GAIN_3 | Default: 38.200 | Linked Fault: ERR-GRO500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500.

PRM_47844 [Grob-Aerospace-G500] — CAX_LOOP_GAIN_4 | Default: 217.600 | Linked Fault: ERR-GRO500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500.

PRM_47845 [Grob-Aerospace-G500] — PLC_LOOP_GAIN_5 | Default: 147.000 | Linked Fault: ERR-GRO500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500.

PRM_47846 [Grob-Aerospace-G500] — VAC_LOOP_GAIN_6 | Default: 76.400 | Linked Fault: ERR-GRO500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G500.

PRM_47847 [Grob-Aerospace-G500] — SPN_LOOP_GAIN_7 | Default: 5.800 | Linked Fault: ERR-GRO500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G500.

PRM_47848 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_8 | Default: 185.200 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

PRM_47849 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_9 | Default: 114.600 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

PRM_47850 [Grob-Aerospace-G500] — ZAX_LOOP_GAIN_10 | Default: 44.000 | Linked Fault: ERR-GRO500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500.

PRM_47851 [Grob-Aerospace-G500] — CAX_LOOP_GAIN_11 | Default: 223.400 | Linked Fault: ERR-GRO500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500.

PRM_47852 [Grob-Aerospace-G500] — PLC_LOOP_GAIN_12 | Default: 152.800 | Linked Fault: ERR-GRO500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500.

PRM_47853 [Grob-Aerospace-G500] — VAC_LOOP_GAIN_13 | Default: 82.200 | Linked Fault: ERR-GRO500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G500.

PRM_47854 [Grob-Aerospace-G500] — SPN_LOOP_GAIN_14 | Default: 11.600 | Linked Fault: ERR-GRO500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G500.

PRM_47855 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_15 | Default: 191.000 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

PRM_47856 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_16 | Default: 120.400 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G500 (BANK 3)

Machine Model: Grob-Aerospace-G500 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G500.

PRM_47861 [Grob-Aerospace-G500] — CLN_LOOP_GAIN_1 | Default: 19.100 | Linked Fault: ERR-GRO500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G500.

PRM_47862 [Grob-Aerospace-G500] — ZAX_LOOP_GAIN_2 | Default: 38.200 | Linked Fault: ERR-GRO500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500.

PRM_47863 [Grob-Aerospace-G500] — OPT_LOOP_GAIN_3 | Default: 57.300 | Linked Fault: ERR-GRO500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G500.

PRM_47864 [Grob-Aerospace-G500] — VAC_LOOP_GAIN_4 | Default: 76.400 | Linked Fault: ERR-GRO500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G500.

PRM_47865 [Grob-Aerospace-G500] — ATC_LOOP_GAIN_5 | Default: 95.500 | Linked Fault: ERR-GRO500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G500.

PRM_47866 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_6 | Default: 114.600 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

PRM_47867 [Grob-Aerospace-G500] — BAX_LOOP_GAIN_7 | Default: 133.700 | Linked Fault: ERR-GRO500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G500.

PRM_47868 [Grob-Aerospace-G500] — PLC_LOOP_GAIN_8 | Default: 152.800 | Linked Fault: ERR-GRO500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500.

PRM_47869 [Grob-Aerospace-G500] — THM_LOOP_GAIN_9 | Default: 171.900 | Linked Fault: ERR-GRO500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G500.

PRM_47870 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_10 | Default: 191.000 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

PRM_47871 [Grob-Aerospace-G500] — YAX_LOOP_GAIN_11 | Default: 210.100 | Linked Fault: ERR-GRO500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G500.

PRM_47872 [Grob-Aerospace-G500] — CAX_LOOP_GAIN_12 | Default: 229.200 | Linked Fault: ERR-GRO500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500.

PRM_47873 [Grob-Aerospace-G500] — LUB_LOOP_GAIN_13 | Default: 248.300 | Linked Fault: ERR-GRO500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G500.

PRM_47874 [Grob-Aerospace-G500] — SPN_LOOP_GAIN_14 | Default: 17.400 | Linked Fault: ERR-GRO500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G500.

PRM_47875 [Grob-Aerospace-G500] — CLN_LOOP_GAIN_15 | Default: 36.500 | Linked Fault: ERR-GRO500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G500.

PRM_47876 [Grob-Aerospace-G500] — ZAX_LOOP_GAIN_16 | Default: 55.600 | Linked Fault: ERR-GRO500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G500 (BANK 4)

Machine Model: Grob-Aerospace-G500 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G500.

PRM_47881 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_1 | Default: 108.800 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

PRM_47882 [Grob-Aerospace-G500] — CAX_LOOP_GAIN_2 | Default: 217.600 | Linked Fault: ERR-GRO500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500.

PRM_47883 [Grob-Aerospace-G500] — VAC_LOOP_GAIN_3 | Default: 76.400 | Linked Fault: ERR-GRO500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G500.

PRM_47884 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_4 | Default: 185.200 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

PRM_47885 [Grob-Aerospace-G500] — ZAX_LOOP_GAIN_5 | Default: 44.000 | Linked Fault: ERR-GRO500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500.

PRM_47886 [Grob-Aerospace-G500] — PLC_LOOP_GAIN_6 | Default: 152.800 | Linked Fault: ERR-GRO500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500.

PRM_47887 [Grob-Aerospace-G500] — SPN_LOOP_GAIN_7 | Default: 11.600 | Linked Fault: ERR-GRO500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G500.

PRM_47888 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_8 | Default: 120.400 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

PRM_47889 [Grob-Aerospace-G500] — CAX_LOOP_GAIN_9 | Default: 229.200 | Linked Fault: ERR-GRO500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500.

PRM_47890 [Grob-Aerospace-G500] — VAC_LOOP_GAIN_10 | Default: 88.000 | Linked Fault: ERR-GRO500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G500.

PRM_47891 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_11 | Default: 196.800 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

PRM_47892 [Grob-Aerospace-G500] — ZAX_LOOP_GAIN_12 | Default: 55.600 | Linked Fault: ERR-GRO500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500.

PRM_47893 [Grob-Aerospace-G500] — PLC_LOOP_GAIN_13 | Default: 164.400 | Linked Fault: ERR-GRO500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500.

PRM_47894 [Grob-Aerospace-G500] — SPN_LOOP_GAIN_14 | Default: 23.200 | Linked Fault: ERR-GRO500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G500.

PRM_47895 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_15 | Default: 132.000 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

PRM_47896 [Grob-Aerospace-G500] — CAX_LOOP_GAIN_16 | Default: 240.800 | Linked Fault: ERR-GRO500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G500 (BANK 5)

Machine Model: Grob-Aerospace-G500 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G500.

PRM_47901 [Grob-Aerospace-G500] — YAX_LOOP_GAIN_1 | Default: 198.500 | Linked Fault: ERR-GRO500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G500.

PRM_47902 [Grob-Aerospace-G500] — PLC_LOOP_GAIN_2 | Default: 147.000 | Linked Fault: ERR-GRO500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500.

PRM_47903 [Grob-Aerospace-G500] — ATC_LOOP_GAIN_3 | Default: 95.500 | Linked Fault: ERR-GRO500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G500.

PRM_47904 [Grob-Aerospace-G500] — ZAX_LOOP_GAIN_4 | Default: 44.000 | Linked Fault: ERR-GRO500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G500.

PRM_47905 [Grob-Aerospace-G500] — LUB_LOOP_GAIN_5 | Default: 242.500 | Linked Fault: ERR-GRO500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G500.

PRM_47906 [Grob-Aerospace-G500] — HYD_LOOP_GAIN_6 | Default: 191.000 | Linked Fault: ERR-GRO500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G500.

PRM_47907 [Grob-Aerospace-G500] — BAX_LOOP_GAIN_7 | Default: 139.500 | Linked Fault: ERR-GRO500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G500.

PRM_47908 [Grob-Aerospace-G500] — VAC_LOOP_GAIN_8 | Default: 88.000 | Linked Fault: ERR-GRO500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G500.

PRM_47909 [Grob-Aerospace-G500] — CLN_LOOP_GAIN_9 | Default: 36.500 | Linked Fault: ERR-GRO500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G500.

PRM_47910 [Grob-Aerospace-G500] — CAX_LOOP_GAIN_10 | Default: 235.000 | Linked Fault: ERR-GRO500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G500.

PRM_47911 [Grob-Aerospace-G500] — THM_LOOP_GAIN_11 | Default: 183.500 | Linked Fault: ERR-GRO500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G500.

PRM_47912 [Grob-Aerospace-G500] — XAX_LOOP_GAIN_12 | Default: 132.000 | Linked Fault: ERR-GRO500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G500.

PRM_47913 [Grob-Aerospace-G500] — OPT_LOOP_GAIN_13 | Default: 80.500 | Linked Fault: ERR-GRO500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G500.

PRM_47914 [Grob-Aerospace-G500] — SPN_LOOP_GAIN_14 | Default: 29.000 | Linked Fault: ERR-GRO500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G500.

PRM_47915 [Grob-Aerospace-G500] — YAX_LOOP_GAIN_15 | Default: 227.500 | Linked Fault: ERR-GRO500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G500.

PRM_47916 [Grob-Aerospace-G500] — PLC_LOOP_GAIN_16 | Default: 176.000 | Linked Fault: ERR-GRO500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G500.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G550 (BANK 1)

Machine Model: Grob-Aerospace-G550 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G550.

PRM_47921 [Grob-Aerospace-G550] — ATC_LOOP_GAIN_1 | Default: 90.850 | Linked Fault: ERR-GRO550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G550.

PRM_47922 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_2 | Default: 181.700 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

PRM_47923 [Grob-Aerospace-G550] — CLN_LOOP_GAIN_3 | Default: 22.550 | Linked Fault: ERR-GRO550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G550.

PRM_47924 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_4 | Default: 113.400 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

PRM_47925 [Grob-Aerospace-G550] — YAX_LOOP_GAIN_5 | Default: 204.250 | Linked Fault: ERR-GRO550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G550.

PRM_47926 [Grob-Aerospace-G550] — ZAX_LOOP_GAIN_6 | Default: 45.100 | Linked Fault: ERR-GRO550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550.

PRM_47927 [Grob-Aerospace-G550] — BAX_LOOP_GAIN_7 | Default: 135.950 | Linked Fault: ERR-GRO550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G550.

PRM_47928 [Grob-Aerospace-G550] — CAX_LOOP_GAIN_8 | Default: 226.800 | Linked Fault: ERR-GRO550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550.

PRM_47929 [Grob-Aerospace-G550] — OPT_LOOP_GAIN_9 | Default: 67.650 | Linked Fault: ERR-GRO550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G550.

PRM_47930 [Grob-Aerospace-G550] — PLC_LOOP_GAIN_10 | Default: 158.500 | Linked Fault: ERR-GRO550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550.

PRM_47931 [Grob-Aerospace-G550] — LUB_LOOP_GAIN_11 | Default: 249.350 | Linked Fault: ERR-GRO550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G550.

PRM_47932 [Grob-Aerospace-G550] — VAC_LOOP_GAIN_12 | Default: 90.200 | Linked Fault: ERR-GRO550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G550.

PRM_47933 [Grob-Aerospace-G550] — THM_LOOP_GAIN_13 | Default: 181.050 | Linked Fault: ERR-GRO550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G550.

PRM_47934 [Grob-Aerospace-G550] — SPN_LOOP_GAIN_14 | Default: 21.900 | Linked Fault: ERR-GRO550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G550.

PRM_47935 [Grob-Aerospace-G550] — ATC_LOOP_GAIN_15 | Default: 112.750 | Linked Fault: ERR-GRO550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G550.

PRM_47936 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_16 | Default: 203.600 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G550 (BANK 2)

Machine Model: Grob-Aerospace-G550 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G550.

PRM_47941 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_1 | Default: 181.700 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

PRM_47942 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_2 | Default: 113.400 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

PRM_47943 [Grob-Aerospace-G550] — ZAX_LOOP_GAIN_3 | Default: 45.100 | Linked Fault: ERR-GRO550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550.

PRM_47944 [Grob-Aerospace-G550] — CAX_LOOP_GAIN_4 | Default: 226.800 | Linked Fault: ERR-GRO550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550.

PRM_47945 [Grob-Aerospace-G550] — PLC_LOOP_GAIN_5 | Default: 158.500 | Linked Fault: ERR-GRO550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550.

PRM_47946 [Grob-Aerospace-G550] — VAC_LOOP_GAIN_6 | Default: 90.200 | Linked Fault: ERR-GRO550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G550.

PRM_47947 [Grob-Aerospace-G550] — SPN_LOOP_GAIN_7 | Default: 21.900 | Linked Fault: ERR-GRO550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G550.

PRM_47948 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_8 | Default: 203.600 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

PRM_47949 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_9 | Default: 135.300 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

PRM_47950 [Grob-Aerospace-G550] — ZAX_LOOP_GAIN_10 | Default: 67.000 | Linked Fault: ERR-GRO550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550.

PRM_47951 [Grob-Aerospace-G550] — CAX_LOOP_GAIN_11 | Default: 248.700 | Linked Fault: ERR-GRO550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550.

PRM_47952 [Grob-Aerospace-G550] — PLC_LOOP_GAIN_12 | Default: 180.400 | Linked Fault: ERR-GRO550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550.

PRM_47953 [Grob-Aerospace-G550] — VAC_LOOP_GAIN_13 | Default: 112.100 | Linked Fault: ERR-GRO550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G550.

PRM_47954 [Grob-Aerospace-G550] — SPN_LOOP_GAIN_14 | Default: 43.800 | Linked Fault: ERR-GRO550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G550.

PRM_47955 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_15 | Default: 225.500 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

PRM_47956 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_16 | Default: 157.200 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G550 (BANK 3)

Machine Model: Grob-Aerospace-G550 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G550.

PRM_47961 [Grob-Aerospace-G550] — CLN_LOOP_GAIN_1 | Default: 22.550 | Linked Fault: ERR-GRO550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G550.

PRM_47962 [Grob-Aerospace-G550] — ZAX_LOOP_GAIN_2 | Default: 45.100 | Linked Fault: ERR-GRO550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550.

PRM_47963 [Grob-Aerospace-G550] — OPT_LOOP_GAIN_3 | Default: 67.650 | Linked Fault: ERR-GRO550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G550.

PRM_47964 [Grob-Aerospace-G550] — VAC_LOOP_GAIN_4 | Default: 90.200 | Linked Fault: ERR-GRO550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G550.

PRM_47965 [Grob-Aerospace-G550] — ATC_LOOP_GAIN_5 | Default: 112.750 | Linked Fault: ERR-GRO550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G550.

PRM_47966 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_6 | Default: 135.300 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

PRM_47967 [Grob-Aerospace-G550] — BAX_LOOP_GAIN_7 | Default: 157.850 | Linked Fault: ERR-GRO550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G550.

PRM_47968 [Grob-Aerospace-G550] — PLC_LOOP_GAIN_8 | Default: 180.400 | Linked Fault: ERR-GRO550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550.

PRM_47969 [Grob-Aerospace-G550] — THM_LOOP_GAIN_9 | Default: 202.950 | Linked Fault: ERR-GRO550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G550.

PRM_47970 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_10 | Default: 225.500 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

PRM_47971 [Grob-Aerospace-G550] — YAX_LOOP_GAIN_11 | Default: 248.050 | Linked Fault: ERR-GRO550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G550.

PRM_47972 [Grob-Aerospace-G550] — CAX_LOOP_GAIN_12 | Default: 20.600 | Linked Fault: ERR-GRO550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550.

PRM_47973 [Grob-Aerospace-G550] — LUB_LOOP_GAIN_13 | Default: 43.150 | Linked Fault: ERR-GRO550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G550.

PRM_47974 [Grob-Aerospace-G550] — SPN_LOOP_GAIN_14 | Default: 65.700 | Linked Fault: ERR-GRO550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G550.

PRM_47975 [Grob-Aerospace-G550] — CLN_LOOP_GAIN_15 | Default: 88.250 | Linked Fault: ERR-GRO550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G550.

PRM_47976 [Grob-Aerospace-G550] — ZAX_LOOP_GAIN_16 | Default: 110.800 | Linked Fault: ERR-GRO550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G550 (BANK 4)

Machine Model: Grob-Aerospace-G550 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G550.

PRM_47981 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_1 | Default: 113.400 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

PRM_47982 [Grob-Aerospace-G550] — CAX_LOOP_GAIN_2 | Default: 226.800 | Linked Fault: ERR-GRO550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550.

PRM_47983 [Grob-Aerospace-G550] — VAC_LOOP_GAIN_3 | Default: 90.200 | Linked Fault: ERR-GRO550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G550.

PRM_47984 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_4 | Default: 203.600 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

PRM_47985 [Grob-Aerospace-G550] — ZAX_LOOP_GAIN_5 | Default: 67.000 | Linked Fault: ERR-GRO550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550.

PRM_47986 [Grob-Aerospace-G550] — PLC_LOOP_GAIN_6 | Default: 180.400 | Linked Fault: ERR-GRO550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550.

PRM_47987 [Grob-Aerospace-G550] — SPN_LOOP_GAIN_7 | Default: 43.800 | Linked Fault: ERR-GRO550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G550.

PRM_47988 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_8 | Default: 157.200 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

PRM_47989 [Grob-Aerospace-G550] — CAX_LOOP_GAIN_9 | Default: 20.600 | Linked Fault: ERR-GRO550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550.

PRM_47990 [Grob-Aerospace-G550] — VAC_LOOP_GAIN_10 | Default: 134.000 | Linked Fault: ERR-GRO550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G550.

PRM_47991 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_11 | Default: 247.400 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

PRM_47992 [Grob-Aerospace-G550] — ZAX_LOOP_GAIN_12 | Default: 110.800 | Linked Fault: ERR-GRO550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550.

PRM_47993 [Grob-Aerospace-G550] — PLC_LOOP_GAIN_13 | Default: 224.200 | Linked Fault: ERR-GRO550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550.

PRM_47994 [Grob-Aerospace-G550] — SPN_LOOP_GAIN_14 | Default: 87.600 | Linked Fault: ERR-GRO550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G550.

PRM_47995 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_15 | Default: 201.000 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

PRM_47996 [Grob-Aerospace-G550] — CAX_LOOP_GAIN_16 | Default: 64.400 | Linked Fault: ERR-GRO550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G550 (BANK 5)

Machine Model: Grob-Aerospace-G550 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G550.

PRM_48001 [Grob-Aerospace-G550] — YAX_LOOP_GAIN_1 | Default: 204.250 | Linked Fault: ERR-GRO550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G550.

PRM_48002 [Grob-Aerospace-G550] — PLC_LOOP_GAIN_2 | Default: 158.500 | Linked Fault: ERR-GRO550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550.

PRM_48003 [Grob-Aerospace-G550] — ATC_LOOP_GAIN_3 | Default: 112.750 | Linked Fault: ERR-GRO550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G550.

PRM_48004 [Grob-Aerospace-G550] — ZAX_LOOP_GAIN_4 | Default: 67.000 | Linked Fault: ERR-GRO550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G550.

PRM_48005 [Grob-Aerospace-G550] — LUB_LOOP_GAIN_5 | Default: 21.250 | Linked Fault: ERR-GRO550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G550.

PRM_48006 [Grob-Aerospace-G550] — HYD_LOOP_GAIN_6 | Default: 225.500 | Linked Fault: ERR-GRO550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G550.

PRM_48007 [Grob-Aerospace-G550] — BAX_LOOP_GAIN_7 | Default: 179.750 | Linked Fault: ERR-GRO550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G550.

PRM_48008 [Grob-Aerospace-G550] — VAC_LOOP_GAIN_8 | Default: 134.000 | Linked Fault: ERR-GRO550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G550.

PRM_48009 [Grob-Aerospace-G550] — CLN_LOOP_GAIN_9 | Default: 88.250 | Linked Fault: ERR-GRO550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G550.

PRM_48010 [Grob-Aerospace-G550] — CAX_LOOP_GAIN_10 | Default: 42.500 | Linked Fault: ERR-GRO550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G550.

PRM_48011 [Grob-Aerospace-G550] — THM_LOOP_GAIN_11 | Default: 246.750 | Linked Fault: ERR-GRO550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G550.

PRM_48012 [Grob-Aerospace-G550] — XAX_LOOP_GAIN_12 | Default: 201.000 | Linked Fault: ERR-GRO550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G550.

PRM_48013 [Grob-Aerospace-G550] — OPT_LOOP_GAIN_13 | Default: 155.250 | Linked Fault: ERR-GRO550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G550.

PRM_48014 [Grob-Aerospace-G550] — SPN_LOOP_GAIN_14 | Default: 109.500 | Linked Fault: ERR-GRO550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G550.

PRM_48015 [Grob-Aerospace-G550] — YAX_LOOP_GAIN_15 | Default: 63.750 | Linked Fault: ERR-GRO550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G550.

PRM_48016 [Grob-Aerospace-G550] — PLC_LOOP_GAIN_16 | Default: 18.000 | Linked Fault: ERR-GRO550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G550.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G600 (BANK 1)

Machine Model: Grob-Aerospace-G600 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G600.

PRM_48021 [Grob-Aerospace-G600] — ATC_LOOP_GAIN_1 | Default: 92.000 | Linked Fault: ERR-GRO600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G600.

PRM_48022 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_2 | Default: 184.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

PRM_48023 [Grob-Aerospace-G600] — CLN_LOOP_GAIN_3 | Default: 26.000 | Linked Fault: ERR-GRO600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G600.

PRM_48024 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_4 | Default: 118.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

PRM_48025 [Grob-Aerospace-G600] — YAX_LOOP_GAIN_5 | Default: 210.000 | Linked Fault: ERR-GRO600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G600.

PRM_48026 [Grob-Aerospace-G600] — ZAX_LOOP_GAIN_6 | Default: 52.000 | Linked Fault: ERR-GRO600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600.

PRM_48027 [Grob-Aerospace-G600] — BAX_LOOP_GAIN_7 | Default: 144.000 | Linked Fault: ERR-GRO600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G600.

PRM_48028 [Grob-Aerospace-G600] — CAX_LOOP_GAIN_8 | Default: 236.000 | Linked Fault: ERR-GRO600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600.

PRM_48029 [Grob-Aerospace-G600] — OPT_LOOP_GAIN_9 | Default: 78.000 | Linked Fault: ERR-GRO600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G600.

PRM_48030 [Grob-Aerospace-G600] — PLC_LOOP_GAIN_10 | Default: 170.000 | Linked Fault: ERR-GRO600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600.

PRM_48031 [Grob-Aerospace-G600] — LUB_LOOP_GAIN_11 | Default: 12.000 | Linked Fault: ERR-GRO600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G600.

PRM_48032 [Grob-Aerospace-G600] — VAC_LOOP_GAIN_12 | Default: 104.000 | Linked Fault: ERR-GRO600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G600.

PRM_48033 [Grob-Aerospace-G600] — THM_LOOP_GAIN_13 | Default: 196.000 | Linked Fault: ERR-GRO600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G600.

PRM_48034 [Grob-Aerospace-G600] — SPN_LOOP_GAIN_14 | Default: 38.000 | Linked Fault: ERR-GRO600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G600.

PRM_48035 [Grob-Aerospace-G600] — ATC_LOOP_GAIN_15 | Default: 130.000 | Linked Fault: ERR-GRO600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G600.

PRM_48036 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_16 | Default: 222.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G600 (BANK 2)

Machine Model: Grob-Aerospace-G600 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G600.

PRM_48041 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_1 | Default: 184.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

PRM_48042 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_2 | Default: 118.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

PRM_48043 [Grob-Aerospace-G600] — ZAX_LOOP_GAIN_3 | Default: 52.000 | Linked Fault: ERR-GRO600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600.

PRM_48044 [Grob-Aerospace-G600] — CAX_LOOP_GAIN_4 | Default: 236.000 | Linked Fault: ERR-GRO600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600.

PRM_48045 [Grob-Aerospace-G600] — PLC_LOOP_GAIN_5 | Default: 170.000 | Linked Fault: ERR-GRO600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600.

PRM_48046 [Grob-Aerospace-G600] — VAC_LOOP_GAIN_6 | Default: 104.000 | Linked Fault: ERR-GRO600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G600.

PRM_48047 [Grob-Aerospace-G600] — SPN_LOOP_GAIN_7 | Default: 38.000 | Linked Fault: ERR-GRO600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G600.

PRM_48048 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_8 | Default: 222.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

PRM_48049 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_9 | Default: 156.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

PRM_48050 [Grob-Aerospace-G600] — ZAX_LOOP_GAIN_10 | Default: 90.000 | Linked Fault: ERR-GRO600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600.

PRM_48051 [Grob-Aerospace-G600] — CAX_LOOP_GAIN_11 | Default: 24.000 | Linked Fault: ERR-GRO600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600.

PRM_48052 [Grob-Aerospace-G600] — PLC_LOOP_GAIN_12 | Default: 208.000 | Linked Fault: ERR-GRO600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600.

PRM_48053 [Grob-Aerospace-G600] — VAC_LOOP_GAIN_13 | Default: 142.000 | Linked Fault: ERR-GRO600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G600.

PRM_48054 [Grob-Aerospace-G600] — SPN_LOOP_GAIN_14 | Default: 76.000 | Linked Fault: ERR-GRO600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G600.

PRM_48055 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_15 | Default: 10.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

PRM_48056 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_16 | Default: 194.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G600 (BANK 3)

Machine Model: Grob-Aerospace-G600 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G600.

PRM_48061 [Grob-Aerospace-G600] — CLN_LOOP_GAIN_1 | Default: 26.000 | Linked Fault: ERR-GRO600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G600.

PRM_48062 [Grob-Aerospace-G600] — ZAX_LOOP_GAIN_2 | Default: 52.000 | Linked Fault: ERR-GRO600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600.

PRM_48063 [Grob-Aerospace-G600] — OPT_LOOP_GAIN_3 | Default: 78.000 | Linked Fault: ERR-GRO600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G600.

PRM_48064 [Grob-Aerospace-G600] — VAC_LOOP_GAIN_4 | Default: 104.000 | Linked Fault: ERR-GRO600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G600.

PRM_48065 [Grob-Aerospace-G600] — ATC_LOOP_GAIN_5 | Default: 130.000 | Linked Fault: ERR-GRO600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G600.

PRM_48066 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_6 | Default: 156.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

PRM_48067 [Grob-Aerospace-G600] — BAX_LOOP_GAIN_7 | Default: 182.000 | Linked Fault: ERR-GRO600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G600.

PRM_48068 [Grob-Aerospace-G600] — PLC_LOOP_GAIN_8 | Default: 208.000 | Linked Fault: ERR-GRO600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600.

PRM_48069 [Grob-Aerospace-G600] — THM_LOOP_GAIN_9 | Default: 234.000 | Linked Fault: ERR-GRO600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G600.

PRM_48070 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_10 | Default: 10.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

PRM_48071 [Grob-Aerospace-G600] — YAX_LOOP_GAIN_11 | Default: 36.000 | Linked Fault: ERR-GRO600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G600.

PRM_48072 [Grob-Aerospace-G600] — CAX_LOOP_GAIN_12 | Default: 62.000 | Linked Fault: ERR-GRO600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600.

PRM_48073 [Grob-Aerospace-G600] — LUB_LOOP_GAIN_13 | Default: 88.000 | Linked Fault: ERR-GRO600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G600.

PRM_48074 [Grob-Aerospace-G600] — SPN_LOOP_GAIN_14 | Default: 114.000 | Linked Fault: ERR-GRO600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G600.

PRM_48075 [Grob-Aerospace-G600] — CLN_LOOP_GAIN_15 | Default: 140.000 | Linked Fault: ERR-GRO600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G600.

PRM_48076 [Grob-Aerospace-G600] — ZAX_LOOP_GAIN_16 | Default: 166.000 | Linked Fault: ERR-GRO600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G600 (BANK 4)

Machine Model: Grob-Aerospace-G600 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G600.

PRM_48081 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_1 | Default: 118.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

PRM_48082 [Grob-Aerospace-G600] — CAX_LOOP_GAIN_2 | Default: 236.000 | Linked Fault: ERR-GRO600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600.

PRM_48083 [Grob-Aerospace-G600] — VAC_LOOP_GAIN_3 | Default: 104.000 | Linked Fault: ERR-GRO600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G600.

PRM_48084 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_4 | Default: 222.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

PRM_48085 [Grob-Aerospace-G600] — ZAX_LOOP_GAIN_5 | Default: 90.000 | Linked Fault: ERR-GRO600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600.

PRM_48086 [Grob-Aerospace-G600] — PLC_LOOP_GAIN_6 | Default: 208.000 | Linked Fault: ERR-GRO600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600.

PRM_48087 [Grob-Aerospace-G600] — SPN_LOOP_GAIN_7 | Default: 76.000 | Linked Fault: ERR-GRO600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G600.

PRM_48088 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_8 | Default: 194.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

PRM_48089 [Grob-Aerospace-G600] — CAX_LOOP_GAIN_9 | Default: 62.000 | Linked Fault: ERR-GRO600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600.

PRM_48090 [Grob-Aerospace-G600] — VAC_LOOP_GAIN_10 | Default: 180.000 | Linked Fault: ERR-GRO600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G600.

PRM_48091 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_11 | Default: 48.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

PRM_48092 [Grob-Aerospace-G600] — ZAX_LOOP_GAIN_12 | Default: 166.000 | Linked Fault: ERR-GRO600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600.

PRM_48093 [Grob-Aerospace-G600] — PLC_LOOP_GAIN_13 | Default: 34.000 | Linked Fault: ERR-GRO600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600.

PRM_48094 [Grob-Aerospace-G600] — SPN_LOOP_GAIN_14 | Default: 152.000 | Linked Fault: ERR-GRO600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G600.

PRM_48095 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_15 | Default: 20.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

PRM_48096 [Grob-Aerospace-G600] — CAX_LOOP_GAIN_16 | Default: 138.000 | Linked Fault: ERR-GRO600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G600 (BANK 5)

Machine Model: Grob-Aerospace-G600 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G600.

PRM_48101 [Grob-Aerospace-G600] — YAX_LOOP_GAIN_1 | Default: 210.000 | Linked Fault: ERR-GRO600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G600.

PRM_48102 [Grob-Aerospace-G600] — PLC_LOOP_GAIN_2 | Default: 170.000 | Linked Fault: ERR-GRO600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600.

PRM_48103 [Grob-Aerospace-G600] — ATC_LOOP_GAIN_3 | Default: 130.000 | Linked Fault: ERR-GRO600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G600.

PRM_48104 [Grob-Aerospace-G600] — ZAX_LOOP_GAIN_4 | Default: 90.000 | Linked Fault: ERR-GRO600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G600.

PRM_48105 [Grob-Aerospace-G600] — LUB_LOOP_GAIN_5 | Default: 50.000 | Linked Fault: ERR-GRO600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G600.

PRM_48106 [Grob-Aerospace-G600] — HYD_LOOP_GAIN_6 | Default: 10.000 | Linked Fault: ERR-GRO600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G600.

PRM_48107 [Grob-Aerospace-G600] — BAX_LOOP_GAIN_7 | Default: 220.000 | Linked Fault: ERR-GRO600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G600.

PRM_48108 [Grob-Aerospace-G600] — VAC_LOOP_GAIN_8 | Default: 180.000 | Linked Fault: ERR-GRO600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G600.

PRM_48109 [Grob-Aerospace-G600] — CLN_LOOP_GAIN_9 | Default: 140.000 | Linked Fault: ERR-GRO600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G600.

PRM_48110 [Grob-Aerospace-G600] — CAX_LOOP_GAIN_10 | Default: 100.000 | Linked Fault: ERR-GRO600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G600.

PRM_48111 [Grob-Aerospace-G600] — THM_LOOP_GAIN_11 | Default: 60.000 | Linked Fault: ERR-GRO600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G600.

PRM_48112 [Grob-Aerospace-G600] — XAX_LOOP_GAIN_12 | Default: 20.000 | Linked Fault: ERR-GRO600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G600.

PRM_48113 [Grob-Aerospace-G600] — OPT_LOOP_GAIN_13 | Default: 230.000 | Linked Fault: ERR-GRO600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G600.

PRM_48114 [Grob-Aerospace-G600] — SPN_LOOP_GAIN_14 | Default: 190.000 | Linked Fault: ERR-GRO600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G600.

PRM_48115 [Grob-Aerospace-G600] — YAX_LOOP_GAIN_15 | Default: 150.000 | Linked Fault: ERR-GRO600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G600.

PRM_48116 [Grob-Aerospace-G600] — PLC_LOOP_GAIN_16 | Default: 110.000 | Linked Fault: ERR-GRO600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G600.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C150 (BANK 1)

Machine Model: Hermle-Performance-C150 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C150.

PRM_48121 [Hermle-Performance-C150] — ATC_LOOP_GAIN_1 | Default: 93.150 | Linked Fault: ERR-HER150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C150.

PRM_48122 [Hermle-Performance-C150] — HYD_LOOP_GAIN_2 | Default: 186.300 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

PRM_48123 [Hermle-Performance-C150] — CLN_LOOP_GAIN_3 | Default: 29.450 | Linked Fault: ERR-HER150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C150.

PRM_48124 [Hermle-Performance-C150] — XAX_LOOP_GAIN_4 | Default: 122.600 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

PRM_48125 [Hermle-Performance-C150] — YAX_LOOP_GAIN_5 | Default: 215.750 | Linked Fault: ERR-HER150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C150.

PRM_48126 [Hermle-Performance-C150] — ZAX_LOOP_GAIN_6 | Default: 58.900 | Linked Fault: ERR-HER150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150.

PRM_48127 [Hermle-Performance-C150] — BAX_LOOP_GAIN_7 | Default: 152.050 | Linked Fault: ERR-HER150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C150.

PRM_48128 [Hermle-Performance-C150] — CAX_LOOP_GAIN_8 | Default: 245.200 | Linked Fault: ERR-HER150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150.

PRM_48129 [Hermle-Performance-C150] — OPT_LOOP_GAIN_9 | Default: 88.350 | Linked Fault: ERR-HER150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C150.

PRM_48130 [Hermle-Performance-C150] — PLC_LOOP_GAIN_10 | Default: 181.500 | Linked Fault: ERR-HER150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150.

PRM_48131 [Hermle-Performance-C150] — LUB_LOOP_GAIN_11 | Default: 24.650 | Linked Fault: ERR-HER150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C150.

PRM_48132 [Hermle-Performance-C150] — VAC_LOOP_GAIN_12 | Default: 117.800 | Linked Fault: ERR-HER150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C150.

PRM_48133 [Hermle-Performance-C150] — THM_LOOP_GAIN_13 | Default: 210.950 | Linked Fault: ERR-HER150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C150.

PRM_48134 [Hermle-Performance-C150] — SPN_LOOP_GAIN_14 | Default: 54.100 | Linked Fault: ERR-HER150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C150.

PRM_48135 [Hermle-Performance-C150] — ATC_LOOP_GAIN_15 | Default: 147.250 | Linked Fault: ERR-HER150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C150.

PRM_48136 [Hermle-Performance-C150] — HYD_LOOP_GAIN_16 | Default: 240.400 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C150 (BANK 2)

Machine Model: Hermle-Performance-C150 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C150.

PRM_48141 [Hermle-Performance-C150] — HYD_LOOP_GAIN_1 | Default: 186.300 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

PRM_48142 [Hermle-Performance-C150] — XAX_LOOP_GAIN_2 | Default: 122.600 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

PRM_48143 [Hermle-Performance-C150] — ZAX_LOOP_GAIN_3 | Default: 58.900 | Linked Fault: ERR-HER150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150.

PRM_48144 [Hermle-Performance-C150] — CAX_LOOP_GAIN_4 | Default: 245.200 | Linked Fault: ERR-HER150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150.

PRM_48145 [Hermle-Performance-C150] — PLC_LOOP_GAIN_5 | Default: 181.500 | Linked Fault: ERR-HER150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150.

PRM_48146 [Hermle-Performance-C150] — VAC_LOOP_GAIN_6 | Default: 117.800 | Linked Fault: ERR-HER150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C150.

PRM_48147 [Hermle-Performance-C150] — SPN_LOOP_GAIN_7 | Default: 54.100 | Linked Fault: ERR-HER150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C150.

PRM_48148 [Hermle-Performance-C150] — HYD_LOOP_GAIN_8 | Default: 240.400 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

PRM_48149 [Hermle-Performance-C150] — XAX_LOOP_GAIN_9 | Default: 176.700 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

PRM_48150 [Hermle-Performance-C150] — ZAX_LOOP_GAIN_10 | Default: 113.000 | Linked Fault: ERR-HER150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150.

PRM_48151 [Hermle-Performance-C150] — CAX_LOOP_GAIN_11 | Default: 49.300 | Linked Fault: ERR-HER150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150.

PRM_48152 [Hermle-Performance-C150] — PLC_LOOP_GAIN_12 | Default: 235.600 | Linked Fault: ERR-HER150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150.

PRM_48153 [Hermle-Performance-C150] — VAC_LOOP_GAIN_13 | Default: 171.900 | Linked Fault: ERR-HER150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C150.

PRM_48154 [Hermle-Performance-C150] — SPN_LOOP_GAIN_14 | Default: 108.200 | Linked Fault: ERR-HER150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C150.

PRM_48155 [Hermle-Performance-C150] — HYD_LOOP_GAIN_15 | Default: 44.500 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

PRM_48156 [Hermle-Performance-C150] — XAX_LOOP_GAIN_16 | Default: 230.800 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C150 (BANK 3)

Machine Model: Hermle-Performance-C150 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C150.

PRM_48161 [Hermle-Performance-C150] — CLN_LOOP_GAIN_1 | Default: 29.450 | Linked Fault: ERR-HER150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C150.

PRM_48162 [Hermle-Performance-C150] — ZAX_LOOP_GAIN_2 | Default: 58.900 | Linked Fault: ERR-HER150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150.

PRM_48163 [Hermle-Performance-C150] — OPT_LOOP_GAIN_3 | Default: 88.350 | Linked Fault: ERR-HER150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C150.

PRM_48164 [Hermle-Performance-C150] — VAC_LOOP_GAIN_4 | Default: 117.800 | Linked Fault: ERR-HER150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C150.

PRM_48165 [Hermle-Performance-C150] — ATC_LOOP_GAIN_5 | Default: 147.250 | Linked Fault: ERR-HER150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C150.

PRM_48166 [Hermle-Performance-C150] — XAX_LOOP_GAIN_6 | Default: 176.700 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

PRM_48167 [Hermle-Performance-C150] — BAX_LOOP_GAIN_7 | Default: 206.150 | Linked Fault: ERR-HER150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C150.

PRM_48168 [Hermle-Performance-C150] — PLC_LOOP_GAIN_8 | Default: 235.600 | Linked Fault: ERR-HER150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150.

PRM_48169 [Hermle-Performance-C150] — THM_LOOP_GAIN_9 | Default: 15.050 | Linked Fault: ERR-HER150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C150.

PRM_48170 [Hermle-Performance-C150] — HYD_LOOP_GAIN_10 | Default: 44.500 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

PRM_48171 [Hermle-Performance-C150] — YAX_LOOP_GAIN_11 | Default: 73.950 | Linked Fault: ERR-HER150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C150.

PRM_48172 [Hermle-Performance-C150] — CAX_LOOP_GAIN_12 | Default: 103.400 | Linked Fault: ERR-HER150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150.

PRM_48173 [Hermle-Performance-C150] — LUB_LOOP_GAIN_13 | Default: 132.850 | Linked Fault: ERR-HER150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C150.

PRM_48174 [Hermle-Performance-C150] — SPN_LOOP_GAIN_14 | Default: 162.300 | Linked Fault: ERR-HER150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C150.

PRM_48175 [Hermle-Performance-C150] — CLN_LOOP_GAIN_15 | Default: 191.750 | Linked Fault: ERR-HER150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C150.

PRM_48176 [Hermle-Performance-C150] — ZAX_LOOP_GAIN_16 | Default: 221.200 | Linked Fault: ERR-HER150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C150 (BANK 4)

Machine Model: Hermle-Performance-C150 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C150.

PRM_48181 [Hermle-Performance-C150] — XAX_LOOP_GAIN_1 | Default: 122.600 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

PRM_48182 [Hermle-Performance-C150] — CAX_LOOP_GAIN_2 | Default: 245.200 | Linked Fault: ERR-HER150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150.

PRM_48183 [Hermle-Performance-C150] — VAC_LOOP_GAIN_3 | Default: 117.800 | Linked Fault: ERR-HER150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C150.

PRM_48184 [Hermle-Performance-C150] — HYD_LOOP_GAIN_4 | Default: 240.400 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

PRM_48185 [Hermle-Performance-C150] — ZAX_LOOP_GAIN_5 | Default: 113.000 | Linked Fault: ERR-HER150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150.

PRM_48186 [Hermle-Performance-C150] — PLC_LOOP_GAIN_6 | Default: 235.600 | Linked Fault: ERR-HER150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150.

PRM_48187 [Hermle-Performance-C150] — SPN_LOOP_GAIN_7 | Default: 108.200 | Linked Fault: ERR-HER150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C150.

PRM_48188 [Hermle-Performance-C150] — XAX_LOOP_GAIN_8 | Default: 230.800 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

PRM_48189 [Hermle-Performance-C150] — CAX_LOOP_GAIN_9 | Default: 103.400 | Linked Fault: ERR-HER150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150.

PRM_48190 [Hermle-Performance-C150] — VAC_LOOP_GAIN_10 | Default: 226.000 | Linked Fault: ERR-HER150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C150.

PRM_48191 [Hermle-Performance-C150] — HYD_LOOP_GAIN_11 | Default: 98.600 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

PRM_48192 [Hermle-Performance-C150] — ZAX_LOOP_GAIN_12 | Default: 221.200 | Linked Fault: ERR-HER150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150.

PRM_48193 [Hermle-Performance-C150] — PLC_LOOP_GAIN_13 | Default: 93.800 | Linked Fault: ERR-HER150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150.

PRM_48194 [Hermle-Performance-C150] — SPN_LOOP_GAIN_14 | Default: 216.400 | Linked Fault: ERR-HER150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C150.

PRM_48195 [Hermle-Performance-C150] — XAX_LOOP_GAIN_15 | Default: 89.000 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

PRM_48196 [Hermle-Performance-C150] — CAX_LOOP_GAIN_16 | Default: 211.600 | Linked Fault: ERR-HER150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C150 (BANK 5)

Machine Model: Hermle-Performance-C150 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C150.

PRM_48201 [Hermle-Performance-C150] — YAX_LOOP_GAIN_1 | Default: 215.750 | Linked Fault: ERR-HER150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C150.

PRM_48202 [Hermle-Performance-C150] — PLC_LOOP_GAIN_2 | Default: 181.500 | Linked Fault: ERR-HER150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150.

PRM_48203 [Hermle-Performance-C150] — ATC_LOOP_GAIN_3 | Default: 147.250 | Linked Fault: ERR-HER150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C150.

PRM_48204 [Hermle-Performance-C150] — ZAX_LOOP_GAIN_4 | Default: 113.000 | Linked Fault: ERR-HER150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C150.

PRM_48205 [Hermle-Performance-C150] — LUB_LOOP_GAIN_5 | Default: 78.750 | Linked Fault: ERR-HER150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C150.

PRM_48206 [Hermle-Performance-C150] — HYD_LOOP_GAIN_6 | Default: 44.500 | Linked Fault: ERR-HER150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C150.

PRM_48207 [Hermle-Performance-C150] — BAX_LOOP_GAIN_7 | Default: 10.250 | Linked Fault: ERR-HER150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C150.

PRM_48208 [Hermle-Performance-C150] — VAC_LOOP_GAIN_8 | Default: 226.000 | Linked Fault: ERR-HER150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C150.

PRM_48209 [Hermle-Performance-C150] — CLN_LOOP_GAIN_9 | Default: 191.750 | Linked Fault: ERR-HER150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C150.

PRM_48210 [Hermle-Performance-C150] — CAX_LOOP_GAIN_10 | Default: 157.500 | Linked Fault: ERR-HER150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C150.

PRM_48211 [Hermle-Performance-C150] — THM_LOOP_GAIN_11 | Default: 123.250 | Linked Fault: ERR-HER150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C150.

PRM_48212 [Hermle-Performance-C150] — XAX_LOOP_GAIN_12 | Default: 89.000 | Linked Fault: ERR-HER150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C150.

PRM_48213 [Hermle-Performance-C150] — OPT_LOOP_GAIN_13 | Default: 54.750 | Linked Fault: ERR-HER150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C150.

PRM_48214 [Hermle-Performance-C150] — SPN_LOOP_GAIN_14 | Default: 20.500 | Linked Fault: ERR-HER150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C150.

PRM_48215 [Hermle-Performance-C150] — YAX_LOOP_GAIN_15 | Default: 236.250 | Linked Fault: ERR-HER150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C150.

PRM_48216 [Hermle-Performance-C150] — PLC_LOOP_GAIN_16 | Default: 202.000 | Linked Fault: ERR-HER150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C150.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C200 (BANK 1)

Machine Model: Hermle-Performance-C200 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C200.

PRM_48221 [Hermle-Performance-C200] — ATC_LOOP_GAIN_1 | Default: 94.300 | Linked Fault: ERR-HER200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C200.

PRM_48222 [Hermle-Performance-C200] — HYD_LOOP_GAIN_2 | Default: 188.600 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

PRM_48223 [Hermle-Performance-C200] — CLN_LOOP_GAIN_3 | Default: 32.900 | Linked Fault: ERR-HER200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C200.

PRM_48224 [Hermle-Performance-C200] — XAX_LOOP_GAIN_4 | Default: 127.200 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

PRM_48225 [Hermle-Performance-C200] — YAX_LOOP_GAIN_5 | Default: 221.500 | Linked Fault: ERR-HER200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C200.

PRM_48226 [Hermle-Performance-C200] — ZAX_LOOP_GAIN_6 | Default: 65.800 | Linked Fault: ERR-HER200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200.

PRM_48227 [Hermle-Performance-C200] — BAX_LOOP_GAIN_7 | Default: 160.100 | Linked Fault: ERR-HER200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C200.

PRM_48228 [Hermle-Performance-C200] — CAX_LOOP_GAIN_8 | Default: 4.400 | Linked Fault: ERR-HER200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200.

PRM_48229 [Hermle-Performance-C200] — OPT_LOOP_GAIN_9 | Default: 98.700 | Linked Fault: ERR-HER200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C200.

PRM_48230 [Hermle-Performance-C200] — PLC_LOOP_GAIN_10 | Default: 193.000 | Linked Fault: ERR-HER200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200.

PRM_48231 [Hermle-Performance-C200] — LUB_LOOP_GAIN_11 | Default: 37.300 | Linked Fault: ERR-HER200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C200.

PRM_48232 [Hermle-Performance-C200] — VAC_LOOP_GAIN_12 | Default: 131.600 | Linked Fault: ERR-HER200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C200.

PRM_48233 [Hermle-Performance-C200] — THM_LOOP_GAIN_13 | Default: 225.900 | Linked Fault: ERR-HER200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C200.

PRM_48234 [Hermle-Performance-C200] — SPN_LOOP_GAIN_14 | Default: 70.200 | Linked Fault: ERR-HER200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C200.

PRM_48235 [Hermle-Performance-C200] — ATC_LOOP_GAIN_15 | Default: 164.500 | Linked Fault: ERR-HER200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C200.

PRM_48236 [Hermle-Performance-C200] — HYD_LOOP_GAIN_16 | Default: 8.800 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C200 (BANK 2)

Machine Model: Hermle-Performance-C200 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C200.

PRM_48241 [Hermle-Performance-C200] — HYD_LOOP_GAIN_1 | Default: 188.600 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

PRM_48242 [Hermle-Performance-C200] — XAX_LOOP_GAIN_2 | Default: 127.200 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

PRM_48243 [Hermle-Performance-C200] — ZAX_LOOP_GAIN_3 | Default: 65.800 | Linked Fault: ERR-HER200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200.

PRM_48244 [Hermle-Performance-C200] — CAX_LOOP_GAIN_4 | Default: 4.400 | Linked Fault: ERR-HER200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200.

PRM_48245 [Hermle-Performance-C200] — PLC_LOOP_GAIN_5 | Default: 193.000 | Linked Fault: ERR-HER200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200.

PRM_48246 [Hermle-Performance-C200] — VAC_LOOP_GAIN_6 | Default: 131.600 | Linked Fault: ERR-HER200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C200.

PRM_48247 [Hermle-Performance-C200] — SPN_LOOP_GAIN_7 | Default: 70.200 | Linked Fault: ERR-HER200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C200.

PRM_48248 [Hermle-Performance-C200] — HYD_LOOP_GAIN_8 | Default: 8.800 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

PRM_48249 [Hermle-Performance-C200] — XAX_LOOP_GAIN_9 | Default: 197.400 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

PRM_48250 [Hermle-Performance-C200] — ZAX_LOOP_GAIN_10 | Default: 136.000 | Linked Fault: ERR-HER200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200.

PRM_48251 [Hermle-Performance-C200] — CAX_LOOP_GAIN_11 | Default: 74.600 | Linked Fault: ERR-HER200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200.

PRM_48252 [Hermle-Performance-C200] — PLC_LOOP_GAIN_12 | Default: 13.200 | Linked Fault: ERR-HER200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200.

PRM_48253 [Hermle-Performance-C200] — VAC_LOOP_GAIN_13 | Default: 201.800 | Linked Fault: ERR-HER200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C200.

PRM_48254 [Hermle-Performance-C200] — SPN_LOOP_GAIN_14 | Default: 140.400 | Linked Fault: ERR-HER200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C200.

PRM_48255 [Hermle-Performance-C200] — HYD_LOOP_GAIN_15 | Default: 79.000 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

PRM_48256 [Hermle-Performance-C200] — XAX_LOOP_GAIN_16 | Default: 17.600 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C200 (BANK 3)

Machine Model: Hermle-Performance-C200 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C200.

PRM_48261 [Hermle-Performance-C200] — CLN_LOOP_GAIN_1 | Default: 32.900 | Linked Fault: ERR-HER200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C200.

PRM_48262 [Hermle-Performance-C200] — ZAX_LOOP_GAIN_2 | Default: 65.800 | Linked Fault: ERR-HER200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200.

PRM_48263 [Hermle-Performance-C200] — OPT_LOOP_GAIN_3 | Default: 98.700 | Linked Fault: ERR-HER200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C200.

PRM_48264 [Hermle-Performance-C200] — VAC_LOOP_GAIN_4 | Default: 131.600 | Linked Fault: ERR-HER200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C200.

PRM_48265 [Hermle-Performance-C200] — ATC_LOOP_GAIN_5 | Default: 164.500 | Linked Fault: ERR-HER200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C200.

PRM_48266 [Hermle-Performance-C200] — XAX_LOOP_GAIN_6 | Default: 197.400 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

PRM_48267 [Hermle-Performance-C200] — BAX_LOOP_GAIN_7 | Default: 230.300 | Linked Fault: ERR-HER200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C200.

PRM_48268 [Hermle-Performance-C200] — PLC_LOOP_GAIN_8 | Default: 13.200 | Linked Fault: ERR-HER200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200.

PRM_48269 [Hermle-Performance-C200] — THM_LOOP_GAIN_9 | Default: 46.100 | Linked Fault: ERR-HER200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C200.

PRM_48270 [Hermle-Performance-C200] — HYD_LOOP_GAIN_10 | Default: 79.000 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

PRM_48271 [Hermle-Performance-C200] — YAX_LOOP_GAIN_11 | Default: 111.900 | Linked Fault: ERR-HER200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C200.

PRM_48272 [Hermle-Performance-C200] — CAX_LOOP_GAIN_12 | Default: 144.800 | Linked Fault: ERR-HER200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200.

PRM_48273 [Hermle-Performance-C200] — LUB_LOOP_GAIN_13 | Default: 177.700 | Linked Fault: ERR-HER200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C200.

PRM_48274 [Hermle-Performance-C200] — SPN_LOOP_GAIN_14 | Default: 210.600 | Linked Fault: ERR-HER200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C200.

PRM_48275 [Hermle-Performance-C200] — CLN_LOOP_GAIN_15 | Default: 243.500 | Linked Fault: ERR-HER200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C200.

PRM_48276 [Hermle-Performance-C200] — ZAX_LOOP_GAIN_16 | Default: 26.400 | Linked Fault: ERR-HER200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C200 (BANK 4)

Machine Model: Hermle-Performance-C200 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C200.

PRM_48281 [Hermle-Performance-C200] — XAX_LOOP_GAIN_1 | Default: 127.200 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

PRM_48282 [Hermle-Performance-C200] — CAX_LOOP_GAIN_2 | Default: 4.400 | Linked Fault: ERR-HER200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200.

PRM_48283 [Hermle-Performance-C200] — VAC_LOOP_GAIN_3 | Default: 131.600 | Linked Fault: ERR-HER200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C200.

PRM_48284 [Hermle-Performance-C200] — HYD_LOOP_GAIN_4 | Default: 8.800 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

PRM_48285 [Hermle-Performance-C200] — ZAX_LOOP_GAIN_5 | Default: 136.000 | Linked Fault: ERR-HER200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200.

PRM_48286 [Hermle-Performance-C200] — PLC_LOOP_GAIN_6 | Default: 13.200 | Linked Fault: ERR-HER200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200.

PRM_48287 [Hermle-Performance-C200] — SPN_LOOP_GAIN_7 | Default: 140.400 | Linked Fault: ERR-HER200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C200.

PRM_48288 [Hermle-Performance-C200] — XAX_LOOP_GAIN_8 | Default: 17.600 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

PRM_48289 [Hermle-Performance-C200] — CAX_LOOP_GAIN_9 | Default: 144.800 | Linked Fault: ERR-HER200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200.

PRM_48290 [Hermle-Performance-C200] — VAC_LOOP_GAIN_10 | Default: 22.000 | Linked Fault: ERR-HER200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C200.

PRM_48291 [Hermle-Performance-C200] — HYD_LOOP_GAIN_11 | Default: 149.200 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

PRM_48292 [Hermle-Performance-C200] — ZAX_LOOP_GAIN_12 | Default: 26.400 | Linked Fault: ERR-HER200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200.

PRM_48293 [Hermle-Performance-C200] — PLC_LOOP_GAIN_13 | Default: 153.600 | Linked Fault: ERR-HER200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200.

PRM_48294 [Hermle-Performance-C200] — SPN_LOOP_GAIN_14 | Default: 30.800 | Linked Fault: ERR-HER200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C200.

PRM_48295 [Hermle-Performance-C200] — XAX_LOOP_GAIN_15 | Default: 158.000 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

PRM_48296 [Hermle-Performance-C200] — CAX_LOOP_GAIN_16 | Default: 35.200 | Linked Fault: ERR-HER200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C200 (BANK 5)

Machine Model: Hermle-Performance-C200 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C200.

PRM_48301 [Hermle-Performance-C200] — YAX_LOOP_GAIN_1 | Default: 221.500 | Linked Fault: ERR-HER200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C200.

PRM_48302 [Hermle-Performance-C200] — PLC_LOOP_GAIN_2 | Default: 193.000 | Linked Fault: ERR-HER200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200.

PRM_48303 [Hermle-Performance-C200] — ATC_LOOP_GAIN_3 | Default: 164.500 | Linked Fault: ERR-HER200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C200.

PRM_48304 [Hermle-Performance-C200] — ZAX_LOOP_GAIN_4 | Default: 136.000 | Linked Fault: ERR-HER200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C200.

PRM_48305 [Hermle-Performance-C200] — LUB_LOOP_GAIN_5 | Default: 107.500 | Linked Fault: ERR-HER200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C200.

PRM_48306 [Hermle-Performance-C200] — HYD_LOOP_GAIN_6 | Default: 79.000 | Linked Fault: ERR-HER200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C200.

PRM_48307 [Hermle-Performance-C200] — BAX_LOOP_GAIN_7 | Default: 50.500 | Linked Fault: ERR-HER200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C200.

PRM_48308 [Hermle-Performance-C200] — VAC_LOOP_GAIN_8 | Default: 22.000 | Linked Fault: ERR-HER200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C200.

PRM_48309 [Hermle-Performance-C200] — CLN_LOOP_GAIN_9 | Default: 243.500 | Linked Fault: ERR-HER200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C200.

PRM_48310 [Hermle-Performance-C200] — CAX_LOOP_GAIN_10 | Default: 215.000 | Linked Fault: ERR-HER200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C200.

PRM_48311 [Hermle-Performance-C200] — THM_LOOP_GAIN_11 | Default: 186.500 | Linked Fault: ERR-HER200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C200.

PRM_48312 [Hermle-Performance-C200] — XAX_LOOP_GAIN_12 | Default: 158.000 | Linked Fault: ERR-HER200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C200.

PRM_48313 [Hermle-Performance-C200] — OPT_LOOP_GAIN_13 | Default: 129.500 | Linked Fault: ERR-HER200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C200.

PRM_48314 [Hermle-Performance-C200] — SPN_LOOP_GAIN_14 | Default: 101.000 | Linked Fault: ERR-HER200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C200.

PRM_48315 [Hermle-Performance-C200] — YAX_LOOP_GAIN_15 | Default: 72.500 | Linked Fault: ERR-HER200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C200.

PRM_48316 [Hermle-Performance-C200] — PLC_LOOP_GAIN_16 | Default: 44.000 | Linked Fault: ERR-HER200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C200.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C250 (BANK 1)

Machine Model: Hermle-Performance-C250 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C250.

PRM_48321 [Hermle-Performance-C250] — ATC_LOOP_GAIN_1 | Default: 95.450 | Linked Fault: ERR-HER250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C250.

PRM_48322 [Hermle-Performance-C250] — HYD_LOOP_GAIN_2 | Default: 190.900 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

PRM_48323 [Hermle-Performance-C250] — CLN_LOOP_GAIN_3 | Default: 36.350 | Linked Fault: ERR-HER250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C250.

PRM_48324 [Hermle-Performance-C250] — XAX_LOOP_GAIN_4 | Default: 131.800 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

PRM_48325 [Hermle-Performance-C250] — YAX_LOOP_GAIN_5 | Default: 227.250 | Linked Fault: ERR-HER250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C250.

PRM_48326 [Hermle-Performance-C250] — ZAX_LOOP_GAIN_6 | Default: 72.700 | Linked Fault: ERR-HER250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250.

PRM_48327 [Hermle-Performance-C250] — BAX_LOOP_GAIN_7 | Default: 168.150 | Linked Fault: ERR-HER250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C250.

PRM_48328 [Hermle-Performance-C250] — CAX_LOOP_GAIN_8 | Default: 13.600 | Linked Fault: ERR-HER250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250.

PRM_48329 [Hermle-Performance-C250] — OPT_LOOP_GAIN_9 | Default: 109.050 | Linked Fault: ERR-HER250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C250.

PRM_48330 [Hermle-Performance-C250] — PLC_LOOP_GAIN_10 | Default: 204.500 | Linked Fault: ERR-HER250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250.

PRM_48331 [Hermle-Performance-C250] — LUB_LOOP_GAIN_11 | Default: 49.950 | Linked Fault: ERR-HER250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C250.

PRM_48332 [Hermle-Performance-C250] — VAC_LOOP_GAIN_12 | Default: 145.400 | Linked Fault: ERR-HER250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C250.

PRM_48333 [Hermle-Performance-C250] — THM_LOOP_GAIN_13 | Default: 240.850 | Linked Fault: ERR-HER250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C250.

PRM_48334 [Hermle-Performance-C250] — SPN_LOOP_GAIN_14 | Default: 86.300 | Linked Fault: ERR-HER250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C250.

PRM_48335 [Hermle-Performance-C250] — ATC_LOOP_GAIN_15 | Default: 181.750 | Linked Fault: ERR-HER250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C250.

PRM_48336 [Hermle-Performance-C250] — HYD_LOOP_GAIN_16 | Default: 27.200 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C250 (BANK 2)

Machine Model: Hermle-Performance-C250 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C250.

PRM_48341 [Hermle-Performance-C250] — HYD_LOOP_GAIN_1 | Default: 190.900 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

PRM_48342 [Hermle-Performance-C250] — XAX_LOOP_GAIN_2 | Default: 131.800 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

PRM_48343 [Hermle-Performance-C250] — ZAX_LOOP_GAIN_3 | Default: 72.700 | Linked Fault: ERR-HER250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250.

PRM_48344 [Hermle-Performance-C250] — CAX_LOOP_GAIN_4 | Default: 13.600 | Linked Fault: ERR-HER250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250.

PRM_48345 [Hermle-Performance-C250] — PLC_LOOP_GAIN_5 | Default: 204.500 | Linked Fault: ERR-HER250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250.

PRM_48346 [Hermle-Performance-C250] — VAC_LOOP_GAIN_6 | Default: 145.400 | Linked Fault: ERR-HER250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C250.

PRM_48347 [Hermle-Performance-C250] — SPN_LOOP_GAIN_7 | Default: 86.300 | Linked Fault: ERR-HER250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C250.

PRM_48348 [Hermle-Performance-C250] — HYD_LOOP_GAIN_8 | Default: 27.200 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

PRM_48349 [Hermle-Performance-C250] — XAX_LOOP_GAIN_9 | Default: 218.100 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

PRM_48350 [Hermle-Performance-C250] — ZAX_LOOP_GAIN_10 | Default: 159.000 | Linked Fault: ERR-HER250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250.

PRM_48351 [Hermle-Performance-C250] — CAX_LOOP_GAIN_11 | Default: 99.900 | Linked Fault: ERR-HER250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250.

PRM_48352 [Hermle-Performance-C250] — PLC_LOOP_GAIN_12 | Default: 40.800 | Linked Fault: ERR-HER250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250.

PRM_48353 [Hermle-Performance-C250] — VAC_LOOP_GAIN_13 | Default: 231.700 | Linked Fault: ERR-HER250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C250.

PRM_48354 [Hermle-Performance-C250] — SPN_LOOP_GAIN_14 | Default: 172.600 | Linked Fault: ERR-HER250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C250.

PRM_48355 [Hermle-Performance-C250] — HYD_LOOP_GAIN_15 | Default: 113.500 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

PRM_48356 [Hermle-Performance-C250] — XAX_LOOP_GAIN_16 | Default: 54.400 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C250 (BANK 3)

Machine Model: Hermle-Performance-C250 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C250.

PRM_48361 [Hermle-Performance-C250] — CLN_LOOP_GAIN_1 | Default: 36.350 | Linked Fault: ERR-HER250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C250.

PRM_48362 [Hermle-Performance-C250] — ZAX_LOOP_GAIN_2 | Default: 72.700 | Linked Fault: ERR-HER250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250.

PRM_48363 [Hermle-Performance-C250] — OPT_LOOP_GAIN_3 | Default: 109.050 | Linked Fault: ERR-HER250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C250.

PRM_48364 [Hermle-Performance-C250] — VAC_LOOP_GAIN_4 | Default: 145.400 | Linked Fault: ERR-HER250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C250.

PRM_48365 [Hermle-Performance-C250] — ATC_LOOP_GAIN_5 | Default: 181.750 | Linked Fault: ERR-HER250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C250.

PRM_48366 [Hermle-Performance-C250] — XAX_LOOP_GAIN_6 | Default: 218.100 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

PRM_48367 [Hermle-Performance-C250] — BAX_LOOP_GAIN_7 | Default: 4.450 | Linked Fault: ERR-HER250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C250.

PRM_48368 [Hermle-Performance-C250] — PLC_LOOP_GAIN_8 | Default: 40.800 | Linked Fault: ERR-HER250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250.

PRM_48369 [Hermle-Performance-C250] — THM_LOOP_GAIN_9 | Default: 77.150 | Linked Fault: ERR-HER250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C250.

PRM_48370 [Hermle-Performance-C250] — HYD_LOOP_GAIN_10 | Default: 113.500 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

PRM_48371 [Hermle-Performance-C250] — YAX_LOOP_GAIN_11 | Default: 149.850 | Linked Fault: ERR-HER250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C250.

PRM_48372 [Hermle-Performance-C250] — CAX_LOOP_GAIN_12 | Default: 186.200 | Linked Fault: ERR-HER250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250.

PRM_48373 [Hermle-Performance-C250] — LUB_LOOP_GAIN_13 | Default: 222.550 | Linked Fault: ERR-HER250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C250.

PRM_48374 [Hermle-Performance-C250] — SPN_LOOP_GAIN_14 | Default: 8.900 | Linked Fault: ERR-HER250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C250.

PRM_48375 [Hermle-Performance-C250] — CLN_LOOP_GAIN_15 | Default: 45.250 | Linked Fault: ERR-HER250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C250.

PRM_48376 [Hermle-Performance-C250] — ZAX_LOOP_GAIN_16 | Default: 81.600 | Linked Fault: ERR-HER250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C250 (BANK 4)

Machine Model: Hermle-Performance-C250 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C250.

PRM_48381 [Hermle-Performance-C250] — XAX_LOOP_GAIN_1 | Default: 131.800 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

PRM_48382 [Hermle-Performance-C250] — CAX_LOOP_GAIN_2 | Default: 13.600 | Linked Fault: ERR-HER250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250.

PRM_48383 [Hermle-Performance-C250] — VAC_LOOP_GAIN_3 | Default: 145.400 | Linked Fault: ERR-HER250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C250.

PRM_48384 [Hermle-Performance-C250] — HYD_LOOP_GAIN_4 | Default: 27.200 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

PRM_48385 [Hermle-Performance-C250] — ZAX_LOOP_GAIN_5 | Default: 159.000 | Linked Fault: ERR-HER250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250.

PRM_48386 [Hermle-Performance-C250] — PLC_LOOP_GAIN_6 | Default: 40.800 | Linked Fault: ERR-HER250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250.

PRM_48387 [Hermle-Performance-C250] — SPN_LOOP_GAIN_7 | Default: 172.600 | Linked Fault: ERR-HER250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C250.

PRM_48388 [Hermle-Performance-C250] — XAX_LOOP_GAIN_8 | Default: 54.400 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

PRM_48389 [Hermle-Performance-C250] — CAX_LOOP_GAIN_9 | Default: 186.200 | Linked Fault: ERR-HER250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250.

PRM_48390 [Hermle-Performance-C250] — VAC_LOOP_GAIN_10 | Default: 68.000 | Linked Fault: ERR-HER250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C250.

PRM_48391 [Hermle-Performance-C250] — HYD_LOOP_GAIN_11 | Default: 199.800 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

PRM_48392 [Hermle-Performance-C250] — ZAX_LOOP_GAIN_12 | Default: 81.600 | Linked Fault: ERR-HER250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250.

PRM_48393 [Hermle-Performance-C250] — PLC_LOOP_GAIN_13 | Default: 213.400 | Linked Fault: ERR-HER250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250.

PRM_48394 [Hermle-Performance-C250] — SPN_LOOP_GAIN_14 | Default: 95.200 | Linked Fault: ERR-HER250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C250.

PRM_48395 [Hermle-Performance-C250] — XAX_LOOP_GAIN_15 | Default: 227.000 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

PRM_48396 [Hermle-Performance-C250] — CAX_LOOP_GAIN_16 | Default: 108.800 | Linked Fault: ERR-HER250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C250 (BANK 5)

Machine Model: Hermle-Performance-C250 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C250.

PRM_48401 [Hermle-Performance-C250] — YAX_LOOP_GAIN_1 | Default: 227.250 | Linked Fault: ERR-HER250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C250.

PRM_48402 [Hermle-Performance-C250] — PLC_LOOP_GAIN_2 | Default: 204.500 | Linked Fault: ERR-HER250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250.

PRM_48403 [Hermle-Performance-C250] — ATC_LOOP_GAIN_3 | Default: 181.750 | Linked Fault: ERR-HER250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C250.

PRM_48404 [Hermle-Performance-C250] — ZAX_LOOP_GAIN_4 | Default: 159.000 | Linked Fault: ERR-HER250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C250.

PRM_48405 [Hermle-Performance-C250] — LUB_LOOP_GAIN_5 | Default: 136.250 | Linked Fault: ERR-HER250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C250.

PRM_48406 [Hermle-Performance-C250] — HYD_LOOP_GAIN_6 | Default: 113.500 | Linked Fault: ERR-HER250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C250.

PRM_48407 [Hermle-Performance-C250] — BAX_LOOP_GAIN_7 | Default: 90.750 | Linked Fault: ERR-HER250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C250.

PRM_48408 [Hermle-Performance-C250] — VAC_LOOP_GAIN_8 | Default: 68.000 | Linked Fault: ERR-HER250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C250.

PRM_48409 [Hermle-Performance-C250] — CLN_LOOP_GAIN_9 | Default: 45.250 | Linked Fault: ERR-HER250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C250.

PRM_48410 [Hermle-Performance-C250] — CAX_LOOP_GAIN_10 | Default: 22.500 | Linked Fault: ERR-HER250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C250.

PRM_48411 [Hermle-Performance-C250] — THM_LOOP_GAIN_11 | Default: 249.750 | Linked Fault: ERR-HER250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C250.

PRM_48412 [Hermle-Performance-C250] — XAX_LOOP_GAIN_12 | Default: 227.000 | Linked Fault: ERR-HER250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C250.

PRM_48413 [Hermle-Performance-C250] — OPT_LOOP_GAIN_13 | Default: 204.250 | Linked Fault: ERR-HER250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C250.

PRM_48414 [Hermle-Performance-C250] — SPN_LOOP_GAIN_14 | Default: 181.500 | Linked Fault: ERR-HER250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C250.

PRM_48415 [Hermle-Performance-C250] — YAX_LOOP_GAIN_15 | Default: 158.750 | Linked Fault: ERR-HER250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C250.

PRM_48416 [Hermle-Performance-C250] — PLC_LOOP_GAIN_16 | Default: 136.000 | Linked Fault: ERR-HER250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C250.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C300 (BANK 1)

Machine Model: Hermle-Performance-C300 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C300.

PRM_48421 [Hermle-Performance-C300] — ATC_LOOP_GAIN_1 | Default: 96.600 | Linked Fault: ERR-HER300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C300.

PRM_48422 [Hermle-Performance-C300] — HYD_LOOP_GAIN_2 | Default: 193.200 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

PRM_48423 [Hermle-Performance-C300] — CLN_LOOP_GAIN_3 | Default: 39.800 | Linked Fault: ERR-HER300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C300.

PRM_48424 [Hermle-Performance-C300] — XAX_LOOP_GAIN_4 | Default: 136.400 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

PRM_48425 [Hermle-Performance-C300] — YAX_LOOP_GAIN_5 | Default: 233.000 | Linked Fault: ERR-HER300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C300.

PRM_48426 [Hermle-Performance-C300] — ZAX_LOOP_GAIN_6 | Default: 79.600 | Linked Fault: ERR-HER300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300.

PRM_48427 [Hermle-Performance-C300] — BAX_LOOP_GAIN_7 | Default: 176.200 | Linked Fault: ERR-HER300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C300.

PRM_48428 [Hermle-Performance-C300] — CAX_LOOP_GAIN_8 | Default: 22.800 | Linked Fault: ERR-HER300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300.

PRM_48429 [Hermle-Performance-C300] — OPT_LOOP_GAIN_9 | Default: 119.400 | Linked Fault: ERR-HER300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C300.

PRM_48430 [Hermle-Performance-C300] — PLC_LOOP_GAIN_10 | Default: 216.000 | Linked Fault: ERR-HER300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300.

PRM_48431 [Hermle-Performance-C300] — LUB_LOOP_GAIN_11 | Default: 62.600 | Linked Fault: ERR-HER300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C300.

PRM_48432 [Hermle-Performance-C300] — VAC_LOOP_GAIN_12 | Default: 159.200 | Linked Fault: ERR-HER300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C300.

PRM_48433 [Hermle-Performance-C300] — THM_LOOP_GAIN_13 | Default: 5.800 | Linked Fault: ERR-HER300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C300.

PRM_48434 [Hermle-Performance-C300] — SPN_LOOP_GAIN_14 | Default: 102.400 | Linked Fault: ERR-HER300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C300.

PRM_48435 [Hermle-Performance-C300] — ATC_LOOP_GAIN_15 | Default: 199.000 | Linked Fault: ERR-HER300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C300.

PRM_48436 [Hermle-Performance-C300] — HYD_LOOP_GAIN_16 | Default: 45.600 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C300 (BANK 2)

Machine Model: Hermle-Performance-C300 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C300.

PRM_48441 [Hermle-Performance-C300] — HYD_LOOP_GAIN_1 | Default: 193.200 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

PRM_48442 [Hermle-Performance-C300] — XAX_LOOP_GAIN_2 | Default: 136.400 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

PRM_48443 [Hermle-Performance-C300] — ZAX_LOOP_GAIN_3 | Default: 79.600 | Linked Fault: ERR-HER300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300.

PRM_48444 [Hermle-Performance-C300] — CAX_LOOP_GAIN_4 | Default: 22.800 | Linked Fault: ERR-HER300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300.

PRM_48445 [Hermle-Performance-C300] — PLC_LOOP_GAIN_5 | Default: 216.000 | Linked Fault: ERR-HER300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300.

PRM_48446 [Hermle-Performance-C300] — VAC_LOOP_GAIN_6 | Default: 159.200 | Linked Fault: ERR-HER300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C300.

PRM_48447 [Hermle-Performance-C300] — SPN_LOOP_GAIN_7 | Default: 102.400 | Linked Fault: ERR-HER300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C300.

PRM_48448 [Hermle-Performance-C300] — HYD_LOOP_GAIN_8 | Default: 45.600 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

PRM_48449 [Hermle-Performance-C300] — XAX_LOOP_GAIN_9 | Default: 238.800 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

PRM_48450 [Hermle-Performance-C300] — ZAX_LOOP_GAIN_10 | Default: 182.000 | Linked Fault: ERR-HER300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300.

PRM_48451 [Hermle-Performance-C300] — CAX_LOOP_GAIN_11 | Default: 125.200 | Linked Fault: ERR-HER300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300.

PRM_48452 [Hermle-Performance-C300] — PLC_LOOP_GAIN_12 | Default: 68.400 | Linked Fault: ERR-HER300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300.

PRM_48453 [Hermle-Performance-C300] — VAC_LOOP_GAIN_13 | Default: 11.600 | Linked Fault: ERR-HER300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C300.

PRM_48454 [Hermle-Performance-C300] — SPN_LOOP_GAIN_14 | Default: 204.800 | Linked Fault: ERR-HER300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C300.

PRM_48455 [Hermle-Performance-C300] — HYD_LOOP_GAIN_15 | Default: 148.000 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

PRM_48456 [Hermle-Performance-C300] — XAX_LOOP_GAIN_16 | Default: 91.200 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C300 (BANK 3)

Machine Model: Hermle-Performance-C300 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C300.

PRM_48461 [Hermle-Performance-C300] — CLN_LOOP_GAIN_1 | Default: 39.800 | Linked Fault: ERR-HER300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C300.

PRM_48462 [Hermle-Performance-C300] — ZAX_LOOP_GAIN_2 | Default: 79.600 | Linked Fault: ERR-HER300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300.

PRM_48463 [Hermle-Performance-C300] — OPT_LOOP_GAIN_3 | Default: 119.400 | Linked Fault: ERR-HER300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C300.

PRM_48464 [Hermle-Performance-C300] — VAC_LOOP_GAIN_4 | Default: 159.200 | Linked Fault: ERR-HER300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C300.

PRM_48465 [Hermle-Performance-C300] — ATC_LOOP_GAIN_5 | Default: 199.000 | Linked Fault: ERR-HER300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C300.

PRM_48466 [Hermle-Performance-C300] — XAX_LOOP_GAIN_6 | Default: 238.800 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

PRM_48467 [Hermle-Performance-C300] — BAX_LOOP_GAIN_7 | Default: 28.600 | Linked Fault: ERR-HER300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C300.

PRM_48468 [Hermle-Performance-C300] — PLC_LOOP_GAIN_8 | Default: 68.400 | Linked Fault: ERR-HER300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300.

PRM_48469 [Hermle-Performance-C300] — THM_LOOP_GAIN_9 | Default: 108.200 | Linked Fault: ERR-HER300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C300.

PRM_48470 [Hermle-Performance-C300] — HYD_LOOP_GAIN_10 | Default: 148.000 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

PRM_48471 [Hermle-Performance-C300] — YAX_LOOP_GAIN_11 | Default: 187.800 | Linked Fault: ERR-HER300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C300.

PRM_48472 [Hermle-Performance-C300] — CAX_LOOP_GAIN_12 | Default: 227.600 | Linked Fault: ERR-HER300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300.

PRM_48473 [Hermle-Performance-C300] — LUB_LOOP_GAIN_13 | Default: 17.400 | Linked Fault: ERR-HER300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C300.

PRM_48474 [Hermle-Performance-C300] — SPN_LOOP_GAIN_14 | Default: 57.200 | Linked Fault: ERR-HER300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C300.

PRM_48475 [Hermle-Performance-C300] — CLN_LOOP_GAIN_15 | Default: 97.000 | Linked Fault: ERR-HER300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C300.

PRM_48476 [Hermle-Performance-C300] — ZAX_LOOP_GAIN_16 | Default: 136.800 | Linked Fault: ERR-HER300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C300 (BANK 4)

Machine Model: Hermle-Performance-C300 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C300.

PRM_48481 [Hermle-Performance-C300] — XAX_LOOP_GAIN_1 | Default: 136.400 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

PRM_48482 [Hermle-Performance-C300] — CAX_LOOP_GAIN_2 | Default: 22.800 | Linked Fault: ERR-HER300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300.

PRM_48483 [Hermle-Performance-C300] — VAC_LOOP_GAIN_3 | Default: 159.200 | Linked Fault: ERR-HER300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C300.

PRM_48484 [Hermle-Performance-C300] — HYD_LOOP_GAIN_4 | Default: 45.600 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

PRM_48485 [Hermle-Performance-C300] — ZAX_LOOP_GAIN_5 | Default: 182.000 | Linked Fault: ERR-HER300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300.

PRM_48486 [Hermle-Performance-C300] — PLC_LOOP_GAIN_6 | Default: 68.400 | Linked Fault: ERR-HER300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300.

PRM_48487 [Hermle-Performance-C300] — SPN_LOOP_GAIN_7 | Default: 204.800 | Linked Fault: ERR-HER300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C300.

PRM_48488 [Hermle-Performance-C300] — XAX_LOOP_GAIN_8 | Default: 91.200 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

PRM_48489 [Hermle-Performance-C300] — CAX_LOOP_GAIN_9 | Default: 227.600 | Linked Fault: ERR-HER300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300.

PRM_48490 [Hermle-Performance-C300] — VAC_LOOP_GAIN_10 | Default: 114.000 | Linked Fault: ERR-HER300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C300.

PRM_48491 [Hermle-Performance-C300] — HYD_LOOP_GAIN_11 | Default: 0.400 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

PRM_48492 [Hermle-Performance-C300] — ZAX_LOOP_GAIN_12 | Default: 136.800 | Linked Fault: ERR-HER300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300.

PRM_48493 [Hermle-Performance-C300] — PLC_LOOP_GAIN_13 | Default: 23.200 | Linked Fault: ERR-HER300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300.

PRM_48494 [Hermle-Performance-C300] — SPN_LOOP_GAIN_14 | Default: 159.600 | Linked Fault: ERR-HER300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C300.

PRM_48495 [Hermle-Performance-C300] — XAX_LOOP_GAIN_15 | Default: 46.000 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

PRM_48496 [Hermle-Performance-C300] — CAX_LOOP_GAIN_16 | Default: 182.400 | Linked Fault: ERR-HER300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C300 (BANK 5)

Machine Model: Hermle-Performance-C300 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C300.

PRM_48501 [Hermle-Performance-C300] — YAX_LOOP_GAIN_1 | Default: 233.000 | Linked Fault: ERR-HER300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C300.

PRM_48502 [Hermle-Performance-C300] — PLC_LOOP_GAIN_2 | Default: 216.000 | Linked Fault: ERR-HER300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300.

PRM_48503 [Hermle-Performance-C300] — ATC_LOOP_GAIN_3 | Default: 199.000 | Linked Fault: ERR-HER300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C300.

PRM_48504 [Hermle-Performance-C300] — ZAX_LOOP_GAIN_4 | Default: 182.000 | Linked Fault: ERR-HER300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C300.

PRM_48505 [Hermle-Performance-C300] — LUB_LOOP_GAIN_5 | Default: 165.000 | Linked Fault: ERR-HER300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C300.

PRM_48506 [Hermle-Performance-C300] — HYD_LOOP_GAIN_6 | Default: 148.000 | Linked Fault: ERR-HER300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C300.

PRM_48507 [Hermle-Performance-C300] — BAX_LOOP_GAIN_7 | Default: 131.000 | Linked Fault: ERR-HER300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C300.

PRM_48508 [Hermle-Performance-C300] — VAC_LOOP_GAIN_8 | Default: 114.000 | Linked Fault: ERR-HER300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C300.

PRM_48509 [Hermle-Performance-C300] — CLN_LOOP_GAIN_9 | Default: 97.000 | Linked Fault: ERR-HER300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C300.

PRM_48510 [Hermle-Performance-C300] — CAX_LOOP_GAIN_10 | Default: 80.000 | Linked Fault: ERR-HER300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C300.

PRM_48511 [Hermle-Performance-C300] — THM_LOOP_GAIN_11 | Default: 63.000 | Linked Fault: ERR-HER300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C300.

PRM_48512 [Hermle-Performance-C300] — XAX_LOOP_GAIN_12 | Default: 46.000 | Linked Fault: ERR-HER300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C300.

PRM_48513 [Hermle-Performance-C300] — OPT_LOOP_GAIN_13 | Default: 29.000 | Linked Fault: ERR-HER300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C300.

PRM_48514 [Hermle-Performance-C300] — SPN_LOOP_GAIN_14 | Default: 12.000 | Linked Fault: ERR-HER300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C300.

PRM_48515 [Hermle-Performance-C300] — YAX_LOOP_GAIN_15 | Default: 245.000 | Linked Fault: ERR-HER300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C300.

PRM_48516 [Hermle-Performance-C300] — PLC_LOOP_GAIN_16 | Default: 228.000 | Linked Fault: ERR-HER300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C300.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C350 (BANK 1)

Machine Model: Hermle-Performance-C350 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C350.

PRM_48521 [Hermle-Performance-C350] — ATC_LOOP_GAIN_1 | Default: 97.750 | Linked Fault: ERR-HER350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C350.

PRM_48522 [Hermle-Performance-C350] — HYD_LOOP_GAIN_2 | Default: 195.500 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

PRM_48523 [Hermle-Performance-C350] — CLN_LOOP_GAIN_3 | Default: 43.250 | Linked Fault: ERR-HER350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C350.

PRM_48524 [Hermle-Performance-C350] — XAX_LOOP_GAIN_4 | Default: 141.000 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

PRM_48525 [Hermle-Performance-C350] — YAX_LOOP_GAIN_5 | Default: 238.750 | Linked Fault: ERR-HER350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C350.

PRM_48526 [Hermle-Performance-C350] — ZAX_LOOP_GAIN_6 | Default: 86.500 | Linked Fault: ERR-HER350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350.

PRM_48527 [Hermle-Performance-C350] — BAX_LOOP_GAIN_7 | Default: 184.250 | Linked Fault: ERR-HER350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C350.

PRM_48528 [Hermle-Performance-C350] — CAX_LOOP_GAIN_8 | Default: 32.000 | Linked Fault: ERR-HER350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350.

PRM_48529 [Hermle-Performance-C350] — OPT_LOOP_GAIN_9 | Default: 129.750 | Linked Fault: ERR-HER350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C350.

PRM_48530 [Hermle-Performance-C350] — PLC_LOOP_GAIN_10 | Default: 227.500 | Linked Fault: ERR-HER350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350.

PRM_48531 [Hermle-Performance-C350] — LUB_LOOP_GAIN_11 | Default: 75.250 | Linked Fault: ERR-HER350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C350.

PRM_48532 [Hermle-Performance-C350] — VAC_LOOP_GAIN_12 | Default: 173.000 | Linked Fault: ERR-HER350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C350.

PRM_48533 [Hermle-Performance-C350] — THM_LOOP_GAIN_13 | Default: 20.750 | Linked Fault: ERR-HER350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C350.

PRM_48534 [Hermle-Performance-C350] — SPN_LOOP_GAIN_14 | Default: 118.500 | Linked Fault: ERR-HER350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C350.

PRM_48535 [Hermle-Performance-C350] — ATC_LOOP_GAIN_15 | Default: 216.250 | Linked Fault: ERR-HER350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C350.

PRM_48536 [Hermle-Performance-C350] — HYD_LOOP_GAIN_16 | Default: 64.000 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C350 (BANK 2)

Machine Model: Hermle-Performance-C350 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C350.

PRM_48541 [Hermle-Performance-C350] — HYD_LOOP_GAIN_1 | Default: 195.500 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

PRM_48542 [Hermle-Performance-C350] — XAX_LOOP_GAIN_2 | Default: 141.000 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

PRM_48543 [Hermle-Performance-C350] — ZAX_LOOP_GAIN_3 | Default: 86.500 | Linked Fault: ERR-HER350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350.

PRM_48544 [Hermle-Performance-C350] — CAX_LOOP_GAIN_4 | Default: 32.000 | Linked Fault: ERR-HER350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350.

PRM_48545 [Hermle-Performance-C350] — PLC_LOOP_GAIN_5 | Default: 227.500 | Linked Fault: ERR-HER350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350.

PRM_48546 [Hermle-Performance-C350] — VAC_LOOP_GAIN_6 | Default: 173.000 | Linked Fault: ERR-HER350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C350.

PRM_48547 [Hermle-Performance-C350] — SPN_LOOP_GAIN_7 | Default: 118.500 | Linked Fault: ERR-HER350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C350.

PRM_48548 [Hermle-Performance-C350] — HYD_LOOP_GAIN_8 | Default: 64.000 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

PRM_48549 [Hermle-Performance-C350] — XAX_LOOP_GAIN_9 | Default: 9.500 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

PRM_48550 [Hermle-Performance-C350] — ZAX_LOOP_GAIN_10 | Default: 205.000 | Linked Fault: ERR-HER350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350.

PRM_48551 [Hermle-Performance-C350] — CAX_LOOP_GAIN_11 | Default: 150.500 | Linked Fault: ERR-HER350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350.

PRM_48552 [Hermle-Performance-C350] — PLC_LOOP_GAIN_12 | Default: 96.000 | Linked Fault: ERR-HER350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350.

PRM_48553 [Hermle-Performance-C350] — VAC_LOOP_GAIN_13 | Default: 41.500 | Linked Fault: ERR-HER350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C350.

PRM_48554 [Hermle-Performance-C350] — SPN_LOOP_GAIN_14 | Default: 237.000 | Linked Fault: ERR-HER350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C350.

PRM_48555 [Hermle-Performance-C350] — HYD_LOOP_GAIN_15 | Default: 182.500 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

PRM_48556 [Hermle-Performance-C350] — XAX_LOOP_GAIN_16 | Default: 128.000 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C350 (BANK 3)

Machine Model: Hermle-Performance-C350 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C350.

PRM_48561 [Hermle-Performance-C350] — CLN_LOOP_GAIN_1 | Default: 43.250 | Linked Fault: ERR-HER350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C350.

PRM_48562 [Hermle-Performance-C350] — ZAX_LOOP_GAIN_2 | Default: 86.500 | Linked Fault: ERR-HER350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350.

PRM_48563 [Hermle-Performance-C350] — OPT_LOOP_GAIN_3 | Default: 129.750 | Linked Fault: ERR-HER350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C350.

PRM_48564 [Hermle-Performance-C350] — VAC_LOOP_GAIN_4 | Default: 173.000 | Linked Fault: ERR-HER350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C350.

PRM_48565 [Hermle-Performance-C350] — ATC_LOOP_GAIN_5 | Default: 216.250 | Linked Fault: ERR-HER350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C350.

PRM_48566 [Hermle-Performance-C350] — XAX_LOOP_GAIN_6 | Default: 9.500 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

PRM_48567 [Hermle-Performance-C350] — BAX_LOOP_GAIN_7 | Default: 52.750 | Linked Fault: ERR-HER350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C350.

PRM_48568 [Hermle-Performance-C350] — PLC_LOOP_GAIN_8 | Default: 96.000 | Linked Fault: ERR-HER350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350.

PRM_48569 [Hermle-Performance-C350] — THM_LOOP_GAIN_9 | Default: 139.250 | Linked Fault: ERR-HER350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C350.

PRM_48570 [Hermle-Performance-C350] — HYD_LOOP_GAIN_10 | Default: 182.500 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

PRM_48571 [Hermle-Performance-C350] — YAX_LOOP_GAIN_11 | Default: 225.750 | Linked Fault: ERR-HER350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C350.

PRM_48572 [Hermle-Performance-C350] — CAX_LOOP_GAIN_12 | Default: 19.000 | Linked Fault: ERR-HER350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350.

PRM_48573 [Hermle-Performance-C350] — LUB_LOOP_GAIN_13 | Default: 62.250 | Linked Fault: ERR-HER350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C350.

PRM_48574 [Hermle-Performance-C350] — SPN_LOOP_GAIN_14 | Default: 105.500 | Linked Fault: ERR-HER350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C350.

PRM_48575 [Hermle-Performance-C350] — CLN_LOOP_GAIN_15 | Default: 148.750 | Linked Fault: ERR-HER350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C350.

PRM_48576 [Hermle-Performance-C350] — ZAX_LOOP_GAIN_16 | Default: 192.000 | Linked Fault: ERR-HER350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C350 (BANK 4)

Machine Model: Hermle-Performance-C350 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C350.

PRM_48581 [Hermle-Performance-C350] — XAX_LOOP_GAIN_1 | Default: 141.000 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

PRM_48582 [Hermle-Performance-C350] — CAX_LOOP_GAIN_2 | Default: 32.000 | Linked Fault: ERR-HER350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350.

PRM_48583 [Hermle-Performance-C350] — VAC_LOOP_GAIN_3 | Default: 173.000 | Linked Fault: ERR-HER350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C350.

PRM_48584 [Hermle-Performance-C350] — HYD_LOOP_GAIN_4 | Default: 64.000 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

PRM_48585 [Hermle-Performance-C350] — ZAX_LOOP_GAIN_5 | Default: 205.000 | Linked Fault: ERR-HER350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350.

PRM_48586 [Hermle-Performance-C350] — PLC_LOOP_GAIN_6 | Default: 96.000 | Linked Fault: ERR-HER350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350.

PRM_48587 [Hermle-Performance-C350] — SPN_LOOP_GAIN_7 | Default: 237.000 | Linked Fault: ERR-HER350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C350.

PRM_48588 [Hermle-Performance-C350] — XAX_LOOP_GAIN_8 | Default: 128.000 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

PRM_48589 [Hermle-Performance-C350] — CAX_LOOP_GAIN_9 | Default: 19.000 | Linked Fault: ERR-HER350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350.

PRM_48590 [Hermle-Performance-C350] — VAC_LOOP_GAIN_10 | Default: 160.000 | Linked Fault: ERR-HER350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C350.

PRM_48591 [Hermle-Performance-C350] — HYD_LOOP_GAIN_11 | Default: 51.000 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

PRM_48592 [Hermle-Performance-C350] — ZAX_LOOP_GAIN_12 | Default: 192.000 | Linked Fault: ERR-HER350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350.

PRM_48593 [Hermle-Performance-C350] — PLC_LOOP_GAIN_13 | Default: 83.000 | Linked Fault: ERR-HER350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350.

PRM_48594 [Hermle-Performance-C350] — SPN_LOOP_GAIN_14 | Default: 224.000 | Linked Fault: ERR-HER350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C350.

PRM_48595 [Hermle-Performance-C350] — XAX_LOOP_GAIN_15 | Default: 115.000 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

PRM_48596 [Hermle-Performance-C350] — CAX_LOOP_GAIN_16 | Default: 6.000 | Linked Fault: ERR-HER350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C350 (BANK 5)

Machine Model: Hermle-Performance-C350 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C350.

PRM_48601 [Hermle-Performance-C350] — YAX_LOOP_GAIN_1 | Default: 238.750 | Linked Fault: ERR-HER350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C350.

PRM_48602 [Hermle-Performance-C350] — PLC_LOOP_GAIN_2 | Default: 227.500 | Linked Fault: ERR-HER350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350.

PRM_48603 [Hermle-Performance-C350] — ATC_LOOP_GAIN_3 | Default: 216.250 | Linked Fault: ERR-HER350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C350.

PRM_48604 [Hermle-Performance-C350] — ZAX_LOOP_GAIN_4 | Default: 205.000 | Linked Fault: ERR-HER350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C350.

PRM_48605 [Hermle-Performance-C350] — LUB_LOOP_GAIN_5 | Default: 193.750 | Linked Fault: ERR-HER350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C350.

PRM_48606 [Hermle-Performance-C350] — HYD_LOOP_GAIN_6 | Default: 182.500 | Linked Fault: ERR-HER350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C350.

PRM_48607 [Hermle-Performance-C350] — BAX_LOOP_GAIN_7 | Default: 171.250 | Linked Fault: ERR-HER350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C350.

PRM_48608 [Hermle-Performance-C350] — VAC_LOOP_GAIN_8 | Default: 160.000 | Linked Fault: ERR-HER350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C350.

PRM_48609 [Hermle-Performance-C350] — CLN_LOOP_GAIN_9 | Default: 148.750 | Linked Fault: ERR-HER350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C350.

PRM_48610 [Hermle-Performance-C350] — CAX_LOOP_GAIN_10 | Default: 137.500 | Linked Fault: ERR-HER350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C350.

PRM_48611 [Hermle-Performance-C350] — THM_LOOP_GAIN_11 | Default: 126.250 | Linked Fault: ERR-HER350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C350.

PRM_48612 [Hermle-Performance-C350] — XAX_LOOP_GAIN_12 | Default: 115.000 | Linked Fault: ERR-HER350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C350.

PRM_48613 [Hermle-Performance-C350] — OPT_LOOP_GAIN_13 | Default: 103.750 | Linked Fault: ERR-HER350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C350.

PRM_48614 [Hermle-Performance-C350] — SPN_LOOP_GAIN_14 | Default: 92.500 | Linked Fault: ERR-HER350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C350.

PRM_48615 [Hermle-Performance-C350] — YAX_LOOP_GAIN_15 | Default: 81.250 | Linked Fault: ERR-HER350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C350.

PRM_48616 [Hermle-Performance-C350] — PLC_LOOP_GAIN_16 | Default: 70.000 | Linked Fault: ERR-HER350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C350.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C400 (BANK 1)

Machine Model: Hermle-Performance-C400 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C400.

PRM_48621 [Hermle-Performance-C400] — ATC_LOOP_GAIN_1 | Default: 98.900 | Linked Fault: ERR-HER400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C400.

PRM_48622 [Hermle-Performance-C400] — HYD_LOOP_GAIN_2 | Default: 197.800 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

PRM_48623 [Hermle-Performance-C400] — CLN_LOOP_GAIN_3 | Default: 46.700 | Linked Fault: ERR-HER400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C400.

PRM_48624 [Hermle-Performance-C400] — XAX_LOOP_GAIN_4 | Default: 145.600 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

PRM_48625 [Hermle-Performance-C400] — YAX_LOOP_GAIN_5 | Default: 244.500 | Linked Fault: ERR-HER400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C400.

PRM_48626 [Hermle-Performance-C400] — ZAX_LOOP_GAIN_6 | Default: 93.400 | Linked Fault: ERR-HER400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400.

PRM_48627 [Hermle-Performance-C400] — BAX_LOOP_GAIN_7 | Default: 192.300 | Linked Fault: ERR-HER400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C400.

PRM_48628 [Hermle-Performance-C400] — CAX_LOOP_GAIN_8 | Default: 41.200 | Linked Fault: ERR-HER400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400.

PRM_48629 [Hermle-Performance-C400] — OPT_LOOP_GAIN_9 | Default: 140.100 | Linked Fault: ERR-HER400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C400.

PRM_48630 [Hermle-Performance-C400] — PLC_LOOP_GAIN_10 | Default: 239.000 | Linked Fault: ERR-HER400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400.

PRM_48631 [Hermle-Performance-C400] — LUB_LOOP_GAIN_11 | Default: 87.900 | Linked Fault: ERR-HER400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C400.

PRM_48632 [Hermle-Performance-C400] — VAC_LOOP_GAIN_12 | Default: 186.800 | Linked Fault: ERR-HER400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C400.

PRM_48633 [Hermle-Performance-C400] — THM_LOOP_GAIN_13 | Default: 35.700 | Linked Fault: ERR-HER400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C400.

PRM_48634 [Hermle-Performance-C400] — SPN_LOOP_GAIN_14 | Default: 134.600 | Linked Fault: ERR-HER400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C400.

PRM_48635 [Hermle-Performance-C400] — ATC_LOOP_GAIN_15 | Default: 233.500 | Linked Fault: ERR-HER400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C400.

PRM_48636 [Hermle-Performance-C400] — HYD_LOOP_GAIN_16 | Default: 82.400 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C400 (BANK 2)

Machine Model: Hermle-Performance-C400 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C400.

PRM_48641 [Hermle-Performance-C400] — HYD_LOOP_GAIN_1 | Default: 197.800 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

PRM_48642 [Hermle-Performance-C400] — XAX_LOOP_GAIN_2 | Default: 145.600 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

PRM_48643 [Hermle-Performance-C400] — ZAX_LOOP_GAIN_3 | Default: 93.400 | Linked Fault: ERR-HER400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400.

PRM_48644 [Hermle-Performance-C400] — CAX_LOOP_GAIN_4 | Default: 41.200 | Linked Fault: ERR-HER400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400.

PRM_48645 [Hermle-Performance-C400] — PLC_LOOP_GAIN_5 | Default: 239.000 | Linked Fault: ERR-HER400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400.

PRM_48646 [Hermle-Performance-C400] — VAC_LOOP_GAIN_6 | Default: 186.800 | Linked Fault: ERR-HER400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C400.

PRM_48647 [Hermle-Performance-C400] — SPN_LOOP_GAIN_7 | Default: 134.600 | Linked Fault: ERR-HER400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C400.

PRM_48648 [Hermle-Performance-C400] — HYD_LOOP_GAIN_8 | Default: 82.400 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

PRM_48649 [Hermle-Performance-C400] — XAX_LOOP_GAIN_9 | Default: 30.200 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

PRM_48650 [Hermle-Performance-C400] — ZAX_LOOP_GAIN_10 | Default: 228.000 | Linked Fault: ERR-HER400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400.

PRM_48651 [Hermle-Performance-C400] — CAX_LOOP_GAIN_11 | Default: 175.800 | Linked Fault: ERR-HER400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400.

PRM_48652 [Hermle-Performance-C400] — PLC_LOOP_GAIN_12 | Default: 123.600 | Linked Fault: ERR-HER400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400.

PRM_48653 [Hermle-Performance-C400] — VAC_LOOP_GAIN_13 | Default: 71.400 | Linked Fault: ERR-HER400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C400.

PRM_48654 [Hermle-Performance-C400] — SPN_LOOP_GAIN_14 | Default: 19.200 | Linked Fault: ERR-HER400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C400.

PRM_48655 [Hermle-Performance-C400] — HYD_LOOP_GAIN_15 | Default: 217.000 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

PRM_48656 [Hermle-Performance-C400] — XAX_LOOP_GAIN_16 | Default: 164.800 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C400 (BANK 3)

Machine Model: Hermle-Performance-C400 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C400.

PRM_48661 [Hermle-Performance-C400] — CLN_LOOP_GAIN_1 | Default: 46.700 | Linked Fault: ERR-HER400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C400.

PRM_48662 [Hermle-Performance-C400] — ZAX_LOOP_GAIN_2 | Default: 93.400 | Linked Fault: ERR-HER400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400.

PRM_48663 [Hermle-Performance-C400] — OPT_LOOP_GAIN_3 | Default: 140.100 | Linked Fault: ERR-HER400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C400.

PRM_48664 [Hermle-Performance-C400] — VAC_LOOP_GAIN_4 | Default: 186.800 | Linked Fault: ERR-HER400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C400.

PRM_48665 [Hermle-Performance-C400] — ATC_LOOP_GAIN_5 | Default: 233.500 | Linked Fault: ERR-HER400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C400.

PRM_48666 [Hermle-Performance-C400] — XAX_LOOP_GAIN_6 | Default: 30.200 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

PRM_48667 [Hermle-Performance-C400] — BAX_LOOP_GAIN_7 | Default: 76.900 | Linked Fault: ERR-HER400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C400.

PRM_48668 [Hermle-Performance-C400] — PLC_LOOP_GAIN_8 | Default: 123.600 | Linked Fault: ERR-HER400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400.

PRM_48669 [Hermle-Performance-C400] — THM_LOOP_GAIN_9 | Default: 170.300 | Linked Fault: ERR-HER400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C400.

PRM_48670 [Hermle-Performance-C400] — HYD_LOOP_GAIN_10 | Default: 217.000 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

PRM_48671 [Hermle-Performance-C400] — YAX_LOOP_GAIN_11 | Default: 13.700 | Linked Fault: ERR-HER400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C400.

PRM_48672 [Hermle-Performance-C400] — CAX_LOOP_GAIN_12 | Default: 60.400 | Linked Fault: ERR-HER400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400.

PRM_48673 [Hermle-Performance-C400] — LUB_LOOP_GAIN_13 | Default: 107.100 | Linked Fault: ERR-HER400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C400.

PRM_48674 [Hermle-Performance-C400] — SPN_LOOP_GAIN_14 | Default: 153.800 | Linked Fault: ERR-HER400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C400.

PRM_48675 [Hermle-Performance-C400] — CLN_LOOP_GAIN_15 | Default: 200.500 | Linked Fault: ERR-HER400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C400.

PRM_48676 [Hermle-Performance-C400] — ZAX_LOOP_GAIN_16 | Default: 247.200 | Linked Fault: ERR-HER400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C400 (BANK 4)

Machine Model: Hermle-Performance-C400 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C400.

PRM_48681 [Hermle-Performance-C400] — XAX_LOOP_GAIN_1 | Default: 145.600 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

PRM_48682 [Hermle-Performance-C400] — CAX_LOOP_GAIN_2 | Default: 41.200 | Linked Fault: ERR-HER400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400.

PRM_48683 [Hermle-Performance-C400] — VAC_LOOP_GAIN_3 | Default: 186.800 | Linked Fault: ERR-HER400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C400.

PRM_48684 [Hermle-Performance-C400] — HYD_LOOP_GAIN_4 | Default: 82.400 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

PRM_48685 [Hermle-Performance-C400] — ZAX_LOOP_GAIN_5 | Default: 228.000 | Linked Fault: ERR-HER400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400.

PRM_48686 [Hermle-Performance-C400] — PLC_LOOP_GAIN_6 | Default: 123.600 | Linked Fault: ERR-HER400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400.

PRM_48687 [Hermle-Performance-C400] — SPN_LOOP_GAIN_7 | Default: 19.200 | Linked Fault: ERR-HER400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C400.

PRM_48688 [Hermle-Performance-C400] — XAX_LOOP_GAIN_8 | Default: 164.800 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

PRM_48689 [Hermle-Performance-C400] — CAX_LOOP_GAIN_9 | Default: 60.400 | Linked Fault: ERR-HER400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400.

PRM_48690 [Hermle-Performance-C400] — VAC_LOOP_GAIN_10 | Default: 206.000 | Linked Fault: ERR-HER400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C400.

PRM_48691 [Hermle-Performance-C400] — HYD_LOOP_GAIN_11 | Default: 101.600 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

PRM_48692 [Hermle-Performance-C400] — ZAX_LOOP_GAIN_12 | Default: 247.200 | Linked Fault: ERR-HER400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400.

PRM_48693 [Hermle-Performance-C400] — PLC_LOOP_GAIN_13 | Default: 142.800 | Linked Fault: ERR-HER400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400.

PRM_48694 [Hermle-Performance-C400] — SPN_LOOP_GAIN_14 | Default: 38.400 | Linked Fault: ERR-HER400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C400.

PRM_48695 [Hermle-Performance-C400] — XAX_LOOP_GAIN_15 | Default: 184.000 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

PRM_48696 [Hermle-Performance-C400] — CAX_LOOP_GAIN_16 | Default: 79.600 | Linked Fault: ERR-HER400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C400 (BANK 5)

Machine Model: Hermle-Performance-C400 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C400.

PRM_48701 [Hermle-Performance-C400] — YAX_LOOP_GAIN_1 | Default: 244.500 | Linked Fault: ERR-HER400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C400.

PRM_48702 [Hermle-Performance-C400] — PLC_LOOP_GAIN_2 | Default: 239.000 | Linked Fault: ERR-HER400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400.

PRM_48703 [Hermle-Performance-C400] — ATC_LOOP_GAIN_3 | Default: 233.500 | Linked Fault: ERR-HER400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C400.

PRM_48704 [Hermle-Performance-C400] — ZAX_LOOP_GAIN_4 | Default: 228.000 | Linked Fault: ERR-HER400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C400.

PRM_48705 [Hermle-Performance-C400] — LUB_LOOP_GAIN_5 | Default: 222.500 | Linked Fault: ERR-HER400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C400.

PRM_48706 [Hermle-Performance-C400] — HYD_LOOP_GAIN_6 | Default: 217.000 | Linked Fault: ERR-HER400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C400.

PRM_48707 [Hermle-Performance-C400] — BAX_LOOP_GAIN_7 | Default: 211.500 | Linked Fault: ERR-HER400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C400.

PRM_48708 [Hermle-Performance-C400] — VAC_LOOP_GAIN_8 | Default: 206.000 | Linked Fault: ERR-HER400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C400.

PRM_48709 [Hermle-Performance-C400] — CLN_LOOP_GAIN_9 | Default: 200.500 | Linked Fault: ERR-HER400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C400.

PRM_48710 [Hermle-Performance-C400] — CAX_LOOP_GAIN_10 | Default: 195.000 | Linked Fault: ERR-HER400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C400.

PRM_48711 [Hermle-Performance-C400] — THM_LOOP_GAIN_11 | Default: 189.500 | Linked Fault: ERR-HER400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C400.

PRM_48712 [Hermle-Performance-C400] — XAX_LOOP_GAIN_12 | Default: 184.000 | Linked Fault: ERR-HER400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C400.

PRM_48713 [Hermle-Performance-C400] — OPT_LOOP_GAIN_13 | Default: 178.500 | Linked Fault: ERR-HER400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C400.

PRM_48714 [Hermle-Performance-C400] — SPN_LOOP_GAIN_14 | Default: 173.000 | Linked Fault: ERR-HER400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C400.

PRM_48715 [Hermle-Performance-C400] — YAX_LOOP_GAIN_15 | Default: 167.500 | Linked Fault: ERR-HER400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C400.

PRM_48716 [Hermle-Performance-C400] — PLC_LOOP_GAIN_16 | Default: 162.000 | Linked Fault: ERR-HER400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C400.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C450 (BANK 1)

Machine Model: Hermle-Performance-C450 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C450.

PRM_48721 [Hermle-Performance-C450] — ATC_LOOP_GAIN_1 | Default: 100.050 | Linked Fault: ERR-HER450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C450.

PRM_48722 [Hermle-Performance-C450] — HYD_LOOP_GAIN_2 | Default: 200.100 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

PRM_48723 [Hermle-Performance-C450] — CLN_LOOP_GAIN_3 | Default: 50.150 | Linked Fault: ERR-HER450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C450.

PRM_48724 [Hermle-Performance-C450] — XAX_LOOP_GAIN_4 | Default: 150.200 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

PRM_48725 [Hermle-Performance-C450] — YAX_LOOP_GAIN_5 | Default: 0.250 | Linked Fault: ERR-HER450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C450.

PRM_48726 [Hermle-Performance-C450] — ZAX_LOOP_GAIN_6 | Default: 100.300 | Linked Fault: ERR-HER450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450.

PRM_48727 [Hermle-Performance-C450] — BAX_LOOP_GAIN_7 | Default: 200.350 | Linked Fault: ERR-HER450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C450.

PRM_48728 [Hermle-Performance-C450] — CAX_LOOP_GAIN_8 | Default: 50.400 | Linked Fault: ERR-HER450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450.

PRM_48729 [Hermle-Performance-C450] — OPT_LOOP_GAIN_9 | Default: 150.450 | Linked Fault: ERR-HER450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C450.

PRM_48730 [Hermle-Performance-C450] — PLC_LOOP_GAIN_10 | Default: 0.500 | Linked Fault: ERR-HER450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450.

PRM_48731 [Hermle-Performance-C450] — LUB_LOOP_GAIN_11 | Default: 100.550 | Linked Fault: ERR-HER450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C450.

PRM_48732 [Hermle-Performance-C450] — VAC_LOOP_GAIN_12 | Default: 200.600 | Linked Fault: ERR-HER450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C450.

PRM_48733 [Hermle-Performance-C450] — THM_LOOP_GAIN_13 | Default: 50.650 | Linked Fault: ERR-HER450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C450.

PRM_48734 [Hermle-Performance-C450] — SPN_LOOP_GAIN_14 | Default: 150.700 | Linked Fault: ERR-HER450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C450.

PRM_48735 [Hermle-Performance-C450] — ATC_LOOP_GAIN_15 | Default: 0.750 | Linked Fault: ERR-HER450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C450.

PRM_48736 [Hermle-Performance-C450] — HYD_LOOP_GAIN_16 | Default: 100.800 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C450 (BANK 2)

Machine Model: Hermle-Performance-C450 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C450.

PRM_48741 [Hermle-Performance-C450] — HYD_LOOP_GAIN_1 | Default: 200.100 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

PRM_48742 [Hermle-Performance-C450] — XAX_LOOP_GAIN_2 | Default: 150.200 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

PRM_48743 [Hermle-Performance-C450] — ZAX_LOOP_GAIN_3 | Default: 100.300 | Linked Fault: ERR-HER450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450.

PRM_48744 [Hermle-Performance-C450] — CAX_LOOP_GAIN_4 | Default: 50.400 | Linked Fault: ERR-HER450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450.

PRM_48745 [Hermle-Performance-C450] — PLC_LOOP_GAIN_5 | Default: 0.500 | Linked Fault: ERR-HER450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450.

PRM_48746 [Hermle-Performance-C450] — VAC_LOOP_GAIN_6 | Default: 200.600 | Linked Fault: ERR-HER450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C450.

PRM_48747 [Hermle-Performance-C450] — SPN_LOOP_GAIN_7 | Default: 150.700 | Linked Fault: ERR-HER450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C450.

PRM_48748 [Hermle-Performance-C450] — HYD_LOOP_GAIN_8 | Default: 100.800 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

PRM_48749 [Hermle-Performance-C450] — XAX_LOOP_GAIN_9 | Default: 50.900 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

PRM_48750 [Hermle-Performance-C450] — ZAX_LOOP_GAIN_10 | Default: 1.000 | Linked Fault: ERR-HER450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450.

PRM_48751 [Hermle-Performance-C450] — CAX_LOOP_GAIN_11 | Default: 201.100 | Linked Fault: ERR-HER450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450.

PRM_48752 [Hermle-Performance-C450] — PLC_LOOP_GAIN_12 | Default: 151.200 | Linked Fault: ERR-HER450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450.

PRM_48753 [Hermle-Performance-C450] — VAC_LOOP_GAIN_13 | Default: 101.300 | Linked Fault: ERR-HER450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C450.

PRM_48754 [Hermle-Performance-C450] — SPN_LOOP_GAIN_14 | Default: 51.400 | Linked Fault: ERR-HER450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C450.

PRM_48755 [Hermle-Performance-C450] — HYD_LOOP_GAIN_15 | Default: 1.500 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

PRM_48756 [Hermle-Performance-C450] — XAX_LOOP_GAIN_16 | Default: 201.600 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C450 (BANK 3)

Machine Model: Hermle-Performance-C450 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C450.

PRM_48761 [Hermle-Performance-C450] — CLN_LOOP_GAIN_1 | Default: 50.150 | Linked Fault: ERR-HER450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C450.

PRM_48762 [Hermle-Performance-C450] — ZAX_LOOP_GAIN_2 | Default: 100.300 | Linked Fault: ERR-HER450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450.

PRM_48763 [Hermle-Performance-C450] — OPT_LOOP_GAIN_3 | Default: 150.450 | Linked Fault: ERR-HER450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C450.

PRM_48764 [Hermle-Performance-C450] — VAC_LOOP_GAIN_4 | Default: 200.600 | Linked Fault: ERR-HER450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C450.

PRM_48765 [Hermle-Performance-C450] — ATC_LOOP_GAIN_5 | Default: 0.750 | Linked Fault: ERR-HER450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C450.

PRM_48766 [Hermle-Performance-C450] — XAX_LOOP_GAIN_6 | Default: 50.900 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

PRM_48767 [Hermle-Performance-C450] — BAX_LOOP_GAIN_7 | Default: 101.050 | Linked Fault: ERR-HER450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C450.

PRM_48768 [Hermle-Performance-C450] — PLC_LOOP_GAIN_8 | Default: 151.200 | Linked Fault: ERR-HER450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450.

PRM_48769 [Hermle-Performance-C450] — THM_LOOP_GAIN_9 | Default: 201.350 | Linked Fault: ERR-HER450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C450.

PRM_48770 [Hermle-Performance-C450] — HYD_LOOP_GAIN_10 | Default: 1.500 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

PRM_48771 [Hermle-Performance-C450] — YAX_LOOP_GAIN_11 | Default: 51.650 | Linked Fault: ERR-HER450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C450.

PRM_48772 [Hermle-Performance-C450] — CAX_LOOP_GAIN_12 | Default: 101.800 | Linked Fault: ERR-HER450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450.

PRM_48773 [Hermle-Performance-C450] — LUB_LOOP_GAIN_13 | Default: 151.950 | Linked Fault: ERR-HER450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C450.

PRM_48774 [Hermle-Performance-C450] — SPN_LOOP_GAIN_14 | Default: 202.100 | Linked Fault: ERR-HER450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C450.

PRM_48775 [Hermle-Performance-C450] — CLN_LOOP_GAIN_15 | Default: 2.250 | Linked Fault: ERR-HER450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C450.

PRM_48776 [Hermle-Performance-C450] — ZAX_LOOP_GAIN_16 | Default: 52.400 | Linked Fault: ERR-HER450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C450 (BANK 4)

Machine Model: Hermle-Performance-C450 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C450.

PRM_48781 [Hermle-Performance-C450] — XAX_LOOP_GAIN_1 | Default: 150.200 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

PRM_48782 [Hermle-Performance-C450] — CAX_LOOP_GAIN_2 | Default: 50.400 | Linked Fault: ERR-HER450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450.

PRM_48783 [Hermle-Performance-C450] — VAC_LOOP_GAIN_3 | Default: 200.600 | Linked Fault: ERR-HER450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C450.

PRM_48784 [Hermle-Performance-C450] — HYD_LOOP_GAIN_4 | Default: 100.800 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

PRM_48785 [Hermle-Performance-C450] — ZAX_LOOP_GAIN_5 | Default: 1.000 | Linked Fault: ERR-HER450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450.

PRM_48786 [Hermle-Performance-C450] — PLC_LOOP_GAIN_6 | Default: 151.200 | Linked Fault: ERR-HER450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450.

PRM_48787 [Hermle-Performance-C450] — SPN_LOOP_GAIN_7 | Default: 51.400 | Linked Fault: ERR-HER450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C450.

PRM_48788 [Hermle-Performance-C450] — XAX_LOOP_GAIN_8 | Default: 201.600 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

PRM_48789 [Hermle-Performance-C450] — CAX_LOOP_GAIN_9 | Default: 101.800 | Linked Fault: ERR-HER450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450.

PRM_48790 [Hermle-Performance-C450] — VAC_LOOP_GAIN_10 | Default: 2.000 | Linked Fault: ERR-HER450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C450.

PRM_48791 [Hermle-Performance-C450] — HYD_LOOP_GAIN_11 | Default: 152.200 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

PRM_48792 [Hermle-Performance-C450] — ZAX_LOOP_GAIN_12 | Default: 52.400 | Linked Fault: ERR-HER450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450.

PRM_48793 [Hermle-Performance-C450] — PLC_LOOP_GAIN_13 | Default: 202.600 | Linked Fault: ERR-HER450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450.

PRM_48794 [Hermle-Performance-C450] — SPN_LOOP_GAIN_14 | Default: 102.800 | Linked Fault: ERR-HER450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C450.

PRM_48795 [Hermle-Performance-C450] — XAX_LOOP_GAIN_15 | Default: 3.000 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

PRM_48796 [Hermle-Performance-C450] — CAX_LOOP_GAIN_16 | Default: 153.200 | Linked Fault: ERR-HER450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C450 (BANK 5)

Machine Model: Hermle-Performance-C450 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C450.

PRM_48801 [Hermle-Performance-C450] — YAX_LOOP_GAIN_1 | Default: 0.250 | Linked Fault: ERR-HER450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C450.

PRM_48802 [Hermle-Performance-C450] — PLC_LOOP_GAIN_2 | Default: 0.500 | Linked Fault: ERR-HER450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450.

PRM_48803 [Hermle-Performance-C450] — ATC_LOOP_GAIN_3 | Default: 0.750 | Linked Fault: ERR-HER450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C450.

PRM_48804 [Hermle-Performance-C450] — ZAX_LOOP_GAIN_4 | Default: 1.000 | Linked Fault: ERR-HER450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C450.

PRM_48805 [Hermle-Performance-C450] — LUB_LOOP_GAIN_5 | Default: 1.250 | Linked Fault: ERR-HER450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C450.

PRM_48806 [Hermle-Performance-C450] — HYD_LOOP_GAIN_6 | Default: 1.500 | Linked Fault: ERR-HER450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C450.

PRM_48807 [Hermle-Performance-C450] — BAX_LOOP_GAIN_7 | Default: 1.750 | Linked Fault: ERR-HER450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C450.

PRM_48808 [Hermle-Performance-C450] — VAC_LOOP_GAIN_8 | Default: 2.000 | Linked Fault: ERR-HER450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C450.

PRM_48809 [Hermle-Performance-C450] — CLN_LOOP_GAIN_9 | Default: 2.250 | Linked Fault: ERR-HER450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C450.

PRM_48810 [Hermle-Performance-C450] — CAX_LOOP_GAIN_10 | Default: 2.500 | Linked Fault: ERR-HER450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C450.

PRM_48811 [Hermle-Performance-C450] — THM_LOOP_GAIN_11 | Default: 2.750 | Linked Fault: ERR-HER450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C450.

PRM_48812 [Hermle-Performance-C450] — XAX_LOOP_GAIN_12 | Default: 3.000 | Linked Fault: ERR-HER450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C450.

PRM_48813 [Hermle-Performance-C450] — OPT_LOOP_GAIN_13 | Default: 3.250 | Linked Fault: ERR-HER450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C450.

PRM_48814 [Hermle-Performance-C450] — SPN_LOOP_GAIN_14 | Default: 3.500 | Linked Fault: ERR-HER450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C450.

PRM_48815 [Hermle-Performance-C450] — YAX_LOOP_GAIN_15 | Default: 3.750 | Linked Fault: ERR-HER450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C450.

PRM_48816 [Hermle-Performance-C450] — PLC_LOOP_GAIN_16 | Default: 4.000 | Linked Fault: ERR-HER450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C450.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C500 (BANK 1)

Machine Model: Hermle-Performance-C500 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C500.

PRM_48821 [Hermle-Performance-C500] — ATC_LOOP_GAIN_1 | Default: 101.200 | Linked Fault: ERR-HER500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C500.

PRM_48822 [Hermle-Performance-C500] — HYD_LOOP_GAIN_2 | Default: 202.400 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

PRM_48823 [Hermle-Performance-C500] — CLN_LOOP_GAIN_3 | Default: 53.600 | Linked Fault: ERR-HER500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C500.

PRM_48824 [Hermle-Performance-C500] — XAX_LOOP_GAIN_4 | Default: 154.800 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

PRM_48825 [Hermle-Performance-C500] — YAX_LOOP_GAIN_5 | Default: 6.000 | Linked Fault: ERR-HER500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C500.

PRM_48826 [Hermle-Performance-C500] — ZAX_LOOP_GAIN_6 | Default: 107.200 | Linked Fault: ERR-HER500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500.

PRM_48827 [Hermle-Performance-C500] — BAX_LOOP_GAIN_7 | Default: 208.400 | Linked Fault: ERR-HER500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C500.

PRM_48828 [Hermle-Performance-C500] — CAX_LOOP_GAIN_8 | Default: 59.600 | Linked Fault: ERR-HER500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500.

PRM_48829 [Hermle-Performance-C500] — OPT_LOOP_GAIN_9 | Default: 160.800 | Linked Fault: ERR-HER500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C500.

PRM_48830 [Hermle-Performance-C500] — PLC_LOOP_GAIN_10 | Default: 12.000 | Linked Fault: ERR-HER500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500.

PRM_48831 [Hermle-Performance-C500] — LUB_LOOP_GAIN_11 | Default: 113.200 | Linked Fault: ERR-HER500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C500.

PRM_48832 [Hermle-Performance-C500] — VAC_LOOP_GAIN_12 | Default: 214.400 | Linked Fault: ERR-HER500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C500.

PRM_48833 [Hermle-Performance-C500] — THM_LOOP_GAIN_13 | Default: 65.600 | Linked Fault: ERR-HER500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C500.

PRM_48834 [Hermle-Performance-C500] — SPN_LOOP_GAIN_14 | Default: 166.800 | Linked Fault: ERR-HER500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C500.

PRM_48835 [Hermle-Performance-C500] — ATC_LOOP_GAIN_15 | Default: 18.000 | Linked Fault: ERR-HER500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C500.

PRM_48836 [Hermle-Performance-C500] — HYD_LOOP_GAIN_16 | Default: 119.200 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C500 (BANK 2)

Machine Model: Hermle-Performance-C500 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C500.

PRM_48841 [Hermle-Performance-C500] — HYD_LOOP_GAIN_1 | Default: 202.400 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

PRM_48842 [Hermle-Performance-C500] — XAX_LOOP_GAIN_2 | Default: 154.800 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

PRM_48843 [Hermle-Performance-C500] — ZAX_LOOP_GAIN_3 | Default: 107.200 | Linked Fault: ERR-HER500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500.

PRM_48844 [Hermle-Performance-C500] — CAX_LOOP_GAIN_4 | Default: 59.600 | Linked Fault: ERR-HER500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500.

PRM_48845 [Hermle-Performance-C500] — PLC_LOOP_GAIN_5 | Default: 12.000 | Linked Fault: ERR-HER500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500.

PRM_48846 [Hermle-Performance-C500] — VAC_LOOP_GAIN_6 | Default: 214.400 | Linked Fault: ERR-HER500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C500.

PRM_48847 [Hermle-Performance-C500] — SPN_LOOP_GAIN_7 | Default: 166.800 | Linked Fault: ERR-HER500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C500.

PRM_48848 [Hermle-Performance-C500] — HYD_LOOP_GAIN_8 | Default: 119.200 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

PRM_48849 [Hermle-Performance-C500] — XAX_LOOP_GAIN_9 | Default: 71.600 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

PRM_48850 [Hermle-Performance-C500] — ZAX_LOOP_GAIN_10 | Default: 24.000 | Linked Fault: ERR-HER500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500.

PRM_48851 [Hermle-Performance-C500] — CAX_LOOP_GAIN_11 | Default: 226.400 | Linked Fault: ERR-HER500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500.

PRM_48852 [Hermle-Performance-C500] — PLC_LOOP_GAIN_12 | Default: 178.800 | Linked Fault: ERR-HER500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500.

PRM_48853 [Hermle-Performance-C500] — VAC_LOOP_GAIN_13 | Default: 131.200 | Linked Fault: ERR-HER500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C500.

PRM_48854 [Hermle-Performance-C500] — SPN_LOOP_GAIN_14 | Default: 83.600 | Linked Fault: ERR-HER500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C500.

PRM_48855 [Hermle-Performance-C500] — HYD_LOOP_GAIN_15 | Default: 36.000 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

PRM_48856 [Hermle-Performance-C500] — XAX_LOOP_GAIN_16 | Default: 238.400 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C500 (BANK 3)

Machine Model: Hermle-Performance-C500 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C500.

PRM_48861 [Hermle-Performance-C500] — CLN_LOOP_GAIN_1 | Default: 53.600 | Linked Fault: ERR-HER500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C500.

PRM_48862 [Hermle-Performance-C500] — ZAX_LOOP_GAIN_2 | Default: 107.200 | Linked Fault: ERR-HER500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500.

PRM_48863 [Hermle-Performance-C500] — OPT_LOOP_GAIN_3 | Default: 160.800 | Linked Fault: ERR-HER500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C500.

PRM_48864 [Hermle-Performance-C500] — VAC_LOOP_GAIN_4 | Default: 214.400 | Linked Fault: ERR-HER500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C500.

PRM_48865 [Hermle-Performance-C500] — ATC_LOOP_GAIN_5 | Default: 18.000 | Linked Fault: ERR-HER500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C500.

PRM_48866 [Hermle-Performance-C500] — XAX_LOOP_GAIN_6 | Default: 71.600 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

PRM_48867 [Hermle-Performance-C500] — BAX_LOOP_GAIN_7 | Default: 125.200 | Linked Fault: ERR-HER500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C500.

PRM_48868 [Hermle-Performance-C500] — PLC_LOOP_GAIN_8 | Default: 178.800 | Linked Fault: ERR-HER500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500.

PRM_48869 [Hermle-Performance-C500] — THM_LOOP_GAIN_9 | Default: 232.400 | Linked Fault: ERR-HER500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C500.

PRM_48870 [Hermle-Performance-C500] — HYD_LOOP_GAIN_10 | Default: 36.000 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

PRM_48871 [Hermle-Performance-C500] — YAX_LOOP_GAIN_11 | Default: 89.600 | Linked Fault: ERR-HER500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C500.

PRM_48872 [Hermle-Performance-C500] — CAX_LOOP_GAIN_12 | Default: 143.200 | Linked Fault: ERR-HER500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500.

PRM_48873 [Hermle-Performance-C500] — LUB_LOOP_GAIN_13 | Default: 196.800 | Linked Fault: ERR-HER500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C500.

PRM_48874 [Hermle-Performance-C500] — SPN_LOOP_GAIN_14 | Default: 0.400 | Linked Fault: ERR-HER500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C500.

PRM_48875 [Hermle-Performance-C500] — CLN_LOOP_GAIN_15 | Default: 54.000 | Linked Fault: ERR-HER500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C500.

PRM_48876 [Hermle-Performance-C500] — ZAX_LOOP_GAIN_16 | Default: 107.600 | Linked Fault: ERR-HER500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C500 (BANK 4)

Machine Model: Hermle-Performance-C500 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C500.

PRM_48881 [Hermle-Performance-C500] — XAX_LOOP_GAIN_1 | Default: 154.800 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

PRM_48882 [Hermle-Performance-C500] — CAX_LOOP_GAIN_2 | Default: 59.600 | Linked Fault: ERR-HER500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500.

PRM_48883 [Hermle-Performance-C500] — VAC_LOOP_GAIN_3 | Default: 214.400 | Linked Fault: ERR-HER500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C500.

PRM_48884 [Hermle-Performance-C500] — HYD_LOOP_GAIN_4 | Default: 119.200 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

PRM_48885 [Hermle-Performance-C500] — ZAX_LOOP_GAIN_5 | Default: 24.000 | Linked Fault: ERR-HER500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500.

PRM_48886 [Hermle-Performance-C500] — PLC_LOOP_GAIN_6 | Default: 178.800 | Linked Fault: ERR-HER500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500.

PRM_48887 [Hermle-Performance-C500] — SPN_LOOP_GAIN_7 | Default: 83.600 | Linked Fault: ERR-HER500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C500.

PRM_48888 [Hermle-Performance-C500] — XAX_LOOP_GAIN_8 | Default: 238.400 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

PRM_48889 [Hermle-Performance-C500] — CAX_LOOP_GAIN_9 | Default: 143.200 | Linked Fault: ERR-HER500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500.

PRM_48890 [Hermle-Performance-C500] — VAC_LOOP_GAIN_10 | Default: 48.000 | Linked Fault: ERR-HER500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C500.

PRM_48891 [Hermle-Performance-C500] — HYD_LOOP_GAIN_11 | Default: 202.800 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

PRM_48892 [Hermle-Performance-C500] — ZAX_LOOP_GAIN_12 | Default: 107.600 | Linked Fault: ERR-HER500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500.

PRM_48893 [Hermle-Performance-C500] — PLC_LOOP_GAIN_13 | Default: 12.400 | Linked Fault: ERR-HER500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500.

PRM_48894 [Hermle-Performance-C500] — SPN_LOOP_GAIN_14 | Default: 167.200 | Linked Fault: ERR-HER500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C500.

PRM_48895 [Hermle-Performance-C500] — XAX_LOOP_GAIN_15 | Default: 72.000 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

PRM_48896 [Hermle-Performance-C500] — CAX_LOOP_GAIN_16 | Default: 226.800 | Linked Fault: ERR-HER500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C500 (BANK 5)

Machine Model: Hermle-Performance-C500 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C500.

PRM_48901 [Hermle-Performance-C500] — YAX_LOOP_GAIN_1 | Default: 6.000 | Linked Fault: ERR-HER500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C500.

PRM_48902 [Hermle-Performance-C500] — PLC_LOOP_GAIN_2 | Default: 12.000 | Linked Fault: ERR-HER500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500.

PRM_48903 [Hermle-Performance-C500] — ATC_LOOP_GAIN_3 | Default: 18.000 | Linked Fault: ERR-HER500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C500.

PRM_48904 [Hermle-Performance-C500] — ZAX_LOOP_GAIN_4 | Default: 24.000 | Linked Fault: ERR-HER500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C500.

PRM_48905 [Hermle-Performance-C500] — LUB_LOOP_GAIN_5 | Default: 30.000 | Linked Fault: ERR-HER500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C500.

PRM_48906 [Hermle-Performance-C500] — HYD_LOOP_GAIN_6 | Default: 36.000 | Linked Fault: ERR-HER500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C500.

PRM_48907 [Hermle-Performance-C500] — BAX_LOOP_GAIN_7 | Default: 42.000 | Linked Fault: ERR-HER500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C500.

PRM_48908 [Hermle-Performance-C500] — VAC_LOOP_GAIN_8 | Default: 48.000 | Linked Fault: ERR-HER500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C500.

PRM_48909 [Hermle-Performance-C500] — CLN_LOOP_GAIN_9 | Default: 54.000 | Linked Fault: ERR-HER500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C500.

PRM_48910 [Hermle-Performance-C500] — CAX_LOOP_GAIN_10 | Default: 60.000 | Linked Fault: ERR-HER500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C500.

PRM_48911 [Hermle-Performance-C500] — THM_LOOP_GAIN_11 | Default: 66.000 | Linked Fault: ERR-HER500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C500.

PRM_48912 [Hermle-Performance-C500] — XAX_LOOP_GAIN_12 | Default: 72.000 | Linked Fault: ERR-HER500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C500.

PRM_48913 [Hermle-Performance-C500] — OPT_LOOP_GAIN_13 | Default: 78.000 | Linked Fault: ERR-HER500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C500.

PRM_48914 [Hermle-Performance-C500] — SPN_LOOP_GAIN_14 | Default: 84.000 | Linked Fault: ERR-HER500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C500.

PRM_48915 [Hermle-Performance-C500] — YAX_LOOP_GAIN_15 | Default: 90.000 | Linked Fault: ERR-HER500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C500.

PRM_48916 [Hermle-Performance-C500] — PLC_LOOP_GAIN_16 | Default: 96.000 | Linked Fault: ERR-HER500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C500.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C550 (BANK 1)

Machine Model: Hermle-Performance-C550 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C550.

PRM_48921 [Hermle-Performance-C550] — ATC_LOOP_GAIN_1 | Default: 102.350 | Linked Fault: ERR-HER550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C550.

PRM_48922 [Hermle-Performance-C550] — HYD_LOOP_GAIN_2 | Default: 204.700 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

PRM_48923 [Hermle-Performance-C550] — CLN_LOOP_GAIN_3 | Default: 57.050 | Linked Fault: ERR-HER550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C550.

PRM_48924 [Hermle-Performance-C550] — XAX_LOOP_GAIN_4 | Default: 159.400 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

PRM_48925 [Hermle-Performance-C550] — YAX_LOOP_GAIN_5 | Default: 11.750 | Linked Fault: ERR-HER550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C550.

PRM_48926 [Hermle-Performance-C550] — ZAX_LOOP_GAIN_6 | Default: 114.100 | Linked Fault: ERR-HER550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550.

PRM_48927 [Hermle-Performance-C550] — BAX_LOOP_GAIN_7 | Default: 216.450 | Linked Fault: ERR-HER550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C550.

PRM_48928 [Hermle-Performance-C550] — CAX_LOOP_GAIN_8 | Default: 68.800 | Linked Fault: ERR-HER550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550.

PRM_48929 [Hermle-Performance-C550] — OPT_LOOP_GAIN_9 | Default: 171.150 | Linked Fault: ERR-HER550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C550.

PRM_48930 [Hermle-Performance-C550] — PLC_LOOP_GAIN_10 | Default: 23.500 | Linked Fault: ERR-HER550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550.

PRM_48931 [Hermle-Performance-C550] — LUB_LOOP_GAIN_11 | Default: 125.850 | Linked Fault: ERR-HER550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C550.

PRM_48932 [Hermle-Performance-C550] — VAC_LOOP_GAIN_12 | Default: 228.200 | Linked Fault: ERR-HER550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C550.

PRM_48933 [Hermle-Performance-C550] — THM_LOOP_GAIN_13 | Default: 80.550 | Linked Fault: ERR-HER550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C550.

PRM_48934 [Hermle-Performance-C550] — SPN_LOOP_GAIN_14 | Default: 182.900 | Linked Fault: ERR-HER550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C550.

PRM_48935 [Hermle-Performance-C550] — ATC_LOOP_GAIN_15 | Default: 35.250 | Linked Fault: ERR-HER550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C550.

PRM_48936 [Hermle-Performance-C550] — HYD_LOOP_GAIN_16 | Default: 137.600 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C550 (BANK 2)

Machine Model: Hermle-Performance-C550 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C550.

PRM_48941 [Hermle-Performance-C550] — HYD_LOOP_GAIN_1 | Default: 204.700 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

PRM_48942 [Hermle-Performance-C550] — XAX_LOOP_GAIN_2 | Default: 159.400 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

PRM_48943 [Hermle-Performance-C550] — ZAX_LOOP_GAIN_3 | Default: 114.100 | Linked Fault: ERR-HER550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550.

PRM_48944 [Hermle-Performance-C550] — CAX_LOOP_GAIN_4 | Default: 68.800 | Linked Fault: ERR-HER550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550.

PRM_48945 [Hermle-Performance-C550] — PLC_LOOP_GAIN_5 | Default: 23.500 | Linked Fault: ERR-HER550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550.

PRM_48946 [Hermle-Performance-C550] — VAC_LOOP_GAIN_6 | Default: 228.200 | Linked Fault: ERR-HER550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C550.

PRM_48947 [Hermle-Performance-C550] — SPN_LOOP_GAIN_7 | Default: 182.900 | Linked Fault: ERR-HER550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C550.

PRM_48948 [Hermle-Performance-C550] — HYD_LOOP_GAIN_8 | Default: 137.600 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

PRM_48949 [Hermle-Performance-C550] — XAX_LOOP_GAIN_9 | Default: 92.300 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

PRM_48950 [Hermle-Performance-C550] — ZAX_LOOP_GAIN_10 | Default: 47.000 | Linked Fault: ERR-HER550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550.

PRM_48951 [Hermle-Performance-C550] — CAX_LOOP_GAIN_11 | Default: 1.700 | Linked Fault: ERR-HER550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550.

PRM_48952 [Hermle-Performance-C550] — PLC_LOOP_GAIN_12 | Default: 206.400 | Linked Fault: ERR-HER550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550.

PRM_48953 [Hermle-Performance-C550] — VAC_LOOP_GAIN_13 | Default: 161.100 | Linked Fault: ERR-HER550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C550.

PRM_48954 [Hermle-Performance-C550] — SPN_LOOP_GAIN_14 | Default: 115.800 | Linked Fault: ERR-HER550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C550.

PRM_48955 [Hermle-Performance-C550] — HYD_LOOP_GAIN_15 | Default: 70.500 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

PRM_48956 [Hermle-Performance-C550] — XAX_LOOP_GAIN_16 | Default: 25.200 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C550 (BANK 3)

Machine Model: Hermle-Performance-C550 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C550.

PRM_48961 [Hermle-Performance-C550] — CLN_LOOP_GAIN_1 | Default: 57.050 | Linked Fault: ERR-HER550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C550.

PRM_48962 [Hermle-Performance-C550] — ZAX_LOOP_GAIN_2 | Default: 114.100 | Linked Fault: ERR-HER550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550.

PRM_48963 [Hermle-Performance-C550] — OPT_LOOP_GAIN_3 | Default: 171.150 | Linked Fault: ERR-HER550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C550.

PRM_48964 [Hermle-Performance-C550] — VAC_LOOP_GAIN_4 | Default: 228.200 | Linked Fault: ERR-HER550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C550.

PRM_48965 [Hermle-Performance-C550] — ATC_LOOP_GAIN_5 | Default: 35.250 | Linked Fault: ERR-HER550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C550.

PRM_48966 [Hermle-Performance-C550] — XAX_LOOP_GAIN_6 | Default: 92.300 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

PRM_48967 [Hermle-Performance-C550] — BAX_LOOP_GAIN_7 | Default: 149.350 | Linked Fault: ERR-HER550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C550.

PRM_48968 [Hermle-Performance-C550] — PLC_LOOP_GAIN_8 | Default: 206.400 | Linked Fault: ERR-HER550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550.

PRM_48969 [Hermle-Performance-C550] — THM_LOOP_GAIN_9 | Default: 13.450 | Linked Fault: ERR-HER550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C550.

PRM_48970 [Hermle-Performance-C550] — HYD_LOOP_GAIN_10 | Default: 70.500 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

PRM_48971 [Hermle-Performance-C550] — YAX_LOOP_GAIN_11 | Default: 127.550 | Linked Fault: ERR-HER550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C550.

PRM_48972 [Hermle-Performance-C550] — CAX_LOOP_GAIN_12 | Default: 184.600 | Linked Fault: ERR-HER550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550.

PRM_48973 [Hermle-Performance-C550] — LUB_LOOP_GAIN_13 | Default: 241.650 | Linked Fault: ERR-HER550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C550.

PRM_48974 [Hermle-Performance-C550] — SPN_LOOP_GAIN_14 | Default: 48.700 | Linked Fault: ERR-HER550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C550.

PRM_48975 [Hermle-Performance-C550] — CLN_LOOP_GAIN_15 | Default: 105.750 | Linked Fault: ERR-HER550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C550.

PRM_48976 [Hermle-Performance-C550] — ZAX_LOOP_GAIN_16 | Default: 162.800 | Linked Fault: ERR-HER550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C550 (BANK 4)

Machine Model: Hermle-Performance-C550 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C550.

PRM_48981 [Hermle-Performance-C550] — XAX_LOOP_GAIN_1 | Default: 159.400 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

PRM_48982 [Hermle-Performance-C550] — CAX_LOOP_GAIN_2 | Default: 68.800 | Linked Fault: ERR-HER550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550.

PRM_48983 [Hermle-Performance-C550] — VAC_LOOP_GAIN_3 | Default: 228.200 | Linked Fault: ERR-HER550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C550.

PRM_48984 [Hermle-Performance-C550] — HYD_LOOP_GAIN_4 | Default: 137.600 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

PRM_48985 [Hermle-Performance-C550] — ZAX_LOOP_GAIN_5 | Default: 47.000 | Linked Fault: ERR-HER550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550.

PRM_48986 [Hermle-Performance-C550] — PLC_LOOP_GAIN_6 | Default: 206.400 | Linked Fault: ERR-HER550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550.

PRM_48987 [Hermle-Performance-C550] — SPN_LOOP_GAIN_7 | Default: 115.800 | Linked Fault: ERR-HER550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C550.

PRM_48988 [Hermle-Performance-C550] — XAX_LOOP_GAIN_8 | Default: 25.200 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

PRM_48989 [Hermle-Performance-C550] — CAX_LOOP_GAIN_9 | Default: 184.600 | Linked Fault: ERR-HER550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550.

PRM_48990 [Hermle-Performance-C550] — VAC_LOOP_GAIN_10 | Default: 94.000 | Linked Fault: ERR-HER550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C550.

PRM_48991 [Hermle-Performance-C550] — HYD_LOOP_GAIN_11 | Default: 3.400 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

PRM_48992 [Hermle-Performance-C550] — ZAX_LOOP_GAIN_12 | Default: 162.800 | Linked Fault: ERR-HER550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550.

PRM_48993 [Hermle-Performance-C550] — PLC_LOOP_GAIN_13 | Default: 72.200 | Linked Fault: ERR-HER550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550.

PRM_48994 [Hermle-Performance-C550] — SPN_LOOP_GAIN_14 | Default: 231.600 | Linked Fault: ERR-HER550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C550.

PRM_48995 [Hermle-Performance-C550] — XAX_LOOP_GAIN_15 | Default: 141.000 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

PRM_48996 [Hermle-Performance-C550] — CAX_LOOP_GAIN_16 | Default: 50.400 | Linked Fault: ERR-HER550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C550 (BANK 5)

Machine Model: Hermle-Performance-C550 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C550.

PRM_49001 [Hermle-Performance-C550] — YAX_LOOP_GAIN_1 | Default: 11.750 | Linked Fault: ERR-HER550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C550.

PRM_49002 [Hermle-Performance-C550] — PLC_LOOP_GAIN_2 | Default: 23.500 | Linked Fault: ERR-HER550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550.

PRM_49003 [Hermle-Performance-C550] — ATC_LOOP_GAIN_3 | Default: 35.250 | Linked Fault: ERR-HER550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C550.

PRM_49004 [Hermle-Performance-C550] — ZAX_LOOP_GAIN_4 | Default: 47.000 | Linked Fault: ERR-HER550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C550.

PRM_49005 [Hermle-Performance-C550] — LUB_LOOP_GAIN_5 | Default: 58.750 | Linked Fault: ERR-HER550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C550.

PRM_49006 [Hermle-Performance-C550] — HYD_LOOP_GAIN_6 | Default: 70.500 | Linked Fault: ERR-HER550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C550.

PRM_49007 [Hermle-Performance-C550] — BAX_LOOP_GAIN_7 | Default: 82.250 | Linked Fault: ERR-HER550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C550.

PRM_49008 [Hermle-Performance-C550] — VAC_LOOP_GAIN_8 | Default: 94.000 | Linked Fault: ERR-HER550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C550.

PRM_49009 [Hermle-Performance-C550] — CLN_LOOP_GAIN_9 | Default: 105.750 | Linked Fault: ERR-HER550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C550.

PRM_49010 [Hermle-Performance-C550] — CAX_LOOP_GAIN_10 | Default: 117.500 | Linked Fault: ERR-HER550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C550.

PRM_49011 [Hermle-Performance-C550] — THM_LOOP_GAIN_11 | Default: 129.250 | Linked Fault: ERR-HER550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C550.

PRM_49012 [Hermle-Performance-C550] — XAX_LOOP_GAIN_12 | Default: 141.000 | Linked Fault: ERR-HER550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C550.

PRM_49013 [Hermle-Performance-C550] — OPT_LOOP_GAIN_13 | Default: 152.750 | Linked Fault: ERR-HER550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C550.

PRM_49014 [Hermle-Performance-C550] — SPN_LOOP_GAIN_14 | Default: 164.500 | Linked Fault: ERR-HER550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C550.

PRM_49015 [Hermle-Performance-C550] — YAX_LOOP_GAIN_15 | Default: 176.250 | Linked Fault: ERR-HER550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C550.

PRM_49016 [Hermle-Performance-C550] — PLC_LOOP_GAIN_16 | Default: 188.000 | Linked Fault: ERR-HER550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C550.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C600 (BANK 1)

Machine Model: Hermle-Performance-C600 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C600.

PRM_49021 [Hermle-Performance-C600] — ATC_LOOP_GAIN_1 | Default: 103.500 | Linked Fault: ERR-HER600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C600.

PRM_49022 [Hermle-Performance-C600] — HYD_LOOP_GAIN_2 | Default: 207.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

PRM_49023 [Hermle-Performance-C600] — CLN_LOOP_GAIN_3 | Default: 60.500 | Linked Fault: ERR-HER600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C600.

PRM_49024 [Hermle-Performance-C600] — XAX_LOOP_GAIN_4 | Default: 164.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

PRM_49025 [Hermle-Performance-C600] — YAX_LOOP_GAIN_5 | Default: 17.500 | Linked Fault: ERR-HER600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C600.

PRM_49026 [Hermle-Performance-C600] — ZAX_LOOP_GAIN_6 | Default: 121.000 | Linked Fault: ERR-HER600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600.

PRM_49027 [Hermle-Performance-C600] — BAX_LOOP_GAIN_7 | Default: 224.500 | Linked Fault: ERR-HER600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C600.

PRM_49028 [Hermle-Performance-C600] — CAX_LOOP_GAIN_8 | Default: 78.000 | Linked Fault: ERR-HER600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600.

PRM_49029 [Hermle-Performance-C600] — OPT_LOOP_GAIN_9 | Default: 181.500 | Linked Fault: ERR-HER600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C600.

PRM_49030 [Hermle-Performance-C600] — PLC_LOOP_GAIN_10 | Default: 35.000 | Linked Fault: ERR-HER600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600.

PRM_49031 [Hermle-Performance-C600] — LUB_LOOP_GAIN_11 | Default: 138.500 | Linked Fault: ERR-HER600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C600.

PRM_49032 [Hermle-Performance-C600] — VAC_LOOP_GAIN_12 | Default: 242.000 | Linked Fault: ERR-HER600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C600.

PRM_49033 [Hermle-Performance-C600] — THM_LOOP_GAIN_13 | Default: 95.500 | Linked Fault: ERR-HER600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C600.

PRM_49034 [Hermle-Performance-C600] — SPN_LOOP_GAIN_14 | Default: 199.000 | Linked Fault: ERR-HER600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C600.

PRM_49035 [Hermle-Performance-C600] — ATC_LOOP_GAIN_15 | Default: 52.500 | Linked Fault: ERR-HER600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C600.

PRM_49036 [Hermle-Performance-C600] — HYD_LOOP_GAIN_16 | Default: 156.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C600 (BANK 2)

Machine Model: Hermle-Performance-C600 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C600.

PRM_49041 [Hermle-Performance-C600] — HYD_LOOP_GAIN_1 | Default: 207.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

PRM_49042 [Hermle-Performance-C600] — XAX_LOOP_GAIN_2 | Default: 164.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

PRM_49043 [Hermle-Performance-C600] — ZAX_LOOP_GAIN_3 | Default: 121.000 | Linked Fault: ERR-HER600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600.

PRM_49044 [Hermle-Performance-C600] — CAX_LOOP_GAIN_4 | Default: 78.000 | Linked Fault: ERR-HER600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600.

PRM_49045 [Hermle-Performance-C600] — PLC_LOOP_GAIN_5 | Default: 35.000 | Linked Fault: ERR-HER600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600.

PRM_49046 [Hermle-Performance-C600] — VAC_LOOP_GAIN_6 | Default: 242.000 | Linked Fault: ERR-HER600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C600.

PRM_49047 [Hermle-Performance-C600] — SPN_LOOP_GAIN_7 | Default: 199.000 | Linked Fault: ERR-HER600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C600.

PRM_49048 [Hermle-Performance-C600] — HYD_LOOP_GAIN_8 | Default: 156.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

PRM_49049 [Hermle-Performance-C600] — XAX_LOOP_GAIN_9 | Default: 113.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

PRM_49050 [Hermle-Performance-C600] — ZAX_LOOP_GAIN_10 | Default: 70.000 | Linked Fault: ERR-HER600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600.

PRM_49051 [Hermle-Performance-C600] — CAX_LOOP_GAIN_11 | Default: 27.000 | Linked Fault: ERR-HER600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600.

PRM_49052 [Hermle-Performance-C600] — PLC_LOOP_GAIN_12 | Default: 234.000 | Linked Fault: ERR-HER600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600.

PRM_49053 [Hermle-Performance-C600] — VAC_LOOP_GAIN_13 | Default: 191.000 | Linked Fault: ERR-HER600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C600.

PRM_49054 [Hermle-Performance-C600] — SPN_LOOP_GAIN_14 | Default: 148.000 | Linked Fault: ERR-HER600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C600.

PRM_49055 [Hermle-Performance-C600] — HYD_LOOP_GAIN_15 | Default: 105.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

PRM_49056 [Hermle-Performance-C600] — XAX_LOOP_GAIN_16 | Default: 62.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C600 (BANK 3)

Machine Model: Hermle-Performance-C600 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C600.

PRM_49061 [Hermle-Performance-C600] — CLN_LOOP_GAIN_1 | Default: 60.500 | Linked Fault: ERR-HER600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C600.

PRM_49062 [Hermle-Performance-C600] — ZAX_LOOP_GAIN_2 | Default: 121.000 | Linked Fault: ERR-HER600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600.

PRM_49063 [Hermle-Performance-C600] — OPT_LOOP_GAIN_3 | Default: 181.500 | Linked Fault: ERR-HER600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C600.

PRM_49064 [Hermle-Performance-C600] — VAC_LOOP_GAIN_4 | Default: 242.000 | Linked Fault: ERR-HER600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C600.

PRM_49065 [Hermle-Performance-C600] — ATC_LOOP_GAIN_5 | Default: 52.500 | Linked Fault: ERR-HER600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C600.

PRM_49066 [Hermle-Performance-C600] — XAX_LOOP_GAIN_6 | Default: 113.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

PRM_49067 [Hermle-Performance-C600] — BAX_LOOP_GAIN_7 | Default: 173.500 | Linked Fault: ERR-HER600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C600.

PRM_49068 [Hermle-Performance-C600] — PLC_LOOP_GAIN_8 | Default: 234.000 | Linked Fault: ERR-HER600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600.

PRM_49069 [Hermle-Performance-C600] — THM_LOOP_GAIN_9 | Default: 44.500 | Linked Fault: ERR-HER600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C600.

PRM_49070 [Hermle-Performance-C600] — HYD_LOOP_GAIN_10 | Default: 105.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

PRM_49071 [Hermle-Performance-C600] — YAX_LOOP_GAIN_11 | Default: 165.500 | Linked Fault: ERR-HER600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C600.

PRM_49072 [Hermle-Performance-C600] — CAX_LOOP_GAIN_12 | Default: 226.000 | Linked Fault: ERR-HER600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600.

PRM_49073 [Hermle-Performance-C600] — LUB_LOOP_GAIN_13 | Default: 36.500 | Linked Fault: ERR-HER600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C600.

PRM_49074 [Hermle-Performance-C600] — SPN_LOOP_GAIN_14 | Default: 97.000 | Linked Fault: ERR-HER600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C600.

PRM_49075 [Hermle-Performance-C600] — CLN_LOOP_GAIN_15 | Default: 157.500 | Linked Fault: ERR-HER600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C600.

PRM_49076 [Hermle-Performance-C600] — ZAX_LOOP_GAIN_16 | Default: 218.000 | Linked Fault: ERR-HER600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C600 (BANK 4)

Machine Model: Hermle-Performance-C600 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C600.

PRM_49081 [Hermle-Performance-C600] — XAX_LOOP_GAIN_1 | Default: 164.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

PRM_49082 [Hermle-Performance-C600] — CAX_LOOP_GAIN_2 | Default: 78.000 | Linked Fault: ERR-HER600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600.

PRM_49083 [Hermle-Performance-C600] — VAC_LOOP_GAIN_3 | Default: 242.000 | Linked Fault: ERR-HER600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C600.

PRM_49084 [Hermle-Performance-C600] — HYD_LOOP_GAIN_4 | Default: 156.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

PRM_49085 [Hermle-Performance-C600] — ZAX_LOOP_GAIN_5 | Default: 70.000 | Linked Fault: ERR-HER600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600.

PRM_49086 [Hermle-Performance-C600] — PLC_LOOP_GAIN_6 | Default: 234.000 | Linked Fault: ERR-HER600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600.

PRM_49087 [Hermle-Performance-C600] — SPN_LOOP_GAIN_7 | Default: 148.000 | Linked Fault: ERR-HER600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C600.

PRM_49088 [Hermle-Performance-C600] — XAX_LOOP_GAIN_8 | Default: 62.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

PRM_49089 [Hermle-Performance-C600] — CAX_LOOP_GAIN_9 | Default: 226.000 | Linked Fault: ERR-HER600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600.

PRM_49090 [Hermle-Performance-C600] — VAC_LOOP_GAIN_10 | Default: 140.000 | Linked Fault: ERR-HER600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C600.

PRM_49091 [Hermle-Performance-C600] — HYD_LOOP_GAIN_11 | Default: 54.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

PRM_49092 [Hermle-Performance-C600] — ZAX_LOOP_GAIN_12 | Default: 218.000 | Linked Fault: ERR-HER600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600.

PRM_49093 [Hermle-Performance-C600] — PLC_LOOP_GAIN_13 | Default: 132.000 | Linked Fault: ERR-HER600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600.

PRM_49094 [Hermle-Performance-C600] — SPN_LOOP_GAIN_14 | Default: 46.000 | Linked Fault: ERR-HER600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C600.

PRM_49095 [Hermle-Performance-C600] — XAX_LOOP_GAIN_15 | Default: 210.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

PRM_49096 [Hermle-Performance-C600] — CAX_LOOP_GAIN_16 | Default: 124.000 | Linked Fault: ERR-HER600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600.

CNC/PLC SERVO PARAMETER DICTIONARY — Hermle-Performance-C600 (BANK 5)

Machine Model: Hermle-Performance-C600 | Controller: Heidenhain TNC7. Use these parameter registers when diagnosing servo following errors or thermal drift on Hermle-Performance-C600.

PRM_49101 [Hermle-Performance-C600] — YAX_LOOP_GAIN_1 | Default: 17.500 | Linked Fault: ERR-HER600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C600.

PRM_49102 [Hermle-Performance-C600] — PLC_LOOP_GAIN_2 | Default: 35.000 | Linked Fault: ERR-HER600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600.

PRM_49103 [Hermle-Performance-C600] — ATC_LOOP_GAIN_3 | Default: 52.500 | Linked Fault: ERR-HER600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Hermle-Performance-C600.

PRM_49104 [Hermle-Performance-C600] — ZAX_LOOP_GAIN_4 | Default: 70.000 | Linked Fault: ERR-HER600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Hermle-Performance-C600.

PRM_49105 [Hermle-Performance-C600] — LUB_LOOP_GAIN_5 | Default: 87.500 | Linked Fault: ERR-HER600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Hermle-Performance-C600.

PRM_49106 [Hermle-Performance-C600] — HYD_LOOP_GAIN_6 | Default: 105.000 | Linked Fault: ERR-HER600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Hermle-Performance-C600.

PRM_49107 [Hermle-Performance-C600] — BAX_LOOP_GAIN_7 | Default: 122.500 | Linked Fault: ERR-HER600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Hermle-Performance-C600.

PRM_49108 [Hermle-Performance-C600] — VAC_LOOP_GAIN_8 | Default: 140.000 | Linked Fault: ERR-HER600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Hermle-Performance-C600.

PRM_49109 [Hermle-Performance-C600] — CLN_LOOP_GAIN_9 | Default: 157.500 | Linked Fault: ERR-HER600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Hermle-Performance-C600.

PRM_49110 [Hermle-Performance-C600] — CAX_LOOP_GAIN_10 | Default: 175.000 | Linked Fault: ERR-HER600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Hermle-Performance-C600.

PRM_49111 [Hermle-Performance-C600] — THM_LOOP_GAIN_11 | Default: 192.500 | Linked Fault: ERR-HER600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Hermle-Performance-C600.

PRM_49112 [Hermle-Performance-C600] — XAX_LOOP_GAIN_12 | Default: 210.000 | Linked Fault: ERR-HER600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Hermle-Performance-C600.

PRM_49113 [Hermle-Performance-C600] — OPT_LOOP_GAIN_13 | Default: 227.500 | Linked Fault: ERR-HER600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Hermle-Performance-C600.

PRM_49114 [Hermle-Performance-C600] — SPN_LOOP_GAIN_14 | Default: 245.000 | Linked Fault: ERR-HER600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Hermle-Performance-C600.

PRM_49115 [Hermle-Performance-C600] — YAX_LOOP_GAIN_15 | Default: 12.500 | Linked Fault: ERR-HER600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Hermle-Performance-C600.

PRM_49116 [Hermle-Performance-C600] — PLC_LOOP_GAIN_16 | Default: 30.000 | Linked Fault: ERR-HER600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Hermle-Performance-C600.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z150 (BANK 1)

Machine Model: Zeiss-Prismo-Z150 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z150.

PRM_49121 [Zeiss-Prismo-Z150] — ATC_LOOP_GAIN_1 | Default: 104.650 | Linked Fault: ERR-ZEI150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z150.

PRM_49122 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_2 | Default: 209.300 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

PRM_49123 [Zeiss-Prismo-Z150] — CLN_LOOP_GAIN_3 | Default: 63.950 | Linked Fault: ERR-ZEI150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z150.

PRM_49124 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_4 | Default: 168.600 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

PRM_49125 [Zeiss-Prismo-Z150] — YAX_LOOP_GAIN_5 | Default: 23.250 | Linked Fault: ERR-ZEI150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z150.

PRM_49126 [Zeiss-Prismo-Z150] — ZAX_LOOP_GAIN_6 | Default: 127.900 | Linked Fault: ERR-ZEI150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150.

PRM_49127 [Zeiss-Prismo-Z150] — BAX_LOOP_GAIN_7 | Default: 232.550 | Linked Fault: ERR-ZEI150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z150.

PRM_49128 [Zeiss-Prismo-Z150] — CAX_LOOP_GAIN_8 | Default: 87.200 | Linked Fault: ERR-ZEI150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150.

PRM_49129 [Zeiss-Prismo-Z150] — OPT_LOOP_GAIN_9 | Default: 191.850 | Linked Fault: ERR-ZEI150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z150.

PRM_49130 [Zeiss-Prismo-Z150] — PLC_LOOP_GAIN_10 | Default: 46.500 | Linked Fault: ERR-ZEI150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150.

PRM_49131 [Zeiss-Prismo-Z150] — LUB_LOOP_GAIN_11 | Default: 151.150 | Linked Fault: ERR-ZEI150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z150.

PRM_49132 [Zeiss-Prismo-Z150] — VAC_LOOP_GAIN_12 | Default: 5.800 | Linked Fault: ERR-ZEI150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z150.

PRM_49133 [Zeiss-Prismo-Z150] — THM_LOOP_GAIN_13 | Default: 110.450 | Linked Fault: ERR-ZEI150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z150.

PRM_49134 [Zeiss-Prismo-Z150] — SPN_LOOP_GAIN_14 | Default: 215.100 | Linked Fault: ERR-ZEI150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z150.

PRM_49135 [Zeiss-Prismo-Z150] — ATC_LOOP_GAIN_15 | Default: 69.750 | Linked Fault: ERR-ZEI150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z150.

PRM_49136 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_16 | Default: 174.400 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z150 (BANK 2)

Machine Model: Zeiss-Prismo-Z150 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z150.

PRM_49141 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_1 | Default: 209.300 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

PRM_49142 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_2 | Default: 168.600 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

PRM_49143 [Zeiss-Prismo-Z150] — ZAX_LOOP_GAIN_3 | Default: 127.900 | Linked Fault: ERR-ZEI150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150.

PRM_49144 [Zeiss-Prismo-Z150] — CAX_LOOP_GAIN_4 | Default: 87.200 | Linked Fault: ERR-ZEI150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150.

PRM_49145 [Zeiss-Prismo-Z150] — PLC_LOOP_GAIN_5 | Default: 46.500 | Linked Fault: ERR-ZEI150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150.

PRM_49146 [Zeiss-Prismo-Z150] — VAC_LOOP_GAIN_6 | Default: 5.800 | Linked Fault: ERR-ZEI150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z150.

PRM_49147 [Zeiss-Prismo-Z150] — SPN_LOOP_GAIN_7 | Default: 215.100 | Linked Fault: ERR-ZEI150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z150.

PRM_49148 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_8 | Default: 174.400 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

PRM_49149 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_9 | Default: 133.700 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

PRM_49150 [Zeiss-Prismo-Z150] — ZAX_LOOP_GAIN_10 | Default: 93.000 | Linked Fault: ERR-ZEI150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150.

PRM_49151 [Zeiss-Prismo-Z150] — CAX_LOOP_GAIN_11 | Default: 52.300 | Linked Fault: ERR-ZEI150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150.

PRM_49152 [Zeiss-Prismo-Z150] — PLC_LOOP_GAIN_12 | Default: 11.600 | Linked Fault: ERR-ZEI150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150.

PRM_49153 [Zeiss-Prismo-Z150] — VAC_LOOP_GAIN_13 | Default: 220.900 | Linked Fault: ERR-ZEI150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z150.

PRM_49154 [Zeiss-Prismo-Z150] — SPN_LOOP_GAIN_14 | Default: 180.200 | Linked Fault: ERR-ZEI150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z150.

PRM_49155 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_15 | Default: 139.500 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

PRM_49156 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_16 | Default: 98.800 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z150 (BANK 3)

Machine Model: Zeiss-Prismo-Z150 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z150.

PRM_49161 [Zeiss-Prismo-Z150] — CLN_LOOP_GAIN_1 | Default: 63.950 | Linked Fault: ERR-ZEI150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z150.

PRM_49162 [Zeiss-Prismo-Z150] — ZAX_LOOP_GAIN_2 | Default: 127.900 | Linked Fault: ERR-ZEI150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150.

PRM_49163 [Zeiss-Prismo-Z150] — OPT_LOOP_GAIN_3 | Default: 191.850 | Linked Fault: ERR-ZEI150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z150.

PRM_49164 [Zeiss-Prismo-Z150] — VAC_LOOP_GAIN_4 | Default: 5.800 | Linked Fault: ERR-ZEI150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z150.

PRM_49165 [Zeiss-Prismo-Z150] — ATC_LOOP_GAIN_5 | Default: 69.750 | Linked Fault: ERR-ZEI150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z150.

PRM_49166 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_6 | Default: 133.700 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

PRM_49167 [Zeiss-Prismo-Z150] — BAX_LOOP_GAIN_7 | Default: 197.650 | Linked Fault: ERR-ZEI150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z150.

PRM_49168 [Zeiss-Prismo-Z150] — PLC_LOOP_GAIN_8 | Default: 11.600 | Linked Fault: ERR-ZEI150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150.

PRM_49169 [Zeiss-Prismo-Z150] — THM_LOOP_GAIN_9 | Default: 75.550 | Linked Fault: ERR-ZEI150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z150.

PRM_49170 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_10 | Default: 139.500 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

PRM_49171 [Zeiss-Prismo-Z150] — YAX_LOOP_GAIN_11 | Default: 203.450 | Linked Fault: ERR-ZEI150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z150.

PRM_49172 [Zeiss-Prismo-Z150] — CAX_LOOP_GAIN_12 | Default: 17.400 | Linked Fault: ERR-ZEI150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150.

PRM_49173 [Zeiss-Prismo-Z150] — LUB_LOOP_GAIN_13 | Default: 81.350 | Linked Fault: ERR-ZEI150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z150.

PRM_49174 [Zeiss-Prismo-Z150] — SPN_LOOP_GAIN_14 | Default: 145.300 | Linked Fault: ERR-ZEI150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z150.

PRM_49175 [Zeiss-Prismo-Z150] — CLN_LOOP_GAIN_15 | Default: 209.250 | Linked Fault: ERR-ZEI150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z150.

PRM_49176 [Zeiss-Prismo-Z150] — ZAX_LOOP_GAIN_16 | Default: 23.200 | Linked Fault: ERR-ZEI150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z150 (BANK 4)

Machine Model: Zeiss-Prismo-Z150 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z150.

PRM_49181 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_1 | Default: 168.600 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

PRM_49182 [Zeiss-Prismo-Z150] — CAX_LOOP_GAIN_2 | Default: 87.200 | Linked Fault: ERR-ZEI150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150.

PRM_49183 [Zeiss-Prismo-Z150] — VAC_LOOP_GAIN_3 | Default: 5.800 | Linked Fault: ERR-ZEI150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z150.

PRM_49184 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_4 | Default: 174.400 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

PRM_49185 [Zeiss-Prismo-Z150] — ZAX_LOOP_GAIN_5 | Default: 93.000 | Linked Fault: ERR-ZEI150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150.

PRM_49186 [Zeiss-Prismo-Z150] — PLC_LOOP_GAIN_6 | Default: 11.600 | Linked Fault: ERR-ZEI150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150.

PRM_49187 [Zeiss-Prismo-Z150] — SPN_LOOP_GAIN_7 | Default: 180.200 | Linked Fault: ERR-ZEI150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z150.

PRM_49188 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_8 | Default: 98.800 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

PRM_49189 [Zeiss-Prismo-Z150] — CAX_LOOP_GAIN_9 | Default: 17.400 | Linked Fault: ERR-ZEI150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150.

PRM_49190 [Zeiss-Prismo-Z150] — VAC_LOOP_GAIN_10 | Default: 186.000 | Linked Fault: ERR-ZEI150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z150.

PRM_49191 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_11 | Default: 104.600 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

PRM_49192 [Zeiss-Prismo-Z150] — ZAX_LOOP_GAIN_12 | Default: 23.200 | Linked Fault: ERR-ZEI150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150.

PRM_49193 [Zeiss-Prismo-Z150] — PLC_LOOP_GAIN_13 | Default: 191.800 | Linked Fault: ERR-ZEI150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150.

PRM_49194 [Zeiss-Prismo-Z150] — SPN_LOOP_GAIN_14 | Default: 110.400 | Linked Fault: ERR-ZEI150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z150.

PRM_49195 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_15 | Default: 29.000 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

PRM_49196 [Zeiss-Prismo-Z150] — CAX_LOOP_GAIN_16 | Default: 197.600 | Linked Fault: ERR-ZEI150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z150 (BANK 5)

Machine Model: Zeiss-Prismo-Z150 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z150.

PRM_49201 [Zeiss-Prismo-Z150] — YAX_LOOP_GAIN_1 | Default: 23.250 | Linked Fault: ERR-ZEI150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z150.

PRM_49202 [Zeiss-Prismo-Z150] — PLC_LOOP_GAIN_2 | Default: 46.500 | Linked Fault: ERR-ZEI150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150.

PRM_49203 [Zeiss-Prismo-Z150] — ATC_LOOP_GAIN_3 | Default: 69.750 | Linked Fault: ERR-ZEI150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z150.

PRM_49204 [Zeiss-Prismo-Z150] — ZAX_LOOP_GAIN_4 | Default: 93.000 | Linked Fault: ERR-ZEI150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z150.

PRM_49205 [Zeiss-Prismo-Z150] — LUB_LOOP_GAIN_5 | Default: 116.250 | Linked Fault: ERR-ZEI150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z150.

PRM_49206 [Zeiss-Prismo-Z150] — HYD_LOOP_GAIN_6 | Default: 139.500 | Linked Fault: ERR-ZEI150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z150.

PRM_49207 [Zeiss-Prismo-Z150] — BAX_LOOP_GAIN_7 | Default: 162.750 | Linked Fault: ERR-ZEI150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z150.

PRM_49208 [Zeiss-Prismo-Z150] — VAC_LOOP_GAIN_8 | Default: 186.000 | Linked Fault: ERR-ZEI150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z150.

PRM_49209 [Zeiss-Prismo-Z150] — CLN_LOOP_GAIN_9 | Default: 209.250 | Linked Fault: ERR-ZEI150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z150.

PRM_49210 [Zeiss-Prismo-Z150] — CAX_LOOP_GAIN_10 | Default: 232.500 | Linked Fault: ERR-ZEI150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z150.

PRM_49211 [Zeiss-Prismo-Z150] — THM_LOOP_GAIN_11 | Default: 5.750 | Linked Fault: ERR-ZEI150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z150.

PRM_49212 [Zeiss-Prismo-Z150] — XAX_LOOP_GAIN_12 | Default: 29.000 | Linked Fault: ERR-ZEI150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z150.

PRM_49213 [Zeiss-Prismo-Z150] — OPT_LOOP_GAIN_13 | Default: 52.250 | Linked Fault: ERR-ZEI150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z150.

PRM_49214 [Zeiss-Prismo-Z150] — SPN_LOOP_GAIN_14 | Default: 75.500 | Linked Fault: ERR-ZEI150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z150.

PRM_49215 [Zeiss-Prismo-Z150] — YAX_LOOP_GAIN_15 | Default: 98.750 | Linked Fault: ERR-ZEI150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z150.

PRM_49216 [Zeiss-Prismo-Z150] — PLC_LOOP_GAIN_16 | Default: 122.000 | Linked Fault: ERR-ZEI150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z150.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z200 (BANK 1)

Machine Model: Zeiss-Prismo-Z200 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z200.

PRM_49221 [Zeiss-Prismo-Z200] — ATC_LOOP_GAIN_1 | Default: 105.800 | Linked Fault: ERR-ZEI200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z200.

PRM_49222 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_2 | Default: 211.600 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

PRM_49223 [Zeiss-Prismo-Z200] — CLN_LOOP_GAIN_3 | Default: 67.400 | Linked Fault: ERR-ZEI200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z200.

PRM_49224 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_4 | Default: 173.200 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

PRM_49225 [Zeiss-Prismo-Z200] — YAX_LOOP_GAIN_5 | Default: 29.000 | Linked Fault: ERR-ZEI200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z200.

PRM_49226 [Zeiss-Prismo-Z200] — ZAX_LOOP_GAIN_6 | Default: 134.800 | Linked Fault: ERR-ZEI200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200.

PRM_49227 [Zeiss-Prismo-Z200] — BAX_LOOP_GAIN_7 | Default: 240.600 | Linked Fault: ERR-ZEI200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z200.

PRM_49228 [Zeiss-Prismo-Z200] — CAX_LOOP_GAIN_8 | Default: 96.400 | Linked Fault: ERR-ZEI200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200.

PRM_49229 [Zeiss-Prismo-Z200] — OPT_LOOP_GAIN_9 | Default: 202.200 | Linked Fault: ERR-ZEI200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z200.

PRM_49230 [Zeiss-Prismo-Z200] — PLC_LOOP_GAIN_10 | Default: 58.000 | Linked Fault: ERR-ZEI200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200.

PRM_49231 [Zeiss-Prismo-Z200] — LUB_LOOP_GAIN_11 | Default: 163.800 | Linked Fault: ERR-ZEI200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z200.

PRM_49232 [Zeiss-Prismo-Z200] — VAC_LOOP_GAIN_12 | Default: 19.600 | Linked Fault: ERR-ZEI200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z200.

PRM_49233 [Zeiss-Prismo-Z200] — THM_LOOP_GAIN_13 | Default: 125.400 | Linked Fault: ERR-ZEI200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z200.

PRM_49234 [Zeiss-Prismo-Z200] — SPN_LOOP_GAIN_14 | Default: 231.200 | Linked Fault: ERR-ZEI200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z200.

PRM_49235 [Zeiss-Prismo-Z200] — ATC_LOOP_GAIN_15 | Default: 87.000 | Linked Fault: ERR-ZEI200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z200.

PRM_49236 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_16 | Default: 192.800 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z200 (BANK 2)

Machine Model: Zeiss-Prismo-Z200 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z200.

PRM_49241 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_1 | Default: 211.600 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

PRM_49242 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_2 | Default: 173.200 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

PRM_49243 [Zeiss-Prismo-Z200] — ZAX_LOOP_GAIN_3 | Default: 134.800 | Linked Fault: ERR-ZEI200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200.

PRM_49244 [Zeiss-Prismo-Z200] — CAX_LOOP_GAIN_4 | Default: 96.400 | Linked Fault: ERR-ZEI200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200.

PRM_49245 [Zeiss-Prismo-Z200] — PLC_LOOP_GAIN_5 | Default: 58.000 | Linked Fault: ERR-ZEI200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200.

PRM_49246 [Zeiss-Prismo-Z200] — VAC_LOOP_GAIN_6 | Default: 19.600 | Linked Fault: ERR-ZEI200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z200.

PRM_49247 [Zeiss-Prismo-Z200] — SPN_LOOP_GAIN_7 | Default: 231.200 | Linked Fault: ERR-ZEI200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z200.

PRM_49248 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_8 | Default: 192.800 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

PRM_49249 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_9 | Default: 154.400 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

PRM_49250 [Zeiss-Prismo-Z200] — ZAX_LOOP_GAIN_10 | Default: 116.000 | Linked Fault: ERR-ZEI200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200.

PRM_49251 [Zeiss-Prismo-Z200] — CAX_LOOP_GAIN_11 | Default: 77.600 | Linked Fault: ERR-ZEI200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200.

PRM_49252 [Zeiss-Prismo-Z200] — PLC_LOOP_GAIN_12 | Default: 39.200 | Linked Fault: ERR-ZEI200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200.

PRM_49253 [Zeiss-Prismo-Z200] — VAC_LOOP_GAIN_13 | Default: 0.800 | Linked Fault: ERR-ZEI200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z200.

PRM_49254 [Zeiss-Prismo-Z200] — SPN_LOOP_GAIN_14 | Default: 212.400 | Linked Fault: ERR-ZEI200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z200.

PRM_49255 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_15 | Default: 174.000 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

PRM_49256 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_16 | Default: 135.600 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z200 (BANK 3)

Machine Model: Zeiss-Prismo-Z200 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z200.

PRM_49261 [Zeiss-Prismo-Z200] — CLN_LOOP_GAIN_1 | Default: 67.400 | Linked Fault: ERR-ZEI200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z200.

PRM_49262 [Zeiss-Prismo-Z200] — ZAX_LOOP_GAIN_2 | Default: 134.800 | Linked Fault: ERR-ZEI200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200.

PRM_49263 [Zeiss-Prismo-Z200] — OPT_LOOP_GAIN_3 | Default: 202.200 | Linked Fault: ERR-ZEI200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z200.

PRM_49264 [Zeiss-Prismo-Z200] — VAC_LOOP_GAIN_4 | Default: 19.600 | Linked Fault: ERR-ZEI200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z200.

PRM_49265 [Zeiss-Prismo-Z200] — ATC_LOOP_GAIN_5 | Default: 87.000 | Linked Fault: ERR-ZEI200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z200.

PRM_49266 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_6 | Default: 154.400 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

PRM_49267 [Zeiss-Prismo-Z200] — BAX_LOOP_GAIN_7 | Default: 221.800 | Linked Fault: ERR-ZEI200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z200.

PRM_49268 [Zeiss-Prismo-Z200] — PLC_LOOP_GAIN_8 | Default: 39.200 | Linked Fault: ERR-ZEI200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200.

PRM_49269 [Zeiss-Prismo-Z200] — THM_LOOP_GAIN_9 | Default: 106.600 | Linked Fault: ERR-ZEI200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z200.

PRM_49270 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_10 | Default: 174.000 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

PRM_49271 [Zeiss-Prismo-Z200] — YAX_LOOP_GAIN_11 | Default: 241.400 | Linked Fault: ERR-ZEI200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z200.

PRM_49272 [Zeiss-Prismo-Z200] — CAX_LOOP_GAIN_12 | Default: 58.800 | Linked Fault: ERR-ZEI200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200.

PRM_49273 [Zeiss-Prismo-Z200] — LUB_LOOP_GAIN_13 | Default: 126.200 | Linked Fault: ERR-ZEI200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z200.

PRM_49274 [Zeiss-Prismo-Z200] — SPN_LOOP_GAIN_14 | Default: 193.600 | Linked Fault: ERR-ZEI200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z200.

PRM_49275 [Zeiss-Prismo-Z200] — CLN_LOOP_GAIN_15 | Default: 11.000 | Linked Fault: ERR-ZEI200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z200.

PRM_49276 [Zeiss-Prismo-Z200] — ZAX_LOOP_GAIN_16 | Default: 78.400 | Linked Fault: ERR-ZEI200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z200 (BANK 4)

Machine Model: Zeiss-Prismo-Z200 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z200.

PRM_49281 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_1 | Default: 173.200 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

PRM_49282 [Zeiss-Prismo-Z200] — CAX_LOOP_GAIN_2 | Default: 96.400 | Linked Fault: ERR-ZEI200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200.

PRM_49283 [Zeiss-Prismo-Z200] — VAC_LOOP_GAIN_3 | Default: 19.600 | Linked Fault: ERR-ZEI200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z200.

PRM_49284 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_4 | Default: 192.800 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

PRM_49285 [Zeiss-Prismo-Z200] — ZAX_LOOP_GAIN_5 | Default: 116.000 | Linked Fault: ERR-ZEI200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200.

PRM_49286 [Zeiss-Prismo-Z200] — PLC_LOOP_GAIN_6 | Default: 39.200 | Linked Fault: ERR-ZEI200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200.

PRM_49287 [Zeiss-Prismo-Z200] — SPN_LOOP_GAIN_7 | Default: 212.400 | Linked Fault: ERR-ZEI200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z200.

PRM_49288 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_8 | Default: 135.600 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

PRM_49289 [Zeiss-Prismo-Z200] — CAX_LOOP_GAIN_9 | Default: 58.800 | Linked Fault: ERR-ZEI200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200.

PRM_49290 [Zeiss-Prismo-Z200] — VAC_LOOP_GAIN_10 | Default: 232.000 | Linked Fault: ERR-ZEI200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z200.

PRM_49291 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_11 | Default: 155.200 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

PRM_49292 [Zeiss-Prismo-Z200] — ZAX_LOOP_GAIN_12 | Default: 78.400 | Linked Fault: ERR-ZEI200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200.

PRM_49293 [Zeiss-Prismo-Z200] — PLC_LOOP_GAIN_13 | Default: 1.600 | Linked Fault: ERR-ZEI200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200.

PRM_49294 [Zeiss-Prismo-Z200] — SPN_LOOP_GAIN_14 | Default: 174.800 | Linked Fault: ERR-ZEI200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z200.

PRM_49295 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_15 | Default: 98.000 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

PRM_49296 [Zeiss-Prismo-Z200] — CAX_LOOP_GAIN_16 | Default: 21.200 | Linked Fault: ERR-ZEI200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z200 (BANK 5)

Machine Model: Zeiss-Prismo-Z200 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z200.

PRM_49301 [Zeiss-Prismo-Z200] — YAX_LOOP_GAIN_1 | Default: 29.000 | Linked Fault: ERR-ZEI200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z200.

PRM_49302 [Zeiss-Prismo-Z200] — PLC_LOOP_GAIN_2 | Default: 58.000 | Linked Fault: ERR-ZEI200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200.

PRM_49303 [Zeiss-Prismo-Z200] — ATC_LOOP_GAIN_3 | Default: 87.000 | Linked Fault: ERR-ZEI200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z200.

PRM_49304 [Zeiss-Prismo-Z200] — ZAX_LOOP_GAIN_4 | Default: 116.000 | Linked Fault: ERR-ZEI200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z200.

PRM_49305 [Zeiss-Prismo-Z200] — LUB_LOOP_GAIN_5 | Default: 145.000 | Linked Fault: ERR-ZEI200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z200.

PRM_49306 [Zeiss-Prismo-Z200] — HYD_LOOP_GAIN_6 | Default: 174.000 | Linked Fault: ERR-ZEI200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z200.

PRM_49307 [Zeiss-Prismo-Z200] — BAX_LOOP_GAIN_7 | Default: 203.000 | Linked Fault: ERR-ZEI200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z200.

PRM_49308 [Zeiss-Prismo-Z200] — VAC_LOOP_GAIN_8 | Default: 232.000 | Linked Fault: ERR-ZEI200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z200.

PRM_49309 [Zeiss-Prismo-Z200] — CLN_LOOP_GAIN_9 | Default: 11.000 | Linked Fault: ERR-ZEI200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z200.

PRM_49310 [Zeiss-Prismo-Z200] — CAX_LOOP_GAIN_10 | Default: 40.000 | Linked Fault: ERR-ZEI200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z200.

PRM_49311 [Zeiss-Prismo-Z200] — THM_LOOP_GAIN_11 | Default: 69.000 | Linked Fault: ERR-ZEI200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z200.

PRM_49312 [Zeiss-Prismo-Z200] — XAX_LOOP_GAIN_12 | Default: 98.000 | Linked Fault: ERR-ZEI200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z200.

PRM_49313 [Zeiss-Prismo-Z200] — OPT_LOOP_GAIN_13 | Default: 127.000 | Linked Fault: ERR-ZEI200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z200.

PRM_49314 [Zeiss-Prismo-Z200] — SPN_LOOP_GAIN_14 | Default: 156.000 | Linked Fault: ERR-ZEI200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z200.

PRM_49315 [Zeiss-Prismo-Z200] — YAX_LOOP_GAIN_15 | Default: 185.000 | Linked Fault: ERR-ZEI200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z200.

PRM_49316 [Zeiss-Prismo-Z200] — PLC_LOOP_GAIN_16 | Default: 214.000 | Linked Fault: ERR-ZEI200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z200.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z250 (BANK 1)

Machine Model: Zeiss-Prismo-Z250 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z250.

PRM_49321 [Zeiss-Prismo-Z250] — ATC_LOOP_GAIN_1 | Default: 106.950 | Linked Fault: ERR-ZEI250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z250.

PRM_49322 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_2 | Default: 213.900 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

PRM_49323 [Zeiss-Prismo-Z250] — CLN_LOOP_GAIN_3 | Default: 70.850 | Linked Fault: ERR-ZEI250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z250.

PRM_49324 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_4 | Default: 177.800 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

PRM_49325 [Zeiss-Prismo-Z250] — YAX_LOOP_GAIN_5 | Default: 34.750 | Linked Fault: ERR-ZEI250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z250.

PRM_49326 [Zeiss-Prismo-Z250] — ZAX_LOOP_GAIN_6 | Default: 141.700 | Linked Fault: ERR-ZEI250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250.

PRM_49327 [Zeiss-Prismo-Z250] — BAX_LOOP_GAIN_7 | Default: 248.650 | Linked Fault: ERR-ZEI250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z250.

PRM_49328 [Zeiss-Prismo-Z250] — CAX_LOOP_GAIN_8 | Default: 105.600 | Linked Fault: ERR-ZEI250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250.

PRM_49329 [Zeiss-Prismo-Z250] — OPT_LOOP_GAIN_9 | Default: 212.550 | Linked Fault: ERR-ZEI250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z250.

PRM_49330 [Zeiss-Prismo-Z250] — PLC_LOOP_GAIN_10 | Default: 69.500 | Linked Fault: ERR-ZEI250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250.

PRM_49331 [Zeiss-Prismo-Z250] — LUB_LOOP_GAIN_11 | Default: 176.450 | Linked Fault: ERR-ZEI250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z250.

PRM_49332 [Zeiss-Prismo-Z250] — VAC_LOOP_GAIN_12 | Default: 33.400 | Linked Fault: ERR-ZEI250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z250.

PRM_49333 [Zeiss-Prismo-Z250] — THM_LOOP_GAIN_13 | Default: 140.350 | Linked Fault: ERR-ZEI250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z250.

PRM_49334 [Zeiss-Prismo-Z250] — SPN_LOOP_GAIN_14 | Default: 247.300 | Linked Fault: ERR-ZEI250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z250.

PRM_49335 [Zeiss-Prismo-Z250] — ATC_LOOP_GAIN_15 | Default: 104.250 | Linked Fault: ERR-ZEI250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z250.

PRM_49336 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_16 | Default: 211.200 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z250 (BANK 2)

Machine Model: Zeiss-Prismo-Z250 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z250.

PRM_49341 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_1 | Default: 213.900 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

PRM_49342 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_2 | Default: 177.800 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

PRM_49343 [Zeiss-Prismo-Z250] — ZAX_LOOP_GAIN_3 | Default: 141.700 | Linked Fault: ERR-ZEI250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250.

PRM_49344 [Zeiss-Prismo-Z250] — CAX_LOOP_GAIN_4 | Default: 105.600 | Linked Fault: ERR-ZEI250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250.

PRM_49345 [Zeiss-Prismo-Z250] — PLC_LOOP_GAIN_5 | Default: 69.500 | Linked Fault: ERR-ZEI250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250.

PRM_49346 [Zeiss-Prismo-Z250] — VAC_LOOP_GAIN_6 | Default: 33.400 | Linked Fault: ERR-ZEI250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z250.

PRM_49347 [Zeiss-Prismo-Z250] — SPN_LOOP_GAIN_7 | Default: 247.300 | Linked Fault: ERR-ZEI250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z250.

PRM_49348 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_8 | Default: 211.200 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

PRM_49349 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_9 | Default: 175.100 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

PRM_49350 [Zeiss-Prismo-Z250] — ZAX_LOOP_GAIN_10 | Default: 139.000 | Linked Fault: ERR-ZEI250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250.

PRM_49351 [Zeiss-Prismo-Z250] — CAX_LOOP_GAIN_11 | Default: 102.900 | Linked Fault: ERR-ZEI250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250.

PRM_49352 [Zeiss-Prismo-Z250] — PLC_LOOP_GAIN_12 | Default: 66.800 | Linked Fault: ERR-ZEI250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250.

PRM_49353 [Zeiss-Prismo-Z250] — VAC_LOOP_GAIN_13 | Default: 30.700 | Linked Fault: ERR-ZEI250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z250.

PRM_49354 [Zeiss-Prismo-Z250] — SPN_LOOP_GAIN_14 | Default: 244.600 | Linked Fault: ERR-ZEI250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z250.

PRM_49355 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_15 | Default: 208.500 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

PRM_49356 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_16 | Default: 172.400 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z250 (BANK 3)

Machine Model: Zeiss-Prismo-Z250 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z250.

PRM_49361 [Zeiss-Prismo-Z250] — CLN_LOOP_GAIN_1 | Default: 70.850 | Linked Fault: ERR-ZEI250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z250.

PRM_49362 [Zeiss-Prismo-Z250] — ZAX_LOOP_GAIN_2 | Default: 141.700 | Linked Fault: ERR-ZEI250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250.

PRM_49363 [Zeiss-Prismo-Z250] — OPT_LOOP_GAIN_3 | Default: 212.550 | Linked Fault: ERR-ZEI250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z250.

PRM_49364 [Zeiss-Prismo-Z250] — VAC_LOOP_GAIN_4 | Default: 33.400 | Linked Fault: ERR-ZEI250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z250.

PRM_49365 [Zeiss-Prismo-Z250] — ATC_LOOP_GAIN_5 | Default: 104.250 | Linked Fault: ERR-ZEI250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z250.

PRM_49366 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_6 | Default: 175.100 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

PRM_49367 [Zeiss-Prismo-Z250] — BAX_LOOP_GAIN_7 | Default: 245.950 | Linked Fault: ERR-ZEI250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z250.

PRM_49368 [Zeiss-Prismo-Z250] — PLC_LOOP_GAIN_8 | Default: 66.800 | Linked Fault: ERR-ZEI250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250.

PRM_49369 [Zeiss-Prismo-Z250] — THM_LOOP_GAIN_9 | Default: 137.650 | Linked Fault: ERR-ZEI250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z250.

PRM_49370 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_10 | Default: 208.500 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

PRM_49371 [Zeiss-Prismo-Z250] — YAX_LOOP_GAIN_11 | Default: 29.350 | Linked Fault: ERR-ZEI250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z250.

PRM_49372 [Zeiss-Prismo-Z250] — CAX_LOOP_GAIN_12 | Default: 100.200 | Linked Fault: ERR-ZEI250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250.

PRM_49373 [Zeiss-Prismo-Z250] — LUB_LOOP_GAIN_13 | Default: 171.050 | Linked Fault: ERR-ZEI250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z250.

PRM_49374 [Zeiss-Prismo-Z250] — SPN_LOOP_GAIN_14 | Default: 241.900 | Linked Fault: ERR-ZEI250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z250.

PRM_49375 [Zeiss-Prismo-Z250] — CLN_LOOP_GAIN_15 | Default: 62.750 | Linked Fault: ERR-ZEI250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z250.

PRM_49376 [Zeiss-Prismo-Z250] — ZAX_LOOP_GAIN_16 | Default: 133.600 | Linked Fault: ERR-ZEI250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z250 (BANK 4)

Machine Model: Zeiss-Prismo-Z250 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z250.

PRM_49381 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_1 | Default: 177.800 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

PRM_49382 [Zeiss-Prismo-Z250] — CAX_LOOP_GAIN_2 | Default: 105.600 | Linked Fault: ERR-ZEI250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250.

PRM_49383 [Zeiss-Prismo-Z250] — VAC_LOOP_GAIN_3 | Default: 33.400 | Linked Fault: ERR-ZEI250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z250.

PRM_49384 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_4 | Default: 211.200 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

PRM_49385 [Zeiss-Prismo-Z250] — ZAX_LOOP_GAIN_5 | Default: 139.000 | Linked Fault: ERR-ZEI250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250.

PRM_49386 [Zeiss-Prismo-Z250] — PLC_LOOP_GAIN_6 | Default: 66.800 | Linked Fault: ERR-ZEI250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250.

PRM_49387 [Zeiss-Prismo-Z250] — SPN_LOOP_GAIN_7 | Default: 244.600 | Linked Fault: ERR-ZEI250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z250.

PRM_49388 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_8 | Default: 172.400 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

PRM_49389 [Zeiss-Prismo-Z250] — CAX_LOOP_GAIN_9 | Default: 100.200 | Linked Fault: ERR-ZEI250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250.

PRM_49390 [Zeiss-Prismo-Z250] — VAC_LOOP_GAIN_10 | Default: 28.000 | Linked Fault: ERR-ZEI250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z250.

PRM_49391 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_11 | Default: 205.800 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

PRM_49392 [Zeiss-Prismo-Z250] — ZAX_LOOP_GAIN_12 | Default: 133.600 | Linked Fault: ERR-ZEI250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250.

PRM_49393 [Zeiss-Prismo-Z250] — PLC_LOOP_GAIN_13 | Default: 61.400 | Linked Fault: ERR-ZEI250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250.

PRM_49394 [Zeiss-Prismo-Z250] — SPN_LOOP_GAIN_14 | Default: 239.200 | Linked Fault: ERR-ZEI250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z250.

PRM_49395 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_15 | Default: 167.000 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

PRM_49396 [Zeiss-Prismo-Z250] — CAX_LOOP_GAIN_16 | Default: 94.800 | Linked Fault: ERR-ZEI250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z250 (BANK 5)

Machine Model: Zeiss-Prismo-Z250 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z250.

PRM_49401 [Zeiss-Prismo-Z250] — YAX_LOOP_GAIN_1 | Default: 34.750 | Linked Fault: ERR-ZEI250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z250.

PRM_49402 [Zeiss-Prismo-Z250] — PLC_LOOP_GAIN_2 | Default: 69.500 | Linked Fault: ERR-ZEI250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250.

PRM_49403 [Zeiss-Prismo-Z250] — ATC_LOOP_GAIN_3 | Default: 104.250 | Linked Fault: ERR-ZEI250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z250.

PRM_49404 [Zeiss-Prismo-Z250] — ZAX_LOOP_GAIN_4 | Default: 139.000 | Linked Fault: ERR-ZEI250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z250.

PRM_49405 [Zeiss-Prismo-Z250] — LUB_LOOP_GAIN_5 | Default: 173.750 | Linked Fault: ERR-ZEI250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z250.

PRM_49406 [Zeiss-Prismo-Z250] — HYD_LOOP_GAIN_6 | Default: 208.500 | Linked Fault: ERR-ZEI250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z250.

PRM_49407 [Zeiss-Prismo-Z250] — BAX_LOOP_GAIN_7 | Default: 243.250 | Linked Fault: ERR-ZEI250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z250.

PRM_49408 [Zeiss-Prismo-Z250] — VAC_LOOP_GAIN_8 | Default: 28.000 | Linked Fault: ERR-ZEI250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z250.

PRM_49409 [Zeiss-Prismo-Z250] — CLN_LOOP_GAIN_9 | Default: 62.750 | Linked Fault: ERR-ZEI250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z250.

PRM_49410 [Zeiss-Prismo-Z250] — CAX_LOOP_GAIN_10 | Default: 97.500 | Linked Fault: ERR-ZEI250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z250.

PRM_49411 [Zeiss-Prismo-Z250] — THM_LOOP_GAIN_11 | Default: 132.250 | Linked Fault: ERR-ZEI250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z250.

PRM_49412 [Zeiss-Prismo-Z250] — XAX_LOOP_GAIN_12 | Default: 167.000 | Linked Fault: ERR-ZEI250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z250.

PRM_49413 [Zeiss-Prismo-Z250] — OPT_LOOP_GAIN_13 | Default: 201.750 | Linked Fault: ERR-ZEI250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z250.

PRM_49414 [Zeiss-Prismo-Z250] — SPN_LOOP_GAIN_14 | Default: 236.500 | Linked Fault: ERR-ZEI250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z250.

PRM_49415 [Zeiss-Prismo-Z250] — YAX_LOOP_GAIN_15 | Default: 21.250 | Linked Fault: ERR-ZEI250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z250.

PRM_49416 [Zeiss-Prismo-Z250] — PLC_LOOP_GAIN_16 | Default: 56.000 | Linked Fault: ERR-ZEI250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z250.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z300 (BANK 1)

Machine Model: Zeiss-Prismo-Z300 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z300.

PRM_49421 [Zeiss-Prismo-Z300] — ATC_LOOP_GAIN_1 | Default: 108.100 | Linked Fault: ERR-ZEI300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z300.

PRM_49422 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_2 | Default: 216.200 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

PRM_49423 [Zeiss-Prismo-Z300] — CLN_LOOP_GAIN_3 | Default: 74.300 | Linked Fault: ERR-ZEI300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z300.

PRM_49424 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_4 | Default: 182.400 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

PRM_49425 [Zeiss-Prismo-Z300] — YAX_LOOP_GAIN_5 | Default: 40.500 | Linked Fault: ERR-ZEI300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z300.

PRM_49426 [Zeiss-Prismo-Z300] — ZAX_LOOP_GAIN_6 | Default: 148.600 | Linked Fault: ERR-ZEI300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300.

PRM_49427 [Zeiss-Prismo-Z300] — BAX_LOOP_GAIN_7 | Default: 6.700 | Linked Fault: ERR-ZEI300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z300.

PRM_49428 [Zeiss-Prismo-Z300] — CAX_LOOP_GAIN_8 | Default: 114.800 | Linked Fault: ERR-ZEI300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300.

PRM_49429 [Zeiss-Prismo-Z300] — OPT_LOOP_GAIN_9 | Default: 222.900 | Linked Fault: ERR-ZEI300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z300.

PRM_49430 [Zeiss-Prismo-Z300] — PLC_LOOP_GAIN_10 | Default: 81.000 | Linked Fault: ERR-ZEI300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300.

PRM_49431 [Zeiss-Prismo-Z300] — LUB_LOOP_GAIN_11 | Default: 189.100 | Linked Fault: ERR-ZEI300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z300.

PRM_49432 [Zeiss-Prismo-Z300] — VAC_LOOP_GAIN_12 | Default: 47.200 | Linked Fault: ERR-ZEI300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z300.

PRM_49433 [Zeiss-Prismo-Z300] — THM_LOOP_GAIN_13 | Default: 155.300 | Linked Fault: ERR-ZEI300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z300.

PRM_49434 [Zeiss-Prismo-Z300] — SPN_LOOP_GAIN_14 | Default: 13.400 | Linked Fault: ERR-ZEI300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z300.

PRM_49435 [Zeiss-Prismo-Z300] — ATC_LOOP_GAIN_15 | Default: 121.500 | Linked Fault: ERR-ZEI300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z300.

PRM_49436 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_16 | Default: 229.600 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z300 (BANK 2)

Machine Model: Zeiss-Prismo-Z300 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z300.

PRM_49441 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_1 | Default: 216.200 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

PRM_49442 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_2 | Default: 182.400 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

PRM_49443 [Zeiss-Prismo-Z300] — ZAX_LOOP_GAIN_3 | Default: 148.600 | Linked Fault: ERR-ZEI300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300.

PRM_49444 [Zeiss-Prismo-Z300] — CAX_LOOP_GAIN_4 | Default: 114.800 | Linked Fault: ERR-ZEI300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300.

PRM_49445 [Zeiss-Prismo-Z300] — PLC_LOOP_GAIN_5 | Default: 81.000 | Linked Fault: ERR-ZEI300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300.

PRM_49446 [Zeiss-Prismo-Z300] — VAC_LOOP_GAIN_6 | Default: 47.200 | Linked Fault: ERR-ZEI300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z300.

PRM_49447 [Zeiss-Prismo-Z300] — SPN_LOOP_GAIN_7 | Default: 13.400 | Linked Fault: ERR-ZEI300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z300.

PRM_49448 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_8 | Default: 229.600 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

PRM_49449 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_9 | Default: 195.800 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

PRM_49450 [Zeiss-Prismo-Z300] — ZAX_LOOP_GAIN_10 | Default: 162.000 | Linked Fault: ERR-ZEI300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300.

PRM_49451 [Zeiss-Prismo-Z300] — CAX_LOOP_GAIN_11 | Default: 128.200 | Linked Fault: ERR-ZEI300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300.

PRM_49452 [Zeiss-Prismo-Z300] — PLC_LOOP_GAIN_12 | Default: 94.400 | Linked Fault: ERR-ZEI300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300.

PRM_49453 [Zeiss-Prismo-Z300] — VAC_LOOP_GAIN_13 | Default: 60.600 | Linked Fault: ERR-ZEI300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z300.

PRM_49454 [Zeiss-Prismo-Z300] — SPN_LOOP_GAIN_14 | Default: 26.800 | Linked Fault: ERR-ZEI300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z300.

PRM_49455 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_15 | Default: 243.000 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

PRM_49456 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_16 | Default: 209.200 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z300 (BANK 3)

Machine Model: Zeiss-Prismo-Z300 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z300.

PRM_49461 [Zeiss-Prismo-Z300] — CLN_LOOP_GAIN_1 | Default: 74.300 | Linked Fault: ERR-ZEI300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z300.

PRM_49462 [Zeiss-Prismo-Z300] — ZAX_LOOP_GAIN_2 | Default: 148.600 | Linked Fault: ERR-ZEI300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300.

PRM_49463 [Zeiss-Prismo-Z300] — OPT_LOOP_GAIN_3 | Default: 222.900 | Linked Fault: ERR-ZEI300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z300.

PRM_49464 [Zeiss-Prismo-Z300] — VAC_LOOP_GAIN_4 | Default: 47.200 | Linked Fault: ERR-ZEI300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z300.

PRM_49465 [Zeiss-Prismo-Z300] — ATC_LOOP_GAIN_5 | Default: 121.500 | Linked Fault: ERR-ZEI300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z300.

PRM_49466 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_6 | Default: 195.800 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

PRM_49467 [Zeiss-Prismo-Z300] — BAX_LOOP_GAIN_7 | Default: 20.100 | Linked Fault: ERR-ZEI300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z300.

PRM_49468 [Zeiss-Prismo-Z300] — PLC_LOOP_GAIN_8 | Default: 94.400 | Linked Fault: ERR-ZEI300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300.

PRM_49469 [Zeiss-Prismo-Z300] — THM_LOOP_GAIN_9 | Default: 168.700 | Linked Fault: ERR-ZEI300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z300.

PRM_49470 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_10 | Default: 243.000 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

PRM_49471 [Zeiss-Prismo-Z300] — YAX_LOOP_GAIN_11 | Default: 67.300 | Linked Fault: ERR-ZEI300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z300.

PRM_49472 [Zeiss-Prismo-Z300] — CAX_LOOP_GAIN_12 | Default: 141.600 | Linked Fault: ERR-ZEI300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300.

PRM_49473 [Zeiss-Prismo-Z300] — LUB_LOOP_GAIN_13 | Default: 215.900 | Linked Fault: ERR-ZEI300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z300.

PRM_49474 [Zeiss-Prismo-Z300] — SPN_LOOP_GAIN_14 | Default: 40.200 | Linked Fault: ERR-ZEI300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z300.

PRM_49475 [Zeiss-Prismo-Z300] — CLN_LOOP_GAIN_15 | Default: 114.500 | Linked Fault: ERR-ZEI300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z300.

PRM_49476 [Zeiss-Prismo-Z300] — ZAX_LOOP_GAIN_16 | Default: 188.800 | Linked Fault: ERR-ZEI300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z300 (BANK 4)

Machine Model: Zeiss-Prismo-Z300 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z300.

PRM_49481 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_1 | Default: 182.400 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

PRM_49482 [Zeiss-Prismo-Z300] — CAX_LOOP_GAIN_2 | Default: 114.800 | Linked Fault: ERR-ZEI300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300.

PRM_49483 [Zeiss-Prismo-Z300] — VAC_LOOP_GAIN_3 | Default: 47.200 | Linked Fault: ERR-ZEI300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z300.

PRM_49484 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_4 | Default: 229.600 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

PRM_49485 [Zeiss-Prismo-Z300] — ZAX_LOOP_GAIN_5 | Default: 162.000 | Linked Fault: ERR-ZEI300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300.

PRM_49486 [Zeiss-Prismo-Z300] — PLC_LOOP_GAIN_6 | Default: 94.400 | Linked Fault: ERR-ZEI300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300.

PRM_49487 [Zeiss-Prismo-Z300] — SPN_LOOP_GAIN_7 | Default: 26.800 | Linked Fault: ERR-ZEI300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z300.

PRM_49488 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_8 | Default: 209.200 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

PRM_49489 [Zeiss-Prismo-Z300] — CAX_LOOP_GAIN_9 | Default: 141.600 | Linked Fault: ERR-ZEI300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300.

PRM_49490 [Zeiss-Prismo-Z300] — VAC_LOOP_GAIN_10 | Default: 74.000 | Linked Fault: ERR-ZEI300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z300.

PRM_49491 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_11 | Default: 6.400 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

PRM_49492 [Zeiss-Prismo-Z300] — ZAX_LOOP_GAIN_12 | Default: 188.800 | Linked Fault: ERR-ZEI300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300.

PRM_49493 [Zeiss-Prismo-Z300] — PLC_LOOP_GAIN_13 | Default: 121.200 | Linked Fault: ERR-ZEI300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300.

PRM_49494 [Zeiss-Prismo-Z300] — SPN_LOOP_GAIN_14 | Default: 53.600 | Linked Fault: ERR-ZEI300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z300.

PRM_49495 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_15 | Default: 236.000 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

PRM_49496 [Zeiss-Prismo-Z300] — CAX_LOOP_GAIN_16 | Default: 168.400 | Linked Fault: ERR-ZEI300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z300 (BANK 5)

Machine Model: Zeiss-Prismo-Z300 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z300.

PRM_49501 [Zeiss-Prismo-Z300] — YAX_LOOP_GAIN_1 | Default: 40.500 | Linked Fault: ERR-ZEI300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z300.

PRM_49502 [Zeiss-Prismo-Z300] — PLC_LOOP_GAIN_2 | Default: 81.000 | Linked Fault: ERR-ZEI300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300.

PRM_49503 [Zeiss-Prismo-Z300] — ATC_LOOP_GAIN_3 | Default: 121.500 | Linked Fault: ERR-ZEI300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z300.

PRM_49504 [Zeiss-Prismo-Z300] — ZAX_LOOP_GAIN_4 | Default: 162.000 | Linked Fault: ERR-ZEI300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z300.

PRM_49505 [Zeiss-Prismo-Z300] — LUB_LOOP_GAIN_5 | Default: 202.500 | Linked Fault: ERR-ZEI300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z300.

PRM_49506 [Zeiss-Prismo-Z300] — HYD_LOOP_GAIN_6 | Default: 243.000 | Linked Fault: ERR-ZEI300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z300.

PRM_49507 [Zeiss-Prismo-Z300] — BAX_LOOP_GAIN_7 | Default: 33.500 | Linked Fault: ERR-ZEI300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z300.

PRM_49508 [Zeiss-Prismo-Z300] — VAC_LOOP_GAIN_8 | Default: 74.000 | Linked Fault: ERR-ZEI300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z300.

PRM_49509 [Zeiss-Prismo-Z300] — CLN_LOOP_GAIN_9 | Default: 114.500 | Linked Fault: ERR-ZEI300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z300.

PRM_49510 [Zeiss-Prismo-Z300] — CAX_LOOP_GAIN_10 | Default: 155.000 | Linked Fault: ERR-ZEI300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z300.

PRM_49511 [Zeiss-Prismo-Z300] — THM_LOOP_GAIN_11 | Default: 195.500 | Linked Fault: ERR-ZEI300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z300.

PRM_49512 [Zeiss-Prismo-Z300] — XAX_LOOP_GAIN_12 | Default: 236.000 | Linked Fault: ERR-ZEI300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z300.

PRM_49513 [Zeiss-Prismo-Z300] — OPT_LOOP_GAIN_13 | Default: 26.500 | Linked Fault: ERR-ZEI300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z300.

PRM_49514 [Zeiss-Prismo-Z300] — SPN_LOOP_GAIN_14 | Default: 67.000 | Linked Fault: ERR-ZEI300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z300.

PRM_49515 [Zeiss-Prismo-Z300] — YAX_LOOP_GAIN_15 | Default: 107.500 | Linked Fault: ERR-ZEI300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z300.

PRM_49516 [Zeiss-Prismo-Z300] — PLC_LOOP_GAIN_16 | Default: 148.000 | Linked Fault: ERR-ZEI300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z300.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z350 (BANK 1)

Machine Model: Zeiss-Prismo-Z350 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z350.

PRM_49521 [Zeiss-Prismo-Z350] — ATC_LOOP_GAIN_1 | Default: 109.250 | Linked Fault: ERR-ZEI350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z350.

PRM_49522 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_2 | Default: 218.500 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

PRM_49523 [Zeiss-Prismo-Z350] — CLN_LOOP_GAIN_3 | Default: 77.750 | Linked Fault: ERR-ZEI350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z350.

PRM_49524 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_4 | Default: 187.000 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

PRM_49525 [Zeiss-Prismo-Z350] — YAX_LOOP_GAIN_5 | Default: 46.250 | Linked Fault: ERR-ZEI350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z350.

PRM_49526 [Zeiss-Prismo-Z350] — ZAX_LOOP_GAIN_6 | Default: 155.500 | Linked Fault: ERR-ZEI350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350.

PRM_49527 [Zeiss-Prismo-Z350] — BAX_LOOP_GAIN_7 | Default: 14.750 | Linked Fault: ERR-ZEI350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z350.

PRM_49528 [Zeiss-Prismo-Z350] — CAX_LOOP_GAIN_8 | Default: 124.000 | Linked Fault: ERR-ZEI350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350.

PRM_49529 [Zeiss-Prismo-Z350] — OPT_LOOP_GAIN_9 | Default: 233.250 | Linked Fault: ERR-ZEI350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z350.

PRM_49530 [Zeiss-Prismo-Z350] — PLC_LOOP_GAIN_10 | Default: 92.500 | Linked Fault: ERR-ZEI350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350.

PRM_49531 [Zeiss-Prismo-Z350] — LUB_LOOP_GAIN_11 | Default: 201.750 | Linked Fault: ERR-ZEI350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z350.

PRM_49532 [Zeiss-Prismo-Z350] — VAC_LOOP_GAIN_12 | Default: 61.000 | Linked Fault: ERR-ZEI350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z350.

PRM_49533 [Zeiss-Prismo-Z350] — THM_LOOP_GAIN_13 | Default: 170.250 | Linked Fault: ERR-ZEI350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z350.

PRM_49534 [Zeiss-Prismo-Z350] — SPN_LOOP_GAIN_14 | Default: 29.500 | Linked Fault: ERR-ZEI350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z350.

PRM_49535 [Zeiss-Prismo-Z350] — ATC_LOOP_GAIN_15 | Default: 138.750 | Linked Fault: ERR-ZEI350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z350.

PRM_49536 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_16 | Default: 248.000 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z350 (BANK 2)

Machine Model: Zeiss-Prismo-Z350 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z350.

PRM_49541 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_1 | Default: 218.500 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

PRM_49542 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_2 | Default: 187.000 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

PRM_49543 [Zeiss-Prismo-Z350] — ZAX_LOOP_GAIN_3 | Default: 155.500 | Linked Fault: ERR-ZEI350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350.

PRM_49544 [Zeiss-Prismo-Z350] — CAX_LOOP_GAIN_4 | Default: 124.000 | Linked Fault: ERR-ZEI350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350.

PRM_49545 [Zeiss-Prismo-Z350] — PLC_LOOP_GAIN_5 | Default: 92.500 | Linked Fault: ERR-ZEI350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350.

PRM_49546 [Zeiss-Prismo-Z350] — VAC_LOOP_GAIN_6 | Default: 61.000 | Linked Fault: ERR-ZEI350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z350.

PRM_49547 [Zeiss-Prismo-Z350] — SPN_LOOP_GAIN_7 | Default: 29.500 | Linked Fault: ERR-ZEI350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z350.

PRM_49548 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_8 | Default: 248.000 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

PRM_49549 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_9 | Default: 216.500 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

PRM_49550 [Zeiss-Prismo-Z350] — ZAX_LOOP_GAIN_10 | Default: 185.000 | Linked Fault: ERR-ZEI350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350.

PRM_49551 [Zeiss-Prismo-Z350] — CAX_LOOP_GAIN_11 | Default: 153.500 | Linked Fault: ERR-ZEI350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350.

PRM_49552 [Zeiss-Prismo-Z350] — PLC_LOOP_GAIN_12 | Default: 122.000 | Linked Fault: ERR-ZEI350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350.

PRM_49553 [Zeiss-Prismo-Z350] — VAC_LOOP_GAIN_13 | Default: 90.500 | Linked Fault: ERR-ZEI350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z350.

PRM_49554 [Zeiss-Prismo-Z350] — SPN_LOOP_GAIN_14 | Default: 59.000 | Linked Fault: ERR-ZEI350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z350.

PRM_49555 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_15 | Default: 27.500 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

PRM_49556 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_16 | Default: 246.000 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z350 (BANK 3)

Machine Model: Zeiss-Prismo-Z350 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z350.

PRM_49561 [Zeiss-Prismo-Z350] — CLN_LOOP_GAIN_1 | Default: 77.750 | Linked Fault: ERR-ZEI350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z350.

PRM_49562 [Zeiss-Prismo-Z350] — ZAX_LOOP_GAIN_2 | Default: 155.500 | Linked Fault: ERR-ZEI350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350.

PRM_49563 [Zeiss-Prismo-Z350] — OPT_LOOP_GAIN_3 | Default: 233.250 | Linked Fault: ERR-ZEI350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z350.

PRM_49564 [Zeiss-Prismo-Z350] — VAC_LOOP_GAIN_4 | Default: 61.000 | Linked Fault: ERR-ZEI350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z350.

PRM_49565 [Zeiss-Prismo-Z350] — ATC_LOOP_GAIN_5 | Default: 138.750 | Linked Fault: ERR-ZEI350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z350.

PRM_49566 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_6 | Default: 216.500 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

PRM_49567 [Zeiss-Prismo-Z350] — BAX_LOOP_GAIN_7 | Default: 44.250 | Linked Fault: ERR-ZEI350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z350.

PRM_49568 [Zeiss-Prismo-Z350] — PLC_LOOP_GAIN_8 | Default: 122.000 | Linked Fault: ERR-ZEI350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350.

PRM_49569 [Zeiss-Prismo-Z350] — THM_LOOP_GAIN_9 | Default: 199.750 | Linked Fault: ERR-ZEI350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z350.

PRM_49570 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_10 | Default: 27.500 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

PRM_49571 [Zeiss-Prismo-Z350] — YAX_LOOP_GAIN_11 | Default: 105.250 | Linked Fault: ERR-ZEI350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z350.

PRM_49572 [Zeiss-Prismo-Z350] — CAX_LOOP_GAIN_12 | Default: 183.000 | Linked Fault: ERR-ZEI350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350.

PRM_49573 [Zeiss-Prismo-Z350] — LUB_LOOP_GAIN_13 | Default: 10.750 | Linked Fault: ERR-ZEI350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z350.

PRM_49574 [Zeiss-Prismo-Z350] — SPN_LOOP_GAIN_14 | Default: 88.500 | Linked Fault: ERR-ZEI350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z350.

PRM_49575 [Zeiss-Prismo-Z350] — CLN_LOOP_GAIN_15 | Default: 166.250 | Linked Fault: ERR-ZEI350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z350.

PRM_49576 [Zeiss-Prismo-Z350] — ZAX_LOOP_GAIN_16 | Default: 244.000 | Linked Fault: ERR-ZEI350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z350 (BANK 4)

Machine Model: Zeiss-Prismo-Z350 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z350.

PRM_49581 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_1 | Default: 187.000 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

PRM_49582 [Zeiss-Prismo-Z350] — CAX_LOOP_GAIN_2 | Default: 124.000 | Linked Fault: ERR-ZEI350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350.

PRM_49583 [Zeiss-Prismo-Z350] — VAC_LOOP_GAIN_3 | Default: 61.000 | Linked Fault: ERR-ZEI350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z350.

PRM_49584 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_4 | Default: 248.000 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

PRM_49585 [Zeiss-Prismo-Z350] — ZAX_LOOP_GAIN_5 | Default: 185.000 | Linked Fault: ERR-ZEI350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350.

PRM_49586 [Zeiss-Prismo-Z350] — PLC_LOOP_GAIN_6 | Default: 122.000 | Linked Fault: ERR-ZEI350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350.

PRM_49587 [Zeiss-Prismo-Z350] — SPN_LOOP_GAIN_7 | Default: 59.000 | Linked Fault: ERR-ZEI350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z350.

PRM_49588 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_8 | Default: 246.000 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

PRM_49589 [Zeiss-Prismo-Z350] — CAX_LOOP_GAIN_9 | Default: 183.000 | Linked Fault: ERR-ZEI350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350.

PRM_49590 [Zeiss-Prismo-Z350] — VAC_LOOP_GAIN_10 | Default: 120.000 | Linked Fault: ERR-ZEI350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z350.

PRM_49591 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_11 | Default: 57.000 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

PRM_49592 [Zeiss-Prismo-Z350] — ZAX_LOOP_GAIN_12 | Default: 244.000 | Linked Fault: ERR-ZEI350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350.

PRM_49593 [Zeiss-Prismo-Z350] — PLC_LOOP_GAIN_13 | Default: 181.000 | Linked Fault: ERR-ZEI350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350.

PRM_49594 [Zeiss-Prismo-Z350] — SPN_LOOP_GAIN_14 | Default: 118.000 | Linked Fault: ERR-ZEI350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z350.

PRM_49595 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_15 | Default: 55.000 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

PRM_49596 [Zeiss-Prismo-Z350] — CAX_LOOP_GAIN_16 | Default: 242.000 | Linked Fault: ERR-ZEI350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z350 (BANK 5)

Machine Model: Zeiss-Prismo-Z350 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z350.

PRM_49601 [Zeiss-Prismo-Z350] — YAX_LOOP_GAIN_1 | Default: 46.250 | Linked Fault: ERR-ZEI350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z350.

PRM_49602 [Zeiss-Prismo-Z350] — PLC_LOOP_GAIN_2 | Default: 92.500 | Linked Fault: ERR-ZEI350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350.

PRM_49603 [Zeiss-Prismo-Z350] — ATC_LOOP_GAIN_3 | Default: 138.750 | Linked Fault: ERR-ZEI350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z350.

PRM_49604 [Zeiss-Prismo-Z350] — ZAX_LOOP_GAIN_4 | Default: 185.000 | Linked Fault: ERR-ZEI350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z350.

PRM_49605 [Zeiss-Prismo-Z350] — LUB_LOOP_GAIN_5 | Default: 231.250 | Linked Fault: ERR-ZEI350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z350.

PRM_49606 [Zeiss-Prismo-Z350] — HYD_LOOP_GAIN_6 | Default: 27.500 | Linked Fault: ERR-ZEI350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z350.

PRM_49607 [Zeiss-Prismo-Z350] — BAX_LOOP_GAIN_7 | Default: 73.750 | Linked Fault: ERR-ZEI350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z350.

PRM_49608 [Zeiss-Prismo-Z350] — VAC_LOOP_GAIN_8 | Default: 120.000 | Linked Fault: ERR-ZEI350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z350.

PRM_49609 [Zeiss-Prismo-Z350] — CLN_LOOP_GAIN_9 | Default: 166.250 | Linked Fault: ERR-ZEI350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z350.

PRM_49610 [Zeiss-Prismo-Z350] — CAX_LOOP_GAIN_10 | Default: 212.500 | Linked Fault: ERR-ZEI350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z350.

PRM_49611 [Zeiss-Prismo-Z350] — THM_LOOP_GAIN_11 | Default: 8.750 | Linked Fault: ERR-ZEI350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z350.

PRM_49612 [Zeiss-Prismo-Z350] — XAX_LOOP_GAIN_12 | Default: 55.000 | Linked Fault: ERR-ZEI350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z350.

PRM_49613 [Zeiss-Prismo-Z350] — OPT_LOOP_GAIN_13 | Default: 101.250 | Linked Fault: ERR-ZEI350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z350.

PRM_49614 [Zeiss-Prismo-Z350] — SPN_LOOP_GAIN_14 | Default: 147.500 | Linked Fault: ERR-ZEI350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z350.

PRM_49615 [Zeiss-Prismo-Z350] — YAX_LOOP_GAIN_15 | Default: 193.750 | Linked Fault: ERR-ZEI350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z350.

PRM_49616 [Zeiss-Prismo-Z350] — PLC_LOOP_GAIN_16 | Default: 240.000 | Linked Fault: ERR-ZEI350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z350.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z400 (BANK 1)

Machine Model: Zeiss-Prismo-Z400 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z400.

PRM_49621 [Zeiss-Prismo-Z400] — ATC_LOOP_GAIN_1 | Default: 110.400 | Linked Fault: ERR-ZEI400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z400.

PRM_49622 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_2 | Default: 220.800 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

PRM_49623 [Zeiss-Prismo-Z400] — CLN_LOOP_GAIN_3 | Default: 81.200 | Linked Fault: ERR-ZEI400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z400.

PRM_49624 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_4 | Default: 191.600 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

PRM_49625 [Zeiss-Prismo-Z400] — YAX_LOOP_GAIN_5 | Default: 52.000 | Linked Fault: ERR-ZEI400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z400.

PRM_49626 [Zeiss-Prismo-Z400] — ZAX_LOOP_GAIN_6 | Default: 162.400 | Linked Fault: ERR-ZEI400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400.

PRM_49627 [Zeiss-Prismo-Z400] — BAX_LOOP_GAIN_7 | Default: 22.800 | Linked Fault: ERR-ZEI400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z400.

PRM_49628 [Zeiss-Prismo-Z400] — CAX_LOOP_GAIN_8 | Default: 133.200 | Linked Fault: ERR-ZEI400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400.

PRM_49629 [Zeiss-Prismo-Z400] — OPT_LOOP_GAIN_9 | Default: 243.600 | Linked Fault: ERR-ZEI400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z400.

PRM_49630 [Zeiss-Prismo-Z400] — PLC_LOOP_GAIN_10 | Default: 104.000 | Linked Fault: ERR-ZEI400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400.

PRM_49631 [Zeiss-Prismo-Z400] — LUB_LOOP_GAIN_11 | Default: 214.400 | Linked Fault: ERR-ZEI400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z400.

PRM_49632 [Zeiss-Prismo-Z400] — VAC_LOOP_GAIN_12 | Default: 74.800 | Linked Fault: ERR-ZEI400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z400.

PRM_49633 [Zeiss-Prismo-Z400] — THM_LOOP_GAIN_13 | Default: 185.200 | Linked Fault: ERR-ZEI400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z400.

PRM_49634 [Zeiss-Prismo-Z400] — SPN_LOOP_GAIN_14 | Default: 45.600 | Linked Fault: ERR-ZEI400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z400.

PRM_49635 [Zeiss-Prismo-Z400] — ATC_LOOP_GAIN_15 | Default: 156.000 | Linked Fault: ERR-ZEI400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z400.

PRM_49636 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_16 | Default: 16.400 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z400 (BANK 2)

Machine Model: Zeiss-Prismo-Z400 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z400.

PRM_49641 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_1 | Default: 220.800 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

PRM_49642 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_2 | Default: 191.600 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

PRM_49643 [Zeiss-Prismo-Z400] — ZAX_LOOP_GAIN_3 | Default: 162.400 | Linked Fault: ERR-ZEI400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400.

PRM_49644 [Zeiss-Prismo-Z400] — CAX_LOOP_GAIN_4 | Default: 133.200 | Linked Fault: ERR-ZEI400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400.

PRM_49645 [Zeiss-Prismo-Z400] — PLC_LOOP_GAIN_5 | Default: 104.000 | Linked Fault: ERR-ZEI400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400.

PRM_49646 [Zeiss-Prismo-Z400] — VAC_LOOP_GAIN_6 | Default: 74.800 | Linked Fault: ERR-ZEI400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z400.

PRM_49647 [Zeiss-Prismo-Z400] — SPN_LOOP_GAIN_7 | Default: 45.600 | Linked Fault: ERR-ZEI400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z400.

PRM_49648 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_8 | Default: 16.400 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

PRM_49649 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_9 | Default: 237.200 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

PRM_49650 [Zeiss-Prismo-Z400] — ZAX_LOOP_GAIN_10 | Default: 208.000 | Linked Fault: ERR-ZEI400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400.

PRM_49651 [Zeiss-Prismo-Z400] — CAX_LOOP_GAIN_11 | Default: 178.800 | Linked Fault: ERR-ZEI400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400.

PRM_49652 [Zeiss-Prismo-Z400] — PLC_LOOP_GAIN_12 | Default: 149.600 | Linked Fault: ERR-ZEI400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400.

PRM_49653 [Zeiss-Prismo-Z400] — VAC_LOOP_GAIN_13 | Default: 120.400 | Linked Fault: ERR-ZEI400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z400.

PRM_49654 [Zeiss-Prismo-Z400] — SPN_LOOP_GAIN_14 | Default: 91.200 | Linked Fault: ERR-ZEI400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z400.

PRM_49655 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_15 | Default: 62.000 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

PRM_49656 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_16 | Default: 32.800 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z400 (BANK 3)

Machine Model: Zeiss-Prismo-Z400 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z400.

PRM_49661 [Zeiss-Prismo-Z400] — CLN_LOOP_GAIN_1 | Default: 81.200 | Linked Fault: ERR-ZEI400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z400.

PRM_49662 [Zeiss-Prismo-Z400] — ZAX_LOOP_GAIN_2 | Default: 162.400 | Linked Fault: ERR-ZEI400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400.

PRM_49663 [Zeiss-Prismo-Z400] — OPT_LOOP_GAIN_3 | Default: 243.600 | Linked Fault: ERR-ZEI400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z400.

PRM_49664 [Zeiss-Prismo-Z400] — VAC_LOOP_GAIN_4 | Default: 74.800 | Linked Fault: ERR-ZEI400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z400.

PRM_49665 [Zeiss-Prismo-Z400] — ATC_LOOP_GAIN_5 | Default: 156.000 | Linked Fault: ERR-ZEI400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z400.

PRM_49666 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_6 | Default: 237.200 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

PRM_49667 [Zeiss-Prismo-Z400] — BAX_LOOP_GAIN_7 | Default: 68.400 | Linked Fault: ERR-ZEI400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z400.

PRM_49668 [Zeiss-Prismo-Z400] — PLC_LOOP_GAIN_8 | Default: 149.600 | Linked Fault: ERR-ZEI400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400.

PRM_49669 [Zeiss-Prismo-Z400] — THM_LOOP_GAIN_9 | Default: 230.800 | Linked Fault: ERR-ZEI400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z400.

PRM_49670 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_10 | Default: 62.000 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

PRM_49671 [Zeiss-Prismo-Z400] — YAX_LOOP_GAIN_11 | Default: 143.200 | Linked Fault: ERR-ZEI400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z400.

PRM_49672 [Zeiss-Prismo-Z400] — CAX_LOOP_GAIN_12 | Default: 224.400 | Linked Fault: ERR-ZEI400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400.

PRM_49673 [Zeiss-Prismo-Z400] — LUB_LOOP_GAIN_13 | Default: 55.600 | Linked Fault: ERR-ZEI400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z400.

PRM_49674 [Zeiss-Prismo-Z400] — SPN_LOOP_GAIN_14 | Default: 136.800 | Linked Fault: ERR-ZEI400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z400.

PRM_49675 [Zeiss-Prismo-Z400] — CLN_LOOP_GAIN_15 | Default: 218.000 | Linked Fault: ERR-ZEI400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z400.

PRM_49676 [Zeiss-Prismo-Z400] — ZAX_LOOP_GAIN_16 | Default: 49.200 | Linked Fault: ERR-ZEI400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z400 (BANK 4)

Machine Model: Zeiss-Prismo-Z400 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z400.

PRM_49681 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_1 | Default: 191.600 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

PRM_49682 [Zeiss-Prismo-Z400] — CAX_LOOP_GAIN_2 | Default: 133.200 | Linked Fault: ERR-ZEI400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400.

PRM_49683 [Zeiss-Prismo-Z400] — VAC_LOOP_GAIN_3 | Default: 74.800 | Linked Fault: ERR-ZEI400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z400.

PRM_49684 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_4 | Default: 16.400 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

PRM_49685 [Zeiss-Prismo-Z400] — ZAX_LOOP_GAIN_5 | Default: 208.000 | Linked Fault: ERR-ZEI400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400.

PRM_49686 [Zeiss-Prismo-Z400] — PLC_LOOP_GAIN_6 | Default: 149.600 | Linked Fault: ERR-ZEI400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400.

PRM_49687 [Zeiss-Prismo-Z400] — SPN_LOOP_GAIN_7 | Default: 91.200 | Linked Fault: ERR-ZEI400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z400.

PRM_49688 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_8 | Default: 32.800 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

PRM_49689 [Zeiss-Prismo-Z400] — CAX_LOOP_GAIN_9 | Default: 224.400 | Linked Fault: ERR-ZEI400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400.

PRM_49690 [Zeiss-Prismo-Z400] — VAC_LOOP_GAIN_10 | Default: 166.000 | Linked Fault: ERR-ZEI400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z400.

PRM_49691 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_11 | Default: 107.600 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

PRM_49692 [Zeiss-Prismo-Z400] — ZAX_LOOP_GAIN_12 | Default: 49.200 | Linked Fault: ERR-ZEI400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400.

PRM_49693 [Zeiss-Prismo-Z400] — PLC_LOOP_GAIN_13 | Default: 240.800 | Linked Fault: ERR-ZEI400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400.

PRM_49694 [Zeiss-Prismo-Z400] — SPN_LOOP_GAIN_14 | Default: 182.400 | Linked Fault: ERR-ZEI400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z400.

PRM_49695 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_15 | Default: 124.000 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

PRM_49696 [Zeiss-Prismo-Z400] — CAX_LOOP_GAIN_16 | Default: 65.600 | Linked Fault: ERR-ZEI400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z400 (BANK 5)

Machine Model: Zeiss-Prismo-Z400 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z400.

PRM_49701 [Zeiss-Prismo-Z400] — YAX_LOOP_GAIN_1 | Default: 52.000 | Linked Fault: ERR-ZEI400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z400.

PRM_49702 [Zeiss-Prismo-Z400] — PLC_LOOP_GAIN_2 | Default: 104.000 | Linked Fault: ERR-ZEI400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400.

PRM_49703 [Zeiss-Prismo-Z400] — ATC_LOOP_GAIN_3 | Default: 156.000 | Linked Fault: ERR-ZEI400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z400.

PRM_49704 [Zeiss-Prismo-Z400] — ZAX_LOOP_GAIN_4 | Default: 208.000 | Linked Fault: ERR-ZEI400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z400.

PRM_49705 [Zeiss-Prismo-Z400] — LUB_LOOP_GAIN_5 | Default: 10.000 | Linked Fault: ERR-ZEI400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z400.

PRM_49706 [Zeiss-Prismo-Z400] — HYD_LOOP_GAIN_6 | Default: 62.000 | Linked Fault: ERR-ZEI400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z400.

PRM_49707 [Zeiss-Prismo-Z400] — BAX_LOOP_GAIN_7 | Default: 114.000 | Linked Fault: ERR-ZEI400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z400.

PRM_49708 [Zeiss-Prismo-Z400] — VAC_LOOP_GAIN_8 | Default: 166.000 | Linked Fault: ERR-ZEI400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z400.

PRM_49709 [Zeiss-Prismo-Z400] — CLN_LOOP_GAIN_9 | Default: 218.000 | Linked Fault: ERR-ZEI400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z400.

PRM_49710 [Zeiss-Prismo-Z400] — CAX_LOOP_GAIN_10 | Default: 20.000 | Linked Fault: ERR-ZEI400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z400.

PRM_49711 [Zeiss-Prismo-Z400] — THM_LOOP_GAIN_11 | Default: 72.000 | Linked Fault: ERR-ZEI400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z400.

PRM_49712 [Zeiss-Prismo-Z400] — XAX_LOOP_GAIN_12 | Default: 124.000 | Linked Fault: ERR-ZEI400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z400.

PRM_49713 [Zeiss-Prismo-Z400] — OPT_LOOP_GAIN_13 | Default: 176.000 | Linked Fault: ERR-ZEI400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z400.

PRM_49714 [Zeiss-Prismo-Z400] — SPN_LOOP_GAIN_14 | Default: 228.000 | Linked Fault: ERR-ZEI400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z400.

PRM_49715 [Zeiss-Prismo-Z400] — YAX_LOOP_GAIN_15 | Default: 30.000 | Linked Fault: ERR-ZEI400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z400.

PRM_49716 [Zeiss-Prismo-Z400] — PLC_LOOP_GAIN_16 | Default: 82.000 | Linked Fault: ERR-ZEI400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z400.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z450 (BANK 1)

Machine Model: Zeiss-Prismo-Z450 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z450.

PRM_49721 [Zeiss-Prismo-Z450] — ATC_LOOP_GAIN_1 | Default: 111.550 | Linked Fault: ERR-ZEI450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z450.

PRM_49722 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_2 | Default: 223.100 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

PRM_49723 [Zeiss-Prismo-Z450] — CLN_LOOP_GAIN_3 | Default: 84.650 | Linked Fault: ERR-ZEI450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z450.

PRM_49724 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_4 | Default: 196.200 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

PRM_49725 [Zeiss-Prismo-Z450] — YAX_LOOP_GAIN_5 | Default: 57.750 | Linked Fault: ERR-ZEI450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z450.

PRM_49726 [Zeiss-Prismo-Z450] — ZAX_LOOP_GAIN_6 | Default: 169.300 | Linked Fault: ERR-ZEI450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450.

PRM_49727 [Zeiss-Prismo-Z450] — BAX_LOOP_GAIN_7 | Default: 30.850 | Linked Fault: ERR-ZEI450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z450.

PRM_49728 [Zeiss-Prismo-Z450] — CAX_LOOP_GAIN_8 | Default: 142.400 | Linked Fault: ERR-ZEI450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450.

PRM_49729 [Zeiss-Prismo-Z450] — OPT_LOOP_GAIN_9 | Default: 3.950 | Linked Fault: ERR-ZEI450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z450.

PRM_49730 [Zeiss-Prismo-Z450] — PLC_LOOP_GAIN_10 | Default: 115.500 | Linked Fault: ERR-ZEI450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450.

PRM_49731 [Zeiss-Prismo-Z450] — LUB_LOOP_GAIN_11 | Default: 227.050 | Linked Fault: ERR-ZEI450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z450.

PRM_49732 [Zeiss-Prismo-Z450] — VAC_LOOP_GAIN_12 | Default: 88.600 | Linked Fault: ERR-ZEI450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z450.

PRM_49733 [Zeiss-Prismo-Z450] — THM_LOOP_GAIN_13 | Default: 200.150 | Linked Fault: ERR-ZEI450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z450.

PRM_49734 [Zeiss-Prismo-Z450] — SPN_LOOP_GAIN_14 | Default: 61.700 | Linked Fault: ERR-ZEI450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z450.

PRM_49735 [Zeiss-Prismo-Z450] — ATC_LOOP_GAIN_15 | Default: 173.250 | Linked Fault: ERR-ZEI450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z450.

PRM_49736 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_16 | Default: 34.800 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z450 (BANK 2)

Machine Model: Zeiss-Prismo-Z450 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z450.

PRM_49741 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_1 | Default: 223.100 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

PRM_49742 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_2 | Default: 196.200 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

PRM_49743 [Zeiss-Prismo-Z450] — ZAX_LOOP_GAIN_3 | Default: 169.300 | Linked Fault: ERR-ZEI450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450.

PRM_49744 [Zeiss-Prismo-Z450] — CAX_LOOP_GAIN_4 | Default: 142.400 | Linked Fault: ERR-ZEI450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450.

PRM_49745 [Zeiss-Prismo-Z450] — PLC_LOOP_GAIN_5 | Default: 115.500 | Linked Fault: ERR-ZEI450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450.

PRM_49746 [Zeiss-Prismo-Z450] — VAC_LOOP_GAIN_6 | Default: 88.600 | Linked Fault: ERR-ZEI450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z450.

PRM_49747 [Zeiss-Prismo-Z450] — SPN_LOOP_GAIN_7 | Default: 61.700 | Linked Fault: ERR-ZEI450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z450.

PRM_49748 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_8 | Default: 34.800 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

PRM_49749 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_9 | Default: 7.900 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

PRM_49750 [Zeiss-Prismo-Z450] — ZAX_LOOP_GAIN_10 | Default: 231.000 | Linked Fault: ERR-ZEI450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450.

PRM_49751 [Zeiss-Prismo-Z450] — CAX_LOOP_GAIN_11 | Default: 204.100 | Linked Fault: ERR-ZEI450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450.

PRM_49752 [Zeiss-Prismo-Z450] — PLC_LOOP_GAIN_12 | Default: 177.200 | Linked Fault: ERR-ZEI450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450.

PRM_49753 [Zeiss-Prismo-Z450] — VAC_LOOP_GAIN_13 | Default: 150.300 | Linked Fault: ERR-ZEI450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z450.

PRM_49754 [Zeiss-Prismo-Z450] — SPN_LOOP_GAIN_14 | Default: 123.400 | Linked Fault: ERR-ZEI450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z450.

PRM_49755 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_15 | Default: 96.500 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

PRM_49756 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_16 | Default: 69.600 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z450 (BANK 3)

Machine Model: Zeiss-Prismo-Z450 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z450.

PRM_49761 [Zeiss-Prismo-Z450] — CLN_LOOP_GAIN_1 | Default: 84.650 | Linked Fault: ERR-ZEI450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z450.

PRM_49762 [Zeiss-Prismo-Z450] — ZAX_LOOP_GAIN_2 | Default: 169.300 | Linked Fault: ERR-ZEI450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450.

PRM_49763 [Zeiss-Prismo-Z450] — OPT_LOOP_GAIN_3 | Default: 3.950 | Linked Fault: ERR-ZEI450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z450.

PRM_49764 [Zeiss-Prismo-Z450] — VAC_LOOP_GAIN_4 | Default: 88.600 | Linked Fault: ERR-ZEI450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z450.

PRM_49765 [Zeiss-Prismo-Z450] — ATC_LOOP_GAIN_5 | Default: 173.250 | Linked Fault: ERR-ZEI450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z450.

PRM_49766 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_6 | Default: 7.900 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

PRM_49767 [Zeiss-Prismo-Z450] — BAX_LOOP_GAIN_7 | Default: 92.550 | Linked Fault: ERR-ZEI450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z450.

PRM_49768 [Zeiss-Prismo-Z450] — PLC_LOOP_GAIN_8 | Default: 177.200 | Linked Fault: ERR-ZEI450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450.

PRM_49769 [Zeiss-Prismo-Z450] — THM_LOOP_GAIN_9 | Default: 11.850 | Linked Fault: ERR-ZEI450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z450.

PRM_49770 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_10 | Default: 96.500 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

PRM_49771 [Zeiss-Prismo-Z450] — YAX_LOOP_GAIN_11 | Default: 181.150 | Linked Fault: ERR-ZEI450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z450.

PRM_49772 [Zeiss-Prismo-Z450] — CAX_LOOP_GAIN_12 | Default: 15.800 | Linked Fault: ERR-ZEI450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450.

PRM_49773 [Zeiss-Prismo-Z450] — LUB_LOOP_GAIN_13 | Default: 100.450 | Linked Fault: ERR-ZEI450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z450.

PRM_49774 [Zeiss-Prismo-Z450] — SPN_LOOP_GAIN_14 | Default: 185.100 | Linked Fault: ERR-ZEI450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z450.

PRM_49775 [Zeiss-Prismo-Z450] — CLN_LOOP_GAIN_15 | Default: 19.750 | Linked Fault: ERR-ZEI450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z450.

PRM_49776 [Zeiss-Prismo-Z450] — ZAX_LOOP_GAIN_16 | Default: 104.400 | Linked Fault: ERR-ZEI450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z450 (BANK 4)

Machine Model: Zeiss-Prismo-Z450 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z450.

PRM_49781 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_1 | Default: 196.200 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

PRM_49782 [Zeiss-Prismo-Z450] — CAX_LOOP_GAIN_2 | Default: 142.400 | Linked Fault: ERR-ZEI450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450.

PRM_49783 [Zeiss-Prismo-Z450] — VAC_LOOP_GAIN_3 | Default: 88.600 | Linked Fault: ERR-ZEI450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z450.

PRM_49784 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_4 | Default: 34.800 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

PRM_49785 [Zeiss-Prismo-Z450] — ZAX_LOOP_GAIN_5 | Default: 231.000 | Linked Fault: ERR-ZEI450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450.

PRM_49786 [Zeiss-Prismo-Z450] — PLC_LOOP_GAIN_6 | Default: 177.200 | Linked Fault: ERR-ZEI450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450.

PRM_49787 [Zeiss-Prismo-Z450] — SPN_LOOP_GAIN_7 | Default: 123.400 | Linked Fault: ERR-ZEI450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z450.

PRM_49788 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_8 | Default: 69.600 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

PRM_49789 [Zeiss-Prismo-Z450] — CAX_LOOP_GAIN_9 | Default: 15.800 | Linked Fault: ERR-ZEI450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450.

PRM_49790 [Zeiss-Prismo-Z450] — VAC_LOOP_GAIN_10 | Default: 212.000 | Linked Fault: ERR-ZEI450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z450.

PRM_49791 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_11 | Default: 158.200 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

PRM_49792 [Zeiss-Prismo-Z450] — ZAX_LOOP_GAIN_12 | Default: 104.400 | Linked Fault: ERR-ZEI450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450.

PRM_49793 [Zeiss-Prismo-Z450] — PLC_LOOP_GAIN_13 | Default: 50.600 | Linked Fault: ERR-ZEI450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450.

PRM_49794 [Zeiss-Prismo-Z450] — SPN_LOOP_GAIN_14 | Default: 246.800 | Linked Fault: ERR-ZEI450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z450.

PRM_49795 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_15 | Default: 193.000 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

PRM_49796 [Zeiss-Prismo-Z450] — CAX_LOOP_GAIN_16 | Default: 139.200 | Linked Fault: ERR-ZEI450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z450 (BANK 5)

Machine Model: Zeiss-Prismo-Z450 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z450.

PRM_49801 [Zeiss-Prismo-Z450] — YAX_LOOP_GAIN_1 | Default: 57.750 | Linked Fault: ERR-ZEI450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z450.

PRM_49802 [Zeiss-Prismo-Z450] — PLC_LOOP_GAIN_2 | Default: 115.500 | Linked Fault: ERR-ZEI450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450.

PRM_49803 [Zeiss-Prismo-Z450] — ATC_LOOP_GAIN_3 | Default: 173.250 | Linked Fault: ERR-ZEI450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z450.

PRM_49804 [Zeiss-Prismo-Z450] — ZAX_LOOP_GAIN_4 | Default: 231.000 | Linked Fault: ERR-ZEI450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z450.

PRM_49805 [Zeiss-Prismo-Z450] — LUB_LOOP_GAIN_5 | Default: 38.750 | Linked Fault: ERR-ZEI450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z450.

PRM_49806 [Zeiss-Prismo-Z450] — HYD_LOOP_GAIN_6 | Default: 96.500 | Linked Fault: ERR-ZEI450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z450.

PRM_49807 [Zeiss-Prismo-Z450] — BAX_LOOP_GAIN_7 | Default: 154.250 | Linked Fault: ERR-ZEI450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z450.

PRM_49808 [Zeiss-Prismo-Z450] — VAC_LOOP_GAIN_8 | Default: 212.000 | Linked Fault: ERR-ZEI450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z450.

PRM_49809 [Zeiss-Prismo-Z450] — CLN_LOOP_GAIN_9 | Default: 19.750 | Linked Fault: ERR-ZEI450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z450.

PRM_49810 [Zeiss-Prismo-Z450] — CAX_LOOP_GAIN_10 | Default: 77.500 | Linked Fault: ERR-ZEI450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z450.

PRM_49811 [Zeiss-Prismo-Z450] — THM_LOOP_GAIN_11 | Default: 135.250 | Linked Fault: ERR-ZEI450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z450.

PRM_49812 [Zeiss-Prismo-Z450] — XAX_LOOP_GAIN_12 | Default: 193.000 | Linked Fault: ERR-ZEI450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z450.

PRM_49813 [Zeiss-Prismo-Z450] — OPT_LOOP_GAIN_13 | Default: 0.750 | Linked Fault: ERR-ZEI450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z450.

PRM_49814 [Zeiss-Prismo-Z450] — SPN_LOOP_GAIN_14 | Default: 58.500 | Linked Fault: ERR-ZEI450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z450.

PRM_49815 [Zeiss-Prismo-Z450] — YAX_LOOP_GAIN_15 | Default: 116.250 | Linked Fault: ERR-ZEI450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z450.

PRM_49816 [Zeiss-Prismo-Z450] — PLC_LOOP_GAIN_16 | Default: 174.000 | Linked Fault: ERR-ZEI450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z450.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z500 (BANK 1)

Machine Model: Zeiss-Prismo-Z500 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z500.

PRM_49821 [Zeiss-Prismo-Z500] — ATC_LOOP_GAIN_1 | Default: 112.700 | Linked Fault: ERR-ZEI500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z500.

PRM_49822 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_2 | Default: 225.400 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

PRM_49823 [Zeiss-Prismo-Z500] — CLN_LOOP_GAIN_3 | Default: 88.100 | Linked Fault: ERR-ZEI500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z500.

PRM_49824 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_4 | Default: 200.800 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

PRM_49825 [Zeiss-Prismo-Z500] — YAX_LOOP_GAIN_5 | Default: 63.500 | Linked Fault: ERR-ZEI500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z500.

PRM_49826 [Zeiss-Prismo-Z500] — ZAX_LOOP_GAIN_6 | Default: 176.200 | Linked Fault: ERR-ZEI500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500.

PRM_49827 [Zeiss-Prismo-Z500] — BAX_LOOP_GAIN_7 | Default: 38.900 | Linked Fault: ERR-ZEI500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z500.

PRM_49828 [Zeiss-Prismo-Z500] — CAX_LOOP_GAIN_8 | Default: 151.600 | Linked Fault: ERR-ZEI500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500.

PRM_49829 [Zeiss-Prismo-Z500] — OPT_LOOP_GAIN_9 | Default: 14.300 | Linked Fault: ERR-ZEI500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z500.

PRM_49830 [Zeiss-Prismo-Z500] — PLC_LOOP_GAIN_10 | Default: 127.000 | Linked Fault: ERR-ZEI500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500.

PRM_49831 [Zeiss-Prismo-Z500] — LUB_LOOP_GAIN_11 | Default: 239.700 | Linked Fault: ERR-ZEI500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z500.

PRM_49832 [Zeiss-Prismo-Z500] — VAC_LOOP_GAIN_12 | Default: 102.400 | Linked Fault: ERR-ZEI500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z500.

PRM_49833 [Zeiss-Prismo-Z500] — THM_LOOP_GAIN_13 | Default: 215.100 | Linked Fault: ERR-ZEI500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z500.

PRM_49834 [Zeiss-Prismo-Z500] — SPN_LOOP_GAIN_14 | Default: 77.800 | Linked Fault: ERR-ZEI500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z500.

PRM_49835 [Zeiss-Prismo-Z500] — ATC_LOOP_GAIN_15 | Default: 190.500 | Linked Fault: ERR-ZEI500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z500.

PRM_49836 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_16 | Default: 53.200 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z500 (BANK 2)

Machine Model: Zeiss-Prismo-Z500 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z500.

PRM_49841 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_1 | Default: 225.400 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

PRM_49842 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_2 | Default: 200.800 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

PRM_49843 [Zeiss-Prismo-Z500] — ZAX_LOOP_GAIN_3 | Default: 176.200 | Linked Fault: ERR-ZEI500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500.

PRM_49844 [Zeiss-Prismo-Z500] — CAX_LOOP_GAIN_4 | Default: 151.600 | Linked Fault: ERR-ZEI500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500.

PRM_49845 [Zeiss-Prismo-Z500] — PLC_LOOP_GAIN_5 | Default: 127.000 | Linked Fault: ERR-ZEI500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500.

PRM_49846 [Zeiss-Prismo-Z500] — VAC_LOOP_GAIN_6 | Default: 102.400 | Linked Fault: ERR-ZEI500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z500.

PRM_49847 [Zeiss-Prismo-Z500] — SPN_LOOP_GAIN_7 | Default: 77.800 | Linked Fault: ERR-ZEI500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z500.

PRM_49848 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_8 | Default: 53.200 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

PRM_49849 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_9 | Default: 28.600 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

PRM_49850 [Zeiss-Prismo-Z500] — ZAX_LOOP_GAIN_10 | Default: 4.000 | Linked Fault: ERR-ZEI500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500.

PRM_49851 [Zeiss-Prismo-Z500] — CAX_LOOP_GAIN_11 | Default: 229.400 | Linked Fault: ERR-ZEI500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500.

PRM_49852 [Zeiss-Prismo-Z500] — PLC_LOOP_GAIN_12 | Default: 204.800 | Linked Fault: ERR-ZEI500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500.

PRM_49853 [Zeiss-Prismo-Z500] — VAC_LOOP_GAIN_13 | Default: 180.200 | Linked Fault: ERR-ZEI500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z500.

PRM_49854 [Zeiss-Prismo-Z500] — SPN_LOOP_GAIN_14 | Default: 155.600 | Linked Fault: ERR-ZEI500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z500.

PRM_49855 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_15 | Default: 131.000 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

PRM_49856 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_16 | Default: 106.400 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z500 (BANK 3)

Machine Model: Zeiss-Prismo-Z500 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z500.

PRM_49861 [Zeiss-Prismo-Z500] — CLN_LOOP_GAIN_1 | Default: 88.100 | Linked Fault: ERR-ZEI500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z500.

PRM_49862 [Zeiss-Prismo-Z500] — ZAX_LOOP_GAIN_2 | Default: 176.200 | Linked Fault: ERR-ZEI500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500.

PRM_49863 [Zeiss-Prismo-Z500] — OPT_LOOP_GAIN_3 | Default: 14.300 | Linked Fault: ERR-ZEI500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z500.

PRM_49864 [Zeiss-Prismo-Z500] — VAC_LOOP_GAIN_4 | Default: 102.400 | Linked Fault: ERR-ZEI500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z500.

PRM_49865 [Zeiss-Prismo-Z500] — ATC_LOOP_GAIN_5 | Default: 190.500 | Linked Fault: ERR-ZEI500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z500.

PRM_49866 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_6 | Default: 28.600 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

PRM_49867 [Zeiss-Prismo-Z500] — BAX_LOOP_GAIN_7 | Default: 116.700 | Linked Fault: ERR-ZEI500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z500.

PRM_49868 [Zeiss-Prismo-Z500] — PLC_LOOP_GAIN_8 | Default: 204.800 | Linked Fault: ERR-ZEI500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500.

PRM_49869 [Zeiss-Prismo-Z500] — THM_LOOP_GAIN_9 | Default: 42.900 | Linked Fault: ERR-ZEI500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z500.

PRM_49870 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_10 | Default: 131.000 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

PRM_49871 [Zeiss-Prismo-Z500] — YAX_LOOP_GAIN_11 | Default: 219.100 | Linked Fault: ERR-ZEI500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z500.

PRM_49872 [Zeiss-Prismo-Z500] — CAX_LOOP_GAIN_12 | Default: 57.200 | Linked Fault: ERR-ZEI500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500.

PRM_49873 [Zeiss-Prismo-Z500] — LUB_LOOP_GAIN_13 | Default: 145.300 | Linked Fault: ERR-ZEI500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z500.

PRM_49874 [Zeiss-Prismo-Z500] — SPN_LOOP_GAIN_14 | Default: 233.400 | Linked Fault: ERR-ZEI500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z500.

PRM_49875 [Zeiss-Prismo-Z500] — CLN_LOOP_GAIN_15 | Default: 71.500 | Linked Fault: ERR-ZEI500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z500.

PRM_49876 [Zeiss-Prismo-Z500] — ZAX_LOOP_GAIN_16 | Default: 159.600 | Linked Fault: ERR-ZEI500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z500 (BANK 4)

Machine Model: Zeiss-Prismo-Z500 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z500.

PRM_49881 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_1 | Default: 200.800 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

PRM_49882 [Zeiss-Prismo-Z500] — CAX_LOOP_GAIN_2 | Default: 151.600 | Linked Fault: ERR-ZEI500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500.

PRM_49883 [Zeiss-Prismo-Z500] — VAC_LOOP_GAIN_3 | Default: 102.400 | Linked Fault: ERR-ZEI500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z500.

PRM_49884 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_4 | Default: 53.200 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

PRM_49885 [Zeiss-Prismo-Z500] — ZAX_LOOP_GAIN_5 | Default: 4.000 | Linked Fault: ERR-ZEI500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500.

PRM_49886 [Zeiss-Prismo-Z500] — PLC_LOOP_GAIN_6 | Default: 204.800 | Linked Fault: ERR-ZEI500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500.

PRM_49887 [Zeiss-Prismo-Z500] — SPN_LOOP_GAIN_7 | Default: 155.600 | Linked Fault: ERR-ZEI500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z500.

PRM_49888 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_8 | Default: 106.400 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

PRM_49889 [Zeiss-Prismo-Z500] — CAX_LOOP_GAIN_9 | Default: 57.200 | Linked Fault: ERR-ZEI500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500.

PRM_49890 [Zeiss-Prismo-Z500] — VAC_LOOP_GAIN_10 | Default: 8.000 | Linked Fault: ERR-ZEI500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z500.

PRM_49891 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_11 | Default: 208.800 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

PRM_49892 [Zeiss-Prismo-Z500] — ZAX_LOOP_GAIN_12 | Default: 159.600 | Linked Fault: ERR-ZEI500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500.

PRM_49893 [Zeiss-Prismo-Z500] — PLC_LOOP_GAIN_13 | Default: 110.400 | Linked Fault: ERR-ZEI500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500.

PRM_49894 [Zeiss-Prismo-Z500] — SPN_LOOP_GAIN_14 | Default: 61.200 | Linked Fault: ERR-ZEI500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z500.

PRM_49895 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_15 | Default: 12.000 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

PRM_49896 [Zeiss-Prismo-Z500] — CAX_LOOP_GAIN_16 | Default: 212.800 | Linked Fault: ERR-ZEI500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z500 (BANK 5)

Machine Model: Zeiss-Prismo-Z500 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z500.

PRM_49901 [Zeiss-Prismo-Z500] — YAX_LOOP_GAIN_1 | Default: 63.500 | Linked Fault: ERR-ZEI500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z500.

PRM_49902 [Zeiss-Prismo-Z500] — PLC_LOOP_GAIN_2 | Default: 127.000 | Linked Fault: ERR-ZEI500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500.

PRM_49903 [Zeiss-Prismo-Z500] — ATC_LOOP_GAIN_3 | Default: 190.500 | Linked Fault: ERR-ZEI500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z500.

PRM_49904 [Zeiss-Prismo-Z500] — ZAX_LOOP_GAIN_4 | Default: 4.000 | Linked Fault: ERR-ZEI500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z500.

PRM_49905 [Zeiss-Prismo-Z500] — LUB_LOOP_GAIN_5 | Default: 67.500 | Linked Fault: ERR-ZEI500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z500.

PRM_49906 [Zeiss-Prismo-Z500] — HYD_LOOP_GAIN_6 | Default: 131.000 | Linked Fault: ERR-ZEI500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z500.

PRM_49907 [Zeiss-Prismo-Z500] — BAX_LOOP_GAIN_7 | Default: 194.500 | Linked Fault: ERR-ZEI500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z500.

PRM_49908 [Zeiss-Prismo-Z500] — VAC_LOOP_GAIN_8 | Default: 8.000 | Linked Fault: ERR-ZEI500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z500.

PRM_49909 [Zeiss-Prismo-Z500] — CLN_LOOP_GAIN_9 | Default: 71.500 | Linked Fault: ERR-ZEI500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z500.

PRM_49910 [Zeiss-Prismo-Z500] — CAX_LOOP_GAIN_10 | Default: 135.000 | Linked Fault: ERR-ZEI500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z500.

PRM_49911 [Zeiss-Prismo-Z500] — THM_LOOP_GAIN_11 | Default: 198.500 | Linked Fault: ERR-ZEI500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z500.

PRM_49912 [Zeiss-Prismo-Z500] — XAX_LOOP_GAIN_12 | Default: 12.000 | Linked Fault: ERR-ZEI500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z500.

PRM_49913 [Zeiss-Prismo-Z500] — OPT_LOOP_GAIN_13 | Default: 75.500 | Linked Fault: ERR-ZEI500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z500.

PRM_49914 [Zeiss-Prismo-Z500] — SPN_LOOP_GAIN_14 | Default: 139.000 | Linked Fault: ERR-ZEI500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z500.

PRM_49915 [Zeiss-Prismo-Z500] — YAX_LOOP_GAIN_15 | Default: 202.500 | Linked Fault: ERR-ZEI500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z500.

PRM_49916 [Zeiss-Prismo-Z500] — PLC_LOOP_GAIN_16 | Default: 16.000 | Linked Fault: ERR-ZEI500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z500.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z550 (BANK 1)

Machine Model: Zeiss-Prismo-Z550 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z550.

PRM_49921 [Zeiss-Prismo-Z550] — ATC_LOOP_GAIN_1 | Default: 113.850 | Linked Fault: ERR-ZEI550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z550.

PRM_49922 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_2 | Default: 227.700 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

PRM_49923 [Zeiss-Prismo-Z550] — CLN_LOOP_GAIN_3 | Default: 91.550 | Linked Fault: ERR-ZEI550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z550.

PRM_49924 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_4 | Default: 205.400 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

PRM_49925 [Zeiss-Prismo-Z550] — YAX_LOOP_GAIN_5 | Default: 69.250 | Linked Fault: ERR-ZEI550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z550.

PRM_49926 [Zeiss-Prismo-Z550] — ZAX_LOOP_GAIN_6 | Default: 183.100 | Linked Fault: ERR-ZEI550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550.

PRM_49927 [Zeiss-Prismo-Z550] — BAX_LOOP_GAIN_7 | Default: 46.950 | Linked Fault: ERR-ZEI550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z550.

PRM_49928 [Zeiss-Prismo-Z550] — CAX_LOOP_GAIN_8 | Default: 160.800 | Linked Fault: ERR-ZEI550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550.

PRM_49929 [Zeiss-Prismo-Z550] — OPT_LOOP_GAIN_9 | Default: 24.650 | Linked Fault: ERR-ZEI550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z550.

PRM_49930 [Zeiss-Prismo-Z550] — PLC_LOOP_GAIN_10 | Default: 138.500 | Linked Fault: ERR-ZEI550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550.

PRM_49931 [Zeiss-Prismo-Z550] — LUB_LOOP_GAIN_11 | Default: 2.350 | Linked Fault: ERR-ZEI550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z550.

PRM_49932 [Zeiss-Prismo-Z550] — VAC_LOOP_GAIN_12 | Default: 116.200 | Linked Fault: ERR-ZEI550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z550.

PRM_49933 [Zeiss-Prismo-Z550] — THM_LOOP_GAIN_13 | Default: 230.050 | Linked Fault: ERR-ZEI550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z550.

PRM_49934 [Zeiss-Prismo-Z550] — SPN_LOOP_GAIN_14 | Default: 93.900 | Linked Fault: ERR-ZEI550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z550.

PRM_49935 [Zeiss-Prismo-Z550] — ATC_LOOP_GAIN_15 | Default: 207.750 | Linked Fault: ERR-ZEI550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z550.

PRM_49936 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_16 | Default: 71.600 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z550 (BANK 2)

Machine Model: Zeiss-Prismo-Z550 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z550.

PRM_49941 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_1 | Default: 227.700 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

PRM_49942 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_2 | Default: 205.400 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

PRM_49943 [Zeiss-Prismo-Z550] — ZAX_LOOP_GAIN_3 | Default: 183.100 | Linked Fault: ERR-ZEI550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550.

PRM_49944 [Zeiss-Prismo-Z550] — CAX_LOOP_GAIN_4 | Default: 160.800 | Linked Fault: ERR-ZEI550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550.

PRM_49945 [Zeiss-Prismo-Z550] — PLC_LOOP_GAIN_5 | Default: 138.500 | Linked Fault: ERR-ZEI550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550.

PRM_49946 [Zeiss-Prismo-Z550] — VAC_LOOP_GAIN_6 | Default: 116.200 | Linked Fault: ERR-ZEI550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z550.

PRM_49947 [Zeiss-Prismo-Z550] — SPN_LOOP_GAIN_7 | Default: 93.900 | Linked Fault: ERR-ZEI550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z550.

PRM_49948 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_8 | Default: 71.600 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

PRM_49949 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_9 | Default: 49.300 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

PRM_49950 [Zeiss-Prismo-Z550] — ZAX_LOOP_GAIN_10 | Default: 27.000 | Linked Fault: ERR-ZEI550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550.

PRM_49951 [Zeiss-Prismo-Z550] — CAX_LOOP_GAIN_11 | Default: 4.700 | Linked Fault: ERR-ZEI550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550.

PRM_49952 [Zeiss-Prismo-Z550] — PLC_LOOP_GAIN_12 | Default: 232.400 | Linked Fault: ERR-ZEI550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550.

PRM_49953 [Zeiss-Prismo-Z550] — VAC_LOOP_GAIN_13 | Default: 210.100 | Linked Fault: ERR-ZEI550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z550.

PRM_49954 [Zeiss-Prismo-Z550] — SPN_LOOP_GAIN_14 | Default: 187.800 | Linked Fault: ERR-ZEI550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z550.

PRM_49955 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_15 | Default: 165.500 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

PRM_49956 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_16 | Default: 143.200 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z550 (BANK 3)

Machine Model: Zeiss-Prismo-Z550 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z550.

PRM_49961 [Zeiss-Prismo-Z550] — CLN_LOOP_GAIN_1 | Default: 91.550 | Linked Fault: ERR-ZEI550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z550.

PRM_49962 [Zeiss-Prismo-Z550] — ZAX_LOOP_GAIN_2 | Default: 183.100 | Linked Fault: ERR-ZEI550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550.

PRM_49963 [Zeiss-Prismo-Z550] — OPT_LOOP_GAIN_3 | Default: 24.650 | Linked Fault: ERR-ZEI550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z550.

PRM_49964 [Zeiss-Prismo-Z550] — VAC_LOOP_GAIN_4 | Default: 116.200 | Linked Fault: ERR-ZEI550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z550.

PRM_49965 [Zeiss-Prismo-Z550] — ATC_LOOP_GAIN_5 | Default: 207.750 | Linked Fault: ERR-ZEI550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z550.

PRM_49966 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_6 | Default: 49.300 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

PRM_49967 [Zeiss-Prismo-Z550] — BAX_LOOP_GAIN_7 | Default: 140.850 | Linked Fault: ERR-ZEI550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z550.

PRM_49968 [Zeiss-Prismo-Z550] — PLC_LOOP_GAIN_8 | Default: 232.400 | Linked Fault: ERR-ZEI550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550.

PRM_49969 [Zeiss-Prismo-Z550] — THM_LOOP_GAIN_9 | Default: 73.950 | Linked Fault: ERR-ZEI550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z550.

PRM_49970 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_10 | Default: 165.500 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

PRM_49971 [Zeiss-Prismo-Z550] — YAX_LOOP_GAIN_11 | Default: 7.050 | Linked Fault: ERR-ZEI550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z550.

PRM_49972 [Zeiss-Prismo-Z550] — CAX_LOOP_GAIN_12 | Default: 98.600 | Linked Fault: ERR-ZEI550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550.

PRM_49973 [Zeiss-Prismo-Z550] — LUB_LOOP_GAIN_13 | Default: 190.150 | Linked Fault: ERR-ZEI550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z550.

PRM_49974 [Zeiss-Prismo-Z550] — SPN_LOOP_GAIN_14 | Default: 31.700 | Linked Fault: ERR-ZEI550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z550.

PRM_49975 [Zeiss-Prismo-Z550] — CLN_LOOP_GAIN_15 | Default: 123.250 | Linked Fault: ERR-ZEI550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z550.

PRM_49976 [Zeiss-Prismo-Z550] — ZAX_LOOP_GAIN_16 | Default: 214.800 | Linked Fault: ERR-ZEI550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z550 (BANK 4)

Machine Model: Zeiss-Prismo-Z550 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z550.

PRM_49981 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_1 | Default: 205.400 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

PRM_49982 [Zeiss-Prismo-Z550] — CAX_LOOP_GAIN_2 | Default: 160.800 | Linked Fault: ERR-ZEI550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550.

PRM_49983 [Zeiss-Prismo-Z550] — VAC_LOOP_GAIN_3 | Default: 116.200 | Linked Fault: ERR-ZEI550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z550.

PRM_49984 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_4 | Default: 71.600 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

PRM_49985 [Zeiss-Prismo-Z550] — ZAX_LOOP_GAIN_5 | Default: 27.000 | Linked Fault: ERR-ZEI550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550.

PRM_49986 [Zeiss-Prismo-Z550] — PLC_LOOP_GAIN_6 | Default: 232.400 | Linked Fault: ERR-ZEI550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550.

PRM_49987 [Zeiss-Prismo-Z550] — SPN_LOOP_GAIN_7 | Default: 187.800 | Linked Fault: ERR-ZEI550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z550.

PRM_49988 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_8 | Default: 143.200 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

PRM_49989 [Zeiss-Prismo-Z550] — CAX_LOOP_GAIN_9 | Default: 98.600 | Linked Fault: ERR-ZEI550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550.

PRM_49990 [Zeiss-Prismo-Z550] — VAC_LOOP_GAIN_10 | Default: 54.000 | Linked Fault: ERR-ZEI550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z550.

PRM_49991 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_11 | Default: 9.400 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

PRM_49992 [Zeiss-Prismo-Z550] — ZAX_LOOP_GAIN_12 | Default: 214.800 | Linked Fault: ERR-ZEI550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550.

PRM_49993 [Zeiss-Prismo-Z550] — PLC_LOOP_GAIN_13 | Default: 170.200 | Linked Fault: ERR-ZEI550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550.

PRM_49994 [Zeiss-Prismo-Z550] — SPN_LOOP_GAIN_14 | Default: 125.600 | Linked Fault: ERR-ZEI550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z550.

PRM_49995 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_15 | Default: 81.000 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

PRM_49996 [Zeiss-Prismo-Z550] — CAX_LOOP_GAIN_16 | Default: 36.400 | Linked Fault: ERR-ZEI550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z550 (BANK 5)

Machine Model: Zeiss-Prismo-Z550 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z550.

PRM_50001 [Zeiss-Prismo-Z550] — YAX_LOOP_GAIN_1 | Default: 69.250 | Linked Fault: ERR-ZEI550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z550.

PRM_50002 [Zeiss-Prismo-Z550] — PLC_LOOP_GAIN_2 | Default: 138.500 | Linked Fault: ERR-ZEI550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550.

PRM_50003 [Zeiss-Prismo-Z550] — ATC_LOOP_GAIN_3 | Default: 207.750 | Linked Fault: ERR-ZEI550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z550.

PRM_50004 [Zeiss-Prismo-Z550] — ZAX_LOOP_GAIN_4 | Default: 27.000 | Linked Fault: ERR-ZEI550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z550.

PRM_50005 [Zeiss-Prismo-Z550] — LUB_LOOP_GAIN_5 | Default: 96.250 | Linked Fault: ERR-ZEI550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z550.

PRM_50006 [Zeiss-Prismo-Z550] — HYD_LOOP_GAIN_6 | Default: 165.500 | Linked Fault: ERR-ZEI550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z550.

PRM_50007 [Zeiss-Prismo-Z550] — BAX_LOOP_GAIN_7 | Default: 234.750 | Linked Fault: ERR-ZEI550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z550.

PRM_50008 [Zeiss-Prismo-Z550] — VAC_LOOP_GAIN_8 | Default: 54.000 | Linked Fault: ERR-ZEI550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z550.

PRM_50009 [Zeiss-Prismo-Z550] — CLN_LOOP_GAIN_9 | Default: 123.250 | Linked Fault: ERR-ZEI550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z550.

PRM_50010 [Zeiss-Prismo-Z550] — CAX_LOOP_GAIN_10 | Default: 192.500 | Linked Fault: ERR-ZEI550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z550.

PRM_50011 [Zeiss-Prismo-Z550] — THM_LOOP_GAIN_11 | Default: 11.750 | Linked Fault: ERR-ZEI550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z550.

PRM_50012 [Zeiss-Prismo-Z550] — XAX_LOOP_GAIN_12 | Default: 81.000 | Linked Fault: ERR-ZEI550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z550.

PRM_50013 [Zeiss-Prismo-Z550] — OPT_LOOP_GAIN_13 | Default: 150.250 | Linked Fault: ERR-ZEI550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z550.

PRM_50014 [Zeiss-Prismo-Z550] — SPN_LOOP_GAIN_14 | Default: 219.500 | Linked Fault: ERR-ZEI550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z550.

PRM_50015 [Zeiss-Prismo-Z550] — YAX_LOOP_GAIN_15 | Default: 38.750 | Linked Fault: ERR-ZEI550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z550.

PRM_50016 [Zeiss-Prismo-Z550] — PLC_LOOP_GAIN_16 | Default: 108.000 | Linked Fault: ERR-ZEI550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z550.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z600 (BANK 1)

Machine Model: Zeiss-Prismo-Z600 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z600.

PRM_50021 [Zeiss-Prismo-Z600] — ATC_LOOP_GAIN_1 | Default: 115.000 | Linked Fault: ERR-ZEI600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z600.

PRM_50022 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_2 | Default: 230.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

PRM_50023 [Zeiss-Prismo-Z600] — CLN_LOOP_GAIN_3 | Default: 95.000 | Linked Fault: ERR-ZEI600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z600.

PRM_50024 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_4 | Default: 210.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

PRM_50025 [Zeiss-Prismo-Z600] — YAX_LOOP_GAIN_5 | Default: 75.000 | Linked Fault: ERR-ZEI600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z600.

PRM_50026 [Zeiss-Prismo-Z600] — ZAX_LOOP_GAIN_6 | Default: 190.000 | Linked Fault: ERR-ZEI600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600.

PRM_50027 [Zeiss-Prismo-Z600] — BAX_LOOP_GAIN_7 | Default: 55.000 | Linked Fault: ERR-ZEI600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z600.

PRM_50028 [Zeiss-Prismo-Z600] — CAX_LOOP_GAIN_8 | Default: 170.000 | Linked Fault: ERR-ZEI600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600.

PRM_50029 [Zeiss-Prismo-Z600] — OPT_LOOP_GAIN_9 | Default: 35.000 | Linked Fault: ERR-ZEI600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z600.

PRM_50030 [Zeiss-Prismo-Z600] — PLC_LOOP_GAIN_10 | Default: 150.000 | Linked Fault: ERR-ZEI600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600.

PRM_50031 [Zeiss-Prismo-Z600] — LUB_LOOP_GAIN_11 | Default: 15.000 | Linked Fault: ERR-ZEI600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z600.

PRM_50032 [Zeiss-Prismo-Z600] — VAC_LOOP_GAIN_12 | Default: 130.000 | Linked Fault: ERR-ZEI600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z600.

PRM_50033 [Zeiss-Prismo-Z600] — THM_LOOP_GAIN_13 | Default: 245.000 | Linked Fault: ERR-ZEI600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z600.

PRM_50034 [Zeiss-Prismo-Z600] — SPN_LOOP_GAIN_14 | Default: 110.000 | Linked Fault: ERR-ZEI600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z600.

PRM_50035 [Zeiss-Prismo-Z600] — ATC_LOOP_GAIN_15 | Default: 225.000 | Linked Fault: ERR-ZEI600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z600.

PRM_50036 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_16 | Default: 90.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z600 (BANK 2)

Machine Model: Zeiss-Prismo-Z600 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z600.

PRM_50041 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_1 | Default: 230.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

PRM_50042 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_2 | Default: 210.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

PRM_50043 [Zeiss-Prismo-Z600] — ZAX_LOOP_GAIN_3 | Default: 190.000 | Linked Fault: ERR-ZEI600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600.

PRM_50044 [Zeiss-Prismo-Z600] — CAX_LOOP_GAIN_4 | Default: 170.000 | Linked Fault: ERR-ZEI600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600.

PRM_50045 [Zeiss-Prismo-Z600] — PLC_LOOP_GAIN_5 | Default: 150.000 | Linked Fault: ERR-ZEI600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600.

PRM_50046 [Zeiss-Prismo-Z600] — VAC_LOOP_GAIN_6 | Default: 130.000 | Linked Fault: ERR-ZEI600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z600.

PRM_50047 [Zeiss-Prismo-Z600] — SPN_LOOP_GAIN_7 | Default: 110.000 | Linked Fault: ERR-ZEI600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z600.

PRM_50048 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_8 | Default: 90.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

PRM_50049 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_9 | Default: 70.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

PRM_50050 [Zeiss-Prismo-Z600] — ZAX_LOOP_GAIN_10 | Default: 50.000 | Linked Fault: ERR-ZEI600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600.

PRM_50051 [Zeiss-Prismo-Z600] — CAX_LOOP_GAIN_11 | Default: 30.000 | Linked Fault: ERR-ZEI600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600.

PRM_50052 [Zeiss-Prismo-Z600] — PLC_LOOP_GAIN_12 | Default: 10.000 | Linked Fault: ERR-ZEI600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600.

PRM_50053 [Zeiss-Prismo-Z600] — VAC_LOOP_GAIN_13 | Default: 240.000 | Linked Fault: ERR-ZEI600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z600.

PRM_50054 [Zeiss-Prismo-Z600] — SPN_LOOP_GAIN_14 | Default: 220.000 | Linked Fault: ERR-ZEI600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z600.

PRM_50055 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_15 | Default: 200.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

PRM_50056 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_16 | Default: 180.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z600 (BANK 3)

Machine Model: Zeiss-Prismo-Z600 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z600.

PRM_50061 [Zeiss-Prismo-Z600] — CLN_LOOP_GAIN_1 | Default: 95.000 | Linked Fault: ERR-ZEI600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z600.

PRM_50062 [Zeiss-Prismo-Z600] — ZAX_LOOP_GAIN_2 | Default: 190.000 | Linked Fault: ERR-ZEI600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600.

PRM_50063 [Zeiss-Prismo-Z600] — OPT_LOOP_GAIN_3 | Default: 35.000 | Linked Fault: ERR-ZEI600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z600.

PRM_50064 [Zeiss-Prismo-Z600] — VAC_LOOP_GAIN_4 | Default: 130.000 | Linked Fault: ERR-ZEI600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z600.

PRM_50065 [Zeiss-Prismo-Z600] — ATC_LOOP_GAIN_5 | Default: 225.000 | Linked Fault: ERR-ZEI600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z600.

PRM_50066 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_6 | Default: 70.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

PRM_50067 [Zeiss-Prismo-Z600] — BAX_LOOP_GAIN_7 | Default: 165.000 | Linked Fault: ERR-ZEI600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z600.

PRM_50068 [Zeiss-Prismo-Z600] — PLC_LOOP_GAIN_8 | Default: 10.000 | Linked Fault: ERR-ZEI600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600.

PRM_50069 [Zeiss-Prismo-Z600] — THM_LOOP_GAIN_9 | Default: 105.000 | Linked Fault: ERR-ZEI600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z600.

PRM_50070 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_10 | Default: 200.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

PRM_50071 [Zeiss-Prismo-Z600] — YAX_LOOP_GAIN_11 | Default: 45.000 | Linked Fault: ERR-ZEI600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z600.

PRM_50072 [Zeiss-Prismo-Z600] — CAX_LOOP_GAIN_12 | Default: 140.000 | Linked Fault: ERR-ZEI600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600.

PRM_50073 [Zeiss-Prismo-Z600] — LUB_LOOP_GAIN_13 | Default: 235.000 | Linked Fault: ERR-ZEI600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z600.

PRM_50074 [Zeiss-Prismo-Z600] — SPN_LOOP_GAIN_14 | Default: 80.000 | Linked Fault: ERR-ZEI600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z600.

PRM_50075 [Zeiss-Prismo-Z600] — CLN_LOOP_GAIN_15 | Default: 175.000 | Linked Fault: ERR-ZEI600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z600.

PRM_50076 [Zeiss-Prismo-Z600] — ZAX_LOOP_GAIN_16 | Default: 20.000 | Linked Fault: ERR-ZEI600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z600 (BANK 4)

Machine Model: Zeiss-Prismo-Z600 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z600.

PRM_50081 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_1 | Default: 210.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

PRM_50082 [Zeiss-Prismo-Z600] — CAX_LOOP_GAIN_2 | Default: 170.000 | Linked Fault: ERR-ZEI600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600.

PRM_50083 [Zeiss-Prismo-Z600] — VAC_LOOP_GAIN_3 | Default: 130.000 | Linked Fault: ERR-ZEI600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z600.

PRM_50084 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_4 | Default: 90.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

PRM_50085 [Zeiss-Prismo-Z600] — ZAX_LOOP_GAIN_5 | Default: 50.000 | Linked Fault: ERR-ZEI600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600.

PRM_50086 [Zeiss-Prismo-Z600] — PLC_LOOP_GAIN_6 | Default: 10.000 | Linked Fault: ERR-ZEI600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600.

PRM_50087 [Zeiss-Prismo-Z600] — SPN_LOOP_GAIN_7 | Default: 220.000 | Linked Fault: ERR-ZEI600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z600.

PRM_50088 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_8 | Default: 180.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

PRM_50089 [Zeiss-Prismo-Z600] — CAX_LOOP_GAIN_9 | Default: 140.000 | Linked Fault: ERR-ZEI600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600.

PRM_50090 [Zeiss-Prismo-Z600] — VAC_LOOP_GAIN_10 | Default: 100.000 | Linked Fault: ERR-ZEI600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z600.

PRM_50091 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_11 | Default: 60.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

PRM_50092 [Zeiss-Prismo-Z600] — ZAX_LOOP_GAIN_12 | Default: 20.000 | Linked Fault: ERR-ZEI600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600.

PRM_50093 [Zeiss-Prismo-Z600] — PLC_LOOP_GAIN_13 | Default: 230.000 | Linked Fault: ERR-ZEI600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600.

PRM_50094 [Zeiss-Prismo-Z600] — SPN_LOOP_GAIN_14 | Default: 190.000 | Linked Fault: ERR-ZEI600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z600.

PRM_50095 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_15 | Default: 150.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

PRM_50096 [Zeiss-Prismo-Z600] — CAX_LOOP_GAIN_16 | Default: 110.000 | Linked Fault: ERR-ZEI600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600.

CNC/PLC SERVO PARAMETER DICTIONARY — Zeiss-Prismo-Z600 (BANK 5)

Machine Model: Zeiss-Prismo-Z600 | Controller: Zeiss C99m CAA Controller. Use these parameter registers when diagnosing servo following errors or thermal drift on Zeiss-Prismo-Z600.

PRM_50101 [Zeiss-Prismo-Z600] — YAX_LOOP_GAIN_1 | Default: 75.000 | Linked Fault: ERR-ZEI600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z600.

PRM_50102 [Zeiss-Prismo-Z600] — PLC_LOOP_GAIN_2 | Default: 150.000 | Linked Fault: ERR-ZEI600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600.

PRM_50103 [Zeiss-Prismo-Z600] — ATC_LOOP_GAIN_3 | Default: 225.000 | Linked Fault: ERR-ZEI600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Zeiss-Prismo-Z600.

PRM_50104 [Zeiss-Prismo-Z600] — ZAX_LOOP_GAIN_4 | Default: 50.000 | Linked Fault: ERR-ZEI600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Zeiss-Prismo-Z600.

PRM_50105 [Zeiss-Prismo-Z600] — LUB_LOOP_GAIN_5 | Default: 125.000 | Linked Fault: ERR-ZEI600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Zeiss-Prismo-Z600.

PRM_50106 [Zeiss-Prismo-Z600] — HYD_LOOP_GAIN_6 | Default: 200.000 | Linked Fault: ERR-ZEI600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Zeiss-Prismo-Z600.

PRM_50107 [Zeiss-Prismo-Z600] — BAX_LOOP_GAIN_7 | Default: 25.000 | Linked Fault: ERR-ZEI600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Zeiss-Prismo-Z600.

PRM_50108 [Zeiss-Prismo-Z600] — VAC_LOOP_GAIN_8 | Default: 100.000 | Linked Fault: ERR-ZEI600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Zeiss-Prismo-Z600.

PRM_50109 [Zeiss-Prismo-Z600] — CLN_LOOP_GAIN_9 | Default: 175.000 | Linked Fault: ERR-ZEI600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Zeiss-Prismo-Z600.

PRM_50110 [Zeiss-Prismo-Z600] — CAX_LOOP_GAIN_10 | Default: 0.000 | Linked Fault: ERR-ZEI600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Zeiss-Prismo-Z600.

PRM_50111 [Zeiss-Prismo-Z600] — THM_LOOP_GAIN_11 | Default: 75.000 | Linked Fault: ERR-ZEI600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Zeiss-Prismo-Z600.

PRM_50112 [Zeiss-Prismo-Z600] — XAX_LOOP_GAIN_12 | Default: 150.000 | Linked Fault: ERR-ZEI600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Zeiss-Prismo-Z600.

PRM_50113 [Zeiss-Prismo-Z600] — OPT_LOOP_GAIN_13 | Default: 225.000 | Linked Fault: ERR-ZEI600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Zeiss-Prismo-Z600.

PRM_50114 [Zeiss-Prismo-Z600] — SPN_LOOP_GAIN_14 | Default: 50.000 | Linked Fault: ERR-ZEI600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Zeiss-Prismo-Z600.

PRM_50115 [Zeiss-Prismo-Z600] — YAX_LOOP_GAIN_15 | Default: 125.000 | Linked Fault: ERR-ZEI600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Zeiss-Prismo-Z600.

PRM_50116 [Zeiss-Prismo-Z600] — PLC_LOOP_GAIN_16 | Default: 200.000 | Linked Fault: ERR-ZEI600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Zeiss-Prismo-Z600.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G400 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G400. Verify part numbers before closing Work Orders.

BOM-76-1.1: PN-GRO400-CAM-GRIP-7602 / Kit KIT-GRO400-ATC-2 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO400-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-76-1.2: PN-GRO400-VLV-PROP-7603 / Kit KIT-GRO400-HYD-3 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-76-1.3: PN-GRO400-PMP-PLG-7604 / Kit KIT-GRO400-CLN-4 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO400-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-76-1.4: PN-GRO400-LIN-COIL-7605 / Kit KIT-GRO400-XAX-5 (Grob-Aerospace-G400) | PM Interval: 1250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-76-1.5: PN-GRO400-BSW-NUT-7606 / Kit KIT-GRO400-YAX-6 (Grob-Aerospace-G400) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO400-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-76-1.6: PN-GRO400-CYL-SEAL-7607 / Kit KIT-GRO400-ZAX-7 (Grob-Aerospace-G400) | PM Interval: 250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-76-1.7: PN-GRO400-TRQ-STAT-7608 / Kit KIT-GRO400-BAX-8 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO400-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-76-1.8: PN-GRO400-BRK-DISC-7609 / Kit KIT-GRO400-CAX-9 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-76-1.9: PN-GRO400-GLASS-RD-7610 / Kit KIT-GRO400-OPT-10 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO400-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-76-1.10: PN-GRO400-IGBT-MOD-7611 / Kit KIT-GRO400-PLC-11 (Grob-Aerospace-G400) | PM Interval: 1250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-76-1.11: PN-GRO400-INJ-MET-7612 / Kit KIT-GRO400-LUB-12 (Grob-Aerospace-G400) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO400-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-76-1.12: PN-GRO400-GASK-VIT-7613 / Kit KIT-GRO400-VAC-13 (Grob-Aerospace-G400) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-76-1.13: PN-GRO400-HEX-CORE-7614 / Kit KIT-GRO400-THM-14 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO400-THM-14 within tolerance ±0.0040 mm TIR.

BOM-76-1.14: PN-GRO400-BRG-CER-7601 / Kit KIT-GRO400-SPN-1 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-76-1.15: PN-GRO400-CAM-GRIP-7602 / Kit KIT-GRO400-ATC-2 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO400-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G400 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G400. Verify part numbers before closing Work Orders.

BOM-76-2.1: PN-GRO400-VLV-PROP-7603 / Kit KIT-GRO400-HYD-3 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-76-2.2: PN-GRO400-LIN-COIL-7605 / Kit KIT-GRO400-XAX-5 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-76-2.3: PN-GRO400-CYL-SEAL-7607 / Kit KIT-GRO400-ZAX-7 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-76-2.4: PN-GRO400-BRK-DISC-7609 / Kit KIT-GRO400-CAX-9 (Grob-Aerospace-G400) | PM Interval: 1250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-76-2.5: PN-GRO400-IGBT-MOD-7611 / Kit KIT-GRO400-PLC-11 (Grob-Aerospace-G400) | PM Interval: 1500 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-76-2.6: PN-GRO400-GASK-VIT-7613 / Kit KIT-GRO400-VAC-13 (Grob-Aerospace-G400) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-76-2.7: PN-GRO400-BRG-CER-7601 / Kit KIT-GRO400-SPN-1 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-76-2.8: PN-GRO400-VLV-PROP-7603 / Kit KIT-GRO400-HYD-3 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-76-2.9: PN-GRO400-LIN-COIL-7605 / Kit KIT-GRO400-XAX-5 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-76-2.10: PN-GRO400-CYL-SEAL-7607 / Kit KIT-GRO400-ZAX-7 (Grob-Aerospace-G400) | PM Interval: 1250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-76-2.11: PN-GRO400-BRK-DISC-7609 / Kit KIT-GRO400-CAX-9 (Grob-Aerospace-G400) | PM Interval: 1500 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-76-2.12: PN-GRO400-IGBT-MOD-7611 / Kit KIT-GRO400-PLC-11 (Grob-Aerospace-G400) | PM Interval: 250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-76-2.13: PN-GRO400-GASK-VIT-7613 / Kit KIT-GRO400-VAC-13 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-76-2.14: PN-GRO400-BRG-CER-7601 / Kit KIT-GRO400-SPN-1 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-76-2.15: PN-GRO400-VLV-PROP-7603 / Kit KIT-GRO400-HYD-3 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO400-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G400 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G400. Verify part numbers before closing Work Orders.

BOM-76-3.1: PN-GRO400-PMP-PLG-7604 / Kit KIT-GRO400-CLN-4 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO400-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-76-3.2: PN-GRO400-CYL-SEAL-7607 / Kit KIT-GRO400-ZAX-7 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-76-3.3: PN-GRO400-GLASS-RD-7610 / Kit KIT-GRO400-OPT-10 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO400-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-76-3.4: PN-GRO400-GASK-VIT-7613 / Kit KIT-GRO400-VAC-13 (Grob-Aerospace-G400) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-76-3.5: PN-GRO400-CAM-GRIP-7602 / Kit KIT-GRO400-ATC-2 (Grob-Aerospace-G400) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO400-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-76-3.6: PN-GRO400-LIN-COIL-7605 / Kit KIT-GRO400-XAX-5 (Grob-Aerospace-G400) | PM Interval: 250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-76-3.7: PN-GRO400-TRQ-STAT-7608 / Kit KIT-GRO400-BAX-8 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO400-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-76-3.8: PN-GRO400-IGBT-MOD-7611 / Kit KIT-GRO400-PLC-11 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-76-3.9: PN-GRO400-HEX-CORE-7614 / Kit KIT-GRO400-THM-14 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO400-THM-14 within tolerance ±0.0040 mm TIR.

BOM-76-3.10: PN-GRO400-VLV-PROP-7603 / Kit KIT-GRO400-HYD-3 (Grob-Aerospace-G400) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-76-3.11: PN-GRO400-BSW-NUT-7606 / Kit KIT-GRO400-YAX-6 (Grob-Aerospace-G400) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO400-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-76-3.12: PN-GRO400-BRK-DISC-7609 / Kit KIT-GRO400-CAX-9 (Grob-Aerospace-G400) | PM Interval: 250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-76-3.13: PN-GRO400-INJ-MET-7612 / Kit KIT-GRO400-LUB-12 (Grob-Aerospace-G400) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO400-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-76-3.14: PN-GRO400-BRG-CER-7601 / Kit KIT-GRO400-SPN-1 (Grob-Aerospace-G400) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-76-3.15: PN-GRO400-PMP-PLG-7604 / Kit KIT-GRO400-CLN-4 (Grob-Aerospace-G400) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO400-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G450 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G450. Verify part numbers before closing Work Orders.

BOM-77-1.1: PN-GRO450-CAM-GRIP-7702 / Kit KIT-GRO450-ATC-2 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO450-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-77-1.2: PN-GRO450-VLV-PROP-7703 / Kit KIT-GRO450-HYD-3 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-77-1.3: PN-GRO450-PMP-PLG-7704 / Kit KIT-GRO450-CLN-4 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO450-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-77-1.4: PN-GRO450-LIN-COIL-7705 / Kit KIT-GRO450-XAX-5 (Grob-Aerospace-G450) | PM Interval: 1250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-77-1.5: PN-GRO450-BSW-NUT-7706 / Kit KIT-GRO450-YAX-6 (Grob-Aerospace-G450) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO450-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-77-1.6: PN-GRO450-CYL-SEAL-7707 / Kit KIT-GRO450-ZAX-7 (Grob-Aerospace-G450) | PM Interval: 250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-77-1.7: PN-GRO450-TRQ-STAT-7708 / Kit KIT-GRO450-BAX-8 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO450-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-77-1.8: PN-GRO450-BRK-DISC-7709 / Kit KIT-GRO450-CAX-9 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-77-1.9: PN-GRO450-GLASS-RD-7710 / Kit KIT-GRO450-OPT-10 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO450-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-77-1.10: PN-GRO450-IGBT-MOD-7711 / Kit KIT-GRO450-PLC-11 (Grob-Aerospace-G450) | PM Interval: 1250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-77-1.11: PN-GRO450-INJ-MET-7712 / Kit KIT-GRO450-LUB-12 (Grob-Aerospace-G450) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO450-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-77-1.12: PN-GRO450-GASK-VIT-7713 / Kit KIT-GRO450-VAC-13 (Grob-Aerospace-G450) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-77-1.13: PN-GRO450-HEX-CORE-7714 / Kit KIT-GRO450-THM-14 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO450-THM-14 within tolerance ±0.0060 mm TIR.

BOM-77-1.14: PN-GRO450-BRG-CER-7701 / Kit KIT-GRO450-SPN-1 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-77-1.15: PN-GRO450-CAM-GRIP-7702 / Kit KIT-GRO450-ATC-2 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO450-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G450 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G450. Verify part numbers before closing Work Orders.

BOM-77-2.1: PN-GRO450-VLV-PROP-7703 / Kit KIT-GRO450-HYD-3 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-77-2.2: PN-GRO450-LIN-COIL-7705 / Kit KIT-GRO450-XAX-5 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-77-2.3: PN-GRO450-CYL-SEAL-7707 / Kit KIT-GRO450-ZAX-7 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-77-2.4: PN-GRO450-BRK-DISC-7709 / Kit KIT-GRO450-CAX-9 (Grob-Aerospace-G450) | PM Interval: 1250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-77-2.5: PN-GRO450-IGBT-MOD-7711 / Kit KIT-GRO450-PLC-11 (Grob-Aerospace-G450) | PM Interval: 1500 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-77-2.6: PN-GRO450-GASK-VIT-7713 / Kit KIT-GRO450-VAC-13 (Grob-Aerospace-G450) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-77-2.7: PN-GRO450-BRG-CER-7701 / Kit KIT-GRO450-SPN-1 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-77-2.8: PN-GRO450-VLV-PROP-7703 / Kit KIT-GRO450-HYD-3 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-77-2.9: PN-GRO450-LIN-COIL-7705 / Kit KIT-GRO450-XAX-5 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-77-2.10: PN-GRO450-CYL-SEAL-7707 / Kit KIT-GRO450-ZAX-7 (Grob-Aerospace-G450) | PM Interval: 1250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-77-2.11: PN-GRO450-BRK-DISC-7709 / Kit KIT-GRO450-CAX-9 (Grob-Aerospace-G450) | PM Interval: 1500 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-77-2.12: PN-GRO450-IGBT-MOD-7711 / Kit KIT-GRO450-PLC-11 (Grob-Aerospace-G450) | PM Interval: 250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-77-2.13: PN-GRO450-GASK-VIT-7713 / Kit KIT-GRO450-VAC-13 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-77-2.14: PN-GRO450-BRG-CER-7701 / Kit KIT-GRO450-SPN-1 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-77-2.15: PN-GRO450-VLV-PROP-7703 / Kit KIT-GRO450-HYD-3 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO450-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G450 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G450. Verify part numbers before closing Work Orders.

BOM-77-3.1: PN-GRO450-PMP-PLG-7704 / Kit KIT-GRO450-CLN-4 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO450-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-77-3.2: PN-GRO450-CYL-SEAL-7707 / Kit KIT-GRO450-ZAX-7 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-77-3.3: PN-GRO450-GLASS-RD-7710 / Kit KIT-GRO450-OPT-10 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO450-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-77-3.4: PN-GRO450-GASK-VIT-7713 / Kit KIT-GRO450-VAC-13 (Grob-Aerospace-G450) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-77-3.5: PN-GRO450-CAM-GRIP-7702 / Kit KIT-GRO450-ATC-2 (Grob-Aerospace-G450) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO450-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-77-3.6: PN-GRO450-LIN-COIL-7705 / Kit KIT-GRO450-XAX-5 (Grob-Aerospace-G450) | PM Interval: 250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-77-3.7: PN-GRO450-TRQ-STAT-7708 / Kit KIT-GRO450-BAX-8 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO450-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-77-3.8: PN-GRO450-IGBT-MOD-7711 / Kit KIT-GRO450-PLC-11 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-77-3.9: PN-GRO450-HEX-CORE-7714 / Kit KIT-GRO450-THM-14 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO450-THM-14 within tolerance ±0.0060 mm TIR.

BOM-77-3.10: PN-GRO450-VLV-PROP-7703 / Kit KIT-GRO450-HYD-3 (Grob-Aerospace-G450) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-77-3.11: PN-GRO450-BSW-NUT-7706 / Kit KIT-GRO450-YAX-6 (Grob-Aerospace-G450) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO450-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-77-3.12: PN-GRO450-BRK-DISC-7709 / Kit KIT-GRO450-CAX-9 (Grob-Aerospace-G450) | PM Interval: 250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-77-3.13: PN-GRO450-INJ-MET-7712 / Kit KIT-GRO450-LUB-12 (Grob-Aerospace-G450) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO450-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-77-3.14: PN-GRO450-BRG-CER-7701 / Kit KIT-GRO450-SPN-1 (Grob-Aerospace-G450) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-77-3.15: PN-GRO450-PMP-PLG-7704 / Kit KIT-GRO450-CLN-4 (Grob-Aerospace-G450) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO450-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G500 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G500. Verify part numbers before closing Work Orders.

BOM-78-1.1: PN-GRO500-CAM-GRIP-7802 / Kit KIT-GRO500-ATC-2 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO500-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-78-1.2: PN-GRO500-VLV-PROP-7803 / Kit KIT-GRO500-HYD-3 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-78-1.3: PN-GRO500-PMP-PLG-7804 / Kit KIT-GRO500-CLN-4 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO500-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-78-1.4: PN-GRO500-LIN-COIL-7805 / Kit KIT-GRO500-XAX-5 (Grob-Aerospace-G500) | PM Interval: 1250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-78-1.5: PN-GRO500-BSW-NUT-7806 / Kit KIT-GRO500-YAX-6 (Grob-Aerospace-G500) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO500-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-78-1.6: PN-GRO500-CYL-SEAL-7807 / Kit KIT-GRO500-ZAX-7 (Grob-Aerospace-G500) | PM Interval: 250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-78-1.7: PN-GRO500-TRQ-STAT-7808 / Kit KIT-GRO500-BAX-8 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO500-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-78-1.8: PN-GRO500-BRK-DISC-7809 / Kit KIT-GRO500-CAX-9 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-78-1.9: PN-GRO500-GLASS-RD-7810 / Kit KIT-GRO500-OPT-10 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO500-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-78-1.10: PN-GRO500-IGBT-MOD-7811 / Kit KIT-GRO500-PLC-11 (Grob-Aerospace-G500) | PM Interval: 1250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-78-1.11: PN-GRO500-INJ-MET-7812 / Kit KIT-GRO500-LUB-12 (Grob-Aerospace-G500) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO500-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-78-1.12: PN-GRO500-GASK-VIT-7813 / Kit KIT-GRO500-VAC-13 (Grob-Aerospace-G500) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-78-1.13: PN-GRO500-HEX-CORE-7814 / Kit KIT-GRO500-THM-14 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO500-THM-14 within tolerance ±0.0020 mm TIR.

BOM-78-1.14: PN-GRO500-BRG-CER-7801 / Kit KIT-GRO500-SPN-1 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-78-1.15: PN-GRO500-CAM-GRIP-7802 / Kit KIT-GRO500-ATC-2 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO500-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G500 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G500. Verify part numbers before closing Work Orders.

BOM-78-2.1: PN-GRO500-VLV-PROP-7803 / Kit KIT-GRO500-HYD-3 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-78-2.2: PN-GRO500-LIN-COIL-7805 / Kit KIT-GRO500-XAX-5 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-78-2.3: PN-GRO500-CYL-SEAL-7807 / Kit KIT-GRO500-ZAX-7 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-78-2.4: PN-GRO500-BRK-DISC-7809 / Kit KIT-GRO500-CAX-9 (Grob-Aerospace-G500) | PM Interval: 1250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-78-2.5: PN-GRO500-IGBT-MOD-7811 / Kit KIT-GRO500-PLC-11 (Grob-Aerospace-G500) | PM Interval: 1500 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-78-2.6: PN-GRO500-GASK-VIT-7813 / Kit KIT-GRO500-VAC-13 (Grob-Aerospace-G500) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-78-2.7: PN-GRO500-BRG-CER-7801 / Kit KIT-GRO500-SPN-1 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-78-2.8: PN-GRO500-VLV-PROP-7803 / Kit KIT-GRO500-HYD-3 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-78-2.9: PN-GRO500-LIN-COIL-7805 / Kit KIT-GRO500-XAX-5 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-78-2.10: PN-GRO500-CYL-SEAL-7807 / Kit KIT-GRO500-ZAX-7 (Grob-Aerospace-G500) | PM Interval: 1250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-78-2.11: PN-GRO500-BRK-DISC-7809 / Kit KIT-GRO500-CAX-9 (Grob-Aerospace-G500) | PM Interval: 1500 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-78-2.12: PN-GRO500-IGBT-MOD-7811 / Kit KIT-GRO500-PLC-11 (Grob-Aerospace-G500) | PM Interval: 250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-78-2.13: PN-GRO500-GASK-VIT-7813 / Kit KIT-GRO500-VAC-13 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-78-2.14: PN-GRO500-BRG-CER-7801 / Kit KIT-GRO500-SPN-1 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-78-2.15: PN-GRO500-VLV-PROP-7803 / Kit KIT-GRO500-HYD-3 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO500-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G500 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G500. Verify part numbers before closing Work Orders.

BOM-78-3.1: PN-GRO500-PMP-PLG-7804 / Kit KIT-GRO500-CLN-4 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO500-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-78-3.2: PN-GRO500-CYL-SEAL-7807 / Kit KIT-GRO500-ZAX-7 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-78-3.3: PN-GRO500-GLASS-RD-7810 / Kit KIT-GRO500-OPT-10 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO500-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-78-3.4: PN-GRO500-GASK-VIT-7813 / Kit KIT-GRO500-VAC-13 (Grob-Aerospace-G500) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-78-3.5: PN-GRO500-CAM-GRIP-7802 / Kit KIT-GRO500-ATC-2 (Grob-Aerospace-G500) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO500-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-78-3.6: PN-GRO500-LIN-COIL-7805 / Kit KIT-GRO500-XAX-5 (Grob-Aerospace-G500) | PM Interval: 250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-78-3.7: PN-GRO500-TRQ-STAT-7808 / Kit KIT-GRO500-BAX-8 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO500-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-78-3.8: PN-GRO500-IGBT-MOD-7811 / Kit KIT-GRO500-PLC-11 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-78-3.9: PN-GRO500-HEX-CORE-7814 / Kit KIT-GRO500-THM-14 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO500-THM-14 within tolerance ±0.0020 mm TIR.

BOM-78-3.10: PN-GRO500-VLV-PROP-7803 / Kit KIT-GRO500-HYD-3 (Grob-Aerospace-G500) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-78-3.11: PN-GRO500-BSW-NUT-7806 / Kit KIT-GRO500-YAX-6 (Grob-Aerospace-G500) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO500-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-78-3.12: PN-GRO500-BRK-DISC-7809 / Kit KIT-GRO500-CAX-9 (Grob-Aerospace-G500) | PM Interval: 250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-78-3.13: PN-GRO500-INJ-MET-7812 / Kit KIT-GRO500-LUB-12 (Grob-Aerospace-G500) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO500-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-78-3.14: PN-GRO500-BRG-CER-7801 / Kit KIT-GRO500-SPN-1 (Grob-Aerospace-G500) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-78-3.15: PN-GRO500-PMP-PLG-7804 / Kit KIT-GRO500-CLN-4 (Grob-Aerospace-G500) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO500-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G550 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G550. Verify part numbers before closing Work Orders.

BOM-79-1.1: PN-GRO550-CAM-GRIP-7902 / Kit KIT-GRO550-ATC-2 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO550-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-79-1.2: PN-GRO550-VLV-PROP-7903 / Kit KIT-GRO550-HYD-3 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-79-1.3: PN-GRO550-PMP-PLG-7904 / Kit KIT-GRO550-CLN-4 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO550-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-79-1.4: PN-GRO550-LIN-COIL-7905 / Kit KIT-GRO550-XAX-5 (Grob-Aerospace-G550) | PM Interval: 1250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-79-1.5: PN-GRO550-BSW-NUT-7906 / Kit KIT-GRO550-YAX-6 (Grob-Aerospace-G550) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO550-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-79-1.6: PN-GRO550-CYL-SEAL-7907 / Kit KIT-GRO550-ZAX-7 (Grob-Aerospace-G550) | PM Interval: 250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-79-1.7: PN-GRO550-TRQ-STAT-7908 / Kit KIT-GRO550-BAX-8 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO550-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-79-1.8: PN-GRO550-BRK-DISC-7909 / Kit KIT-GRO550-CAX-9 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-79-1.9: PN-GRO550-GLASS-RD-7910 / Kit KIT-GRO550-OPT-10 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO550-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-79-1.10: PN-GRO550-IGBT-MOD-7911 / Kit KIT-GRO550-PLC-11 (Grob-Aerospace-G550) | PM Interval: 1250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-79-1.11: PN-GRO550-INJ-MET-7912 / Kit KIT-GRO550-LUB-12 (Grob-Aerospace-G550) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO550-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-79-1.12: PN-GRO550-GASK-VIT-7913 / Kit KIT-GRO550-VAC-13 (Grob-Aerospace-G550) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-79-1.13: PN-GRO550-HEX-CORE-7914 / Kit KIT-GRO550-THM-14 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO550-THM-14 within tolerance ±0.0040 mm TIR.

BOM-79-1.14: PN-GRO550-BRG-CER-7901 / Kit KIT-GRO550-SPN-1 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-79-1.15: PN-GRO550-CAM-GRIP-7902 / Kit KIT-GRO550-ATC-2 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO550-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G550 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G550. Verify part numbers before closing Work Orders.

BOM-79-2.1: PN-GRO550-VLV-PROP-7903 / Kit KIT-GRO550-HYD-3 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-79-2.2: PN-GRO550-LIN-COIL-7905 / Kit KIT-GRO550-XAX-5 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-79-2.3: PN-GRO550-CYL-SEAL-7907 / Kit KIT-GRO550-ZAX-7 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-79-2.4: PN-GRO550-BRK-DISC-7909 / Kit KIT-GRO550-CAX-9 (Grob-Aerospace-G550) | PM Interval: 1250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-79-2.5: PN-GRO550-IGBT-MOD-7911 / Kit KIT-GRO550-PLC-11 (Grob-Aerospace-G550) | PM Interval: 1500 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-79-2.6: PN-GRO550-GASK-VIT-7913 / Kit KIT-GRO550-VAC-13 (Grob-Aerospace-G550) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-79-2.7: PN-GRO550-BRG-CER-7901 / Kit KIT-GRO550-SPN-1 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-79-2.8: PN-GRO550-VLV-PROP-7903 / Kit KIT-GRO550-HYD-3 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-79-2.9: PN-GRO550-LIN-COIL-7905 / Kit KIT-GRO550-XAX-5 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-79-2.10: PN-GRO550-CYL-SEAL-7907 / Kit KIT-GRO550-ZAX-7 (Grob-Aerospace-G550) | PM Interval: 1250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-79-2.11: PN-GRO550-BRK-DISC-7909 / Kit KIT-GRO550-CAX-9 (Grob-Aerospace-G550) | PM Interval: 1500 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-79-2.12: PN-GRO550-IGBT-MOD-7911 / Kit KIT-GRO550-PLC-11 (Grob-Aerospace-G550) | PM Interval: 250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-79-2.13: PN-GRO550-GASK-VIT-7913 / Kit KIT-GRO550-VAC-13 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-79-2.14: PN-GRO550-BRG-CER-7901 / Kit KIT-GRO550-SPN-1 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-79-2.15: PN-GRO550-VLV-PROP-7903 / Kit KIT-GRO550-HYD-3 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO550-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G550 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G550. Verify part numbers before closing Work Orders.

BOM-79-3.1: PN-GRO550-PMP-PLG-7904 / Kit KIT-GRO550-CLN-4 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO550-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-79-3.2: PN-GRO550-CYL-SEAL-7907 / Kit KIT-GRO550-ZAX-7 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-79-3.3: PN-GRO550-GLASS-RD-7910 / Kit KIT-GRO550-OPT-10 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO550-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-79-3.4: PN-GRO550-GASK-VIT-7913 / Kit KIT-GRO550-VAC-13 (Grob-Aerospace-G550) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-79-3.5: PN-GRO550-CAM-GRIP-7902 / Kit KIT-GRO550-ATC-2 (Grob-Aerospace-G550) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO550-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-79-3.6: PN-GRO550-LIN-COIL-7905 / Kit KIT-GRO550-XAX-5 (Grob-Aerospace-G550) | PM Interval: 250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-79-3.7: PN-GRO550-TRQ-STAT-7908 / Kit KIT-GRO550-BAX-8 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO550-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-79-3.8: PN-GRO550-IGBT-MOD-7911 / Kit KIT-GRO550-PLC-11 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-79-3.9: PN-GRO550-HEX-CORE-7914 / Kit KIT-GRO550-THM-14 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO550-THM-14 within tolerance ±0.0040 mm TIR.

BOM-79-3.10: PN-GRO550-VLV-PROP-7903 / Kit KIT-GRO550-HYD-3 (Grob-Aerospace-G550) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-79-3.11: PN-GRO550-BSW-NUT-7906 / Kit KIT-GRO550-YAX-6 (Grob-Aerospace-G550) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO550-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-79-3.12: PN-GRO550-BRK-DISC-7909 / Kit KIT-GRO550-CAX-9 (Grob-Aerospace-G550) | PM Interval: 250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-79-3.13: PN-GRO550-INJ-MET-7912 / Kit KIT-GRO550-LUB-12 (Grob-Aerospace-G550) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO550-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-79-3.14: PN-GRO550-BRG-CER-7901 / Kit KIT-GRO550-SPN-1 (Grob-Aerospace-G550) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-79-3.15: PN-GRO550-PMP-PLG-7904 / Kit KIT-GRO550-CLN-4 (Grob-Aerospace-G550) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO550-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G600 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G600. Verify part numbers before closing Work Orders.

BOM-80-1.1: PN-GRO600-CAM-GRIP-8002 / Kit KIT-GRO600-ATC-2 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO600-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-80-1.2: PN-GRO600-VLV-PROP-8003 / Kit KIT-GRO600-HYD-3 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-80-1.3: PN-GRO600-PMP-PLG-8004 / Kit KIT-GRO600-CLN-4 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO600-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-80-1.4: PN-GRO600-LIN-COIL-8005 / Kit KIT-GRO600-XAX-5 (Grob-Aerospace-G600) | PM Interval: 1250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-80-1.5: PN-GRO600-BSW-NUT-8006 / Kit KIT-GRO600-YAX-6 (Grob-Aerospace-G600) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO600-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-80-1.6: PN-GRO600-CYL-SEAL-8007 / Kit KIT-GRO600-ZAX-7 (Grob-Aerospace-G600) | PM Interval: 250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-80-1.7: PN-GRO600-TRQ-STAT-8008 / Kit KIT-GRO600-BAX-8 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO600-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-80-1.8: PN-GRO600-BRK-DISC-8009 / Kit KIT-GRO600-CAX-9 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-80-1.9: PN-GRO600-GLASS-RD-8010 / Kit KIT-GRO600-OPT-10 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO600-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-80-1.10: PN-GRO600-IGBT-MOD-8011 / Kit KIT-GRO600-PLC-11 (Grob-Aerospace-G600) | PM Interval: 1250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-80-1.11: PN-GRO600-INJ-MET-8012 / Kit KIT-GRO600-LUB-12 (Grob-Aerospace-G600) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO600-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-80-1.12: PN-GRO600-GASK-VIT-8013 / Kit KIT-GRO600-VAC-13 (Grob-Aerospace-G600) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-80-1.13: PN-GRO600-HEX-CORE-8014 / Kit KIT-GRO600-THM-14 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO600-THM-14 within tolerance ±0.0060 mm TIR.

BOM-80-1.14: PN-GRO600-BRG-CER-8001 / Kit KIT-GRO600-SPN-1 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-80-1.15: PN-GRO600-CAM-GRIP-8002 / Kit KIT-GRO600-ATC-2 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO600-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G600 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G600. Verify part numbers before closing Work Orders.

BOM-80-2.1: PN-GRO600-VLV-PROP-8003 / Kit KIT-GRO600-HYD-3 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-80-2.2: PN-GRO600-LIN-COIL-8005 / Kit KIT-GRO600-XAX-5 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-80-2.3: PN-GRO600-CYL-SEAL-8007 / Kit KIT-GRO600-ZAX-7 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-80-2.4: PN-GRO600-BRK-DISC-8009 / Kit KIT-GRO600-CAX-9 (Grob-Aerospace-G600) | PM Interval: 1250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-80-2.5: PN-GRO600-IGBT-MOD-8011 / Kit KIT-GRO600-PLC-11 (Grob-Aerospace-G600) | PM Interval: 1500 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-80-2.6: PN-GRO600-GASK-VIT-8013 / Kit KIT-GRO600-VAC-13 (Grob-Aerospace-G600) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-80-2.7: PN-GRO600-BRG-CER-8001 / Kit KIT-GRO600-SPN-1 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-80-2.8: PN-GRO600-VLV-PROP-8003 / Kit KIT-GRO600-HYD-3 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-80-2.9: PN-GRO600-LIN-COIL-8005 / Kit KIT-GRO600-XAX-5 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-80-2.10: PN-GRO600-CYL-SEAL-8007 / Kit KIT-GRO600-ZAX-7 (Grob-Aerospace-G600) | PM Interval: 1250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-80-2.11: PN-GRO600-BRK-DISC-8009 / Kit KIT-GRO600-CAX-9 (Grob-Aerospace-G600) | PM Interval: 1500 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-80-2.12: PN-GRO600-IGBT-MOD-8011 / Kit KIT-GRO600-PLC-11 (Grob-Aerospace-G600) | PM Interval: 250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-80-2.13: PN-GRO600-GASK-VIT-8013 / Kit KIT-GRO600-VAC-13 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-80-2.14: PN-GRO600-BRG-CER-8001 / Kit KIT-GRO600-SPN-1 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-80-2.15: PN-GRO600-VLV-PROP-8003 / Kit KIT-GRO600-HYD-3 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO600-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G600 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G600. Verify part numbers before closing Work Orders.

BOM-80-3.1: PN-GRO600-PMP-PLG-8004 / Kit KIT-GRO600-CLN-4 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO600-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-80-3.2: PN-GRO600-CYL-SEAL-8007 / Kit KIT-GRO600-ZAX-7 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-80-3.3: PN-GRO600-GLASS-RD-8010 / Kit KIT-GRO600-OPT-10 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO600-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-80-3.4: PN-GRO600-GASK-VIT-8013 / Kit KIT-GRO600-VAC-13 (Grob-Aerospace-G600) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-80-3.5: PN-GRO600-CAM-GRIP-8002 / Kit KIT-GRO600-ATC-2 (Grob-Aerospace-G600) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO600-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-80-3.6: PN-GRO600-LIN-COIL-8005 / Kit KIT-GRO600-XAX-5 (Grob-Aerospace-G600) | PM Interval: 250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-80-3.7: PN-GRO600-TRQ-STAT-8008 / Kit KIT-GRO600-BAX-8 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO600-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-80-3.8: PN-GRO600-IGBT-MOD-8011 / Kit KIT-GRO600-PLC-11 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-80-3.9: PN-GRO600-HEX-CORE-8014 / Kit KIT-GRO600-THM-14 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO600-THM-14 within tolerance ±0.0060 mm TIR.

BOM-80-3.10: PN-GRO600-VLV-PROP-8003 / Kit KIT-GRO600-HYD-3 (Grob-Aerospace-G600) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-80-3.11: PN-GRO600-BSW-NUT-8006 / Kit KIT-GRO600-YAX-6 (Grob-Aerospace-G600) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO600-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-80-3.12: PN-GRO600-BRK-DISC-8009 / Kit KIT-GRO600-CAX-9 (Grob-Aerospace-G600) | PM Interval: 250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-80-3.13: PN-GRO600-INJ-MET-8012 / Kit KIT-GRO600-LUB-12 (Grob-Aerospace-G600) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO600-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-80-3.14: PN-GRO600-BRG-CER-8001 / Kit KIT-GRO600-SPN-1 (Grob-Aerospace-G600) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-80-3.15: PN-GRO600-PMP-PLG-8004 / Kit KIT-GRO600-CLN-4 (Grob-Aerospace-G600) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO600-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C150 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C150. Verify part numbers before closing Work Orders.

BOM-81-1.1: PN-HER150-CAM-GRIP-8102 / Kit KIT-HER150-ATC-2 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER150-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-81-1.2: PN-HER150-VLV-PROP-8103 / Kit KIT-HER150-HYD-3 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-81-1.3: PN-HER150-PMP-PLG-8104 / Kit KIT-HER150-CLN-4 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER150-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-81-1.4: PN-HER150-LIN-COIL-8105 / Kit KIT-HER150-XAX-5 (Hermle-Performance-C150) | PM Interval: 1250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-81-1.5: PN-HER150-BSW-NUT-8106 / Kit KIT-HER150-YAX-6 (Hermle-Performance-C150) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER150-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-81-1.6: PN-HER150-CYL-SEAL-8107 / Kit KIT-HER150-ZAX-7 (Hermle-Performance-C150) | PM Interval: 250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-81-1.7: PN-HER150-TRQ-STAT-8108 / Kit KIT-HER150-BAX-8 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER150-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-81-1.8: PN-HER150-BRK-DISC-8109 / Kit KIT-HER150-CAX-9 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-81-1.9: PN-HER150-GLASS-RD-8110 / Kit KIT-HER150-OPT-10 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER150-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-81-1.10: PN-HER150-IGBT-MOD-8111 / Kit KIT-HER150-PLC-11 (Hermle-Performance-C150) | PM Interval: 1250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-81-1.11: PN-HER150-INJ-MET-8112 / Kit KIT-HER150-LUB-12 (Hermle-Performance-C150) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER150-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-81-1.12: PN-HER150-GASK-VIT-8113 / Kit KIT-HER150-VAC-13 (Hermle-Performance-C150) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-81-1.13: PN-HER150-HEX-CORE-8114 / Kit KIT-HER150-THM-14 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER150-THM-14 within tolerance ±0.0020 mm TIR.

BOM-81-1.14: PN-HER150-BRG-CER-8101 / Kit KIT-HER150-SPN-1 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-81-1.15: PN-HER150-CAM-GRIP-8102 / Kit KIT-HER150-ATC-2 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER150-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C150 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C150. Verify part numbers before closing Work Orders.

BOM-81-2.1: PN-HER150-VLV-PROP-8103 / Kit KIT-HER150-HYD-3 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-81-2.2: PN-HER150-LIN-COIL-8105 / Kit KIT-HER150-XAX-5 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-81-2.3: PN-HER150-CYL-SEAL-8107 / Kit KIT-HER150-ZAX-7 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-81-2.4: PN-HER150-BRK-DISC-8109 / Kit KIT-HER150-CAX-9 (Hermle-Performance-C150) | PM Interval: 1250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-81-2.5: PN-HER150-IGBT-MOD-8111 / Kit KIT-HER150-PLC-11 (Hermle-Performance-C150) | PM Interval: 1500 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-81-2.6: PN-HER150-GASK-VIT-8113 / Kit KIT-HER150-VAC-13 (Hermle-Performance-C150) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-81-2.7: PN-HER150-BRG-CER-8101 / Kit KIT-HER150-SPN-1 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-81-2.8: PN-HER150-VLV-PROP-8103 / Kit KIT-HER150-HYD-3 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-81-2.9: PN-HER150-LIN-COIL-8105 / Kit KIT-HER150-XAX-5 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-81-2.10: PN-HER150-CYL-SEAL-8107 / Kit KIT-HER150-ZAX-7 (Hermle-Performance-C150) | PM Interval: 1250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-81-2.11: PN-HER150-BRK-DISC-8109 / Kit KIT-HER150-CAX-9 (Hermle-Performance-C150) | PM Interval: 1500 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-81-2.12: PN-HER150-IGBT-MOD-8111 / Kit KIT-HER150-PLC-11 (Hermle-Performance-C150) | PM Interval: 250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-81-2.13: PN-HER150-GASK-VIT-8113 / Kit KIT-HER150-VAC-13 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-81-2.14: PN-HER150-BRG-CER-8101 / Kit KIT-HER150-SPN-1 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-81-2.15: PN-HER150-VLV-PROP-8103 / Kit KIT-HER150-HYD-3 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER150-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C150 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C150. Verify part numbers before closing Work Orders.

BOM-81-3.1: PN-HER150-PMP-PLG-8104 / Kit KIT-HER150-CLN-4 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER150-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-81-3.2: PN-HER150-CYL-SEAL-8107 / Kit KIT-HER150-ZAX-7 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-81-3.3: PN-HER150-GLASS-RD-8110 / Kit KIT-HER150-OPT-10 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER150-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-81-3.4: PN-HER150-GASK-VIT-8113 / Kit KIT-HER150-VAC-13 (Hermle-Performance-C150) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-81-3.5: PN-HER150-CAM-GRIP-8102 / Kit KIT-HER150-ATC-2 (Hermle-Performance-C150) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER150-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-81-3.6: PN-HER150-LIN-COIL-8105 / Kit KIT-HER150-XAX-5 (Hermle-Performance-C150) | PM Interval: 250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-81-3.7: PN-HER150-TRQ-STAT-8108 / Kit KIT-HER150-BAX-8 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER150-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-81-3.8: PN-HER150-IGBT-MOD-8111 / Kit KIT-HER150-PLC-11 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-81-3.9: PN-HER150-HEX-CORE-8114 / Kit KIT-HER150-THM-14 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER150-THM-14 within tolerance ±0.0020 mm TIR.

BOM-81-3.10: PN-HER150-VLV-PROP-8103 / Kit KIT-HER150-HYD-3 (Hermle-Performance-C150) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-81-3.11: PN-HER150-BSW-NUT-8106 / Kit KIT-HER150-YAX-6 (Hermle-Performance-C150) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER150-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-81-3.12: PN-HER150-BRK-DISC-8109 / Kit KIT-HER150-CAX-9 (Hermle-Performance-C150) | PM Interval: 250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-81-3.13: PN-HER150-INJ-MET-8112 / Kit KIT-HER150-LUB-12 (Hermle-Performance-C150) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER150-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-81-3.14: PN-HER150-BRG-CER-8101 / Kit KIT-HER150-SPN-1 (Hermle-Performance-C150) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-81-3.15: PN-HER150-PMP-PLG-8104 / Kit KIT-HER150-CLN-4 (Hermle-Performance-C150) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER150-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C200 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C200. Verify part numbers before closing Work Orders.

BOM-82-1.1: PN-HER200-CAM-GRIP-8202 / Kit KIT-HER200-ATC-2 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER200-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-82-1.2: PN-HER200-VLV-PROP-8203 / Kit KIT-HER200-HYD-3 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-82-1.3: PN-HER200-PMP-PLG-8204 / Kit KIT-HER200-CLN-4 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER200-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-82-1.4: PN-HER200-LIN-COIL-8205 / Kit KIT-HER200-XAX-5 (Hermle-Performance-C200) | PM Interval: 1250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-82-1.5: PN-HER200-BSW-NUT-8206 / Kit KIT-HER200-YAX-6 (Hermle-Performance-C200) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER200-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-82-1.6: PN-HER200-CYL-SEAL-8207 / Kit KIT-HER200-ZAX-7 (Hermle-Performance-C200) | PM Interval: 250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-82-1.7: PN-HER200-TRQ-STAT-8208 / Kit KIT-HER200-BAX-8 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER200-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-82-1.8: PN-HER200-BRK-DISC-8209 / Kit KIT-HER200-CAX-9 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-82-1.9: PN-HER200-GLASS-RD-8210 / Kit KIT-HER200-OPT-10 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER200-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-82-1.10: PN-HER200-IGBT-MOD-8211 / Kit KIT-HER200-PLC-11 (Hermle-Performance-C200) | PM Interval: 1250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-82-1.11: PN-HER200-INJ-MET-8212 / Kit KIT-HER200-LUB-12 (Hermle-Performance-C200) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER200-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-82-1.12: PN-HER200-GASK-VIT-8213 / Kit KIT-HER200-VAC-13 (Hermle-Performance-C200) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-82-1.13: PN-HER200-HEX-CORE-8214 / Kit KIT-HER200-THM-14 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER200-THM-14 within tolerance ±0.0040 mm TIR.

BOM-82-1.14: PN-HER200-BRG-CER-8201 / Kit KIT-HER200-SPN-1 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-82-1.15: PN-HER200-CAM-GRIP-8202 / Kit KIT-HER200-ATC-2 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER200-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C200 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C200. Verify part numbers before closing Work Orders.

BOM-82-2.1: PN-HER200-VLV-PROP-8203 / Kit KIT-HER200-HYD-3 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-82-2.2: PN-HER200-LIN-COIL-8205 / Kit KIT-HER200-XAX-5 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-82-2.3: PN-HER200-CYL-SEAL-8207 / Kit KIT-HER200-ZAX-7 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-82-2.4: PN-HER200-BRK-DISC-8209 / Kit KIT-HER200-CAX-9 (Hermle-Performance-C200) | PM Interval: 1250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-82-2.5: PN-HER200-IGBT-MOD-8211 / Kit KIT-HER200-PLC-11 (Hermle-Performance-C200) | PM Interval: 1500 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-82-2.6: PN-HER200-GASK-VIT-8213 / Kit KIT-HER200-VAC-13 (Hermle-Performance-C200) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-82-2.7: PN-HER200-BRG-CER-8201 / Kit KIT-HER200-SPN-1 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-82-2.8: PN-HER200-VLV-PROP-8203 / Kit KIT-HER200-HYD-3 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-82-2.9: PN-HER200-LIN-COIL-8205 / Kit KIT-HER200-XAX-5 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-82-2.10: PN-HER200-CYL-SEAL-8207 / Kit KIT-HER200-ZAX-7 (Hermle-Performance-C200) | PM Interval: 1250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-82-2.11: PN-HER200-BRK-DISC-8209 / Kit KIT-HER200-CAX-9 (Hermle-Performance-C200) | PM Interval: 1500 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-82-2.12: PN-HER200-IGBT-MOD-8211 / Kit KIT-HER200-PLC-11 (Hermle-Performance-C200) | PM Interval: 250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-82-2.13: PN-HER200-GASK-VIT-8213 / Kit KIT-HER200-VAC-13 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-82-2.14: PN-HER200-BRG-CER-8201 / Kit KIT-HER200-SPN-1 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-82-2.15: PN-HER200-VLV-PROP-8203 / Kit KIT-HER200-HYD-3 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER200-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C200 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C200. Verify part numbers before closing Work Orders.

BOM-82-3.1: PN-HER200-PMP-PLG-8204 / Kit KIT-HER200-CLN-4 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER200-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-82-3.2: PN-HER200-CYL-SEAL-8207 / Kit KIT-HER200-ZAX-7 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-82-3.3: PN-HER200-GLASS-RD-8210 / Kit KIT-HER200-OPT-10 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER200-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-82-3.4: PN-HER200-GASK-VIT-8213 / Kit KIT-HER200-VAC-13 (Hermle-Performance-C200) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-82-3.5: PN-HER200-CAM-GRIP-8202 / Kit KIT-HER200-ATC-2 (Hermle-Performance-C200) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER200-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-82-3.6: PN-HER200-LIN-COIL-8205 / Kit KIT-HER200-XAX-5 (Hermle-Performance-C200) | PM Interval: 250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-82-3.7: PN-HER200-TRQ-STAT-8208 / Kit KIT-HER200-BAX-8 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER200-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-82-3.8: PN-HER200-IGBT-MOD-8211 / Kit KIT-HER200-PLC-11 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-82-3.9: PN-HER200-HEX-CORE-8214 / Kit KIT-HER200-THM-14 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER200-THM-14 within tolerance ±0.0040 mm TIR.

BOM-82-3.10: PN-HER200-VLV-PROP-8203 / Kit KIT-HER200-HYD-3 (Hermle-Performance-C200) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-82-3.11: PN-HER200-BSW-NUT-8206 / Kit KIT-HER200-YAX-6 (Hermle-Performance-C200) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER200-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-82-3.12: PN-HER200-BRK-DISC-8209 / Kit KIT-HER200-CAX-9 (Hermle-Performance-C200) | PM Interval: 250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-82-3.13: PN-HER200-INJ-MET-8212 / Kit KIT-HER200-LUB-12 (Hermle-Performance-C200) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER200-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-82-3.14: PN-HER200-BRG-CER-8201 / Kit KIT-HER200-SPN-1 (Hermle-Performance-C200) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-82-3.15: PN-HER200-PMP-PLG-8204 / Kit KIT-HER200-CLN-4 (Hermle-Performance-C200) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER200-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C250 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C250. Verify part numbers before closing Work Orders.

BOM-83-1.1: PN-HER250-CAM-GRIP-8302 / Kit KIT-HER250-ATC-2 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER250-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-83-1.2: PN-HER250-VLV-PROP-8303 / Kit KIT-HER250-HYD-3 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-83-1.3: PN-HER250-PMP-PLG-8304 / Kit KIT-HER250-CLN-4 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER250-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-83-1.4: PN-HER250-LIN-COIL-8305 / Kit KIT-HER250-XAX-5 (Hermle-Performance-C250) | PM Interval: 1250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-83-1.5: PN-HER250-BSW-NUT-8306 / Kit KIT-HER250-YAX-6 (Hermle-Performance-C250) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER250-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-83-1.6: PN-HER250-CYL-SEAL-8307 / Kit KIT-HER250-ZAX-7 (Hermle-Performance-C250) | PM Interval: 250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-83-1.7: PN-HER250-TRQ-STAT-8308 / Kit KIT-HER250-BAX-8 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER250-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-83-1.8: PN-HER250-BRK-DISC-8309 / Kit KIT-HER250-CAX-9 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-83-1.9: PN-HER250-GLASS-RD-8310 / Kit KIT-HER250-OPT-10 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER250-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-83-1.10: PN-HER250-IGBT-MOD-8311 / Kit KIT-HER250-PLC-11 (Hermle-Performance-C250) | PM Interval: 1250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-83-1.11: PN-HER250-INJ-MET-8312 / Kit KIT-HER250-LUB-12 (Hermle-Performance-C250) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER250-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-83-1.12: PN-HER250-GASK-VIT-8313 / Kit KIT-HER250-VAC-13 (Hermle-Performance-C250) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-83-1.13: PN-HER250-HEX-CORE-8314 / Kit KIT-HER250-THM-14 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER250-THM-14 within tolerance ±0.0060 mm TIR.

BOM-83-1.14: PN-HER250-BRG-CER-8301 / Kit KIT-HER250-SPN-1 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-83-1.15: PN-HER250-CAM-GRIP-8302 / Kit KIT-HER250-ATC-2 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER250-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C250 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C250. Verify part numbers before closing Work Orders.

BOM-83-2.1: PN-HER250-VLV-PROP-8303 / Kit KIT-HER250-HYD-3 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-83-2.2: PN-HER250-LIN-COIL-8305 / Kit KIT-HER250-XAX-5 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-83-2.3: PN-HER250-CYL-SEAL-8307 / Kit KIT-HER250-ZAX-7 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-83-2.4: PN-HER250-BRK-DISC-8309 / Kit KIT-HER250-CAX-9 (Hermle-Performance-C250) | PM Interval: 1250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-83-2.5: PN-HER250-IGBT-MOD-8311 / Kit KIT-HER250-PLC-11 (Hermle-Performance-C250) | PM Interval: 1500 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-83-2.6: PN-HER250-GASK-VIT-8313 / Kit KIT-HER250-VAC-13 (Hermle-Performance-C250) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-83-2.7: PN-HER250-BRG-CER-8301 / Kit KIT-HER250-SPN-1 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-83-2.8: PN-HER250-VLV-PROP-8303 / Kit KIT-HER250-HYD-3 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-83-2.9: PN-HER250-LIN-COIL-8305 / Kit KIT-HER250-XAX-5 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-83-2.10: PN-HER250-CYL-SEAL-8307 / Kit KIT-HER250-ZAX-7 (Hermle-Performance-C250) | PM Interval: 1250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-83-2.11: PN-HER250-BRK-DISC-8309 / Kit KIT-HER250-CAX-9 (Hermle-Performance-C250) | PM Interval: 1500 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-83-2.12: PN-HER250-IGBT-MOD-8311 / Kit KIT-HER250-PLC-11 (Hermle-Performance-C250) | PM Interval: 250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-83-2.13: PN-HER250-GASK-VIT-8313 / Kit KIT-HER250-VAC-13 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-83-2.14: PN-HER250-BRG-CER-8301 / Kit KIT-HER250-SPN-1 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-83-2.15: PN-HER250-VLV-PROP-8303 / Kit KIT-HER250-HYD-3 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER250-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C250 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C250. Verify part numbers before closing Work Orders.

BOM-83-3.1: PN-HER250-PMP-PLG-8304 / Kit KIT-HER250-CLN-4 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER250-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-83-3.2: PN-HER250-CYL-SEAL-8307 / Kit KIT-HER250-ZAX-7 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-83-3.3: PN-HER250-GLASS-RD-8310 / Kit KIT-HER250-OPT-10 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER250-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-83-3.4: PN-HER250-GASK-VIT-8313 / Kit KIT-HER250-VAC-13 (Hermle-Performance-C250) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-83-3.5: PN-HER250-CAM-GRIP-8302 / Kit KIT-HER250-ATC-2 (Hermle-Performance-C250) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER250-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-83-3.6: PN-HER250-LIN-COIL-8305 / Kit KIT-HER250-XAX-5 (Hermle-Performance-C250) | PM Interval: 250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-83-3.7: PN-HER250-TRQ-STAT-8308 / Kit KIT-HER250-BAX-8 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER250-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-83-3.8: PN-HER250-IGBT-MOD-8311 / Kit KIT-HER250-PLC-11 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-83-3.9: PN-HER250-HEX-CORE-8314 / Kit KIT-HER250-THM-14 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER250-THM-14 within tolerance ±0.0060 mm TIR.

BOM-83-3.10: PN-HER250-VLV-PROP-8303 / Kit KIT-HER250-HYD-3 (Hermle-Performance-C250) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-83-3.11: PN-HER250-BSW-NUT-8306 / Kit KIT-HER250-YAX-6 (Hermle-Performance-C250) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER250-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-83-3.12: PN-HER250-BRK-DISC-8309 / Kit KIT-HER250-CAX-9 (Hermle-Performance-C250) | PM Interval: 250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-83-3.13: PN-HER250-INJ-MET-8312 / Kit KIT-HER250-LUB-12 (Hermle-Performance-C250) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER250-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-83-3.14: PN-HER250-BRG-CER-8301 / Kit KIT-HER250-SPN-1 (Hermle-Performance-C250) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-83-3.15: PN-HER250-PMP-PLG-8304 / Kit KIT-HER250-CLN-4 (Hermle-Performance-C250) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER250-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C300 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C300. Verify part numbers before closing Work Orders.

BOM-84-1.1: PN-HER300-CAM-GRIP-8402 / Kit KIT-HER300-ATC-2 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER300-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-84-1.2: PN-HER300-VLV-PROP-8403 / Kit KIT-HER300-HYD-3 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-84-1.3: PN-HER300-PMP-PLG-8404 / Kit KIT-HER300-CLN-4 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER300-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-84-1.4: PN-HER300-LIN-COIL-8405 / Kit KIT-HER300-XAX-5 (Hermle-Performance-C300) | PM Interval: 1250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-84-1.5: PN-HER300-BSW-NUT-8406 / Kit KIT-HER300-YAX-6 (Hermle-Performance-C300) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER300-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-84-1.6: PN-HER300-CYL-SEAL-8407 / Kit KIT-HER300-ZAX-7 (Hermle-Performance-C300) | PM Interval: 250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-84-1.7: PN-HER300-TRQ-STAT-8408 / Kit KIT-HER300-BAX-8 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER300-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-84-1.8: PN-HER300-BRK-DISC-8409 / Kit KIT-HER300-CAX-9 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-84-1.9: PN-HER300-GLASS-RD-8410 / Kit KIT-HER300-OPT-10 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER300-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-84-1.10: PN-HER300-IGBT-MOD-8411 / Kit KIT-HER300-PLC-11 (Hermle-Performance-C300) | PM Interval: 1250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-84-1.11: PN-HER300-INJ-MET-8412 / Kit KIT-HER300-LUB-12 (Hermle-Performance-C300) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER300-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-84-1.12: PN-HER300-GASK-VIT-8413 / Kit KIT-HER300-VAC-13 (Hermle-Performance-C300) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-84-1.13: PN-HER300-HEX-CORE-8414 / Kit KIT-HER300-THM-14 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER300-THM-14 within tolerance ±0.0020 mm TIR.

BOM-84-1.14: PN-HER300-BRG-CER-8401 / Kit KIT-HER300-SPN-1 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-84-1.15: PN-HER300-CAM-GRIP-8402 / Kit KIT-HER300-ATC-2 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER300-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C300 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C300. Verify part numbers before closing Work Orders.

BOM-84-2.1: PN-HER300-VLV-PROP-8403 / Kit KIT-HER300-HYD-3 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-84-2.2: PN-HER300-LIN-COIL-8405 / Kit KIT-HER300-XAX-5 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-84-2.3: PN-HER300-CYL-SEAL-8407 / Kit KIT-HER300-ZAX-7 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-84-2.4: PN-HER300-BRK-DISC-8409 / Kit KIT-HER300-CAX-9 (Hermle-Performance-C300) | PM Interval: 1250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-84-2.5: PN-HER300-IGBT-MOD-8411 / Kit KIT-HER300-PLC-11 (Hermle-Performance-C300) | PM Interval: 1500 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-84-2.6: PN-HER300-GASK-VIT-8413 / Kit KIT-HER300-VAC-13 (Hermle-Performance-C300) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-84-2.7: PN-HER300-BRG-CER-8401 / Kit KIT-HER300-SPN-1 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-84-2.8: PN-HER300-VLV-PROP-8403 / Kit KIT-HER300-HYD-3 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-84-2.9: PN-HER300-LIN-COIL-8405 / Kit KIT-HER300-XAX-5 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-84-2.10: PN-HER300-CYL-SEAL-8407 / Kit KIT-HER300-ZAX-7 (Hermle-Performance-C300) | PM Interval: 1250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-84-2.11: PN-HER300-BRK-DISC-8409 / Kit KIT-HER300-CAX-9 (Hermle-Performance-C300) | PM Interval: 1500 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-84-2.12: PN-HER300-IGBT-MOD-8411 / Kit KIT-HER300-PLC-11 (Hermle-Performance-C300) | PM Interval: 250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-84-2.13: PN-HER300-GASK-VIT-8413 / Kit KIT-HER300-VAC-13 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-84-2.14: PN-HER300-BRG-CER-8401 / Kit KIT-HER300-SPN-1 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-84-2.15: PN-HER300-VLV-PROP-8403 / Kit KIT-HER300-HYD-3 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER300-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C300 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C300. Verify part numbers before closing Work Orders.

BOM-84-3.1: PN-HER300-PMP-PLG-8404 / Kit KIT-HER300-CLN-4 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER300-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-84-3.2: PN-HER300-CYL-SEAL-8407 / Kit KIT-HER300-ZAX-7 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-84-3.3: PN-HER300-GLASS-RD-8410 / Kit KIT-HER300-OPT-10 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER300-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-84-3.4: PN-HER300-GASK-VIT-8413 / Kit KIT-HER300-VAC-13 (Hermle-Performance-C300) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-84-3.5: PN-HER300-CAM-GRIP-8402 / Kit KIT-HER300-ATC-2 (Hermle-Performance-C300) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER300-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-84-3.6: PN-HER300-LIN-COIL-8405 / Kit KIT-HER300-XAX-5 (Hermle-Performance-C300) | PM Interval: 250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-84-3.7: PN-HER300-TRQ-STAT-8408 / Kit KIT-HER300-BAX-8 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER300-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-84-3.8: PN-HER300-IGBT-MOD-8411 / Kit KIT-HER300-PLC-11 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-84-3.9: PN-HER300-HEX-CORE-8414 / Kit KIT-HER300-THM-14 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER300-THM-14 within tolerance ±0.0020 mm TIR.

BOM-84-3.10: PN-HER300-VLV-PROP-8403 / Kit KIT-HER300-HYD-3 (Hermle-Performance-C300) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-84-3.11: PN-HER300-BSW-NUT-8406 / Kit KIT-HER300-YAX-6 (Hermle-Performance-C300) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER300-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-84-3.12: PN-HER300-BRK-DISC-8409 / Kit KIT-HER300-CAX-9 (Hermle-Performance-C300) | PM Interval: 250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-84-3.13: PN-HER300-INJ-MET-8412 / Kit KIT-HER300-LUB-12 (Hermle-Performance-C300) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER300-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-84-3.14: PN-HER300-BRG-CER-8401 / Kit KIT-HER300-SPN-1 (Hermle-Performance-C300) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-84-3.15: PN-HER300-PMP-PLG-8404 / Kit KIT-HER300-CLN-4 (Hermle-Performance-C300) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER300-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C350 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C350. Verify part numbers before closing Work Orders.

BOM-85-1.1: PN-HER350-CAM-GRIP-8502 / Kit KIT-HER350-ATC-2 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER350-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-85-1.2: PN-HER350-VLV-PROP-8503 / Kit KIT-HER350-HYD-3 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-85-1.3: PN-HER350-PMP-PLG-8504 / Kit KIT-HER350-CLN-4 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER350-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-85-1.4: PN-HER350-LIN-COIL-8505 / Kit KIT-HER350-XAX-5 (Hermle-Performance-C350) | PM Interval: 1250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-85-1.5: PN-HER350-BSW-NUT-8506 / Kit KIT-HER350-YAX-6 (Hermle-Performance-C350) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER350-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-85-1.6: PN-HER350-CYL-SEAL-8507 / Kit KIT-HER350-ZAX-7 (Hermle-Performance-C350) | PM Interval: 250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-85-1.7: PN-HER350-TRQ-STAT-8508 / Kit KIT-HER350-BAX-8 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER350-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-85-1.8: PN-HER350-BRK-DISC-8509 / Kit KIT-HER350-CAX-9 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-85-1.9: PN-HER350-GLASS-RD-8510 / Kit KIT-HER350-OPT-10 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER350-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-85-1.10: PN-HER350-IGBT-MOD-8511 / Kit KIT-HER350-PLC-11 (Hermle-Performance-C350) | PM Interval: 1250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-85-1.11: PN-HER350-INJ-MET-8512 / Kit KIT-HER350-LUB-12 (Hermle-Performance-C350) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER350-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-85-1.12: PN-HER350-GASK-VIT-8513 / Kit KIT-HER350-VAC-13 (Hermle-Performance-C350) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-85-1.13: PN-HER350-HEX-CORE-8514 / Kit KIT-HER350-THM-14 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER350-THM-14 within tolerance ±0.0040 mm TIR.

BOM-85-1.14: PN-HER350-BRG-CER-8501 / Kit KIT-HER350-SPN-1 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-85-1.15: PN-HER350-CAM-GRIP-8502 / Kit KIT-HER350-ATC-2 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER350-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C350 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C350. Verify part numbers before closing Work Orders.

BOM-85-2.1: PN-HER350-VLV-PROP-8503 / Kit KIT-HER350-HYD-3 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-85-2.2: PN-HER350-LIN-COIL-8505 / Kit KIT-HER350-XAX-5 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-85-2.3: PN-HER350-CYL-SEAL-8507 / Kit KIT-HER350-ZAX-7 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-85-2.4: PN-HER350-BRK-DISC-8509 / Kit KIT-HER350-CAX-9 (Hermle-Performance-C350) | PM Interval: 1250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-85-2.5: PN-HER350-IGBT-MOD-8511 / Kit KIT-HER350-PLC-11 (Hermle-Performance-C350) | PM Interval: 1500 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-85-2.6: PN-HER350-GASK-VIT-8513 / Kit KIT-HER350-VAC-13 (Hermle-Performance-C350) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-85-2.7: PN-HER350-BRG-CER-8501 / Kit KIT-HER350-SPN-1 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-85-2.8: PN-HER350-VLV-PROP-8503 / Kit KIT-HER350-HYD-3 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-85-2.9: PN-HER350-LIN-COIL-8505 / Kit KIT-HER350-XAX-5 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-85-2.10: PN-HER350-CYL-SEAL-8507 / Kit KIT-HER350-ZAX-7 (Hermle-Performance-C350) | PM Interval: 1250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-85-2.11: PN-HER350-BRK-DISC-8509 / Kit KIT-HER350-CAX-9 (Hermle-Performance-C350) | PM Interval: 1500 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-85-2.12: PN-HER350-IGBT-MOD-8511 / Kit KIT-HER350-PLC-11 (Hermle-Performance-C350) | PM Interval: 250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-85-2.13: PN-HER350-GASK-VIT-8513 / Kit KIT-HER350-VAC-13 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-85-2.14: PN-HER350-BRG-CER-8501 / Kit KIT-HER350-SPN-1 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-85-2.15: PN-HER350-VLV-PROP-8503 / Kit KIT-HER350-HYD-3 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER350-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C350 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C350. Verify part numbers before closing Work Orders.

BOM-85-3.1: PN-HER350-PMP-PLG-8504 / Kit KIT-HER350-CLN-4 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER350-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-85-3.2: PN-HER350-CYL-SEAL-8507 / Kit KIT-HER350-ZAX-7 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-85-3.3: PN-HER350-GLASS-RD-8510 / Kit KIT-HER350-OPT-10 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER350-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-85-3.4: PN-HER350-GASK-VIT-8513 / Kit KIT-HER350-VAC-13 (Hermle-Performance-C350) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-85-3.5: PN-HER350-CAM-GRIP-8502 / Kit KIT-HER350-ATC-2 (Hermle-Performance-C350) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER350-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-85-3.6: PN-HER350-LIN-COIL-8505 / Kit KIT-HER350-XAX-5 (Hermle-Performance-C350) | PM Interval: 250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-85-3.7: PN-HER350-TRQ-STAT-8508 / Kit KIT-HER350-BAX-8 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER350-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-85-3.8: PN-HER350-IGBT-MOD-8511 / Kit KIT-HER350-PLC-11 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-85-3.9: PN-HER350-HEX-CORE-8514 / Kit KIT-HER350-THM-14 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER350-THM-14 within tolerance ±0.0040 mm TIR.

BOM-85-3.10: PN-HER350-VLV-PROP-8503 / Kit KIT-HER350-HYD-3 (Hermle-Performance-C350) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-85-3.11: PN-HER350-BSW-NUT-8506 / Kit KIT-HER350-YAX-6 (Hermle-Performance-C350) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER350-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-85-3.12: PN-HER350-BRK-DISC-8509 / Kit KIT-HER350-CAX-9 (Hermle-Performance-C350) | PM Interval: 250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-85-3.13: PN-HER350-INJ-MET-8512 / Kit KIT-HER350-LUB-12 (Hermle-Performance-C350) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER350-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-85-3.14: PN-HER350-BRG-CER-8501 / Kit KIT-HER350-SPN-1 (Hermle-Performance-C350) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-85-3.15: PN-HER350-PMP-PLG-8504 / Kit KIT-HER350-CLN-4 (Hermle-Performance-C350) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER350-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C400 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C400. Verify part numbers before closing Work Orders.

BOM-86-1.1: PN-HER400-CAM-GRIP-8602 / Kit KIT-HER400-ATC-2 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER400-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-86-1.2: PN-HER400-VLV-PROP-8603 / Kit KIT-HER400-HYD-3 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-86-1.3: PN-HER400-PMP-PLG-8604 / Kit KIT-HER400-CLN-4 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER400-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-86-1.4: PN-HER400-LIN-COIL-8605 / Kit KIT-HER400-XAX-5 (Hermle-Performance-C400) | PM Interval: 1250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-86-1.5: PN-HER400-BSW-NUT-8606 / Kit KIT-HER400-YAX-6 (Hermle-Performance-C400) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER400-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-86-1.6: PN-HER400-CYL-SEAL-8607 / Kit KIT-HER400-ZAX-7 (Hermle-Performance-C400) | PM Interval: 250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-86-1.7: PN-HER400-TRQ-STAT-8608 / Kit KIT-HER400-BAX-8 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER400-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-86-1.8: PN-HER400-BRK-DISC-8609 / Kit KIT-HER400-CAX-9 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-86-1.9: PN-HER400-GLASS-RD-8610 / Kit KIT-HER400-OPT-10 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER400-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-86-1.10: PN-HER400-IGBT-MOD-8611 / Kit KIT-HER400-PLC-11 (Hermle-Performance-C400) | PM Interval: 1250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-86-1.11: PN-HER400-INJ-MET-8612 / Kit KIT-HER400-LUB-12 (Hermle-Performance-C400) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER400-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-86-1.12: PN-HER400-GASK-VIT-8613 / Kit KIT-HER400-VAC-13 (Hermle-Performance-C400) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-86-1.13: PN-HER400-HEX-CORE-8614 / Kit KIT-HER400-THM-14 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER400-THM-14 within tolerance ±0.0060 mm TIR.

BOM-86-1.14: PN-HER400-BRG-CER-8601 / Kit KIT-HER400-SPN-1 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-86-1.15: PN-HER400-CAM-GRIP-8602 / Kit KIT-HER400-ATC-2 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER400-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C400 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C400. Verify part numbers before closing Work Orders.

BOM-86-2.1: PN-HER400-VLV-PROP-8603 / Kit KIT-HER400-HYD-3 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-86-2.2: PN-HER400-LIN-COIL-8605 / Kit KIT-HER400-XAX-5 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-86-2.3: PN-HER400-CYL-SEAL-8607 / Kit KIT-HER400-ZAX-7 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-86-2.4: PN-HER400-BRK-DISC-8609 / Kit KIT-HER400-CAX-9 (Hermle-Performance-C400) | PM Interval: 1250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-86-2.5: PN-HER400-IGBT-MOD-8611 / Kit KIT-HER400-PLC-11 (Hermle-Performance-C400) | PM Interval: 1500 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-86-2.6: PN-HER400-GASK-VIT-8613 / Kit KIT-HER400-VAC-13 (Hermle-Performance-C400) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-86-2.7: PN-HER400-BRG-CER-8601 / Kit KIT-HER400-SPN-1 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-86-2.8: PN-HER400-VLV-PROP-8603 / Kit KIT-HER400-HYD-3 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-86-2.9: PN-HER400-LIN-COIL-8605 / Kit KIT-HER400-XAX-5 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-86-2.10: PN-HER400-CYL-SEAL-8607 / Kit KIT-HER400-ZAX-7 (Hermle-Performance-C400) | PM Interval: 1250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-86-2.11: PN-HER400-BRK-DISC-8609 / Kit KIT-HER400-CAX-9 (Hermle-Performance-C400) | PM Interval: 1500 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-86-2.12: PN-HER400-IGBT-MOD-8611 / Kit KIT-HER400-PLC-11 (Hermle-Performance-C400) | PM Interval: 250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-86-2.13: PN-HER400-GASK-VIT-8613 / Kit KIT-HER400-VAC-13 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-86-2.14: PN-HER400-BRG-CER-8601 / Kit KIT-HER400-SPN-1 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-86-2.15: PN-HER400-VLV-PROP-8603 / Kit KIT-HER400-HYD-3 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER400-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C400 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C400. Verify part numbers before closing Work Orders.

BOM-86-3.1: PN-HER400-PMP-PLG-8604 / Kit KIT-HER400-CLN-4 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER400-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-86-3.2: PN-HER400-CYL-SEAL-8607 / Kit KIT-HER400-ZAX-7 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-86-3.3: PN-HER400-GLASS-RD-8610 / Kit KIT-HER400-OPT-10 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER400-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-86-3.4: PN-HER400-GASK-VIT-8613 / Kit KIT-HER400-VAC-13 (Hermle-Performance-C400) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-86-3.5: PN-HER400-CAM-GRIP-8602 / Kit KIT-HER400-ATC-2 (Hermle-Performance-C400) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER400-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-86-3.6: PN-HER400-LIN-COIL-8605 / Kit KIT-HER400-XAX-5 (Hermle-Performance-C400) | PM Interval: 250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-86-3.7: PN-HER400-TRQ-STAT-8608 / Kit KIT-HER400-BAX-8 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER400-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-86-3.8: PN-HER400-IGBT-MOD-8611 / Kit KIT-HER400-PLC-11 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-86-3.9: PN-HER400-HEX-CORE-8614 / Kit KIT-HER400-THM-14 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER400-THM-14 within tolerance ±0.0060 mm TIR.

BOM-86-3.10: PN-HER400-VLV-PROP-8603 / Kit KIT-HER400-HYD-3 (Hermle-Performance-C400) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-86-3.11: PN-HER400-BSW-NUT-8606 / Kit KIT-HER400-YAX-6 (Hermle-Performance-C400) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER400-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-86-3.12: PN-HER400-BRK-DISC-8609 / Kit KIT-HER400-CAX-9 (Hermle-Performance-C400) | PM Interval: 250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-86-3.13: PN-HER400-INJ-MET-8612 / Kit KIT-HER400-LUB-12 (Hermle-Performance-C400) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER400-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-86-3.14: PN-HER400-BRG-CER-8601 / Kit KIT-HER400-SPN-1 (Hermle-Performance-C400) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-86-3.15: PN-HER400-PMP-PLG-8604 / Kit KIT-HER400-CLN-4 (Hermle-Performance-C400) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER400-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C450 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C450. Verify part numbers before closing Work Orders.

BOM-87-1.1: PN-HER450-CAM-GRIP-8702 / Kit KIT-HER450-ATC-2 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER450-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-87-1.2: PN-HER450-VLV-PROP-8703 / Kit KIT-HER450-HYD-3 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-87-1.3: PN-HER450-PMP-PLG-8704 / Kit KIT-HER450-CLN-4 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER450-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-87-1.4: PN-HER450-LIN-COIL-8705 / Kit KIT-HER450-XAX-5 (Hermle-Performance-C450) | PM Interval: 1250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-87-1.5: PN-HER450-BSW-NUT-8706 / Kit KIT-HER450-YAX-6 (Hermle-Performance-C450) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER450-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-87-1.6: PN-HER450-CYL-SEAL-8707 / Kit KIT-HER450-ZAX-7 (Hermle-Performance-C450) | PM Interval: 250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-87-1.7: PN-HER450-TRQ-STAT-8708 / Kit KIT-HER450-BAX-8 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER450-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-87-1.8: PN-HER450-BRK-DISC-8709 / Kit KIT-HER450-CAX-9 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-87-1.9: PN-HER450-GLASS-RD-8710 / Kit KIT-HER450-OPT-10 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER450-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-87-1.10: PN-HER450-IGBT-MOD-8711 / Kit KIT-HER450-PLC-11 (Hermle-Performance-C450) | PM Interval: 1250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-87-1.11: PN-HER450-INJ-MET-8712 / Kit KIT-HER450-LUB-12 (Hermle-Performance-C450) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER450-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-87-1.12: PN-HER450-GASK-VIT-8713 / Kit KIT-HER450-VAC-13 (Hermle-Performance-C450) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-87-1.13: PN-HER450-HEX-CORE-8714 / Kit KIT-HER450-THM-14 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER450-THM-14 within tolerance ±0.0020 mm TIR.

BOM-87-1.14: PN-HER450-BRG-CER-8701 / Kit KIT-HER450-SPN-1 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-87-1.15: PN-HER450-CAM-GRIP-8702 / Kit KIT-HER450-ATC-2 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER450-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C450 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C450. Verify part numbers before closing Work Orders.

BOM-87-2.1: PN-HER450-VLV-PROP-8703 / Kit KIT-HER450-HYD-3 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-87-2.2: PN-HER450-LIN-COIL-8705 / Kit KIT-HER450-XAX-5 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-87-2.3: PN-HER450-CYL-SEAL-8707 / Kit KIT-HER450-ZAX-7 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-87-2.4: PN-HER450-BRK-DISC-8709 / Kit KIT-HER450-CAX-9 (Hermle-Performance-C450) | PM Interval: 1250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-87-2.5: PN-HER450-IGBT-MOD-8711 / Kit KIT-HER450-PLC-11 (Hermle-Performance-C450) | PM Interval: 1500 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-87-2.6: PN-HER450-GASK-VIT-8713 / Kit KIT-HER450-VAC-13 (Hermle-Performance-C450) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-87-2.7: PN-HER450-BRG-CER-8701 / Kit KIT-HER450-SPN-1 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-87-2.8: PN-HER450-VLV-PROP-8703 / Kit KIT-HER450-HYD-3 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-87-2.9: PN-HER450-LIN-COIL-8705 / Kit KIT-HER450-XAX-5 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-87-2.10: PN-HER450-CYL-SEAL-8707 / Kit KIT-HER450-ZAX-7 (Hermle-Performance-C450) | PM Interval: 1250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-87-2.11: PN-HER450-BRK-DISC-8709 / Kit KIT-HER450-CAX-9 (Hermle-Performance-C450) | PM Interval: 1500 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-87-2.12: PN-HER450-IGBT-MOD-8711 / Kit KIT-HER450-PLC-11 (Hermle-Performance-C450) | PM Interval: 250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-87-2.13: PN-HER450-GASK-VIT-8713 / Kit KIT-HER450-VAC-13 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-87-2.14: PN-HER450-BRG-CER-8701 / Kit KIT-HER450-SPN-1 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-87-2.15: PN-HER450-VLV-PROP-8703 / Kit KIT-HER450-HYD-3 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER450-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C450 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C450. Verify part numbers before closing Work Orders.

BOM-87-3.1: PN-HER450-PMP-PLG-8704 / Kit KIT-HER450-CLN-4 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER450-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-87-3.2: PN-HER450-CYL-SEAL-8707 / Kit KIT-HER450-ZAX-7 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-87-3.3: PN-HER450-GLASS-RD-8710 / Kit KIT-HER450-OPT-10 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER450-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-87-3.4: PN-HER450-GASK-VIT-8713 / Kit KIT-HER450-VAC-13 (Hermle-Performance-C450) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-87-3.5: PN-HER450-CAM-GRIP-8702 / Kit KIT-HER450-ATC-2 (Hermle-Performance-C450) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER450-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-87-3.6: PN-HER450-LIN-COIL-8705 / Kit KIT-HER450-XAX-5 (Hermle-Performance-C450) | PM Interval: 250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-87-3.7: PN-HER450-TRQ-STAT-8708 / Kit KIT-HER450-BAX-8 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER450-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-87-3.8: PN-HER450-IGBT-MOD-8711 / Kit KIT-HER450-PLC-11 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-87-3.9: PN-HER450-HEX-CORE-8714 / Kit KIT-HER450-THM-14 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER450-THM-14 within tolerance ±0.0020 mm TIR.

BOM-87-3.10: PN-HER450-VLV-PROP-8703 / Kit KIT-HER450-HYD-3 (Hermle-Performance-C450) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-87-3.11: PN-HER450-BSW-NUT-8706 / Kit KIT-HER450-YAX-6 (Hermle-Performance-C450) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER450-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-87-3.12: PN-HER450-BRK-DISC-8709 / Kit KIT-HER450-CAX-9 (Hermle-Performance-C450) | PM Interval: 250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-87-3.13: PN-HER450-INJ-MET-8712 / Kit KIT-HER450-LUB-12 (Hermle-Performance-C450) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER450-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-87-3.14: PN-HER450-BRG-CER-8701 / Kit KIT-HER450-SPN-1 (Hermle-Performance-C450) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-87-3.15: PN-HER450-PMP-PLG-8704 / Kit KIT-HER450-CLN-4 (Hermle-Performance-C450) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER450-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C500 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C500. Verify part numbers before closing Work Orders.

BOM-88-1.1: PN-HER500-CAM-GRIP-8802 / Kit KIT-HER500-ATC-2 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER500-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-88-1.2: PN-HER500-VLV-PROP-8803 / Kit KIT-HER500-HYD-3 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-88-1.3: PN-HER500-PMP-PLG-8804 / Kit KIT-HER500-CLN-4 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER500-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-88-1.4: PN-HER500-LIN-COIL-8805 / Kit KIT-HER500-XAX-5 (Hermle-Performance-C500) | PM Interval: 1250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-88-1.5: PN-HER500-BSW-NUT-8806 / Kit KIT-HER500-YAX-6 (Hermle-Performance-C500) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER500-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-88-1.6: PN-HER500-CYL-SEAL-8807 / Kit KIT-HER500-ZAX-7 (Hermle-Performance-C500) | PM Interval: 250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-88-1.7: PN-HER500-TRQ-STAT-8808 / Kit KIT-HER500-BAX-8 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER500-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-88-1.8: PN-HER500-BRK-DISC-8809 / Kit KIT-HER500-CAX-9 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-88-1.9: PN-HER500-GLASS-RD-8810 / Kit KIT-HER500-OPT-10 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER500-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-88-1.10: PN-HER500-IGBT-MOD-8811 / Kit KIT-HER500-PLC-11 (Hermle-Performance-C500) | PM Interval: 1250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-88-1.11: PN-HER500-INJ-MET-8812 / Kit KIT-HER500-LUB-12 (Hermle-Performance-C500) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER500-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-88-1.12: PN-HER500-GASK-VIT-8813 / Kit KIT-HER500-VAC-13 (Hermle-Performance-C500) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-88-1.13: PN-HER500-HEX-CORE-8814 / Kit KIT-HER500-THM-14 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER500-THM-14 within tolerance ±0.0040 mm TIR.

BOM-88-1.14: PN-HER500-BRG-CER-8801 / Kit KIT-HER500-SPN-1 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-88-1.15: PN-HER500-CAM-GRIP-8802 / Kit KIT-HER500-ATC-2 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER500-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C500 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C500. Verify part numbers before closing Work Orders.

BOM-88-2.1: PN-HER500-VLV-PROP-8803 / Kit KIT-HER500-HYD-3 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-88-2.2: PN-HER500-LIN-COIL-8805 / Kit KIT-HER500-XAX-5 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-88-2.3: PN-HER500-CYL-SEAL-8807 / Kit KIT-HER500-ZAX-7 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-88-2.4: PN-HER500-BRK-DISC-8809 / Kit KIT-HER500-CAX-9 (Hermle-Performance-C500) | PM Interval: 1250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-88-2.5: PN-HER500-IGBT-MOD-8811 / Kit KIT-HER500-PLC-11 (Hermle-Performance-C500) | PM Interval: 1500 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-88-2.6: PN-HER500-GASK-VIT-8813 / Kit KIT-HER500-VAC-13 (Hermle-Performance-C500) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-88-2.7: PN-HER500-BRG-CER-8801 / Kit KIT-HER500-SPN-1 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-88-2.8: PN-HER500-VLV-PROP-8803 / Kit KIT-HER500-HYD-3 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-88-2.9: PN-HER500-LIN-COIL-8805 / Kit KIT-HER500-XAX-5 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-88-2.10: PN-HER500-CYL-SEAL-8807 / Kit KIT-HER500-ZAX-7 (Hermle-Performance-C500) | PM Interval: 1250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-88-2.11: PN-HER500-BRK-DISC-8809 / Kit KIT-HER500-CAX-9 (Hermle-Performance-C500) | PM Interval: 1500 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-88-2.12: PN-HER500-IGBT-MOD-8811 / Kit KIT-HER500-PLC-11 (Hermle-Performance-C500) | PM Interval: 250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-88-2.13: PN-HER500-GASK-VIT-8813 / Kit KIT-HER500-VAC-13 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-88-2.14: PN-HER500-BRG-CER-8801 / Kit KIT-HER500-SPN-1 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-88-2.15: PN-HER500-VLV-PROP-8803 / Kit KIT-HER500-HYD-3 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER500-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C500 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C500. Verify part numbers before closing Work Orders.

BOM-88-3.1: PN-HER500-PMP-PLG-8804 / Kit KIT-HER500-CLN-4 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER500-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-88-3.2: PN-HER500-CYL-SEAL-8807 / Kit KIT-HER500-ZAX-7 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-88-3.3: PN-HER500-GLASS-RD-8810 / Kit KIT-HER500-OPT-10 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER500-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-88-3.4: PN-HER500-GASK-VIT-8813 / Kit KIT-HER500-VAC-13 (Hermle-Performance-C500) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-88-3.5: PN-HER500-CAM-GRIP-8802 / Kit KIT-HER500-ATC-2 (Hermle-Performance-C500) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER500-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-88-3.6: PN-HER500-LIN-COIL-8805 / Kit KIT-HER500-XAX-5 (Hermle-Performance-C500) | PM Interval: 250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-88-3.7: PN-HER500-TRQ-STAT-8808 / Kit KIT-HER500-BAX-8 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER500-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-88-3.8: PN-HER500-IGBT-MOD-8811 / Kit KIT-HER500-PLC-11 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-88-3.9: PN-HER500-HEX-CORE-8814 / Kit KIT-HER500-THM-14 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER500-THM-14 within tolerance ±0.0040 mm TIR.

BOM-88-3.10: PN-HER500-VLV-PROP-8803 / Kit KIT-HER500-HYD-3 (Hermle-Performance-C500) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-88-3.11: PN-HER500-BSW-NUT-8806 / Kit KIT-HER500-YAX-6 (Hermle-Performance-C500) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER500-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-88-3.12: PN-HER500-BRK-DISC-8809 / Kit KIT-HER500-CAX-9 (Hermle-Performance-C500) | PM Interval: 250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-88-3.13: PN-HER500-INJ-MET-8812 / Kit KIT-HER500-LUB-12 (Hermle-Performance-C500) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER500-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-88-3.14: PN-HER500-BRG-CER-8801 / Kit KIT-HER500-SPN-1 (Hermle-Performance-C500) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-88-3.15: PN-HER500-PMP-PLG-8804 / Kit KIT-HER500-CLN-4 (Hermle-Performance-C500) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER500-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C550 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C550. Verify part numbers before closing Work Orders.

BOM-89-1.1: PN-HER550-CAM-GRIP-8902 / Kit KIT-HER550-ATC-2 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER550-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-89-1.2: PN-HER550-VLV-PROP-8903 / Kit KIT-HER550-HYD-3 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-89-1.3: PN-HER550-PMP-PLG-8904 / Kit KIT-HER550-CLN-4 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER550-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-89-1.4: PN-HER550-LIN-COIL-8905 / Kit KIT-HER550-XAX-5 (Hermle-Performance-C550) | PM Interval: 1250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-89-1.5: PN-HER550-BSW-NUT-8906 / Kit KIT-HER550-YAX-6 (Hermle-Performance-C550) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER550-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-89-1.6: PN-HER550-CYL-SEAL-8907 / Kit KIT-HER550-ZAX-7 (Hermle-Performance-C550) | PM Interval: 250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-89-1.7: PN-HER550-TRQ-STAT-8908 / Kit KIT-HER550-BAX-8 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER550-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-89-1.8: PN-HER550-BRK-DISC-8909 / Kit KIT-HER550-CAX-9 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-89-1.9: PN-HER550-GLASS-RD-8910 / Kit KIT-HER550-OPT-10 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER550-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-89-1.10: PN-HER550-IGBT-MOD-8911 / Kit KIT-HER550-PLC-11 (Hermle-Performance-C550) | PM Interval: 1250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-89-1.11: PN-HER550-INJ-MET-8912 / Kit KIT-HER550-LUB-12 (Hermle-Performance-C550) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER550-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-89-1.12: PN-HER550-GASK-VIT-8913 / Kit KIT-HER550-VAC-13 (Hermle-Performance-C550) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-89-1.13: PN-HER550-HEX-CORE-8914 / Kit KIT-HER550-THM-14 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER550-THM-14 within tolerance ±0.0060 mm TIR.

BOM-89-1.14: PN-HER550-BRG-CER-8901 / Kit KIT-HER550-SPN-1 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-89-1.15: PN-HER550-CAM-GRIP-8902 / Kit KIT-HER550-ATC-2 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER550-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C550 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C550. Verify part numbers before closing Work Orders.

BOM-89-2.1: PN-HER550-VLV-PROP-8903 / Kit KIT-HER550-HYD-3 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-89-2.2: PN-HER550-LIN-COIL-8905 / Kit KIT-HER550-XAX-5 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-89-2.3: PN-HER550-CYL-SEAL-8907 / Kit KIT-HER550-ZAX-7 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-89-2.4: PN-HER550-BRK-DISC-8909 / Kit KIT-HER550-CAX-9 (Hermle-Performance-C550) | PM Interval: 1250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-89-2.5: PN-HER550-IGBT-MOD-8911 / Kit KIT-HER550-PLC-11 (Hermle-Performance-C550) | PM Interval: 1500 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-89-2.6: PN-HER550-GASK-VIT-8913 / Kit KIT-HER550-VAC-13 (Hermle-Performance-C550) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-89-2.7: PN-HER550-BRG-CER-8901 / Kit KIT-HER550-SPN-1 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-89-2.8: PN-HER550-VLV-PROP-8903 / Kit KIT-HER550-HYD-3 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-89-2.9: PN-HER550-LIN-COIL-8905 / Kit KIT-HER550-XAX-5 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-89-2.10: PN-HER550-CYL-SEAL-8907 / Kit KIT-HER550-ZAX-7 (Hermle-Performance-C550) | PM Interval: 1250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-89-2.11: PN-HER550-BRK-DISC-8909 / Kit KIT-HER550-CAX-9 (Hermle-Performance-C550) | PM Interval: 1500 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-89-2.12: PN-HER550-IGBT-MOD-8911 / Kit KIT-HER550-PLC-11 (Hermle-Performance-C550) | PM Interval: 250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-89-2.13: PN-HER550-GASK-VIT-8913 / Kit KIT-HER550-VAC-13 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-89-2.14: PN-HER550-BRG-CER-8901 / Kit KIT-HER550-SPN-1 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-89-2.15: PN-HER550-VLV-PROP-8903 / Kit KIT-HER550-HYD-3 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER550-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C550 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C550. Verify part numbers before closing Work Orders.

BOM-89-3.1: PN-HER550-PMP-PLG-8904 / Kit KIT-HER550-CLN-4 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER550-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-89-3.2: PN-HER550-CYL-SEAL-8907 / Kit KIT-HER550-ZAX-7 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-89-3.3: PN-HER550-GLASS-RD-8910 / Kit KIT-HER550-OPT-10 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER550-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-89-3.4: PN-HER550-GASK-VIT-8913 / Kit KIT-HER550-VAC-13 (Hermle-Performance-C550) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-89-3.5: PN-HER550-CAM-GRIP-8902 / Kit KIT-HER550-ATC-2 (Hermle-Performance-C550) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER550-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-89-3.6: PN-HER550-LIN-COIL-8905 / Kit KIT-HER550-XAX-5 (Hermle-Performance-C550) | PM Interval: 250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-89-3.7: PN-HER550-TRQ-STAT-8908 / Kit KIT-HER550-BAX-8 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER550-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-89-3.8: PN-HER550-IGBT-MOD-8911 / Kit KIT-HER550-PLC-11 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-89-3.9: PN-HER550-HEX-CORE-8914 / Kit KIT-HER550-THM-14 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER550-THM-14 within tolerance ±0.0060 mm TIR.

BOM-89-3.10: PN-HER550-VLV-PROP-8903 / Kit KIT-HER550-HYD-3 (Hermle-Performance-C550) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-89-3.11: PN-HER550-BSW-NUT-8906 / Kit KIT-HER550-YAX-6 (Hermle-Performance-C550) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER550-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-89-3.12: PN-HER550-BRK-DISC-8909 / Kit KIT-HER550-CAX-9 (Hermle-Performance-C550) | PM Interval: 250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-89-3.13: PN-HER550-INJ-MET-8912 / Kit KIT-HER550-LUB-12 (Hermle-Performance-C550) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER550-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-89-3.14: PN-HER550-BRG-CER-8901 / Kit KIT-HER550-SPN-1 (Hermle-Performance-C550) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-89-3.15: PN-HER550-PMP-PLG-8904 / Kit KIT-HER550-CLN-4 (Hermle-Performance-C550) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER550-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C600 (GROUP 1)

Certified Aerospace Bill of Materials for Hermle-Performance-C600. Verify part numbers before closing Work Orders.

BOM-90-1.1: PN-HER600-CAM-GRIP-9002 / Kit KIT-HER600-ATC-2 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER600-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-90-1.2: PN-HER600-VLV-PROP-9003 / Kit KIT-HER600-HYD-3 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-90-1.3: PN-HER600-PMP-PLG-9004 / Kit KIT-HER600-CLN-4 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER600-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-90-1.4: PN-HER600-LIN-COIL-9005 / Kit KIT-HER600-XAX-5 (Hermle-Performance-C600) | PM Interval: 1250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-90-1.5: PN-HER600-BSW-NUT-9006 / Kit KIT-HER600-YAX-6 (Hermle-Performance-C600) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER600-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-90-1.6: PN-HER600-CYL-SEAL-9007 / Kit KIT-HER600-ZAX-7 (Hermle-Performance-C600) | PM Interval: 250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-90-1.7: PN-HER600-TRQ-STAT-9008 / Kit KIT-HER600-BAX-8 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER600-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-90-1.8: PN-HER600-BRK-DISC-9009 / Kit KIT-HER600-CAX-9 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-90-1.9: PN-HER600-GLASS-RD-9010 / Kit KIT-HER600-OPT-10 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER600-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-90-1.10: PN-HER600-IGBT-MOD-9011 / Kit KIT-HER600-PLC-11 (Hermle-Performance-C600) | PM Interval: 1250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-90-1.11: PN-HER600-INJ-MET-9012 / Kit KIT-HER600-LUB-12 (Hermle-Performance-C600) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER600-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-90-1.12: PN-HER600-GASK-VIT-9013 / Kit KIT-HER600-VAC-13 (Hermle-Performance-C600) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-90-1.13: PN-HER600-HEX-CORE-9014 / Kit KIT-HER600-THM-14 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER600-THM-14 within tolerance ±0.0020 mm TIR.

BOM-90-1.14: PN-HER600-BRG-CER-9001 / Kit KIT-HER600-SPN-1 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-90-1.15: PN-HER600-CAM-GRIP-9002 / Kit KIT-HER600-ATC-2 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER600-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C600 (GROUP 2)

Certified Aerospace Bill of Materials for Hermle-Performance-C600. Verify part numbers before closing Work Orders.

BOM-90-2.1: PN-HER600-VLV-PROP-9003 / Kit KIT-HER600-HYD-3 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-90-2.2: PN-HER600-LIN-COIL-9005 / Kit KIT-HER600-XAX-5 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-90-2.3: PN-HER600-CYL-SEAL-9007 / Kit KIT-HER600-ZAX-7 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-90-2.4: PN-HER600-BRK-DISC-9009 / Kit KIT-HER600-CAX-9 (Hermle-Performance-C600) | PM Interval: 1250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-90-2.5: PN-HER600-IGBT-MOD-9011 / Kit KIT-HER600-PLC-11 (Hermle-Performance-C600) | PM Interval: 1500 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-90-2.6: PN-HER600-GASK-VIT-9013 / Kit KIT-HER600-VAC-13 (Hermle-Performance-C600) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-90-2.7: PN-HER600-BRG-CER-9001 / Kit KIT-HER600-SPN-1 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-90-2.8: PN-HER600-VLV-PROP-9003 / Kit KIT-HER600-HYD-3 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-90-2.9: PN-HER600-LIN-COIL-9005 / Kit KIT-HER600-XAX-5 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-90-2.10: PN-HER600-CYL-SEAL-9007 / Kit KIT-HER600-ZAX-7 (Hermle-Performance-C600) | PM Interval: 1250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-90-2.11: PN-HER600-BRK-DISC-9009 / Kit KIT-HER600-CAX-9 (Hermle-Performance-C600) | PM Interval: 1500 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-90-2.12: PN-HER600-IGBT-MOD-9011 / Kit KIT-HER600-PLC-11 (Hermle-Performance-C600) | PM Interval: 250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-90-2.13: PN-HER600-GASK-VIT-9013 / Kit KIT-HER600-VAC-13 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-90-2.14: PN-HER600-BRG-CER-9001 / Kit KIT-HER600-SPN-1 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-90-2.15: PN-HER600-VLV-PROP-9003 / Kit KIT-HER600-HYD-3 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER600-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Hermle-Performance-C600 (GROUP 3)

Certified Aerospace Bill of Materials for Hermle-Performance-C600. Verify part numbers before closing Work Orders.

BOM-90-3.1: PN-HER600-PMP-PLG-9004 / Kit KIT-HER600-CLN-4 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER600-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-90-3.2: PN-HER600-CYL-SEAL-9007 / Kit KIT-HER600-ZAX-7 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-HER600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-90-3.3: PN-HER600-GLASS-RD-9010 / Kit KIT-HER600-OPT-10 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-HER600-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-90-3.4: PN-HER600-GASK-VIT-9013 / Kit KIT-HER600-VAC-13 (Hermle-Performance-C600) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-HER600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-90-3.5: PN-HER600-CAM-GRIP-9002 / Kit KIT-HER600-ATC-2 (Hermle-Performance-C600) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-HER600-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-90-3.6: PN-HER600-LIN-COIL-9005 / Kit KIT-HER600-XAX-5 (Hermle-Performance-C600) | PM Interval: 250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-HER600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-90-3.7: PN-HER600-TRQ-STAT-9008 / Kit KIT-HER600-BAX-8 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-HER600-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-90-3.8: PN-HER600-IGBT-MOD-9011 / Kit KIT-HER600-PLC-11 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-HER600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-90-3.9: PN-HER600-HEX-CORE-9014 / Kit KIT-HER600-THM-14 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-HER600-THM-14 within tolerance ±0.0020 mm TIR.

BOM-90-3.10: PN-HER600-VLV-PROP-9003 / Kit KIT-HER600-HYD-3 (Hermle-Performance-C600) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-HER600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-90-3.11: PN-HER600-BSW-NUT-9006 / Kit KIT-HER600-YAX-6 (Hermle-Performance-C600) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-HER600-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-90-3.12: PN-HER600-BRK-DISC-9009 / Kit KIT-HER600-CAX-9 (Hermle-Performance-C600) | PM Interval: 250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-HER600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-90-3.13: PN-HER600-INJ-MET-9012 / Kit KIT-HER600-LUB-12 (Hermle-Performance-C600) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-HER600-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-90-3.14: PN-HER600-BRG-CER-9001 / Kit KIT-HER600-SPN-1 (Hermle-Performance-C600) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-HER600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-90-3.15: PN-HER600-PMP-PLG-9004 / Kit KIT-HER600-CLN-4 (Hermle-Performance-C600) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-HER600-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z150 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z150. Verify part numbers before closing Work Orders.

BOM-91-1.1: PN-ZEI150-CAM-GRIP-9102 / Kit KIT-ZEI150-ATC-2 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI150-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-91-1.2: PN-ZEI150-VLV-PROP-9103 / Kit KIT-ZEI150-HYD-3 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-91-1.3: PN-ZEI150-PMP-PLG-9104 / Kit KIT-ZEI150-CLN-4 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI150-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-91-1.4: PN-ZEI150-LIN-COIL-9105 / Kit KIT-ZEI150-XAX-5 (Zeiss-Prismo-Z150) | PM Interval: 1250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-91-1.5: PN-ZEI150-BSW-NUT-9106 / Kit KIT-ZEI150-YAX-6 (Zeiss-Prismo-Z150) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI150-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-91-1.6: PN-ZEI150-CYL-SEAL-9107 / Kit KIT-ZEI150-ZAX-7 (Zeiss-Prismo-Z150) | PM Interval: 250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-91-1.7: PN-ZEI150-TRQ-STAT-9108 / Kit KIT-ZEI150-BAX-8 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI150-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-91-1.8: PN-ZEI150-BRK-DISC-9109 / Kit KIT-ZEI150-CAX-9 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-91-1.9: PN-ZEI150-GLASS-RD-9110 / Kit KIT-ZEI150-OPT-10 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI150-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-91-1.10: PN-ZEI150-IGBT-MOD-9111 / Kit KIT-ZEI150-PLC-11 (Zeiss-Prismo-Z150) | PM Interval: 1250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-91-1.11: PN-ZEI150-INJ-MET-9112 / Kit KIT-ZEI150-LUB-12 (Zeiss-Prismo-Z150) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI150-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-91-1.12: PN-ZEI150-GASK-VIT-9113 / Kit KIT-ZEI150-VAC-13 (Zeiss-Prismo-Z150) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-91-1.13: PN-ZEI150-HEX-CORE-9114 / Kit KIT-ZEI150-THM-14 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI150-THM-14 within tolerance ±0.0040 mm TIR.

BOM-91-1.14: PN-ZEI150-BRG-CER-9101 / Kit KIT-ZEI150-SPN-1 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-91-1.15: PN-ZEI150-CAM-GRIP-9102 / Kit KIT-ZEI150-ATC-2 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI150-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z150 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z150. Verify part numbers before closing Work Orders.

BOM-91-2.1: PN-ZEI150-VLV-PROP-9103 / Kit KIT-ZEI150-HYD-3 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-91-2.2: PN-ZEI150-LIN-COIL-9105 / Kit KIT-ZEI150-XAX-5 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-91-2.3: PN-ZEI150-CYL-SEAL-9107 / Kit KIT-ZEI150-ZAX-7 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-91-2.4: PN-ZEI150-BRK-DISC-9109 / Kit KIT-ZEI150-CAX-9 (Zeiss-Prismo-Z150) | PM Interval: 1250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-91-2.5: PN-ZEI150-IGBT-MOD-9111 / Kit KIT-ZEI150-PLC-11 (Zeiss-Prismo-Z150) | PM Interval: 1500 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-91-2.6: PN-ZEI150-GASK-VIT-9113 / Kit KIT-ZEI150-VAC-13 (Zeiss-Prismo-Z150) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-91-2.7: PN-ZEI150-BRG-CER-9101 / Kit KIT-ZEI150-SPN-1 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-91-2.8: PN-ZEI150-VLV-PROP-9103 / Kit KIT-ZEI150-HYD-3 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-91-2.9: PN-ZEI150-LIN-COIL-9105 / Kit KIT-ZEI150-XAX-5 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-91-2.10: PN-ZEI150-CYL-SEAL-9107 / Kit KIT-ZEI150-ZAX-7 (Zeiss-Prismo-Z150) | PM Interval: 1250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-91-2.11: PN-ZEI150-BRK-DISC-9109 / Kit KIT-ZEI150-CAX-9 (Zeiss-Prismo-Z150) | PM Interval: 1500 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-91-2.12: PN-ZEI150-IGBT-MOD-9111 / Kit KIT-ZEI150-PLC-11 (Zeiss-Prismo-Z150) | PM Interval: 250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-91-2.13: PN-ZEI150-GASK-VIT-9113 / Kit KIT-ZEI150-VAC-13 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-91-2.14: PN-ZEI150-BRG-CER-9101 / Kit KIT-ZEI150-SPN-1 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-91-2.15: PN-ZEI150-VLV-PROP-9103 / Kit KIT-ZEI150-HYD-3 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI150-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z150 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z150. Verify part numbers before closing Work Orders.

BOM-91-3.1: PN-ZEI150-PMP-PLG-9104 / Kit KIT-ZEI150-CLN-4 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI150-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-91-3.2: PN-ZEI150-CYL-SEAL-9107 / Kit KIT-ZEI150-ZAX-7 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-91-3.3: PN-ZEI150-GLASS-RD-9110 / Kit KIT-ZEI150-OPT-10 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI150-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-91-3.4: PN-ZEI150-GASK-VIT-9113 / Kit KIT-ZEI150-VAC-13 (Zeiss-Prismo-Z150) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-91-3.5: PN-ZEI150-CAM-GRIP-9102 / Kit KIT-ZEI150-ATC-2 (Zeiss-Prismo-Z150) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI150-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-91-3.6: PN-ZEI150-LIN-COIL-9105 / Kit KIT-ZEI150-XAX-5 (Zeiss-Prismo-Z150) | PM Interval: 250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-91-3.7: PN-ZEI150-TRQ-STAT-9108 / Kit KIT-ZEI150-BAX-8 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI150-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-91-3.8: PN-ZEI150-IGBT-MOD-9111 / Kit KIT-ZEI150-PLC-11 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-91-3.9: PN-ZEI150-HEX-CORE-9114 / Kit KIT-ZEI150-THM-14 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI150-THM-14 within tolerance ±0.0040 mm TIR.

BOM-91-3.10: PN-ZEI150-VLV-PROP-9103 / Kit KIT-ZEI150-HYD-3 (Zeiss-Prismo-Z150) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-91-3.11: PN-ZEI150-BSW-NUT-9106 / Kit KIT-ZEI150-YAX-6 (Zeiss-Prismo-Z150) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI150-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-91-3.12: PN-ZEI150-BRK-DISC-9109 / Kit KIT-ZEI150-CAX-9 (Zeiss-Prismo-Z150) | PM Interval: 250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-91-3.13: PN-ZEI150-INJ-MET-9112 / Kit KIT-ZEI150-LUB-12 (Zeiss-Prismo-Z150) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI150-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-91-3.14: PN-ZEI150-BRG-CER-9101 / Kit KIT-ZEI150-SPN-1 (Zeiss-Prismo-Z150) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-91-3.15: PN-ZEI150-PMP-PLG-9104 / Kit KIT-ZEI150-CLN-4 (Zeiss-Prismo-Z150) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI150-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z200 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z200. Verify part numbers before closing Work Orders.

BOM-92-1.1: PN-ZEI200-CAM-GRIP-9202 / Kit KIT-ZEI200-ATC-2 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI200-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-92-1.2: PN-ZEI200-VLV-PROP-9203 / Kit KIT-ZEI200-HYD-3 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-92-1.3: PN-ZEI200-PMP-PLG-9204 / Kit KIT-ZEI200-CLN-4 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI200-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-92-1.4: PN-ZEI200-LIN-COIL-9205 / Kit KIT-ZEI200-XAX-5 (Zeiss-Prismo-Z200) | PM Interval: 1250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-92-1.5: PN-ZEI200-BSW-NUT-9206 / Kit KIT-ZEI200-YAX-6 (Zeiss-Prismo-Z200) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI200-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-92-1.6: PN-ZEI200-CYL-SEAL-9207 / Kit KIT-ZEI200-ZAX-7 (Zeiss-Prismo-Z200) | PM Interval: 250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-92-1.7: PN-ZEI200-TRQ-STAT-9208 / Kit KIT-ZEI200-BAX-8 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI200-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-92-1.8: PN-ZEI200-BRK-DISC-9209 / Kit KIT-ZEI200-CAX-9 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-92-1.9: PN-ZEI200-GLASS-RD-9210 / Kit KIT-ZEI200-OPT-10 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI200-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-92-1.10: PN-ZEI200-IGBT-MOD-9211 / Kit KIT-ZEI200-PLC-11 (Zeiss-Prismo-Z200) | PM Interval: 1250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-92-1.11: PN-ZEI200-INJ-MET-9212 / Kit KIT-ZEI200-LUB-12 (Zeiss-Prismo-Z200) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI200-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-92-1.12: PN-ZEI200-GASK-VIT-9213 / Kit KIT-ZEI200-VAC-13 (Zeiss-Prismo-Z200) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-92-1.13: PN-ZEI200-HEX-CORE-9214 / Kit KIT-ZEI200-THM-14 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI200-THM-14 within tolerance ±0.0060 mm TIR.

BOM-92-1.14: PN-ZEI200-BRG-CER-9201 / Kit KIT-ZEI200-SPN-1 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-92-1.15: PN-ZEI200-CAM-GRIP-9202 / Kit KIT-ZEI200-ATC-2 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI200-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z200 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z200. Verify part numbers before closing Work Orders.

BOM-92-2.1: PN-ZEI200-VLV-PROP-9203 / Kit KIT-ZEI200-HYD-3 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-92-2.2: PN-ZEI200-LIN-COIL-9205 / Kit KIT-ZEI200-XAX-5 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-92-2.3: PN-ZEI200-CYL-SEAL-9207 / Kit KIT-ZEI200-ZAX-7 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-92-2.4: PN-ZEI200-BRK-DISC-9209 / Kit KIT-ZEI200-CAX-9 (Zeiss-Prismo-Z200) | PM Interval: 1250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-92-2.5: PN-ZEI200-IGBT-MOD-9211 / Kit KIT-ZEI200-PLC-11 (Zeiss-Prismo-Z200) | PM Interval: 1500 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-92-2.6: PN-ZEI200-GASK-VIT-9213 / Kit KIT-ZEI200-VAC-13 (Zeiss-Prismo-Z200) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-92-2.7: PN-ZEI200-BRG-CER-9201 / Kit KIT-ZEI200-SPN-1 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-92-2.8: PN-ZEI200-VLV-PROP-9203 / Kit KIT-ZEI200-HYD-3 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-92-2.9: PN-ZEI200-LIN-COIL-9205 / Kit KIT-ZEI200-XAX-5 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-92-2.10: PN-ZEI200-CYL-SEAL-9207 / Kit KIT-ZEI200-ZAX-7 (Zeiss-Prismo-Z200) | PM Interval: 1250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-92-2.11: PN-ZEI200-BRK-DISC-9209 / Kit KIT-ZEI200-CAX-9 (Zeiss-Prismo-Z200) | PM Interval: 1500 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-92-2.12: PN-ZEI200-IGBT-MOD-9211 / Kit KIT-ZEI200-PLC-11 (Zeiss-Prismo-Z200) | PM Interval: 250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-92-2.13: PN-ZEI200-GASK-VIT-9213 / Kit KIT-ZEI200-VAC-13 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-92-2.14: PN-ZEI200-BRG-CER-9201 / Kit KIT-ZEI200-SPN-1 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-92-2.15: PN-ZEI200-VLV-PROP-9203 / Kit KIT-ZEI200-HYD-3 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI200-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z200 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z200. Verify part numbers before closing Work Orders.

BOM-92-3.1: PN-ZEI200-PMP-PLG-9204 / Kit KIT-ZEI200-CLN-4 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI200-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-92-3.2: PN-ZEI200-CYL-SEAL-9207 / Kit KIT-ZEI200-ZAX-7 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-92-3.3: PN-ZEI200-GLASS-RD-9210 / Kit KIT-ZEI200-OPT-10 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI200-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-92-3.4: PN-ZEI200-GASK-VIT-9213 / Kit KIT-ZEI200-VAC-13 (Zeiss-Prismo-Z200) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-92-3.5: PN-ZEI200-CAM-GRIP-9202 / Kit KIT-ZEI200-ATC-2 (Zeiss-Prismo-Z200) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI200-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-92-3.6: PN-ZEI200-LIN-COIL-9205 / Kit KIT-ZEI200-XAX-5 (Zeiss-Prismo-Z200) | PM Interval: 250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-92-3.7: PN-ZEI200-TRQ-STAT-9208 / Kit KIT-ZEI200-BAX-8 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI200-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-92-3.8: PN-ZEI200-IGBT-MOD-9211 / Kit KIT-ZEI200-PLC-11 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-92-3.9: PN-ZEI200-HEX-CORE-9214 / Kit KIT-ZEI200-THM-14 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI200-THM-14 within tolerance ±0.0060 mm TIR.

BOM-92-3.10: PN-ZEI200-VLV-PROP-9203 / Kit KIT-ZEI200-HYD-3 (Zeiss-Prismo-Z200) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-92-3.11: PN-ZEI200-BSW-NUT-9206 / Kit KIT-ZEI200-YAX-6 (Zeiss-Prismo-Z200) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI200-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-92-3.12: PN-ZEI200-BRK-DISC-9209 / Kit KIT-ZEI200-CAX-9 (Zeiss-Prismo-Z200) | PM Interval: 250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-92-3.13: PN-ZEI200-INJ-MET-9212 / Kit KIT-ZEI200-LUB-12 (Zeiss-Prismo-Z200) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI200-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-92-3.14: PN-ZEI200-BRG-CER-9201 / Kit KIT-ZEI200-SPN-1 (Zeiss-Prismo-Z200) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-92-3.15: PN-ZEI200-PMP-PLG-9204 / Kit KIT-ZEI200-CLN-4 (Zeiss-Prismo-Z200) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI200-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z250 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z250. Verify part numbers before closing Work Orders.

BOM-93-1.1: PN-ZEI250-CAM-GRIP-9302 / Kit KIT-ZEI250-ATC-2 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI250-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-93-1.2: PN-ZEI250-VLV-PROP-9303 / Kit KIT-ZEI250-HYD-3 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-93-1.3: PN-ZEI250-PMP-PLG-9304 / Kit KIT-ZEI250-CLN-4 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI250-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-93-1.4: PN-ZEI250-LIN-COIL-9305 / Kit KIT-ZEI250-XAX-5 (Zeiss-Prismo-Z250) | PM Interval: 1250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-93-1.5: PN-ZEI250-BSW-NUT-9306 / Kit KIT-ZEI250-YAX-6 (Zeiss-Prismo-Z250) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI250-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-93-1.6: PN-ZEI250-CYL-SEAL-9307 / Kit KIT-ZEI250-ZAX-7 (Zeiss-Prismo-Z250) | PM Interval: 250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-93-1.7: PN-ZEI250-TRQ-STAT-9308 / Kit KIT-ZEI250-BAX-8 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI250-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-93-1.8: PN-ZEI250-BRK-DISC-9309 / Kit KIT-ZEI250-CAX-9 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-93-1.9: PN-ZEI250-GLASS-RD-9310 / Kit KIT-ZEI250-OPT-10 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI250-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-93-1.10: PN-ZEI250-IGBT-MOD-9311 / Kit KIT-ZEI250-PLC-11 (Zeiss-Prismo-Z250) | PM Interval: 1250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-93-1.11: PN-ZEI250-INJ-MET-9312 / Kit KIT-ZEI250-LUB-12 (Zeiss-Prismo-Z250) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI250-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-93-1.12: PN-ZEI250-GASK-VIT-9313 / Kit KIT-ZEI250-VAC-13 (Zeiss-Prismo-Z250) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-93-1.13: PN-ZEI250-HEX-CORE-9314 / Kit KIT-ZEI250-THM-14 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI250-THM-14 within tolerance ±0.0020 mm TIR.

BOM-93-1.14: PN-ZEI250-BRG-CER-9301 / Kit KIT-ZEI250-SPN-1 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-93-1.15: PN-ZEI250-CAM-GRIP-9302 / Kit KIT-ZEI250-ATC-2 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI250-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z250 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z250. Verify part numbers before closing Work Orders.

BOM-93-2.1: PN-ZEI250-VLV-PROP-9303 / Kit KIT-ZEI250-HYD-3 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-93-2.2: PN-ZEI250-LIN-COIL-9305 / Kit KIT-ZEI250-XAX-5 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-93-2.3: PN-ZEI250-CYL-SEAL-9307 / Kit KIT-ZEI250-ZAX-7 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-93-2.4: PN-ZEI250-BRK-DISC-9309 / Kit KIT-ZEI250-CAX-9 (Zeiss-Prismo-Z250) | PM Interval: 1250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-93-2.5: PN-ZEI250-IGBT-MOD-9311 / Kit KIT-ZEI250-PLC-11 (Zeiss-Prismo-Z250) | PM Interval: 1500 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-93-2.6: PN-ZEI250-GASK-VIT-9313 / Kit KIT-ZEI250-VAC-13 (Zeiss-Prismo-Z250) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-93-2.7: PN-ZEI250-BRG-CER-9301 / Kit KIT-ZEI250-SPN-1 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-93-2.8: PN-ZEI250-VLV-PROP-9303 / Kit KIT-ZEI250-HYD-3 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-93-2.9: PN-ZEI250-LIN-COIL-9305 / Kit KIT-ZEI250-XAX-5 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-93-2.10: PN-ZEI250-CYL-SEAL-9307 / Kit KIT-ZEI250-ZAX-7 (Zeiss-Prismo-Z250) | PM Interval: 1250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-93-2.11: PN-ZEI250-BRK-DISC-9309 / Kit KIT-ZEI250-CAX-9 (Zeiss-Prismo-Z250) | PM Interval: 1500 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-93-2.12: PN-ZEI250-IGBT-MOD-9311 / Kit KIT-ZEI250-PLC-11 (Zeiss-Prismo-Z250) | PM Interval: 250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-93-2.13: PN-ZEI250-GASK-VIT-9313 / Kit KIT-ZEI250-VAC-13 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-93-2.14: PN-ZEI250-BRG-CER-9301 / Kit KIT-ZEI250-SPN-1 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-93-2.15: PN-ZEI250-VLV-PROP-9303 / Kit KIT-ZEI250-HYD-3 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI250-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z250 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z250. Verify part numbers before closing Work Orders.

BOM-93-3.1: PN-ZEI250-PMP-PLG-9304 / Kit KIT-ZEI250-CLN-4 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI250-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-93-3.2: PN-ZEI250-CYL-SEAL-9307 / Kit KIT-ZEI250-ZAX-7 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-93-3.3: PN-ZEI250-GLASS-RD-9310 / Kit KIT-ZEI250-OPT-10 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI250-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-93-3.4: PN-ZEI250-GASK-VIT-9313 / Kit KIT-ZEI250-VAC-13 (Zeiss-Prismo-Z250) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-93-3.5: PN-ZEI250-CAM-GRIP-9302 / Kit KIT-ZEI250-ATC-2 (Zeiss-Prismo-Z250) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI250-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-93-3.6: PN-ZEI250-LIN-COIL-9305 / Kit KIT-ZEI250-XAX-5 (Zeiss-Prismo-Z250) | PM Interval: 250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-93-3.7: PN-ZEI250-TRQ-STAT-9308 / Kit KIT-ZEI250-BAX-8 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI250-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-93-3.8: PN-ZEI250-IGBT-MOD-9311 / Kit KIT-ZEI250-PLC-11 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-93-3.9: PN-ZEI250-HEX-CORE-9314 / Kit KIT-ZEI250-THM-14 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI250-THM-14 within tolerance ±0.0020 mm TIR.

BOM-93-3.10: PN-ZEI250-VLV-PROP-9303 / Kit KIT-ZEI250-HYD-3 (Zeiss-Prismo-Z250) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-93-3.11: PN-ZEI250-BSW-NUT-9306 / Kit KIT-ZEI250-YAX-6 (Zeiss-Prismo-Z250) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI250-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-93-3.12: PN-ZEI250-BRK-DISC-9309 / Kit KIT-ZEI250-CAX-9 (Zeiss-Prismo-Z250) | PM Interval: 250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-93-3.13: PN-ZEI250-INJ-MET-9312 / Kit KIT-ZEI250-LUB-12 (Zeiss-Prismo-Z250) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI250-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-93-3.14: PN-ZEI250-BRG-CER-9301 / Kit KIT-ZEI250-SPN-1 (Zeiss-Prismo-Z250) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-93-3.15: PN-ZEI250-PMP-PLG-9304 / Kit KIT-ZEI250-CLN-4 (Zeiss-Prismo-Z250) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI250-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z300 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z300. Verify part numbers before closing Work Orders.

BOM-94-1.1: PN-ZEI300-CAM-GRIP-9402 / Kit KIT-ZEI300-ATC-2 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI300-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-94-1.2: PN-ZEI300-VLV-PROP-9403 / Kit KIT-ZEI300-HYD-3 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-94-1.3: PN-ZEI300-PMP-PLG-9404 / Kit KIT-ZEI300-CLN-4 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI300-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-94-1.4: PN-ZEI300-LIN-COIL-9405 / Kit KIT-ZEI300-XAX-5 (Zeiss-Prismo-Z300) | PM Interval: 1250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-94-1.5: PN-ZEI300-BSW-NUT-9406 / Kit KIT-ZEI300-YAX-6 (Zeiss-Prismo-Z300) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI300-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-94-1.6: PN-ZEI300-CYL-SEAL-9407 / Kit KIT-ZEI300-ZAX-7 (Zeiss-Prismo-Z300) | PM Interval: 250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-94-1.7: PN-ZEI300-TRQ-STAT-9408 / Kit KIT-ZEI300-BAX-8 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI300-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-94-1.8: PN-ZEI300-BRK-DISC-9409 / Kit KIT-ZEI300-CAX-9 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-94-1.9: PN-ZEI300-GLASS-RD-9410 / Kit KIT-ZEI300-OPT-10 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI300-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-94-1.10: PN-ZEI300-IGBT-MOD-9411 / Kit KIT-ZEI300-PLC-11 (Zeiss-Prismo-Z300) | PM Interval: 1250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-94-1.11: PN-ZEI300-INJ-MET-9412 / Kit KIT-ZEI300-LUB-12 (Zeiss-Prismo-Z300) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI300-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-94-1.12: PN-ZEI300-GASK-VIT-9413 / Kit KIT-ZEI300-VAC-13 (Zeiss-Prismo-Z300) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-94-1.13: PN-ZEI300-HEX-CORE-9414 / Kit KIT-ZEI300-THM-14 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI300-THM-14 within tolerance ±0.0040 mm TIR.

BOM-94-1.14: PN-ZEI300-BRG-CER-9401 / Kit KIT-ZEI300-SPN-1 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-94-1.15: PN-ZEI300-CAM-GRIP-9402 / Kit KIT-ZEI300-ATC-2 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI300-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z300 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z300. Verify part numbers before closing Work Orders.

BOM-94-2.1: PN-ZEI300-VLV-PROP-9403 / Kit KIT-ZEI300-HYD-3 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-94-2.2: PN-ZEI300-LIN-COIL-9405 / Kit KIT-ZEI300-XAX-5 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-94-2.3: PN-ZEI300-CYL-SEAL-9407 / Kit KIT-ZEI300-ZAX-7 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-94-2.4: PN-ZEI300-BRK-DISC-9409 / Kit KIT-ZEI300-CAX-9 (Zeiss-Prismo-Z300) | PM Interval: 1250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-94-2.5: PN-ZEI300-IGBT-MOD-9411 / Kit KIT-ZEI300-PLC-11 (Zeiss-Prismo-Z300) | PM Interval: 1500 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-94-2.6: PN-ZEI300-GASK-VIT-9413 / Kit KIT-ZEI300-VAC-13 (Zeiss-Prismo-Z300) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-94-2.7: PN-ZEI300-BRG-CER-9401 / Kit KIT-ZEI300-SPN-1 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-94-2.8: PN-ZEI300-VLV-PROP-9403 / Kit KIT-ZEI300-HYD-3 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-94-2.9: PN-ZEI300-LIN-COIL-9405 / Kit KIT-ZEI300-XAX-5 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-94-2.10: PN-ZEI300-CYL-SEAL-9407 / Kit KIT-ZEI300-ZAX-7 (Zeiss-Prismo-Z300) | PM Interval: 1250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-94-2.11: PN-ZEI300-BRK-DISC-9409 / Kit KIT-ZEI300-CAX-9 (Zeiss-Prismo-Z300) | PM Interval: 1500 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-94-2.12: PN-ZEI300-IGBT-MOD-9411 / Kit KIT-ZEI300-PLC-11 (Zeiss-Prismo-Z300) | PM Interval: 250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-94-2.13: PN-ZEI300-GASK-VIT-9413 / Kit KIT-ZEI300-VAC-13 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-94-2.14: PN-ZEI300-BRG-CER-9401 / Kit KIT-ZEI300-SPN-1 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-94-2.15: PN-ZEI300-VLV-PROP-9403 / Kit KIT-ZEI300-HYD-3 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI300-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z300 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z300. Verify part numbers before closing Work Orders.

BOM-94-3.1: PN-ZEI300-PMP-PLG-9404 / Kit KIT-ZEI300-CLN-4 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI300-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-94-3.2: PN-ZEI300-CYL-SEAL-9407 / Kit KIT-ZEI300-ZAX-7 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-94-3.3: PN-ZEI300-GLASS-RD-9410 / Kit KIT-ZEI300-OPT-10 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI300-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-94-3.4: PN-ZEI300-GASK-VIT-9413 / Kit KIT-ZEI300-VAC-13 (Zeiss-Prismo-Z300) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-94-3.5: PN-ZEI300-CAM-GRIP-9402 / Kit KIT-ZEI300-ATC-2 (Zeiss-Prismo-Z300) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI300-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-94-3.6: PN-ZEI300-LIN-COIL-9405 / Kit KIT-ZEI300-XAX-5 (Zeiss-Prismo-Z300) | PM Interval: 250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-94-3.7: PN-ZEI300-TRQ-STAT-9408 / Kit KIT-ZEI300-BAX-8 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI300-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-94-3.8: PN-ZEI300-IGBT-MOD-9411 / Kit KIT-ZEI300-PLC-11 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-94-3.9: PN-ZEI300-HEX-CORE-9414 / Kit KIT-ZEI300-THM-14 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI300-THM-14 within tolerance ±0.0040 mm TIR.

BOM-94-3.10: PN-ZEI300-VLV-PROP-9403 / Kit KIT-ZEI300-HYD-3 (Zeiss-Prismo-Z300) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-94-3.11: PN-ZEI300-BSW-NUT-9406 / Kit KIT-ZEI300-YAX-6 (Zeiss-Prismo-Z300) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI300-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-94-3.12: PN-ZEI300-BRK-DISC-9409 / Kit KIT-ZEI300-CAX-9 (Zeiss-Prismo-Z300) | PM Interval: 250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-94-3.13: PN-ZEI300-INJ-MET-9412 / Kit KIT-ZEI300-LUB-12 (Zeiss-Prismo-Z300) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI300-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-94-3.14: PN-ZEI300-BRG-CER-9401 / Kit KIT-ZEI300-SPN-1 (Zeiss-Prismo-Z300) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-94-3.15: PN-ZEI300-PMP-PLG-9404 / Kit KIT-ZEI300-CLN-4 (Zeiss-Prismo-Z300) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI300-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z350 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z350. Verify part numbers before closing Work Orders.

BOM-95-1.1: PN-ZEI350-CAM-GRIP-9502 / Kit KIT-ZEI350-ATC-2 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI350-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-95-1.2: PN-ZEI350-VLV-PROP-9503 / Kit KIT-ZEI350-HYD-3 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-95-1.3: PN-ZEI350-PMP-PLG-9504 / Kit KIT-ZEI350-CLN-4 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI350-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-95-1.4: PN-ZEI350-LIN-COIL-9505 / Kit KIT-ZEI350-XAX-5 (Zeiss-Prismo-Z350) | PM Interval: 1250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-95-1.5: PN-ZEI350-BSW-NUT-9506 / Kit KIT-ZEI350-YAX-6 (Zeiss-Prismo-Z350) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI350-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-95-1.6: PN-ZEI350-CYL-SEAL-9507 / Kit KIT-ZEI350-ZAX-7 (Zeiss-Prismo-Z350) | PM Interval: 250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-95-1.7: PN-ZEI350-TRQ-STAT-9508 / Kit KIT-ZEI350-BAX-8 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI350-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-95-1.8: PN-ZEI350-BRK-DISC-9509 / Kit KIT-ZEI350-CAX-9 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-95-1.9: PN-ZEI350-GLASS-RD-9510 / Kit KIT-ZEI350-OPT-10 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI350-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-95-1.10: PN-ZEI350-IGBT-MOD-9511 / Kit KIT-ZEI350-PLC-11 (Zeiss-Prismo-Z350) | PM Interval: 1250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-95-1.11: PN-ZEI350-INJ-MET-9512 / Kit KIT-ZEI350-LUB-12 (Zeiss-Prismo-Z350) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI350-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-95-1.12: PN-ZEI350-GASK-VIT-9513 / Kit KIT-ZEI350-VAC-13 (Zeiss-Prismo-Z350) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-95-1.13: PN-ZEI350-HEX-CORE-9514 / Kit KIT-ZEI350-THM-14 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI350-THM-14 within tolerance ±0.0060 mm TIR.

BOM-95-1.14: PN-ZEI350-BRG-CER-9501 / Kit KIT-ZEI350-SPN-1 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-95-1.15: PN-ZEI350-CAM-GRIP-9502 / Kit KIT-ZEI350-ATC-2 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI350-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z350 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z350. Verify part numbers before closing Work Orders.

BOM-95-2.1: PN-ZEI350-VLV-PROP-9503 / Kit KIT-ZEI350-HYD-3 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-95-2.2: PN-ZEI350-LIN-COIL-9505 / Kit KIT-ZEI350-XAX-5 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-95-2.3: PN-ZEI350-CYL-SEAL-9507 / Kit KIT-ZEI350-ZAX-7 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-95-2.4: PN-ZEI350-BRK-DISC-9509 / Kit KIT-ZEI350-CAX-9 (Zeiss-Prismo-Z350) | PM Interval: 1250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-95-2.5: PN-ZEI350-IGBT-MOD-9511 / Kit KIT-ZEI350-PLC-11 (Zeiss-Prismo-Z350) | PM Interval: 1500 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-95-2.6: PN-ZEI350-GASK-VIT-9513 / Kit KIT-ZEI350-VAC-13 (Zeiss-Prismo-Z350) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-95-2.7: PN-ZEI350-BRG-CER-9501 / Kit KIT-ZEI350-SPN-1 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-95-2.8: PN-ZEI350-VLV-PROP-9503 / Kit KIT-ZEI350-HYD-3 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-95-2.9: PN-ZEI350-LIN-COIL-9505 / Kit KIT-ZEI350-XAX-5 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-95-2.10: PN-ZEI350-CYL-SEAL-9507 / Kit KIT-ZEI350-ZAX-7 (Zeiss-Prismo-Z350) | PM Interval: 1250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-95-2.11: PN-ZEI350-BRK-DISC-9509 / Kit KIT-ZEI350-CAX-9 (Zeiss-Prismo-Z350) | PM Interval: 1500 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-95-2.12: PN-ZEI350-IGBT-MOD-9511 / Kit KIT-ZEI350-PLC-11 (Zeiss-Prismo-Z350) | PM Interval: 250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-95-2.13: PN-ZEI350-GASK-VIT-9513 / Kit KIT-ZEI350-VAC-13 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-95-2.14: PN-ZEI350-BRG-CER-9501 / Kit KIT-ZEI350-SPN-1 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-95-2.15: PN-ZEI350-VLV-PROP-9503 / Kit KIT-ZEI350-HYD-3 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI350-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z350 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z350. Verify part numbers before closing Work Orders.

BOM-95-3.1: PN-ZEI350-PMP-PLG-9504 / Kit KIT-ZEI350-CLN-4 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI350-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-95-3.2: PN-ZEI350-CYL-SEAL-9507 / Kit KIT-ZEI350-ZAX-7 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-95-3.3: PN-ZEI350-GLASS-RD-9510 / Kit KIT-ZEI350-OPT-10 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI350-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-95-3.4: PN-ZEI350-GASK-VIT-9513 / Kit KIT-ZEI350-VAC-13 (Zeiss-Prismo-Z350) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-95-3.5: PN-ZEI350-CAM-GRIP-9502 / Kit KIT-ZEI350-ATC-2 (Zeiss-Prismo-Z350) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI350-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-95-3.6: PN-ZEI350-LIN-COIL-9505 / Kit KIT-ZEI350-XAX-5 (Zeiss-Prismo-Z350) | PM Interval: 250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-95-3.7: PN-ZEI350-TRQ-STAT-9508 / Kit KIT-ZEI350-BAX-8 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI350-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-95-3.8: PN-ZEI350-IGBT-MOD-9511 / Kit KIT-ZEI350-PLC-11 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-95-3.9: PN-ZEI350-HEX-CORE-9514 / Kit KIT-ZEI350-THM-14 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI350-THM-14 within tolerance ±0.0060 mm TIR.

BOM-95-3.10: PN-ZEI350-VLV-PROP-9503 / Kit KIT-ZEI350-HYD-3 (Zeiss-Prismo-Z350) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-95-3.11: PN-ZEI350-BSW-NUT-9506 / Kit KIT-ZEI350-YAX-6 (Zeiss-Prismo-Z350) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI350-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-95-3.12: PN-ZEI350-BRK-DISC-9509 / Kit KIT-ZEI350-CAX-9 (Zeiss-Prismo-Z350) | PM Interval: 250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-95-3.13: PN-ZEI350-INJ-MET-9512 / Kit KIT-ZEI350-LUB-12 (Zeiss-Prismo-Z350) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI350-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-95-3.14: PN-ZEI350-BRG-CER-9501 / Kit KIT-ZEI350-SPN-1 (Zeiss-Prismo-Z350) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-95-3.15: PN-ZEI350-PMP-PLG-9504 / Kit KIT-ZEI350-CLN-4 (Zeiss-Prismo-Z350) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI350-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z400 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z400. Verify part numbers before closing Work Orders.

BOM-96-1.1: PN-ZEI400-CAM-GRIP-9602 / Kit KIT-ZEI400-ATC-2 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI400-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-96-1.2: PN-ZEI400-VLV-PROP-9603 / Kit KIT-ZEI400-HYD-3 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-96-1.3: PN-ZEI400-PMP-PLG-9604 / Kit KIT-ZEI400-CLN-4 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI400-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-96-1.4: PN-ZEI400-LIN-COIL-9605 / Kit KIT-ZEI400-XAX-5 (Zeiss-Prismo-Z400) | PM Interval: 1250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-96-1.5: PN-ZEI400-BSW-NUT-9606 / Kit KIT-ZEI400-YAX-6 (Zeiss-Prismo-Z400) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI400-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-96-1.6: PN-ZEI400-CYL-SEAL-9607 / Kit KIT-ZEI400-ZAX-7 (Zeiss-Prismo-Z400) | PM Interval: 250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-96-1.7: PN-ZEI400-TRQ-STAT-9608 / Kit KIT-ZEI400-BAX-8 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI400-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-96-1.8: PN-ZEI400-BRK-DISC-9609 / Kit KIT-ZEI400-CAX-9 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-96-1.9: PN-ZEI400-GLASS-RD-9610 / Kit KIT-ZEI400-OPT-10 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI400-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-96-1.10: PN-ZEI400-IGBT-MOD-9611 / Kit KIT-ZEI400-PLC-11 (Zeiss-Prismo-Z400) | PM Interval: 1250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-96-1.11: PN-ZEI400-INJ-MET-9612 / Kit KIT-ZEI400-LUB-12 (Zeiss-Prismo-Z400) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI400-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-96-1.12: PN-ZEI400-GASK-VIT-9613 / Kit KIT-ZEI400-VAC-13 (Zeiss-Prismo-Z400) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-96-1.13: PN-ZEI400-HEX-CORE-9614 / Kit KIT-ZEI400-THM-14 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI400-THM-14 within tolerance ±0.0020 mm TIR.

BOM-96-1.14: PN-ZEI400-BRG-CER-9601 / Kit KIT-ZEI400-SPN-1 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-96-1.15: PN-ZEI400-CAM-GRIP-9602 / Kit KIT-ZEI400-ATC-2 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI400-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z400 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z400. Verify part numbers before closing Work Orders.

BOM-96-2.1: PN-ZEI400-VLV-PROP-9603 / Kit KIT-ZEI400-HYD-3 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-96-2.2: PN-ZEI400-LIN-COIL-9605 / Kit KIT-ZEI400-XAX-5 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-96-2.3: PN-ZEI400-CYL-SEAL-9607 / Kit KIT-ZEI400-ZAX-7 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-96-2.4: PN-ZEI400-BRK-DISC-9609 / Kit KIT-ZEI400-CAX-9 (Zeiss-Prismo-Z400) | PM Interval: 1250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-96-2.5: PN-ZEI400-IGBT-MOD-9611 / Kit KIT-ZEI400-PLC-11 (Zeiss-Prismo-Z400) | PM Interval: 1500 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-96-2.6: PN-ZEI400-GASK-VIT-9613 / Kit KIT-ZEI400-VAC-13 (Zeiss-Prismo-Z400) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-96-2.7: PN-ZEI400-BRG-CER-9601 / Kit KIT-ZEI400-SPN-1 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-96-2.8: PN-ZEI400-VLV-PROP-9603 / Kit KIT-ZEI400-HYD-3 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-96-2.9: PN-ZEI400-LIN-COIL-9605 / Kit KIT-ZEI400-XAX-5 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-96-2.10: PN-ZEI400-CYL-SEAL-9607 / Kit KIT-ZEI400-ZAX-7 (Zeiss-Prismo-Z400) | PM Interval: 1250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-96-2.11: PN-ZEI400-BRK-DISC-9609 / Kit KIT-ZEI400-CAX-9 (Zeiss-Prismo-Z400) | PM Interval: 1500 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-96-2.12: PN-ZEI400-IGBT-MOD-9611 / Kit KIT-ZEI400-PLC-11 (Zeiss-Prismo-Z400) | PM Interval: 250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-96-2.13: PN-ZEI400-GASK-VIT-9613 / Kit KIT-ZEI400-VAC-13 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-96-2.14: PN-ZEI400-BRG-CER-9601 / Kit KIT-ZEI400-SPN-1 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-96-2.15: PN-ZEI400-VLV-PROP-9603 / Kit KIT-ZEI400-HYD-3 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI400-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z400 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z400. Verify part numbers before closing Work Orders.

BOM-96-3.1: PN-ZEI400-PMP-PLG-9604 / Kit KIT-ZEI400-CLN-4 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI400-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-96-3.2: PN-ZEI400-CYL-SEAL-9607 / Kit KIT-ZEI400-ZAX-7 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-96-3.3: PN-ZEI400-GLASS-RD-9610 / Kit KIT-ZEI400-OPT-10 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI400-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-96-3.4: PN-ZEI400-GASK-VIT-9613 / Kit KIT-ZEI400-VAC-13 (Zeiss-Prismo-Z400) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-96-3.5: PN-ZEI400-CAM-GRIP-9602 / Kit KIT-ZEI400-ATC-2 (Zeiss-Prismo-Z400) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI400-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-96-3.6: PN-ZEI400-LIN-COIL-9605 / Kit KIT-ZEI400-XAX-5 (Zeiss-Prismo-Z400) | PM Interval: 250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-96-3.7: PN-ZEI400-TRQ-STAT-9608 / Kit KIT-ZEI400-BAX-8 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI400-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-96-3.8: PN-ZEI400-IGBT-MOD-9611 / Kit KIT-ZEI400-PLC-11 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-96-3.9: PN-ZEI400-HEX-CORE-9614 / Kit KIT-ZEI400-THM-14 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI400-THM-14 within tolerance ±0.0020 mm TIR.

BOM-96-3.10: PN-ZEI400-VLV-PROP-9603 / Kit KIT-ZEI400-HYD-3 (Zeiss-Prismo-Z400) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-96-3.11: PN-ZEI400-BSW-NUT-9606 / Kit KIT-ZEI400-YAX-6 (Zeiss-Prismo-Z400) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI400-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-96-3.12: PN-ZEI400-BRK-DISC-9609 / Kit KIT-ZEI400-CAX-9 (Zeiss-Prismo-Z400) | PM Interval: 250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-96-3.13: PN-ZEI400-INJ-MET-9612 / Kit KIT-ZEI400-LUB-12 (Zeiss-Prismo-Z400) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI400-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-96-3.14: PN-ZEI400-BRG-CER-9601 / Kit KIT-ZEI400-SPN-1 (Zeiss-Prismo-Z400) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-96-3.15: PN-ZEI400-PMP-PLG-9604 / Kit KIT-ZEI400-CLN-4 (Zeiss-Prismo-Z400) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI400-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z450 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z450. Verify part numbers before closing Work Orders.

BOM-97-1.1: PN-ZEI450-CAM-GRIP-9702 / Kit KIT-ZEI450-ATC-2 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI450-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-97-1.2: PN-ZEI450-VLV-PROP-9703 / Kit KIT-ZEI450-HYD-3 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-97-1.3: PN-ZEI450-PMP-PLG-9704 / Kit KIT-ZEI450-CLN-4 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI450-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-97-1.4: PN-ZEI450-LIN-COIL-9705 / Kit KIT-ZEI450-XAX-5 (Zeiss-Prismo-Z450) | PM Interval: 1250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-97-1.5: PN-ZEI450-BSW-NUT-9706 / Kit KIT-ZEI450-YAX-6 (Zeiss-Prismo-Z450) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI450-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-97-1.6: PN-ZEI450-CYL-SEAL-9707 / Kit KIT-ZEI450-ZAX-7 (Zeiss-Prismo-Z450) | PM Interval: 250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-97-1.7: PN-ZEI450-TRQ-STAT-9708 / Kit KIT-ZEI450-BAX-8 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI450-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-97-1.8: PN-ZEI450-BRK-DISC-9709 / Kit KIT-ZEI450-CAX-9 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-97-1.9: PN-ZEI450-GLASS-RD-9710 / Kit KIT-ZEI450-OPT-10 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI450-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-97-1.10: PN-ZEI450-IGBT-MOD-9711 / Kit KIT-ZEI450-PLC-11 (Zeiss-Prismo-Z450) | PM Interval: 1250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-97-1.11: PN-ZEI450-INJ-MET-9712 / Kit KIT-ZEI450-LUB-12 (Zeiss-Prismo-Z450) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI450-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-97-1.12: PN-ZEI450-GASK-VIT-9713 / Kit KIT-ZEI450-VAC-13 (Zeiss-Prismo-Z450) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-97-1.13: PN-ZEI450-HEX-CORE-9714 / Kit KIT-ZEI450-THM-14 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI450-THM-14 within tolerance ±0.0040 mm TIR.

BOM-97-1.14: PN-ZEI450-BRG-CER-9701 / Kit KIT-ZEI450-SPN-1 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-97-1.15: PN-ZEI450-CAM-GRIP-9702 / Kit KIT-ZEI450-ATC-2 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI450-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z450 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z450. Verify part numbers before closing Work Orders.

BOM-97-2.1: PN-ZEI450-VLV-PROP-9703 / Kit KIT-ZEI450-HYD-3 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-97-2.2: PN-ZEI450-LIN-COIL-9705 / Kit KIT-ZEI450-XAX-5 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-97-2.3: PN-ZEI450-CYL-SEAL-9707 / Kit KIT-ZEI450-ZAX-7 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-97-2.4: PN-ZEI450-BRK-DISC-9709 / Kit KIT-ZEI450-CAX-9 (Zeiss-Prismo-Z450) | PM Interval: 1250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-97-2.5: PN-ZEI450-IGBT-MOD-9711 / Kit KIT-ZEI450-PLC-11 (Zeiss-Prismo-Z450) | PM Interval: 1500 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-97-2.6: PN-ZEI450-GASK-VIT-9713 / Kit KIT-ZEI450-VAC-13 (Zeiss-Prismo-Z450) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-97-2.7: PN-ZEI450-BRG-CER-9701 / Kit KIT-ZEI450-SPN-1 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-97-2.8: PN-ZEI450-VLV-PROP-9703 / Kit KIT-ZEI450-HYD-3 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-97-2.9: PN-ZEI450-LIN-COIL-9705 / Kit KIT-ZEI450-XAX-5 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-97-2.10: PN-ZEI450-CYL-SEAL-9707 / Kit KIT-ZEI450-ZAX-7 (Zeiss-Prismo-Z450) | PM Interval: 1250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-97-2.11: PN-ZEI450-BRK-DISC-9709 / Kit KIT-ZEI450-CAX-9 (Zeiss-Prismo-Z450) | PM Interval: 1500 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-97-2.12: PN-ZEI450-IGBT-MOD-9711 / Kit KIT-ZEI450-PLC-11 (Zeiss-Prismo-Z450) | PM Interval: 250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-97-2.13: PN-ZEI450-GASK-VIT-9713 / Kit KIT-ZEI450-VAC-13 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-97-2.14: PN-ZEI450-BRG-CER-9701 / Kit KIT-ZEI450-SPN-1 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-97-2.15: PN-ZEI450-VLV-PROP-9703 / Kit KIT-ZEI450-HYD-3 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI450-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z450 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z450. Verify part numbers before closing Work Orders.

BOM-97-3.1: PN-ZEI450-PMP-PLG-9704 / Kit KIT-ZEI450-CLN-4 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI450-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-97-3.2: PN-ZEI450-CYL-SEAL-9707 / Kit KIT-ZEI450-ZAX-7 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-97-3.3: PN-ZEI450-GLASS-RD-9710 / Kit KIT-ZEI450-OPT-10 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI450-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-97-3.4: PN-ZEI450-GASK-VIT-9713 / Kit KIT-ZEI450-VAC-13 (Zeiss-Prismo-Z450) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-97-3.5: PN-ZEI450-CAM-GRIP-9702 / Kit KIT-ZEI450-ATC-2 (Zeiss-Prismo-Z450) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI450-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-97-3.6: PN-ZEI450-LIN-COIL-9705 / Kit KIT-ZEI450-XAX-5 (Zeiss-Prismo-Z450) | PM Interval: 250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-97-3.7: PN-ZEI450-TRQ-STAT-9708 / Kit KIT-ZEI450-BAX-8 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI450-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-97-3.8: PN-ZEI450-IGBT-MOD-9711 / Kit KIT-ZEI450-PLC-11 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-97-3.9: PN-ZEI450-HEX-CORE-9714 / Kit KIT-ZEI450-THM-14 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI450-THM-14 within tolerance ±0.0040 mm TIR.

BOM-97-3.10: PN-ZEI450-VLV-PROP-9703 / Kit KIT-ZEI450-HYD-3 (Zeiss-Prismo-Z450) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-97-3.11: PN-ZEI450-BSW-NUT-9706 / Kit KIT-ZEI450-YAX-6 (Zeiss-Prismo-Z450) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI450-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-97-3.12: PN-ZEI450-BRK-DISC-9709 / Kit KIT-ZEI450-CAX-9 (Zeiss-Prismo-Z450) | PM Interval: 250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-97-3.13: PN-ZEI450-INJ-MET-9712 / Kit KIT-ZEI450-LUB-12 (Zeiss-Prismo-Z450) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI450-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-97-3.14: PN-ZEI450-BRG-CER-9701 / Kit KIT-ZEI450-SPN-1 (Zeiss-Prismo-Z450) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-97-3.15: PN-ZEI450-PMP-PLG-9704 / Kit KIT-ZEI450-CLN-4 (Zeiss-Prismo-Z450) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI450-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z500 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z500. Verify part numbers before closing Work Orders.

BOM-98-1.1: PN-ZEI500-CAM-GRIP-9802 / Kit KIT-ZEI500-ATC-2 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI500-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-98-1.2: PN-ZEI500-VLV-PROP-9803 / Kit KIT-ZEI500-HYD-3 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-98-1.3: PN-ZEI500-PMP-PLG-9804 / Kit KIT-ZEI500-CLN-4 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI500-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-98-1.4: PN-ZEI500-LIN-COIL-9805 / Kit KIT-ZEI500-XAX-5 (Zeiss-Prismo-Z500) | PM Interval: 1250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-98-1.5: PN-ZEI500-BSW-NUT-9806 / Kit KIT-ZEI500-YAX-6 (Zeiss-Prismo-Z500) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI500-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-98-1.6: PN-ZEI500-CYL-SEAL-9807 / Kit KIT-ZEI500-ZAX-7 (Zeiss-Prismo-Z500) | PM Interval: 250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-98-1.7: PN-ZEI500-TRQ-STAT-9808 / Kit KIT-ZEI500-BAX-8 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI500-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-98-1.8: PN-ZEI500-BRK-DISC-9809 / Kit KIT-ZEI500-CAX-9 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-98-1.9: PN-ZEI500-GLASS-RD-9810 / Kit KIT-ZEI500-OPT-10 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI500-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-98-1.10: PN-ZEI500-IGBT-MOD-9811 / Kit KIT-ZEI500-PLC-11 (Zeiss-Prismo-Z500) | PM Interval: 1250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-98-1.11: PN-ZEI500-INJ-MET-9812 / Kit KIT-ZEI500-LUB-12 (Zeiss-Prismo-Z500) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI500-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-98-1.12: PN-ZEI500-GASK-VIT-9813 / Kit KIT-ZEI500-VAC-13 (Zeiss-Prismo-Z500) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-98-1.13: PN-ZEI500-HEX-CORE-9814 / Kit KIT-ZEI500-THM-14 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI500-THM-14 within tolerance ±0.0060 mm TIR.

BOM-98-1.14: PN-ZEI500-BRG-CER-9801 / Kit KIT-ZEI500-SPN-1 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-98-1.15: PN-ZEI500-CAM-GRIP-9802 / Kit KIT-ZEI500-ATC-2 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI500-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z500 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z500. Verify part numbers before closing Work Orders.

BOM-98-2.1: PN-ZEI500-VLV-PROP-9803 / Kit KIT-ZEI500-HYD-3 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-98-2.2: PN-ZEI500-LIN-COIL-9805 / Kit KIT-ZEI500-XAX-5 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-98-2.3: PN-ZEI500-CYL-SEAL-9807 / Kit KIT-ZEI500-ZAX-7 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-98-2.4: PN-ZEI500-BRK-DISC-9809 / Kit KIT-ZEI500-CAX-9 (Zeiss-Prismo-Z500) | PM Interval: 1250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-98-2.5: PN-ZEI500-IGBT-MOD-9811 / Kit KIT-ZEI500-PLC-11 (Zeiss-Prismo-Z500) | PM Interval: 1500 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-98-2.6: PN-ZEI500-GASK-VIT-9813 / Kit KIT-ZEI500-VAC-13 (Zeiss-Prismo-Z500) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-98-2.7: PN-ZEI500-BRG-CER-9801 / Kit KIT-ZEI500-SPN-1 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-98-2.8: PN-ZEI500-VLV-PROP-9803 / Kit KIT-ZEI500-HYD-3 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-98-2.9: PN-ZEI500-LIN-COIL-9805 / Kit KIT-ZEI500-XAX-5 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-98-2.10: PN-ZEI500-CYL-SEAL-9807 / Kit KIT-ZEI500-ZAX-7 (Zeiss-Prismo-Z500) | PM Interval: 1250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-98-2.11: PN-ZEI500-BRK-DISC-9809 / Kit KIT-ZEI500-CAX-9 (Zeiss-Prismo-Z500) | PM Interval: 1500 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-98-2.12: PN-ZEI500-IGBT-MOD-9811 / Kit KIT-ZEI500-PLC-11 (Zeiss-Prismo-Z500) | PM Interval: 250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-98-2.13: PN-ZEI500-GASK-VIT-9813 / Kit KIT-ZEI500-VAC-13 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-98-2.14: PN-ZEI500-BRG-CER-9801 / Kit KIT-ZEI500-SPN-1 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-98-2.15: PN-ZEI500-VLV-PROP-9803 / Kit KIT-ZEI500-HYD-3 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI500-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z500 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z500. Verify part numbers before closing Work Orders.

BOM-98-3.1: PN-ZEI500-PMP-PLG-9804 / Kit KIT-ZEI500-CLN-4 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI500-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-98-3.2: PN-ZEI500-CYL-SEAL-9807 / Kit KIT-ZEI500-ZAX-7 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-98-3.3: PN-ZEI500-GLASS-RD-9810 / Kit KIT-ZEI500-OPT-10 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI500-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-98-3.4: PN-ZEI500-GASK-VIT-9813 / Kit KIT-ZEI500-VAC-13 (Zeiss-Prismo-Z500) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-98-3.5: PN-ZEI500-CAM-GRIP-9802 / Kit KIT-ZEI500-ATC-2 (Zeiss-Prismo-Z500) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI500-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-98-3.6: PN-ZEI500-LIN-COIL-9805 / Kit KIT-ZEI500-XAX-5 (Zeiss-Prismo-Z500) | PM Interval: 250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-98-3.7: PN-ZEI500-TRQ-STAT-9808 / Kit KIT-ZEI500-BAX-8 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI500-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-98-3.8: PN-ZEI500-IGBT-MOD-9811 / Kit KIT-ZEI500-PLC-11 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-98-3.9: PN-ZEI500-HEX-CORE-9814 / Kit KIT-ZEI500-THM-14 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI500-THM-14 within tolerance ±0.0060 mm TIR.

BOM-98-3.10: PN-ZEI500-VLV-PROP-9803 / Kit KIT-ZEI500-HYD-3 (Zeiss-Prismo-Z500) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-98-3.11: PN-ZEI500-BSW-NUT-9806 / Kit KIT-ZEI500-YAX-6 (Zeiss-Prismo-Z500) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI500-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-98-3.12: PN-ZEI500-BRK-DISC-9809 / Kit KIT-ZEI500-CAX-9 (Zeiss-Prismo-Z500) | PM Interval: 250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-98-3.13: PN-ZEI500-INJ-MET-9812 / Kit KIT-ZEI500-LUB-12 (Zeiss-Prismo-Z500) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI500-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-98-3.14: PN-ZEI500-BRG-CER-9801 / Kit KIT-ZEI500-SPN-1 (Zeiss-Prismo-Z500) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-98-3.15: PN-ZEI500-PMP-PLG-9804 / Kit KIT-ZEI500-CLN-4 (Zeiss-Prismo-Z500) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI500-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z550 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z550. Verify part numbers before closing Work Orders.

BOM-99-1.1: PN-ZEI550-CAM-GRIP-9902 / Kit KIT-ZEI550-ATC-2 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI550-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-99-1.2: PN-ZEI550-VLV-PROP-9903 / Kit KIT-ZEI550-HYD-3 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-99-1.3: PN-ZEI550-PMP-PLG-9904 / Kit KIT-ZEI550-CLN-4 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI550-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-99-1.4: PN-ZEI550-LIN-COIL-9905 / Kit KIT-ZEI550-XAX-5 (Zeiss-Prismo-Z550) | PM Interval: 1250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-99-1.5: PN-ZEI550-BSW-NUT-9906 / Kit KIT-ZEI550-YAX-6 (Zeiss-Prismo-Z550) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI550-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-99-1.6: PN-ZEI550-CYL-SEAL-9907 / Kit KIT-ZEI550-ZAX-7 (Zeiss-Prismo-Z550) | PM Interval: 250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-99-1.7: PN-ZEI550-TRQ-STAT-9908 / Kit KIT-ZEI550-BAX-8 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI550-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-99-1.8: PN-ZEI550-BRK-DISC-9909 / Kit KIT-ZEI550-CAX-9 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-99-1.9: PN-ZEI550-GLASS-RD-9910 / Kit KIT-ZEI550-OPT-10 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI550-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-99-1.10: PN-ZEI550-IGBT-MOD-9911 / Kit KIT-ZEI550-PLC-11 (Zeiss-Prismo-Z550) | PM Interval: 1250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-99-1.11: PN-ZEI550-INJ-MET-9912 / Kit KIT-ZEI550-LUB-12 (Zeiss-Prismo-Z550) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI550-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-99-1.12: PN-ZEI550-GASK-VIT-9913 / Kit KIT-ZEI550-VAC-13 (Zeiss-Prismo-Z550) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-99-1.13: PN-ZEI550-HEX-CORE-9914 / Kit KIT-ZEI550-THM-14 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI550-THM-14 within tolerance ±0.0020 mm TIR.

BOM-99-1.14: PN-ZEI550-BRG-CER-9901 / Kit KIT-ZEI550-SPN-1 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-99-1.15: PN-ZEI550-CAM-GRIP-9902 / Kit KIT-ZEI550-ATC-2 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI550-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z550 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z550. Verify part numbers before closing Work Orders.

BOM-99-2.1: PN-ZEI550-VLV-PROP-9903 / Kit KIT-ZEI550-HYD-3 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-99-2.2: PN-ZEI550-LIN-COIL-9905 / Kit KIT-ZEI550-XAX-5 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-99-2.3: PN-ZEI550-CYL-SEAL-9907 / Kit KIT-ZEI550-ZAX-7 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-99-2.4: PN-ZEI550-BRK-DISC-9909 / Kit KIT-ZEI550-CAX-9 (Zeiss-Prismo-Z550) | PM Interval: 1250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-99-2.5: PN-ZEI550-IGBT-MOD-9911 / Kit KIT-ZEI550-PLC-11 (Zeiss-Prismo-Z550) | PM Interval: 1500 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-99-2.6: PN-ZEI550-GASK-VIT-9913 / Kit KIT-ZEI550-VAC-13 (Zeiss-Prismo-Z550) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-99-2.7: PN-ZEI550-BRG-CER-9901 / Kit KIT-ZEI550-SPN-1 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-99-2.8: PN-ZEI550-VLV-PROP-9903 / Kit KIT-ZEI550-HYD-3 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-99-2.9: PN-ZEI550-LIN-COIL-9905 / Kit KIT-ZEI550-XAX-5 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-99-2.10: PN-ZEI550-CYL-SEAL-9907 / Kit KIT-ZEI550-ZAX-7 (Zeiss-Prismo-Z550) | PM Interval: 1250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-99-2.11: PN-ZEI550-BRK-DISC-9909 / Kit KIT-ZEI550-CAX-9 (Zeiss-Prismo-Z550) | PM Interval: 1500 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-99-2.12: PN-ZEI550-IGBT-MOD-9911 / Kit KIT-ZEI550-PLC-11 (Zeiss-Prismo-Z550) | PM Interval: 250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-99-2.13: PN-ZEI550-GASK-VIT-9913 / Kit KIT-ZEI550-VAC-13 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-99-2.14: PN-ZEI550-BRG-CER-9901 / Kit KIT-ZEI550-SPN-1 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-99-2.15: PN-ZEI550-VLV-PROP-9903 / Kit KIT-ZEI550-HYD-3 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI550-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z550 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z550. Verify part numbers before closing Work Orders.

BOM-99-3.1: PN-ZEI550-PMP-PLG-9904 / Kit KIT-ZEI550-CLN-4 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI550-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-99-3.2: PN-ZEI550-CYL-SEAL-9907 / Kit KIT-ZEI550-ZAX-7 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-99-3.3: PN-ZEI550-GLASS-RD-9910 / Kit KIT-ZEI550-OPT-10 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI550-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-99-3.4: PN-ZEI550-GASK-VIT-9913 / Kit KIT-ZEI550-VAC-13 (Zeiss-Prismo-Z550) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-99-3.5: PN-ZEI550-CAM-GRIP-9902 / Kit KIT-ZEI550-ATC-2 (Zeiss-Prismo-Z550) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI550-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-99-3.6: PN-ZEI550-LIN-COIL-9905 / Kit KIT-ZEI550-XAX-5 (Zeiss-Prismo-Z550) | PM Interval: 250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-99-3.7: PN-ZEI550-TRQ-STAT-9908 / Kit KIT-ZEI550-BAX-8 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI550-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-99-3.8: PN-ZEI550-IGBT-MOD-9911 / Kit KIT-ZEI550-PLC-11 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-99-3.9: PN-ZEI550-HEX-CORE-9914 / Kit KIT-ZEI550-THM-14 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI550-THM-14 within tolerance ±0.0020 mm TIR.

BOM-99-3.10: PN-ZEI550-VLV-PROP-9903 / Kit KIT-ZEI550-HYD-3 (Zeiss-Prismo-Z550) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-99-3.11: PN-ZEI550-BSW-NUT-9906 / Kit KIT-ZEI550-YAX-6 (Zeiss-Prismo-Z550) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI550-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-99-3.12: PN-ZEI550-BRK-DISC-9909 / Kit KIT-ZEI550-CAX-9 (Zeiss-Prismo-Z550) | PM Interval: 250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-99-3.13: PN-ZEI550-INJ-MET-9912 / Kit KIT-ZEI550-LUB-12 (Zeiss-Prismo-Z550) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI550-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-99-3.14: PN-ZEI550-BRG-CER-9901 / Kit KIT-ZEI550-SPN-1 (Zeiss-Prismo-Z550) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-99-3.15: PN-ZEI550-PMP-PLG-9904 / Kit KIT-ZEI550-CLN-4 (Zeiss-Prismo-Z550) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI550-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z600 (GROUP 1)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z600. Verify part numbers before closing Work Orders.

BOM-100-1.1: PN-ZEI600-CAM-GRIP-10002 / Kit KIT-ZEI600-ATC-2 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI600-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-100-1.2: PN-ZEI600-VLV-PROP-10003 / Kit KIT-ZEI600-HYD-3 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-100-1.3: PN-ZEI600-PMP-PLG-10004 / Kit KIT-ZEI600-CLN-4 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI600-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-100-1.4: PN-ZEI600-LIN-COIL-10005 / Kit KIT-ZEI600-XAX-5 (Zeiss-Prismo-Z600) | PM Interval: 1250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-100-1.5: PN-ZEI600-BSW-NUT-10006 / Kit KIT-ZEI600-YAX-6 (Zeiss-Prismo-Z600) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI600-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-100-1.6: PN-ZEI600-CYL-SEAL-10007 / Kit KIT-ZEI600-ZAX-7 (Zeiss-Prismo-Z600) | PM Interval: 250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-100-1.7: PN-ZEI600-TRQ-STAT-10008 / Kit KIT-ZEI600-BAX-8 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI600-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-100-1.8: PN-ZEI600-BRK-DISC-10009 / Kit KIT-ZEI600-CAX-9 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-100-1.9: PN-ZEI600-GLASS-RD-10010 / Kit KIT-ZEI600-OPT-10 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI600-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-100-1.10: PN-ZEI600-IGBT-MOD-10011 / Kit KIT-ZEI600-PLC-11 (Zeiss-Prismo-Z600) | PM Interval: 1250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-100-1.11: PN-ZEI600-INJ-MET-10012 / Kit KIT-ZEI600-LUB-12 (Zeiss-Prismo-Z600) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI600-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-100-1.12: PN-ZEI600-GASK-VIT-10013 / Kit KIT-ZEI600-VAC-13 (Zeiss-Prismo-Z600) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-100-1.13: PN-ZEI600-HEX-CORE-10014 / Kit KIT-ZEI600-THM-14 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI600-THM-14 within tolerance ±0.0040 mm TIR.

BOM-100-1.14: PN-ZEI600-BRG-CER-10001 / Kit KIT-ZEI600-SPN-1 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-100-1.15: PN-ZEI600-CAM-GRIP-10002 / Kit KIT-ZEI600-ATC-2 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI600-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z600 (GROUP 2)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z600. Verify part numbers before closing Work Orders.

BOM-100-2.1: PN-ZEI600-VLV-PROP-10003 / Kit KIT-ZEI600-HYD-3 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-100-2.2: PN-ZEI600-LIN-COIL-10005 / Kit KIT-ZEI600-XAX-5 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-100-2.3: PN-ZEI600-CYL-SEAL-10007 / Kit KIT-ZEI600-ZAX-7 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-100-2.4: PN-ZEI600-BRK-DISC-10009 / Kit KIT-ZEI600-CAX-9 (Zeiss-Prismo-Z600) | PM Interval: 1250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-100-2.5: PN-ZEI600-IGBT-MOD-10011 / Kit KIT-ZEI600-PLC-11 (Zeiss-Prismo-Z600) | PM Interval: 1500 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-100-2.6: PN-ZEI600-GASK-VIT-10013 / Kit KIT-ZEI600-VAC-13 (Zeiss-Prismo-Z600) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-100-2.7: PN-ZEI600-BRG-CER-10001 / Kit KIT-ZEI600-SPN-1 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-100-2.8: PN-ZEI600-VLV-PROP-10003 / Kit KIT-ZEI600-HYD-3 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-100-2.9: PN-ZEI600-LIN-COIL-10005 / Kit KIT-ZEI600-XAX-5 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-100-2.10: PN-ZEI600-CYL-SEAL-10007 / Kit KIT-ZEI600-ZAX-7 (Zeiss-Prismo-Z600) | PM Interval: 1250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-100-2.11: PN-ZEI600-BRK-DISC-10009 / Kit KIT-ZEI600-CAX-9 (Zeiss-Prismo-Z600) | PM Interval: 1500 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-100-2.12: PN-ZEI600-IGBT-MOD-10011 / Kit KIT-ZEI600-PLC-11 (Zeiss-Prismo-Z600) | PM Interval: 250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-100-2.13: PN-ZEI600-GASK-VIT-10013 / Kit KIT-ZEI600-VAC-13 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-100-2.14: PN-ZEI600-BRG-CER-10001 / Kit KIT-ZEI600-SPN-1 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-100-2.15: PN-ZEI600-VLV-PROP-10003 / Kit KIT-ZEI600-HYD-3 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI600-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Zeiss-Prismo-Z600 (GROUP 3)

Certified Aerospace Bill of Materials for Zeiss-Prismo-Z600. Verify part numbers before closing Work Orders.

BOM-100-3.1: PN-ZEI600-PMP-PLG-10004 / Kit KIT-ZEI600-CLN-4 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI600-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-100-3.2: PN-ZEI600-CYL-SEAL-10007 / Kit KIT-ZEI600-ZAX-7 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-ZEI600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-100-3.3: PN-ZEI600-GLASS-RD-10010 / Kit KIT-ZEI600-OPT-10 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-ZEI600-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-100-3.4: PN-ZEI600-GASK-VIT-10013 / Kit KIT-ZEI600-VAC-13 (Zeiss-Prismo-Z600) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-ZEI600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-100-3.5: PN-ZEI600-CAM-GRIP-10002 / Kit KIT-ZEI600-ATC-2 (Zeiss-Prismo-Z600) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-ZEI600-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-100-3.6: PN-ZEI600-LIN-COIL-10005 / Kit KIT-ZEI600-XAX-5 (Zeiss-Prismo-Z600) | PM Interval: 250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-ZEI600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-100-3.7: PN-ZEI600-TRQ-STAT-10008 / Kit KIT-ZEI600-BAX-8 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-ZEI600-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-100-3.8: PN-ZEI600-IGBT-MOD-10011 / Kit KIT-ZEI600-PLC-11 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-ZEI600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-100-3.9: PN-ZEI600-HEX-CORE-10014 / Kit KIT-ZEI600-THM-14 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-ZEI600-THM-14 within tolerance ±0.0040 mm TIR.

BOM-100-3.10: PN-ZEI600-VLV-PROP-10003 / Kit KIT-ZEI600-HYD-3 (Zeiss-Prismo-Z600) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-ZEI600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-100-3.11: PN-ZEI600-BSW-NUT-10006 / Kit KIT-ZEI600-YAX-6 (Zeiss-Prismo-Z600) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-ZEI600-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-100-3.12: PN-ZEI600-BRK-DISC-10009 / Kit KIT-ZEI600-CAX-9 (Zeiss-Prismo-Z600) | PM Interval: 250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-ZEI600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-100-3.13: PN-ZEI600-INJ-MET-10012 / Kit KIT-ZEI600-LUB-12 (Zeiss-Prismo-Z600) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-ZEI600-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-100-3.14: PN-ZEI600-BRG-CER-10001 / Kit KIT-ZEI600-SPN-1 (Zeiss-Prismo-Z600) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-ZEI600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-100-3.15: PN-ZEI600-PMP-PLG-10004 / Kit KIT-ZEI600-CLN-4 (Zeiss-Prismo-Z600) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-ZEI600-CLN-4 within tolerance ±0.0050 mm TIR.